UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Engineering Mathematics I — Unit 1 Master

Unit master: every session of Unit 1 · Partial Differentiation in one document, with one module per session

Merged from the session packs Class 001, Class 002, Classes 003–004, Class 005, Class 006, Class 007, Class 008, Class 009, Class 010, Class 011. Each module is a complete session pack; item numbers are renumbered across the unit, so cross-references point within this document. Generated 15 October 2026.

· How to use this pack

One module per class session, in teaching order. Use the module banners and the contents list to jump to a session; the appendices collect every symbol, source and gap of the unit. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·15	Deck 001, slide 15 (rendered .pptx)
T001a 00:14:20	Transcript “Introduction to Partial Differentiation”, segment start
T001b 00:11:41	Transcript “Continuity and differentiability of a function of two independent variables”
CW U1 P7	Unit 1 class-work problem 7
ESA Dec-23 1a	Past paper: Dec 2023, question 1.a
S002·7	Deck 002, slide 7 (rendered .pptx)
T002 00:06:10	AV-summary transcript 002, segment start time
V002 00:07:45	Frame grabbed from the class 002 video
CW U1 P2 · ESA Dec-23 1a	Unit 1 class-work problem 2 · past-paper question
S003·7 · S004·9	Deck 003 / 004, slide number (rendered .pptx)
T003a 00:06:27	Transcript 003 part I: “Total derivative, chain rule and implicit functions”
T003b 00:01:10	Transcript 003 part II: “Problems on total derivatives, chain rule”
T004a 00:13:09	Transcript 004 part I: “Implicit functions and problems”
T004b 00:02:31	Transcript 004 part II: “Problems on composite and implicit functions”
CW U1 P14 · ESA Jan-24 s3 1b	Unit 1 class-work problem 14 · past-paper question
S005·6	Deck 005, slide 6 (rendered .pptx)
T005 00:08:57	AV-summary transcript 005, segment start
CW U1 P20	Unit 1 class-work problem 20
ESA Jan-24 s3 1c	Past paper: Jan 2024, set 3, question 1.c
S006·12	Deck 006, slide 12 (rendered .pptx)
T006a 00:05:18	Transcript 006 part I (slides 3–4, 9.1 min)

T006b 00:08:20	Transcript 006 part II (slides 5–17, 12.9 min)
S007·9	Deck 007, slide 9 (rendered .pptx)
T007 00:18:07	AV-summary transcript 007, segment start
CW U1 P24	Unit 1 class-work problem 24
ESA Dec-23 1b	Past paper: Dec 2023, question 1.b
S008·12	Deck 008, slide 12 (rendered .pptx)
T008 00:07:38	AV-summary transcript 008, segment start
S009·7	Deck 009, slide 7 (rendered .pptx)
T009 00:07:04	AV-summary transcript 009 (“Extremas of functions”), segment start
CW U1 P31	Unit 1 class-work problem 31
ESA Dec-23 1c	Past paper: Dec 2023, question 1.c
S010·8	Deck 010, slide 8 (rendered .pptx)
T010a 00:07:38	AV-summary transcript “Extremas of functions continued” (older lecture), segment start
T010b 00:03:12	AV-summary transcript “Problems on Maxima and Minima for a function of two variables” (newer lecture)
CW U1 P29	Unit 1 class-work problem 29
ESA Jan-24 s2 1a	Past paper: Jan 2024 set 2, question 1.a
S011·5	Deck 011, slide 5 (rendered .pptx)
T011a 00:01:08	Transcript “Lagrange's method of undetermined coefficient” (theory lecture), segment start
T011b 00:05:03	Transcript “Problems on Lagrange's method of undetermined multipliers”
CW U1 P33	Unit 1 class-work problem 33
Extra Q57	Extra question list “lagrange-ques”, question 57
ESA Jul-23 1c	Past paper: Jul 2023, question 1.c

Contents

1. 1 · Functions of two or more variables
2. 2 · Limit of a function of two variables
3. 3 · Continuity of a function of two variables
4. 4 · Partial derivatives: history, definition and notation
5. 5 · Second-order and mixed partial derivatives
6. 6 · Higher-order partial derivatives
7. 7 · The one-variable picture: derivative as slope
8. 8 · A function of two variables is a surface
9. 9 · First-order partials as slopes of tangents
10. 10 · Slope in any other direction: the gradient
11. 11 · Second-order partials: concavity and mixed partials
12. 12 · Problems on partial derivatives

13. 13 · Composite functions and the one-variable chain rule
14. 14 · The total derivative rule
15. 15 · The total differential: approximation and sensitivity
16. 16 · Rates of change problems
17. 17 · The chain rule for composite functions
18. 18 · Implicit functions of two variables
19. 19 · Implicit functions of three variables
20. 20 · Homogeneous functions
21. 21 · Euler's theorem
22. 22 · Second and higher order: the corollary
23. 23 · Extension: when f(u) is homogeneous
24. 24 · Applications of homogeneous functions
25. 25 · Testing homogeneity: the scaling test
26. 26 · Verifying Euler's theorem directly
27. 27 · When u itself is homogeneous: first and second order
28. 28 · When f(u) is homogeneous: the extension
29. 29 · Why Taylor's and Maclaurin's series?
30. 30 · Taylor's and Maclaurin's series in one variable
31. 31 · Taylor's expansion of a function of two variables
32. 32 · Maclaurin's expansion of a function of two variables
33. 33 · Taylor expansion about a point (a, b)
34. 34 · Maclaurin expansion about the origin
35. 35 · Approximate values from a Taylor polynomial
36. 36 · Maxima, minima and saddle points
37. 37 · Definitions: relative maximum, minimum and extreme value
38. 38 · Conditions and the working rule
39. 39 · The method, and the ESA cubic functions
40. 40 · Functions with many stationary points
41. 41 · Applied problems: eliminate one variable first
42. 42 · Lagrange's method: the auxiliary function and the procedure
43. 43 · Geometric problems: inscribed solids, nearest and farthest points
44. 44 · Applied problems: boxes, splitting a number, temperature
45. Appendix A · Symbol glossary
46. Appendix B · Sources, coverage and gaps

M1

Class 001 · Introduction to Partial Differentiation

1 Functions of two or more variables

Most quantities depend on more than one variable: the area of a rectangle on its length and breadth, the volume of a cylinder on r and h, a potential on x, y, z, t. A function of two variables $z = f(x, y)$ assigns to each ordered pair (x, y) in a domain D a unique real number. Its graph is a surface in three dimensions, so tools are used to plot it; this unit works with two variables, and everything extends to n variables.

Key terms and definitions

Function of two variables. A function of two variables denoted by $z = f(x, y)$ is a rule that assigns to each ordered pair of real numbers (x, y) in a set D a unique real number denoted by $f(x, y)$. The set D is the domain of f and its range is the set of values that f takes on: $\{f(x, y) : (x, y) \in D\}$.
S001·8 · T001a 00:05:46

Independent and dependent variables. In $z = f(x, y)$, x and y are independent variables and z is the dependent variable (a bivariate function); with three independent variables one writes $w = f(x, y, z)$, and in general $f(x_1, x_2, \ldots, x_n)$. (transcript only) T001a 00:02:18 · T001a 00:02:54

Graphs and diagrams

$$z = \sin x \cos y$$

$$-\pi \le x,\ y \le \pi;\ \ -1 \le z \le 1$$

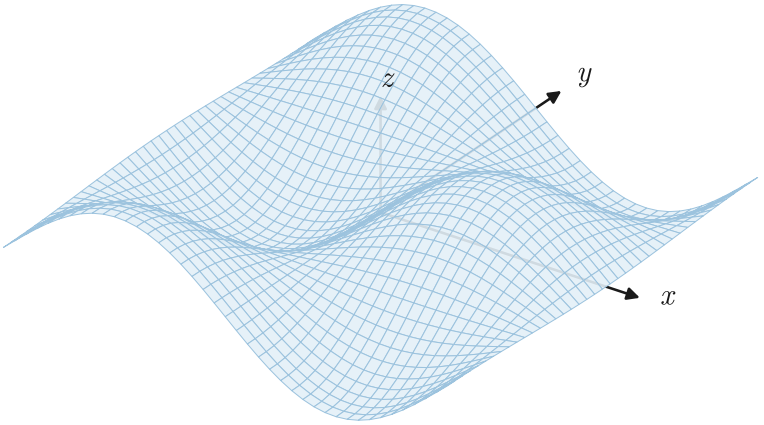


Figure G1.1. Graph of $z = \sin x \cos y$. ESA ★

S001·8 · T001a 00:09:07

How to read it. A function of two variables is a surface over the xy-plane: above each point (x, y) the height is $f(x, y)$. $\sin x \cos y$ is an egg-box of hills and hollows between -1 and 1; the slide's Gnuplot picture shows the same surface.

$z = e^{1 - x^2 - y^2}$

top $(0, 0, e)$

green: level curves $z =$ const (circles)

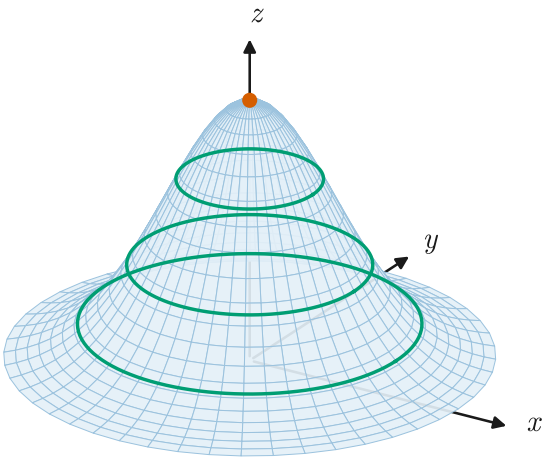


Figure G1.2. Graph of $z = e^{1-x^2-y^2}$. ESA ★

S001·8 · T001a 00:09:07

How to read it. A single bell-shaped hill with its top $z = e$ above the origin, falling to 0 in every direction. The green level curves (drawn on the surface) are circles $x^2 + y^2 =$ constant.

Important points and why

1. Examples on the slides: area of a rectangle $A(l, b)$; volume of a cylinder $v(r, h)$; electric potential in a cathode-ray tube $V = f(x, y, z, t)$; propagation of an electromagnetic wave (magnetic field, electric field, wavelength, direction); weather (topography, pressure, humidity, temperature); spread of a pandemic (area, population, awareness, immunity, age and gender distribution). ESA ★ S001·4 · S001·5 · S001·6 · S001·7 · T001a 00:03:26
2. A single-variable function is enough to understand a concept, but to draw a conclusion about a real situation we almost always need several variables. ESA ★ TRANSCRIPT ONLY T001a 00:05:16
3. f maps an ordered pair to one real number: for $z = x^3y$, the pair $(2, 1)$ is mapped to 8. ESA ★ T001a 00:07:16
4. Graph of $z = f(x, y)$: a surface, which needs three axes; a function of three variables would need four dimensions, so the unit sticks to two variables. The slide shows $\sin x \cos y$ and $e^{1-x^2-y^2}$ (Figures G1.1–G1.2). ESA ★ S001·8 · T001a 00:08:35 · T001a 00:09:07

SHORT-NOTE RECAP

Quantities depending on several variables need functions of several variables (area $A(l, b)$, volume $v(r, h)$, potential $V(x, y, z, t)$, weather, pandemics). $z = f(x, y)$ assigns to each $(x, y) \in D$ a unique real number; D is the domain, the set of values the range. The graph is a surface.

2 Limit of a function of two variables

In one variable x can approach a only from the left or the right. In the plane (x, y) can approach (a, b) along infinitely many paths: lines, parabolas, ellipses. The limit is L only if every path gives L. Proving a limit exists is hard; proving it does not exist is easy: find two paths that give different values.

Key terms and definitions

Limit of a function of two variables. A function $f(x, y)$ is said to approach the limit L as (x, y) approaches (a, b), if the limit remains the same along any path of (x, y) approaching (a, b): $\lim_{(x,y)\to(a,b)} f(x, y) = L$. S001·9 · T001a 00:11:01

Formulas

E2.1 · Limit of f(x, y) ESA ★

$$\lim_{(x,y)\to(a,b)} f(x, y) = L$$

gives	The value f approaches near (a, b).
applies	The same L along every path to (a, b).
note	For a polynomial (or any function continuous at the point) just substitute.

S001·9 · T001a 00:13:59

E2.2 · Two-path test (limit does not exist) ESA ★

$$\lim_{\text{path 1}} f \neq \lim_{\text{path 2}} f \implies \lim_{(x,y)\to(a,b)} f(x, y) \text{ does not exist}$$

gives	A proof that the limit does not exist.
applies	Substitution gives $\frac{0}{0}$.
note	Lines $y = mx$ through the origin are the usual first try: if the answer depends on m, the limit does not exist.

S001·10 · T001a 00:17:42

Graphs and diagrams

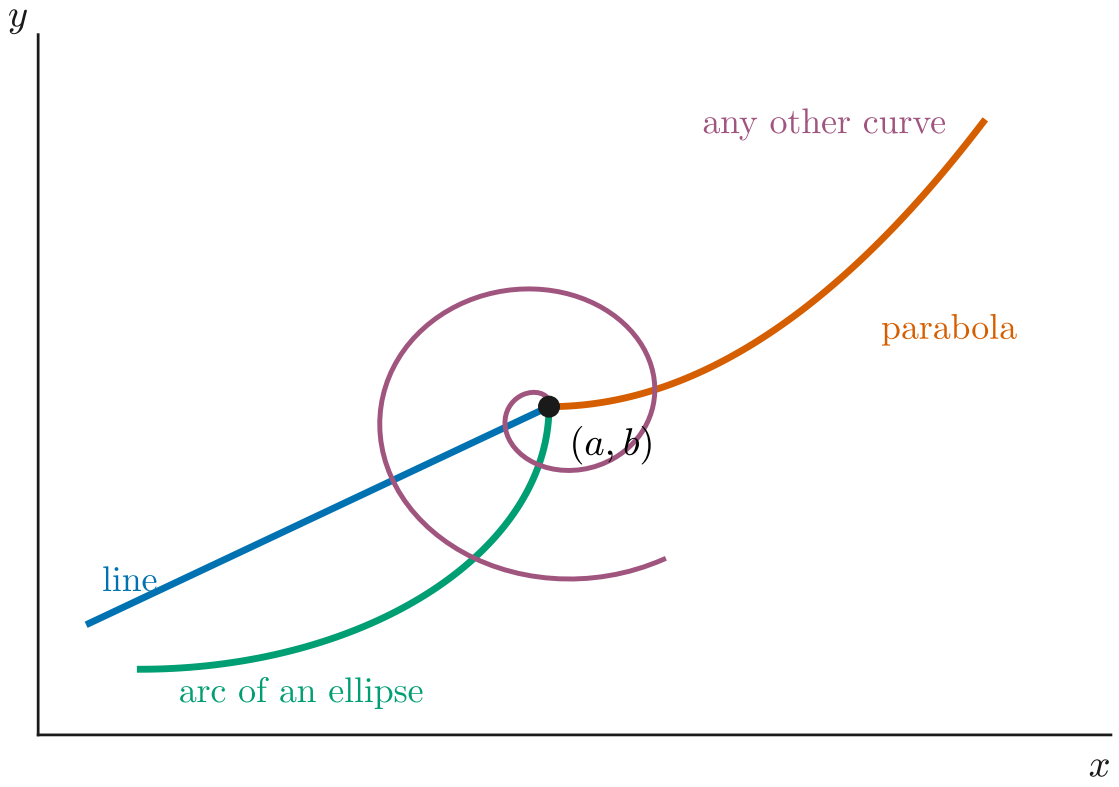


Figure G2.1. Paths along which (x, y) can approach (a, b). ESA ★

S001·9 · T001a 00:11:52 · T001a 00:12:19

How to read it. In one variable x reaches a only from the left or the right. In the plane, (x, y) can reach (a, b) along a line, a parabola, an arc of an ellipse, or any other path. The limit exists only if all of them give the same value.

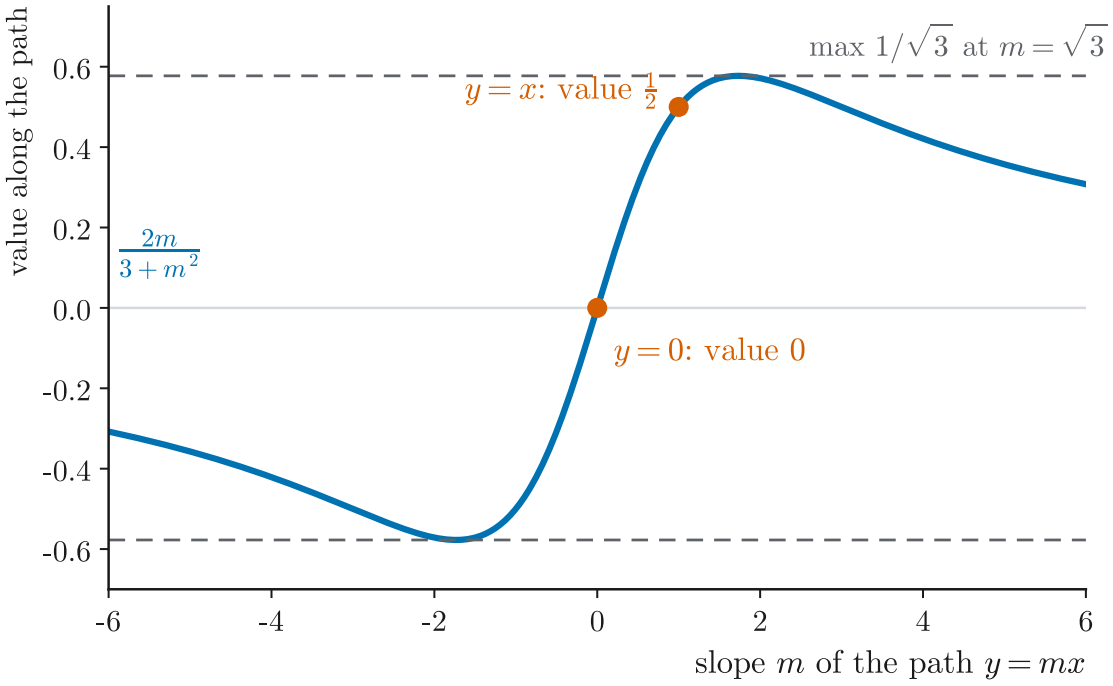
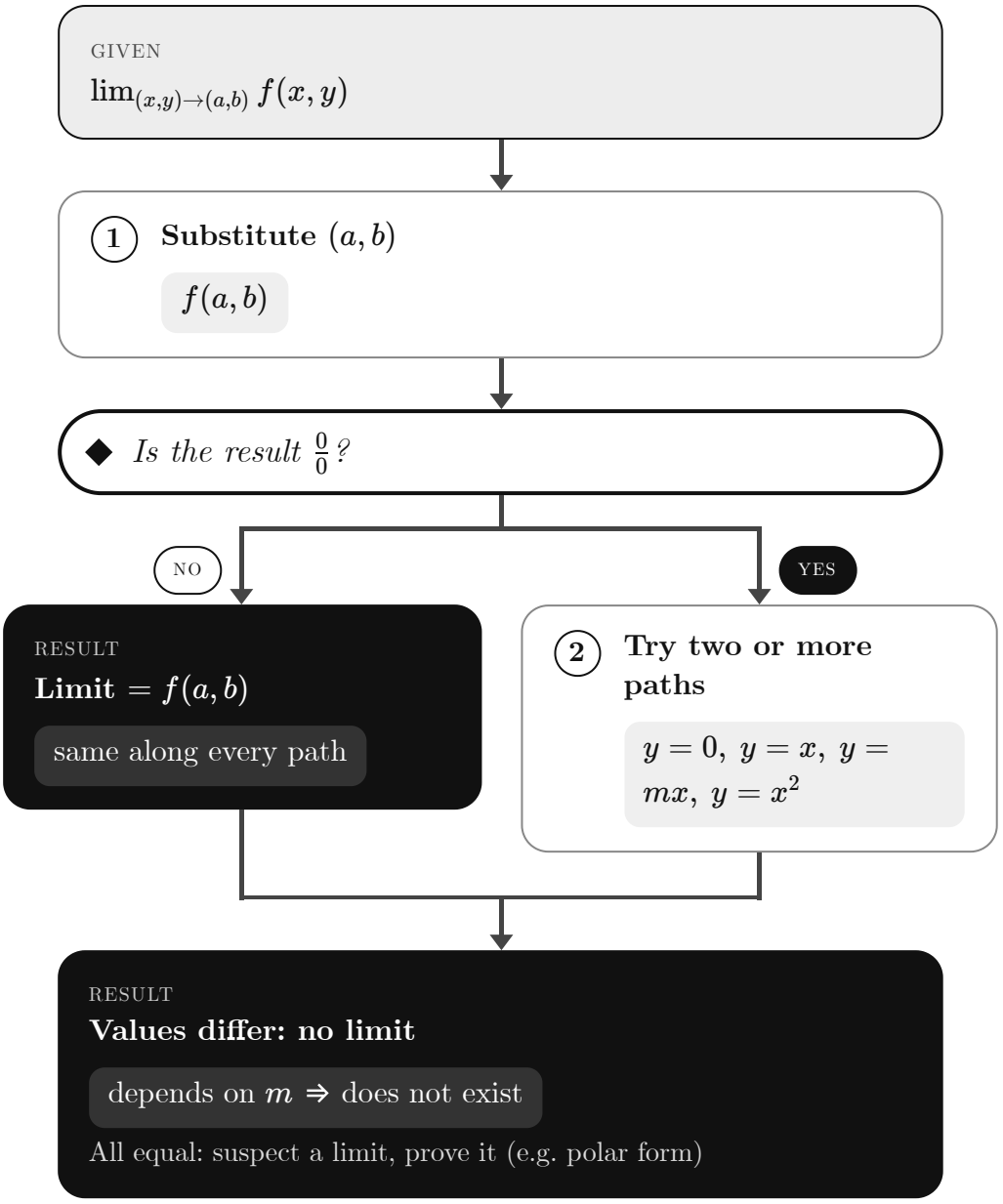


Figure G2.2. $\frac{2xy}{3x^2+y^2}$ along the line $y=mx$. ESA ★★

ADDED FOR CLARITY S001·10 · T001a 00:17:24

How to read it. The value of $\frac{2xy}{3x^2+y^2}$ on the line $y=mx$ is the constant $\frac{2m}{3+m^2}$, plotted against the slope m. $m=0$ gives 0, $m=1$ gives $\frac{1}{2}$, and the largest value is $\frac{1}{\sqrt{3}}$ at $m=\sqrt{3}$. Because the value changes with the path, the limit at the origin does not exist.

Flowcharts



Remark. Agreement along a few paths is only evidence; disagreement is proof.

Exam trap. Read the point carefully: slide $(2,1)$ gives -8, the lecture's $(2,-1)$ gives -6.

Figure F2.1. Finding (or disproving) a limit at a point ESA ★★

S001·9 · S001·10 · T001a 00:13:59 · T001a 00:17:42

One variable and two variables

	$y = f(x)$	$z = f(x, y)$
Domain	Real numbers	Ordered pairs (x, y)
Graph	A curve in the plane	A surface in space
Approach to the point	From the left or the right	Along infinitely many paths
Limit exists if	Left and right limits agree	Every path gives the same value
Continuous if	$\lim_{x \to c} f = f(c)$	$\lim_{(x,y) \to (a,b)} f = f(a, b)$
Derivative	$\frac{dy}{dx}$	$\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$ (others constant)

S001·9 · S001·11 · T001a 00:09:54 · T001a 00:11:52

Important points and why

1. One variable: $x \to a$ from the left or the right only. Two variables: $(x, y) \to (a, b)$ along any of infinitely many paths (Figure G2.1). ESA ★ S001·9 · T001a 00:11:32 · T001a 00:12:34
2. If substitution gives a number (a polynomial, no $\frac{0}{0}$), that number is the limit, whatever the path. ESA ★ T001a 00:13:59
3. It is hard to show that a limit exists but easy to show it does not: two paths with different values are enough. ESA ★★ S001·10 · T001a 00:17:42
4. Agreement along a few paths proves nothing: for $\frac{2xy}{3x^2+y^2}$, the x-axis and the parabola $y = x^2$ both give 0, but $y = x$ gives $\frac{1}{2}$. ESA ★★ T001a 00:16:48 · T001a 00:17:24

Why can one path never prove that a limit exists?
There are infinitely many paths, and a function can agree on any finite number of them and still differ on another: $\frac{2xy}{3x^2+y^2}$ gives 0 on the x-axis and on $y = x^2$ but $\frac{1}{2}$ on $y = x$. To prove existence one needs an estimate for all paths at once (e.g. polar coordinates: $\frac{3x^2y}{x^2+y^2} = 3r\cos^2\theta\sin\theta \to 0$).
ADDED FOR CLARITY T001a 00:17:42 · S001·12

Exam trap. Same value along two or three paths does not prove that the limit exists; different values along two paths does prove it does not. T001a 00:16:48 · T001a 00:17:42

Exam trap. Slide 001·9 uses the point $(2, 1)$ (limit -8); the lecture uses $(2, -1)$ (limit -6). Read the point in the question. ADDED FOR CLARITY S001·9 · T001a 00:12:59

Worked examples

W2.1 · Limit of a polynomial (slide 001·9) ESA ★

GIVEN Evaluate $\lim_{(x,y)\to(2,1)}(x^2 - 2xy + 3y^2 - 4x + 3y - 6)$; the lecture also does the point $(2, -1)$.

STEP 1 A polynomial is continuous everywhere, so substitute.

STEP 2 At $(2,1)$: $4 - 4 + 3 - 8 + 3 - 6 = -8$.

STEP 3 At $(2,-1)$: $4 + 4 + 3 - 8 - 3 - 6 = -6$.

ANSWER -8 at $(2,1)$; -6 at $(2,-1)$

CHECK sympy agrees for both points. The value is reached along every path.

S001·9 · T001a 00:12:59 · T001a 00:13:47

W2.2 · Show that a limit does not exist (slide 001·10) ESA ★★

GIVEN Show that $\lim_{(x,y)\to(0,0)} \dfrac{2xy}{3x^2 + y^2}$ does not exist.

STEP 1 Substitution gives $\frac{0}{0}$: indeterminate.

STEP 2 Along $y = 0$ (x-axis): $\frac{0}{3x^2} = 0 \to 0$.

STEP 3 Along $y = x^2$: $\frac{2x^3}{3x^2+x^4} = \frac{2x}{3+x^2} \to 0$.

STEP 4 Along $y = x$: $\frac{2x^2}{4x^2} = \frac{1}{2}$.

STEP 5 General line $y = mx$: $\frac{2m}{3+m^2}$, which depends on m (Figure G2.2).

ANSWER Different paths give 0 and $\frac{1}{2}$: the limit does not exist

CHECK sympy: $\frac{2m}{m^2+3}$ along $y = mx$.

S001·10 · T001a 00:14:20 · T001a 00:15:04 · T001a 00:17:03

SHORT-NOTE RECAP

$\lim_{(x,y)\to(a,b)} f = L$ if every path to (a,b) gives L. Polynomial: substitute. $\frac{0}{0}$: try paths ($y = 0$, $y = x$, $y = mx$, $y = x^2$); two different values ⇒ no limit. $\frac{2xy}{3x^2+y^2}$: 0 on $y = 0$, $\frac{1}{2}$ on $y = x$.

3 Continuity of a function of two variables

f is continuous at (a,b) when $f(a,b)$ is defined, the limit exists, and $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$, as in one variable. Geometrically the surface has no hole or break there. A function can fail to be continuous even though the limit exists ($\frac{3x^2y}{x^2+y^2}$ at the origin), or because the limit does not exist ($\frac{2xy}{3x^2+y^2}$).

Key terms and definitions

Continuity at a point. $f(x,y)$ is continuous at (a,b) if $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$: $f(a,b)$ must be defined, the limit must exist, and the two must be equal. S001·11 · T001b 00:00:38 · T001a 00:18:46

> *"A function is continuous at a point if the graph doesn't have any holes or breaks at that point. Graphing the function is helpful. If a function is not continuous at (a,b), this does not mean the limit does not exist. We will need to consider the approaching (a,b) from different paths."* S001·11

Formulas

E3.1 · Continuity at (a, b) ESA ★

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$$

gives	The test for continuity.
applies	Three conditions: $f(a,b)$ defined, limit exists, both equal.
note	One variable: $\lim_{x\to c} f(x) = f(c)$.

S001·11 · T001b 00:00:38

Graphs and diagrams

$$z = \frac{3x^2 y}{x^2 + y^2}$$

limit 0 at the origin; $f(0,0)$ undefined

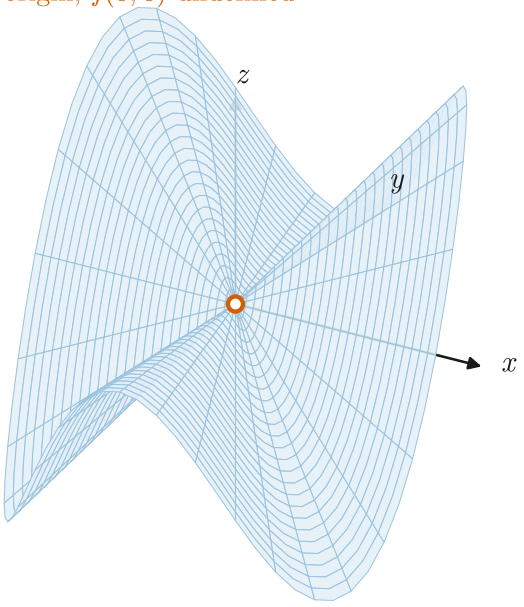


Figure G3.1. $z = \frac{3x^2y}{x^2+y^2}$: the limit at the origin exists. ESA ★

S001·12 · T001b 00:02:41

How to read it. The surface folds near the origin but every direction leads to height 0 there (vermilion point). It is missing only the single point $(0,0)$, where f is undefined: a removable discontinuity.

$z = \dfrac{2xy}{3x^2 + y^2}$

ray $y = x$: height $\frac{1}{2}$

x-axis: height 0

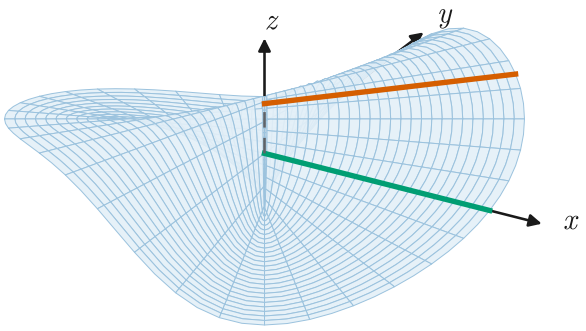
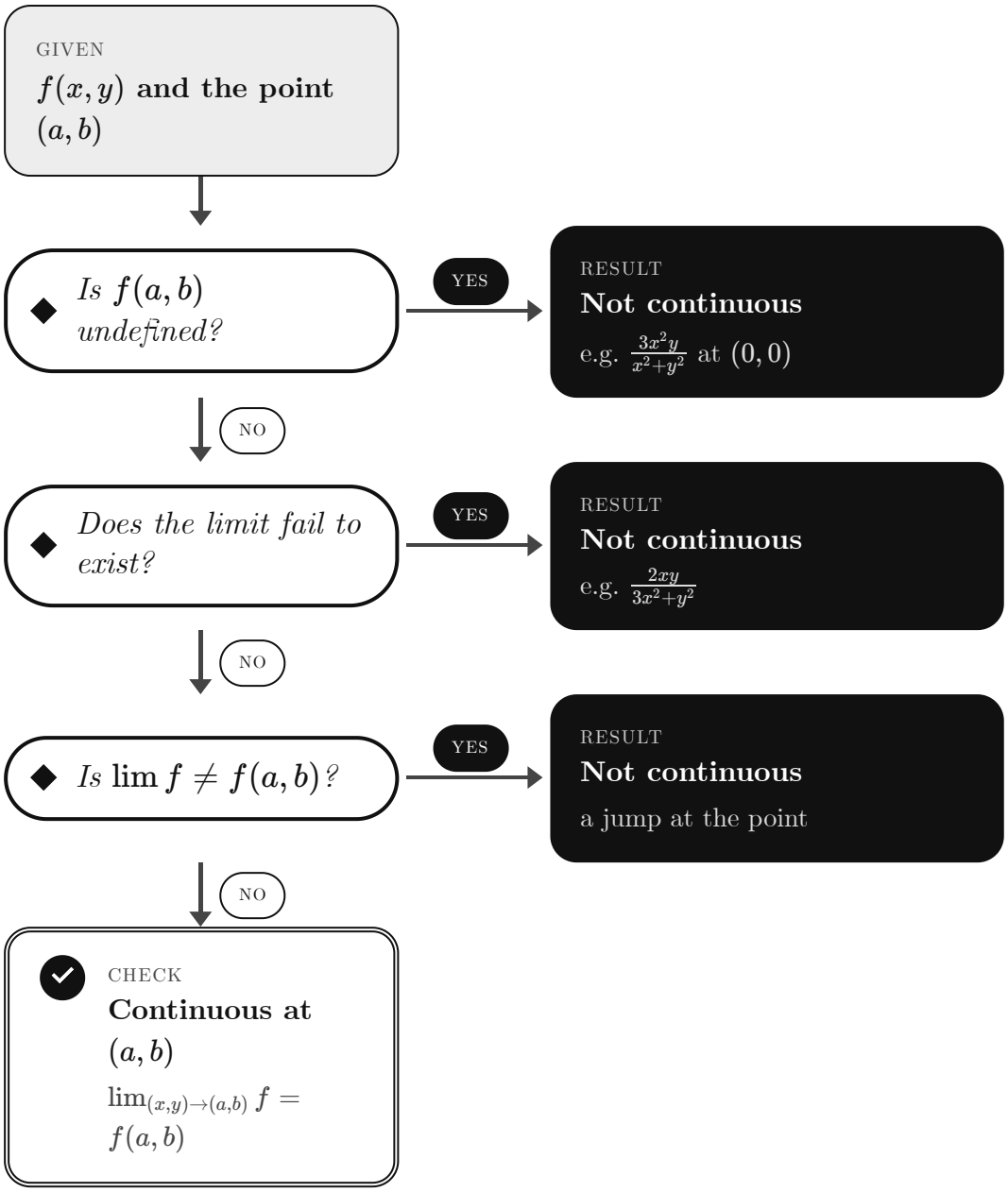


Figure G3.2. $z = \frac{2xy}{3x^2+y^2}$: the limit at the origin does not exist. ESA ★

S001·13 · T001b 00:06:13

How to read it. Each ray from the origin keeps a constant height $\frac{2m}{3+m^2}$, so the surface is a twisted fan of straight lines meeting at the z-axis at different heights (vermilion: the ray $y = x$ at height $\frac{1}{2}$; blue: the x-axis at height 0). There is no single value at the origin.

Flowcharts



Remark. Only the first rung failing with the limit existing is a removable discontinuity: define $f(a,b)$ as the limit.

Figure F3.1. Is f continuous at (a, b)? ESA ★

S001·11 · T001a 00:18:46 · T001b 00:03:17

Important points and why

1. One variable: f is continuous at c if (i) $\lim_{x\to c} f(x)$ exists, (ii) $f(c)$ exists, (iii) they are equal. The same three conditions define continuity for $f(x,y)$. ESA ★ T001a 00:18:46 · S001·11

2. $\frac{3x^2y}{x^2+y^2}$: the limit at the origin exists (it is 0) but $f(0,0)$ is $\frac{0}{0}$, so f is not continuous there. $\frac{2xy}{3x^2+y^2}$: the limit does not exist and $f(0,0)$ is undefined (Figures G3.1–G3.2). ESA ★ S001·12 · S001·13 · T001b 00:04:14 · T001b 00:05:44
3. To decide continuity in higher dimensions, graph the function: a hole, fold or sudden drop in the surface shows a discontinuity. ESA ★ S001·11 · T001b 00:06:45 · T001a 00:18:07

Why is $\frac{3x^2y}{x^2+y^2}$ not continuous at the origin even though its limit is 0?
Continuity needs $f(0,0)$ to be defined and equal to the limit. Here $f(0,0)$ is $\frac{0}{0}$, so the third condition cannot be checked. Defining $f(0,0)=0$ would make it continuous.
ADDED FOR CLARITY S001·12 · T001b 00:04:14

Exam trap. "Not continuous" does not mean "the limit does not exist" (slide 001·11): $\frac{3x^2y}{x^2+y^2}$ has limit 0 at the origin. S001·11 · T001b 00:02:10

Exam trap. Continuous is not the same as smooth: $z=|x|+|y|$ is continuous everywhere but has a corner at the origin. The lecture's "continuous if the surface is smooth" is loose.
ADDED FOR CLARITY T001b 00:06:26

Worked examples

W3.1 · Limit exists, function not continuous (slide 001·12) ESA ★

GIVEN Discuss the continuity of $f(x,y)=\dfrac{3x^2y}{x^2+y^2}$ at the origin.

STEP 1 $f(0,0)=\frac{0}{0}$ is not defined, so condition (ii) fails: f is not continuous at $(0,0)$.

STEP 2 The limit exists: in polar form $x=r\cos\theta$, $y=r\sin\theta$, $f=3r\cos^2\theta\sin\theta$, and $|f|\le 3r\to 0$ for every path (added for clarity).

STEP 3 So $\lim_{(x,y)\to(0,0)} f=0$.

ANSWER Not continuous at $(0,0)$; the limit is 0 (removable: set $f(0,0)=0$)

CHECK sympy: polar form $3r\sin\theta\cos^2\theta$.

S001·12 · T001b 00:02:41 · T001b 00:04:27

W3.2 · Limit does not exist, function not continuous (slide 001·13) ESA ★

GIVEN Discuss the continuity of $f(x,y) = \dfrac{2xy}{3x^2+y^2}$ at the origin.

STEP 1 The limit does not exist (W2.2).

STEP 2 $f(0,0)$ is also undefined.

ANSWER Not continuous at $(0,0)$; no choice of $f(0,0)$ can make it continuous

CHECK Figure G3.2 shows the jump along different directions.

S001·13 · T001b 00:05:05 · T001b 00:05:44

SHORT-NOTE RECAP

Continuous at (a,b): $f(a,b)$ defined, limit exists, $\lim f = f(a,b)$; no hole or break in the surface.
Not continuous ≠ no limit: $\frac{3x^2y}{x^2+y^2}$ has limit 0 but $f(0,0)$ undefined.

4 Partial derivatives: history, definition and notation

A partial derivative is the derivative with respect to one variable with the others held constant. The idea is already in Leibniz's product rule; the symbol ∂ came from Legendre (1786) and was reintroduced by Jacobi (1841). $\frac{\partial z}{\partial x} = z_x = f_x$ is defined by a limit, and every rule of ordinary differentiation (product, quotient, chain) still holds.

Key terms and definitions

Partial derivative. A partial derivative of a function of several variables is the derivative with respect to one of those variables, with the others held constant. S001·15 · T001b 00:12:52

Formulas

E4.1 · Partial derivatives from the definition ESA ★★

$$\frac{\partial z}{\partial x} = \lim_{h \to 0} \frac{f(x+h, y) - f(x, y)}{h}$$
$$\frac{\partial z}{\partial y} = \lim_{h \to 0} \frac{f(x, y+h) - f(x, y)}{h}$$

gives	z_x (y held constant) and z_y (x held constant).
applies	When the limits exist.
note	Only one variable is moved by h; the other stays fixed.

S001·15

E4.2 · Notation for first-order partials ESA ★★

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} = z_x = f_x, \qquad \frac{\partial z}{\partial y} = \frac{\partial f}{\partial y} = z_y = f_y$$

gives	The equivalent symbols.
applies	$z = f(x, y)$.
note	The symbol is ∂ (“partial”, often read “dou”), not δ or d.

S001·15 · T001b 00:11:41 · T001b 00:13:05

Flowcharts

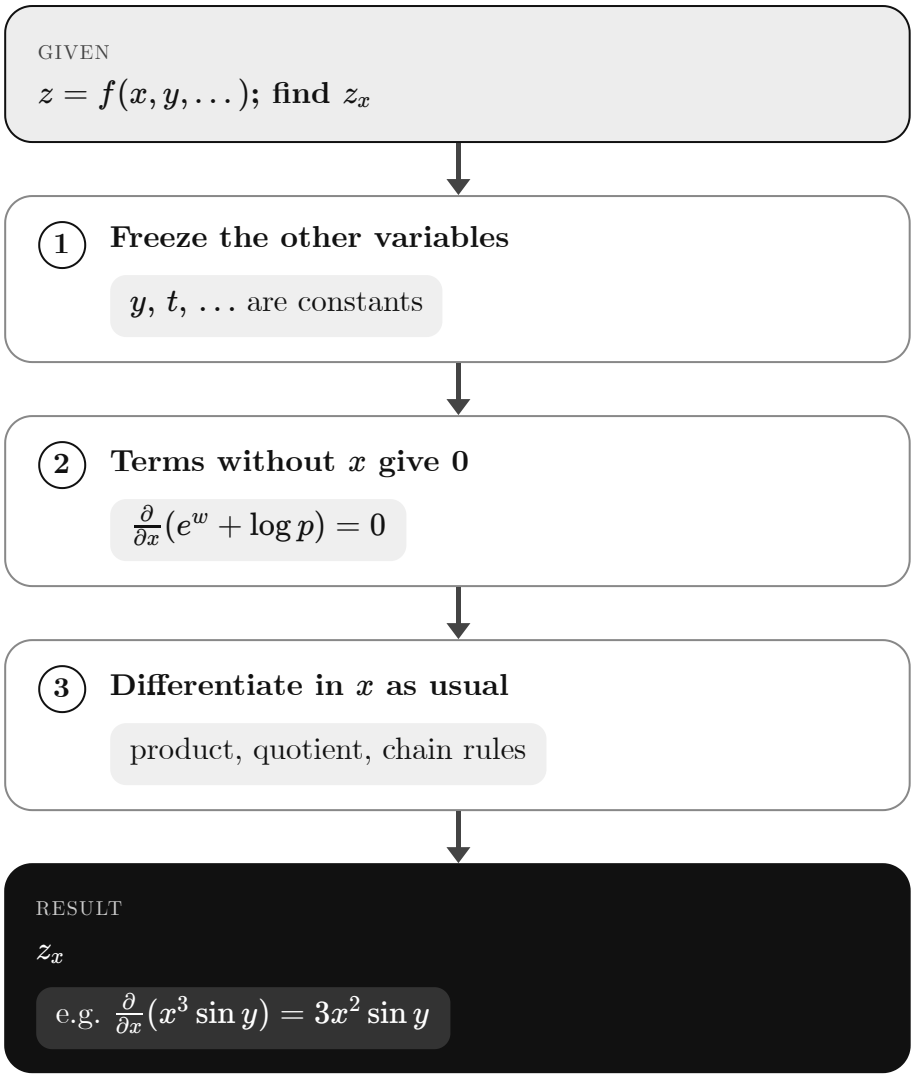


Figure F4.1. Computing a partial derivative ESA ★★

S001·15 · T001b 00:15:08 · T001b 00:13:57

Important points and why

1. The product rule is already partial differentiation: in $\frac{d}{dx}(x^3 \sin x) = x^3 \cos x + 3x^2 \sin x$, each term treats the other factor as a constant k. Leibniz: $d(uv) = u\,dv + v\,du$. ESA ★ S001·14 · T001b 00:07:56 · T001b 00:10:18
2. History: Leibniz first used partial differentiation (product rule); Adrien-Marie Legendre created the modern symbol ∂ in 1786 but later abandoned it; Carl Gustav Jacob Jacobi reintroduced it in 1841. ESA ★ S001·14 · T001b 00:11:23
3. To find z_x, look only at the terms containing x; everything else is a constant, however many other variables there are. ESA ★★ T001b 00:15:08 · T001b 00:15:48
4. A function of x, y, t has three first partials z_x, z_y, z_t; for z_x treat y and t as constants. ESA ★ T001b 00:13:20
5. All rules of ordinary differentiation hold for partial derivatives: product, quotient, chain rule, implicit and composite functions. ESA ★★ T001b 00:13:57 · T001b 00:14:16

Why does every rule of ordinary differentiation still work?
With the other variables frozen, $f(x, y)$ is an ordinary function of x alone, and $\frac{\partial f}{\partial x}$ is its ordinary derivative (E4.1 is the usual limit with y fixed).

T001b 00:13:57 · S001·15

Exam trap. The partial symbol is ∂, not δ (the lecture stresses this) and not d. T001b 00:11:41

Worked examples

W4.1 · A partial derivative from the definition — ESA ★

GIVEN Use E4.1 to find f_x for $f(x, y) = x^2 y$.

STEP 1 $\frac{f(x+h,y)-f(x,y)}{h} = \frac{(x+h)^2 y - x^2 y}{h} = \frac{(2xh+h^2)y}{h} = (2x + h)y.$

STEP 2 Let $h \to 0$: $2xy$.

ANSWER $\boxed{f_x = 2xy}$

CHECK Same as differentiating x^2 with y constant; sympy agrees.

ADDED FOR CLARITY S001·15

W4.2 · Partials of a function of several variables (lecture) — ESA ★★

GIVEN $z = x^3 \sin y + e^w + \log p$ (a function of x, y, w, p). Find z_x, z_p, z_y and z_{xy}.

STEP 1 z_x: only $x^3 \sin y$ contains x; $\sin y$ is a constant: $3x^2 \sin y$.

STEP 2 z_p: only $\log p$ contains p: $\frac{1}{p}$.

STEP 3 z_y: only $\sin y$ varies: $x^3 \cos y$.

STEP 4 z_{xy}: differentiate $z_x = 3x^2 \sin y$ with respect to y: $3x^2 \cos y$.

ANSWER $\boxed{z_x = 3x^2 \sin y,\ z_p = \frac{1}{p},\ z_y = x^3 \cos y,\ z_{xy} = 3x^2 \cos y}$

CHECK sympy agrees; $z_{yx} = \frac{\partial}{\partial x}(x^3 \cos y) = 3x^2 \cos y$ too (Clairaut).

T001b 00:14:40 · T001b 00:15:48 · T001b 00:16:26 · T001b 00:16:44

SHORT-NOTE RECAP

Partial derivative: derivative in one variable, others constant. $z_x = \lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h}$; notation $\frac{\partial z}{\partial x} = z_x = f_x$. Leibniz (product rule), Legendre 1786, Jacobi 1841. All differentiation rules hold.

5 Second-order and mixed partial derivatives

Differentiating again gives z_{xx}, z_{yy} and the mixed partials z_{xy}, z_{yx}. Clairaut's theorem: if the second derivatives are continuous, the order of differentiation does not matter, $z_{xy} = z_{yx}$. Subscripts are read in the order of differentiation; in the ∂ notation the order is read from right to left.

Key terms and definitions

Mixed (crossed) partial derivatives. $z_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)$ and $z_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right)$: differentiate with respect to one variable, then the other. S001·16 · T001b 00:18:24

> *"Note 1: The crossed or mixed partial derivatives are in general equal (Clairaut's theorem): $\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)$. That is, the order of differentiation is immaterial if the second order derivatives involved are continuous."* S001·17 · T001b 00:19:57

> *"Note 2: In the subscript notation, the subscripts are written in same order in which the differentiation is carried out, while in the ∂ notation the order is opposite. For example, $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = f_{xy}$."* S001·17

Formulas

E5.1 · Second-order partials ESA ★★

$$z_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial x^2} = f_{xx}, \qquad z_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial y^2} = f_{yy}$$

gives	Second derivatives in one variable.
applies	Differentiate z_x again in x (or z_y in y).
note	Same rules as ordinary second derivatives.

S001·16 · T001b 00:17:04

E5.2 · Mixed partials ESA ★★

$$z_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial y\,\partial x} = f_{xy}, \qquad z_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial x\,\partial y} = f_{yx}$$

gives	The two crossed derivatives.
applies	Differentiate in one variable, then the other.
note	Subscripts: left to right. ∂ notation: right to left.

S001·16 · S001·17 · T001b 00:18:41

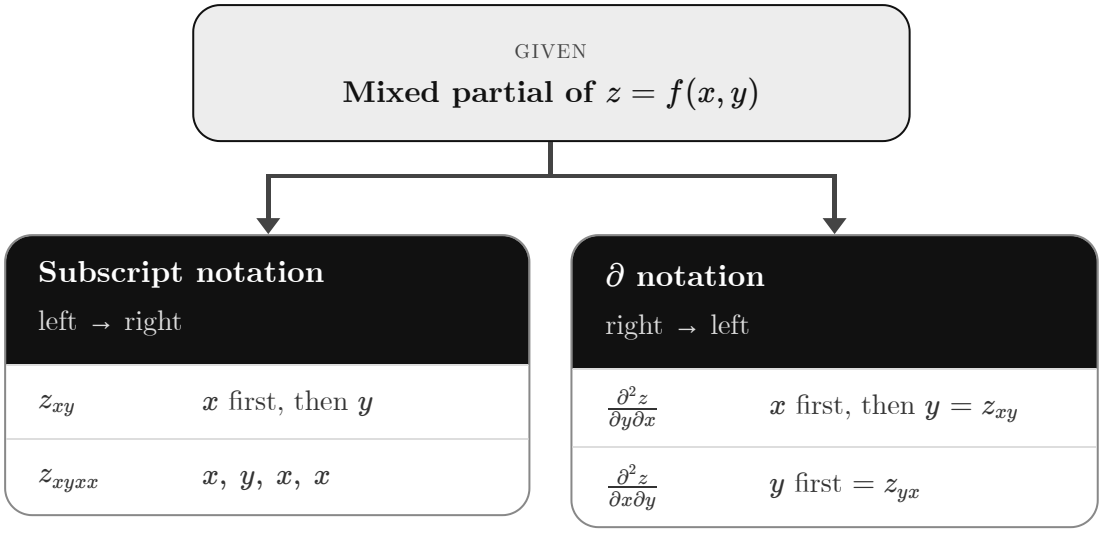
E5.3 · Clairaut's theorem ESA ★★★

$$f_{xy} = f_{yx} \qquad \left(\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 f}{\partial x\,\partial y}\right)$$

GIVES	The order of differentiation does not matter.
APPLIES	f_{xy} and f_{yx} continuous near the point.
NOTE	Called "a very important theorem" in the lecture; it is why Taylor's series has $2hkf_{xy}$.

S001·17 · T001b 00:19:57 · T001b 00:20:10

Flowcharts



Remark. By Clairaut's theorem the order rarely changes the answer, but the notation is still examined.

Figure F5.1. Reading the order of a mixed partial ESA ★★

S001·17 · T001b 00:18:51

Important points and why

1. Read z_{xyxx} left to right: $z \to z_x \to z_{xy} \to z_{xyx} \to z_{xyxx}$ (differentiate in x, then y, then x, then x). ESA ★ T001b 00:18:51 · T001b 00:19:21
2. Clairaut's theorem is "a very important theorem": whenever the second partials are continuous (true for all examination functions away from bad points), compute whichever mixed partial is easier. ESA ★★★ S001·17 · T001b 00:20:10
3. Verifying $f_{xy} = f_{yx}$ is a standard class-work question: x^y (CW P2) and $x^2 \tan^{-1}\frac{y}{x} - y^2 \tan^{-1}\frac{x}{y}$ (CW P8, W5.3). ESA ★★ CW U1 P2 · CW U1 P8

Exam trap. $\frac{\partial^2 z}{\partial y\,\partial x}$ means x first, then y: it is z_{xy}, not z_{yx}. S001·17

Exam trap. Clairaut needs continuous second partials: for $f = \frac{xy(x^2-y^2)}{x^2+y^2}$, $f(0,0)=0$, one gets $f_{xy}(0,0) = -1$ but $f_{yx}(0,0) = 1$. ADDED FOR CLARITY S001·17

Worked examples

W5.1 · Second-order partials of a function of three variables

ESA ★★

GIVEN If $f(x,y,z) = e^{x^2+y^2+z^2}$, find f_{xx}, f_{yy}, f_{zz} at $(-1,1,-1)$.

STEP 1 $f_x = 2xe^{x^2+y^2+z^2}$.

STEP 2 $f_{xx} = (2+4x^2)e^{x^2+y^2+z^2}$ (product rule).

STEP 3 At $(-1,1,-1)$: $x^2+y^2+z^2 = 3$, so $f_{xx} = 6e^3$; by symmetry $f_{yy} = (2+4y^2)e^3 = 6e^3$, $f_{zz} = 6e^3$.

ANSWER $\boxed{f_{xx} = f_{yy} = f_{zz} = 6e^3}$

CHECK CW answer agrees; sympy agrees.

CW U1 P4

W5.2 · Find n so that v satisfies an equation

ESA ★★

GIVEN Find n so that $v = r^n(3\cos^2\theta - 1)$ satisfies $\dfrac{\partial}{\partial r}\left(r^2\dfrac{\partial v}{\partial r}\right) + \dfrac{1}{\sin\theta}\dfrac{\partial}{\partial\theta}\left(\sin\theta\dfrac{\partial v}{\partial\theta}\right) = 0$.

STEP 1 $v_r = nr^{n-1}(3\cos^2\theta - 1)$, so $\frac{\partial}{\partial r}(r^2 v_r) = n(n+1)r^n(3\cos^2\theta - 1)$.

STEP 2 $v_\theta = -6r^n\cos\theta\sin\theta$, so $\sin\theta\, v_\theta = -6r^n\cos\theta\sin^2\theta$.

STEP 3 $\frac{\partial}{\partial\theta}(\cdot) = -6r^n(2\sin\theta\cos^2\theta - \sin^3\theta)$; divide by $\sin\theta$: $-6r^n(3\cos^2\theta - 1)$.

STEP 4 Sum: $[n(n+1) - 6]\, r^n(3\cos^2\theta - 1) = 0 \Rightarrow n^2 + n - 6 = 0$.

ANSWER $\boxed{n = 2 \text{ or } n = -3}$

CHECK CW answer $(-3,\ 2)$ agrees; sympy solves to $[-3, 2]$. Same pattern as ESA Jan-24 s4 1c (worked in the class 002 pack).

CW U1 P7 · ESA Jan-24 s4 1c

W5.3 · Verify that the mixed partials are equal ESA ★★

GIVEN If $u = x^2 \tan^{-1}\dfrac{y}{x} - y^2 \tan^{-1}\dfrac{x}{y}$, show that $\dfrac{\partial^2 u}{\partial x \partial y} = \dfrac{\partial^2 u}{\partial y \partial x} = \dfrac{x^2 - y^2}{x^2 + y^2}$.

STEP 1 $u_x = 2x \tan^{-1}\frac{y}{x} + x^2 \cdot \frac{-y}{x^2+y^2} - y^2 \cdot \frac{y}{x^2+y^2} = 2x \tan^{-1}\frac{y}{x} - y$.

STEP 2 $u_{xy} = \frac{\partial}{\partial y}(u_x) = \frac{2x^2}{x^2+y^2} - 1 = \frac{x^2-y^2}{x^2+y^2}$.

STEP 3 Similarly $u_y = x - 2y \tan^{-1}\frac{x}{y}$, and $u_{yx} = 1 - \frac{2y^2}{x^2+y^2} = \frac{x^2-y^2}{x^2+y^2}$.

ANSWER $\boxed{u_{xy} = u_{yx} = \dfrac{x^2 - y^2}{x^2 + y^2}}$

CHECK sympy: both simplify to $\frac{x^2-y^2}{x^2+y^2}$.

CW U1 P8

SHORT-NOTE RECAP

z_{xx}, z_{yy}, $z_{xy} = \frac{\partial}{\partial y}(z_x) = \frac{\partial^2 z}{\partial y \partial x}$, z_{yx}. Clairaut: $z_{xy} = z_{yx}$ when the second partials are continuous. Subscripts left to right, ∂ right to left.

M2

Class 002 · Geometrical Interpretation of Partial Derivatives

6 Higher-order partial derivatives

Differentiating a partial derivative again gives higher-order partial derivatives. A function of two variables has 2 first-order, 4 second-order and 2^n nth-order partial derivatives; with m variables there are m^n of order n. The two mixed second-order partials are usually equal.

Key terms and definitions

First-order partial derivative. The partial derivative $\partial f/\partial x$ obtained by differentiating once (here, with respect to x while y is held constant). S002·3

Second-order partial derivatives. $\dfrac{\partial^2 f}{\partial x^2}$, $\dfrac{\partial^2 f}{\partial y^2}$, $\dfrac{\partial^2 f}{\partial x \partial y}$, $\dfrac{\partial^2 f}{\partial y \partial x}$, which are obtained by differentiating twice. S002·3

Higher-order partial derivatives. Third, fourth and nth order derivatives involve 3, 4 or n times differentiation respectively. S002·3

Mixed partial derivative. A higher-order partial in which you differentiate with respect to more than one variable, e.g. f_{xy} and f_{yx}. S002·9 · T002 00:00:06

> *"A function of 2 variables has two first order derivatives, four second order derivatives and 2^n of nth order derivatives. A function of m independent variables will have m^n derivatives of order n."* S002·3

Formulas

E6.1 · First-order partial derivatives ESA ★★

$$f_x = \frac{\partial f}{\partial x}, \qquad f_y = \frac{\partial f}{\partial y}$$

gives	Rate of change of f in the x (or y) direction.
applies	Differentiate once; hold the other variable constant.
note	Two of them for a function of two variables.

S002·3 · T002 00:00:44

E6.2 · Second-order partial derivatives ESA ★★

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y\,\partial x}, \quad f_{yx} = \frac{\partial^2 f}{\partial x\,\partial y}$$

GIVES The four second-order partials of $f(x, y)$.

APPLIES Differentiate twice. In f_{xy}, the subscripts are read left to right: first x, then y.

NOTE Notation order varies between books. The two mixed partials are equal when continuous (E6.4).

ADDED FOR CLARITY S002·3 · T002 00:00:58

E6.3 · How many partial derivatives of order n? ESA ★

2 variables: 2^n m variables: m^n

GIVES The number of nth-order partial derivatives, counting each order of differentiation separately.

APPLIES Two variables: 2 first-order, 4 second-order, 8 third-order.

NOTE With equal mixed partials, two variables give only $n + 1$ distinct ones.

S002·3 · T002 00:01:37 · T002 00:02:13

E6.4 · Equality of mixed partials (Clairaut) ESA ★★

$$f_{xy} = f_{yx} \quad \Longleftrightarrow \quad \frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 f}{\partial x\,\partial y}$$

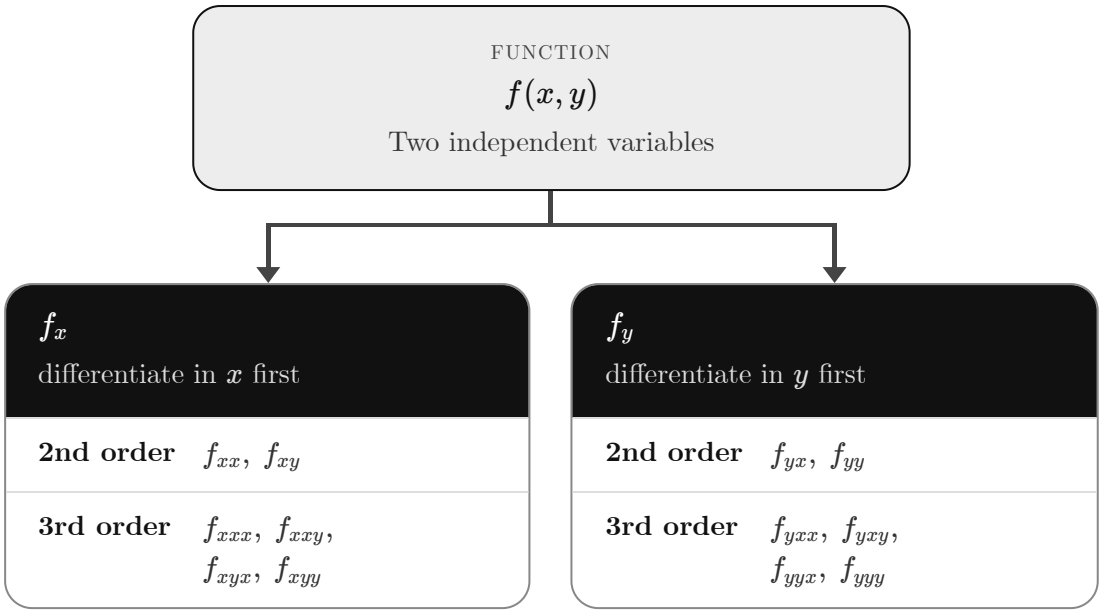
GIVES Order of differentiation does not matter.

APPLIES When f_{xy} and f_{yx} are continuous (true for every function in this unit's problems).

NOTE Verified in CW U1 P2 and P8.

TRANSCRIPT ONLY ADDED FOR CLARITY T002 00:00:24 · T002 00:14:59 · CW U1 P2

Flowcharts



Remark. Totals: 2, 4, 8, ..., 2^n. With m variables there are m^n. When mixed partials are equal, only $n+1$ of the 2^n are different.

Figure F6.1. Counting partial derivatives of each order. ESA ★

S002·3 · T002 00:01:37 · T002 00:02:13

Important points and why

1. A partial derivative is taken with respect to one variable while every other variable is held constant. ESA ★★ T002 00:00:06
2. Differentiating a first-order partial again gives a second-order partial; n differentiations give an nth-order partial. ESA ★★ S002·3
3. For $f(x, y)$: 2 first-order (f_x, f_y), 4 second-order ($f_{xx}, f_{yy}, f_{xy}, f_{yx}$), 2^n of order n. ESA ★★ S002·3 · T002 00:00:44
4. For $f(x_1, \ldots, x_m)$: m^n derivatives of order n; e.g. m^2 second-order partials. ESA ★ S002·3 · T002 00:02:13
5. The two mixed partials "were equal" in the class examples: $f_{xy} = f_{yx}$ for the smooth functions of this unit. ESA ★★ TRANSCRIPT ONLY T002 00:00:24

Exam trap. In f_{xy} the order is x first, then y. Some books write the same derivative as $\partial^2 f / \partial y\, \partial x$. Both mean the same, and for this unit's functions the order does not change the answer. ADDED FOR CLARITY

SHORT-NOTE RECAP

Differentiating $f(x, y)$ once gives the first-order partials f_x, f_y; differentiating again gives four second-order partials f_{xx}, f_{yy}, f_{xy}, f_{yx}. In general n differentiations give 2^n partials of order n for two variables and m^n for m variables. In f_{xy}, differentiate first in x, then in y. When the mixed partials are continuous, $f_{xy} = f_{yx}$ (Clairaut).

7 The one-variable picture: derivative as slope

For $y = f(x)$ the derivative measures change: how fast y changes as you move along the x-axis. Geometrically, $f'(x_0)$ is the slope of the tangent line to the curve at x_0. Partial derivatives extend exactly this idea to surfaces.

Key terms and definitions

Derivative (one variable). For $y = f(x)$, dy/dx is the change in y produced by a change in position along the x-axis; at a point it equals the slope of the tangent to the curve. S002·4 · T002 00:02:50

> *"We know that the derivative of a single valued function gives the slope of the tangent line to the curve at the given point."* S002·4

Formulas

E7.1 · Derivative as slope (one variable) ESA ★

$$\left.\frac{dy}{dx}\right|_{x_0} = f'(x_0) = \text{slope of the tangent to } y = f(x) \text{ at } (x_0, f(x_0))$$

GIVES Slope of the tangent line.

APPLIES f differentiable at x_0.

S002·4 · T002 00:03:13

E7.2 · Tangent line to y = f(x) ESA ★

$$y - f(x_0) = f'(x_0)\,(x - x_0)$$

GIVES Equation of the tangent at x_0.

APPLIES f differentiable at x_0.

NOTE Point-slope form.

ADDED FOR CLARITY S002·4

Graphs and diagrams

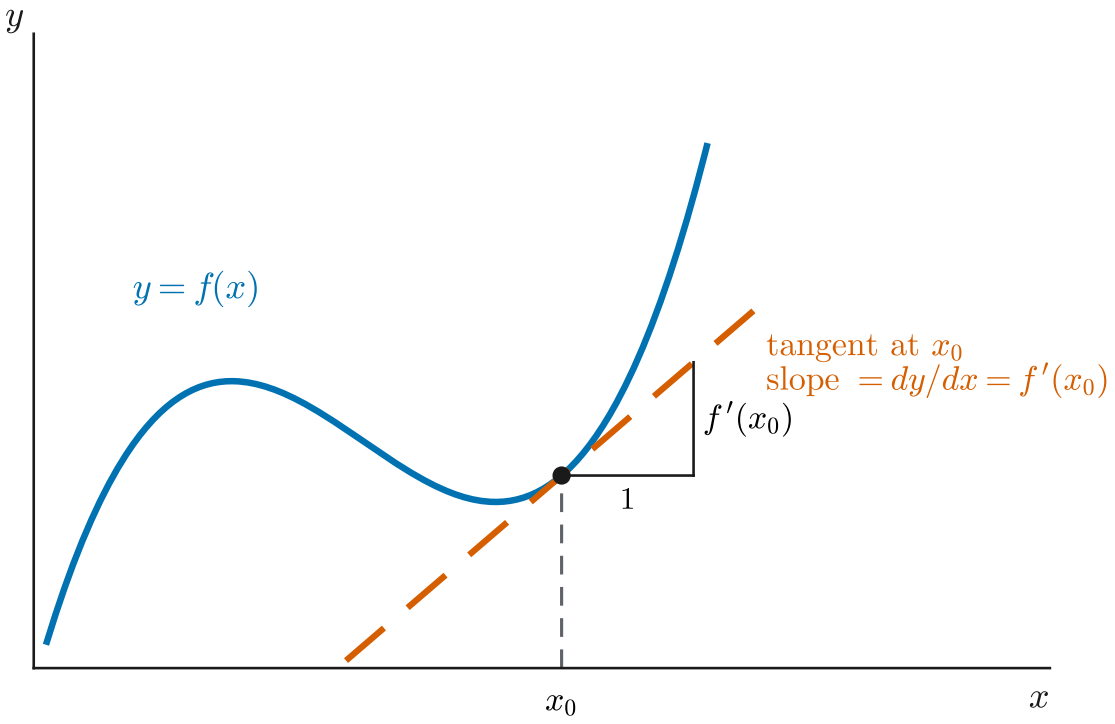


Figure G7.1. Derivative of y = f(x) as the slope of the tangent. ESA ★

S002·4 · T002 00:03:13

How to read it. The blue curve is $y = f(x)$, and the dashed vermilion line is its tangent at x_0. The small triangle shows the slope: a run of 1 gives a rise of $f'(x_0)$. In an exam, label the point, the tangent and the slope dy/dx at x_0.

Important points and why

1. A derivative always represents change: dy/dx is the change on the curve as you change your position on the x-axis. ESA ★ T002 00:02:50 · T002 00:02:59
2. At a particular point, the derivative is the slope of the tangent at that point. ESA ★ S002·4 · T002 00:03:13

SHORT-NOTE RECAP

For $y = f(x)$, dy/dx measures how y changes as x changes. At x_0 it equals the slope of the tangent line to the curve at $(x_0, f(x_0))$, and the tangent is $y - f(x_0) = f'(x_0)(x - x_0)$.

8 A function of two variables is a surface

$z = f(x, y)$ gives one height z for every point (x, y), so its graph is a surface in three dimensions. If (x, y) is restricted to a curve in the xy-plane, such as a circle or a parabola, the matching points of the surface form a curve lying on the surface.

Key terms and definitions

Bivariate function. A function $z = f(x, y)$ of two independent variables. Its graph is a surface in three dimensions. T002 00:04:16

Curve on a surface. If the points (x, y) are restricted to a curve in the xy-plane (circle, parabola, ...), the points $(x, y, f(x, y))$ form a curve that lies on the surface. S002·5

> *"If the points (x,y) lie on a circle in the xy plane, then z = f(x,y) is a curve which looks like the graph shown. If the points (x,y) lie on a parabola in the xy plane, then z = f(x,y) is a curve which looks like the graph shown."* S002·5

Graphs and diagrams

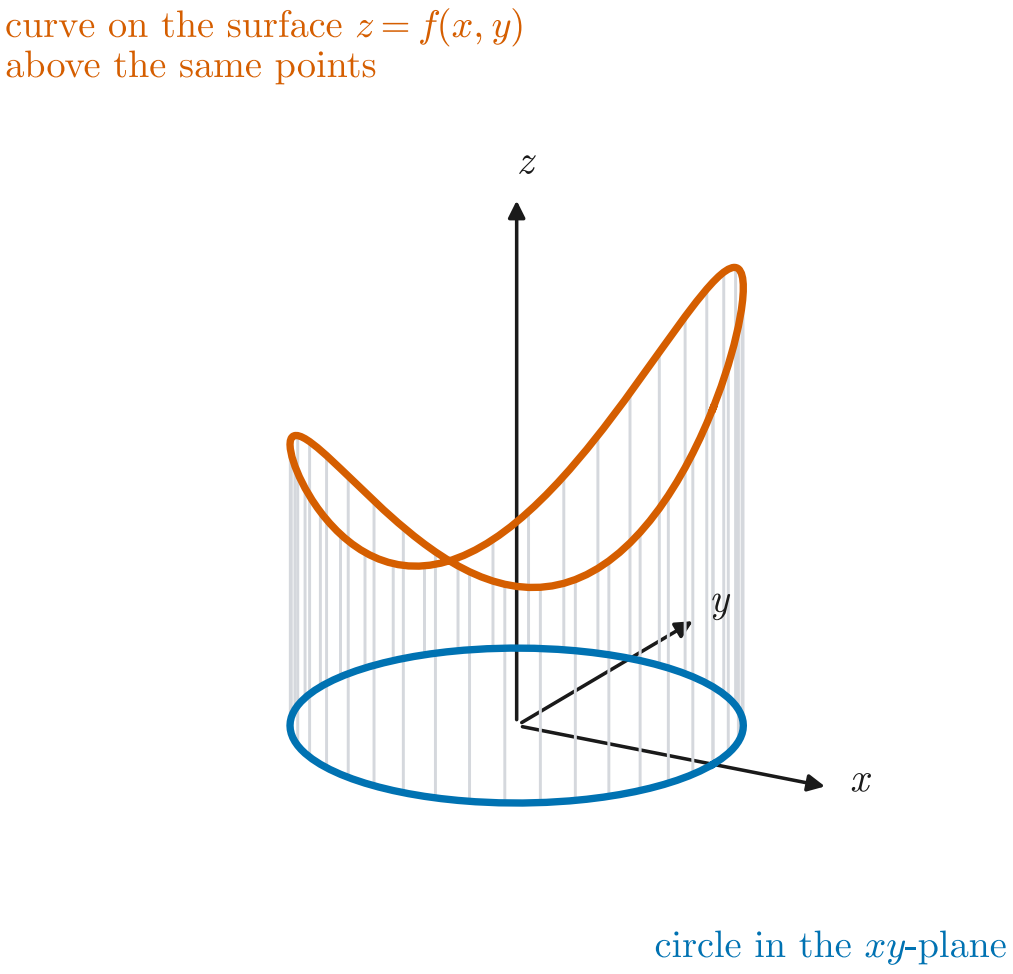


Figure G8.1. Points on a circle in the xy-plane give a curve on the surface. ESA ★

S002·5 · T002 00:03:32

How to read it. The blue circle lies in the xy-plane. Each grey line rises to $z = f(x, y)$, and the vermilion curve joins the tops: a closed curve lying on the surface $z = f(x, y) = 2.2 + 0.8xy + 0.5x$ (an illustrative choice).

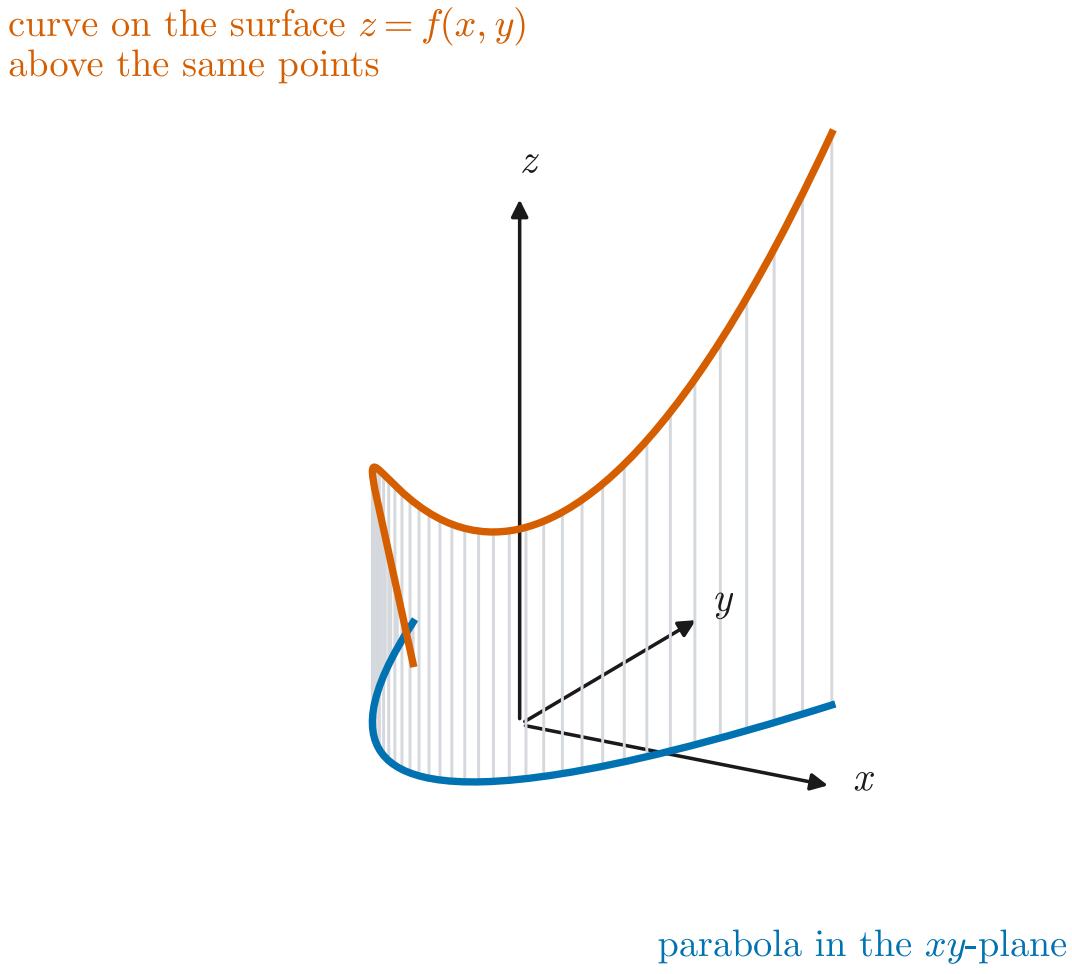


Figure G8.2. Points on a parabola in the xy-plane give a curve on the surface. ESA ★

S002·5 · T002 00:03:47

How to read it. Same idea with a parabola $y = x^2 - 1.2$ in the xy-plane. Every point of the parabola has one height on the surface, so the vermilion curve lies on the surface above it.

Important points and why

1. A function of two independent variables always gives a surface in three dimensions: each (x, y) has one z. ESA ★ T002 00:04:16
2. Restricting (x, y) to a curve in the xy-plane (circle, parabola) picks out a curve on the surface. ESA ★ S002·5 · T002 00:03:32

SHORT-NOTE RECAP

A bivariate function $z = f(x, y)$ assigns one height to each point of the xy-plane, so its graph is a surface in 3-D. Points (x, y) on a circle or parabola in the xy-plane give a curve lying on that surface.

9 First-order partials as slopes of tangents

At a point P on a surface there are infinitely many tangent lines, one for each direction. $\partial f/\partial x$ at (x_0, y_0) is the slope of the one parallel to the x-axis, found by fixing $y = y_0$; $\partial f/\partial y$ is the slope of the one parallel to the y-axis, found by fixing $x = x_0$.

Key terms and definitions

Partial derivative, geometrically. The slope of the tangent line to the surface $z = f(x, y)$ at (x_0, y_0), along the direction of the variable you differentiate with respect to. S002·6

Trace (plane section). The curve cut from the surface by a vertical plane such as $y = y_0$ (giving $z = f(x, y_0)$) or $x = x_0$ (giving $z = f(x_0, y)$). ADDED FOR CLARITY S002·7

> *"Partial derivative of the function f(x,y) at the point (x_0, y_0) can be visualized as the slope of the tangent line along the direction of the independent variable with respect to which we would differentiate the function."* S002·6

Formulas

E9.1 · ∂f/∂x as an ordinary derivative ESA ★★

$$\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} = \left.\frac{d}{dx}\big(f(x, y_0)\big)\right|_{x=x_0}$$

GIVES	Slope of the tangent to the surface at $(x_0, y_0, f(x_0, y_0))$ that is parallel to the x-axis.
APPLIES	Fix $y = y_0$ first; $f(x, y_0)$ depends only on x.
NOTE	The tangent lies in the plane $y = y_0$.

S002·7 · V002 00:07:45 · T002 00:06:10

E9.2 · ∂f/∂y as an ordinary derivative ESA ★★

$$\left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} = \left.\frac{d}{dy}\big(f(x_0, y)\big)\right|_{y=y_0}$$

GIVES	Slope of the tangent to the surface that is parallel to the y-axis.
APPLIES	Fix $x = x_0$ first; $f(x_0, y)$ "just depends on y".
NOTE	The tangent lies in the plane $x = x_0$.

S002·7 · V002 00:07:45 · T002 00:07:10

E9.3 · The two tangent lines at P

ESA ★

$$x\text{-direction: } z - z_0 = f_x(x_0, y_0)\,(x - x_0),\quad y = y_0$$
$$y\text{-direction: } z - z_0 = f_y(x_0, y_0)\,(y - y_0),\quad x = x_0$$

GIVES Equations of the tangents whose slopes are f_x and f_y.

APPLIES $z_0 = f(x_0, y_0)$.

NOTE Point-slope form inside each plane section.

ADDED FOR CLARITY S002·7

Graphs and diagrams

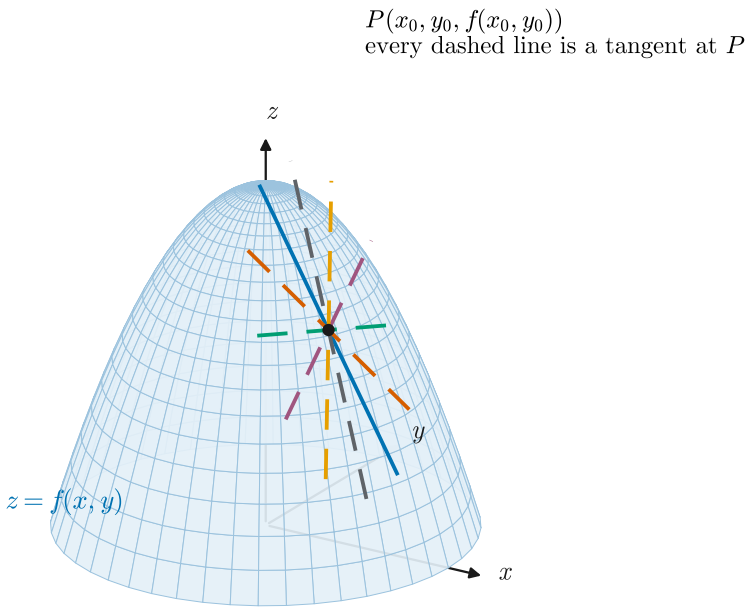


Figure G9.1. A surface has infinitely many tangents at a point. ESA ★★

S002·6 · T002 00:04:36 · T002 00:05:24

How to read it. The dome is $z = 4 - x^2 - y^2$ and $P = (1, -0.5, 2.75)$. Each coloured line is a tangent to the surface at P in a different direction. Partial differentiation picks out two of them: the ones parallel to the x- and y-axes.

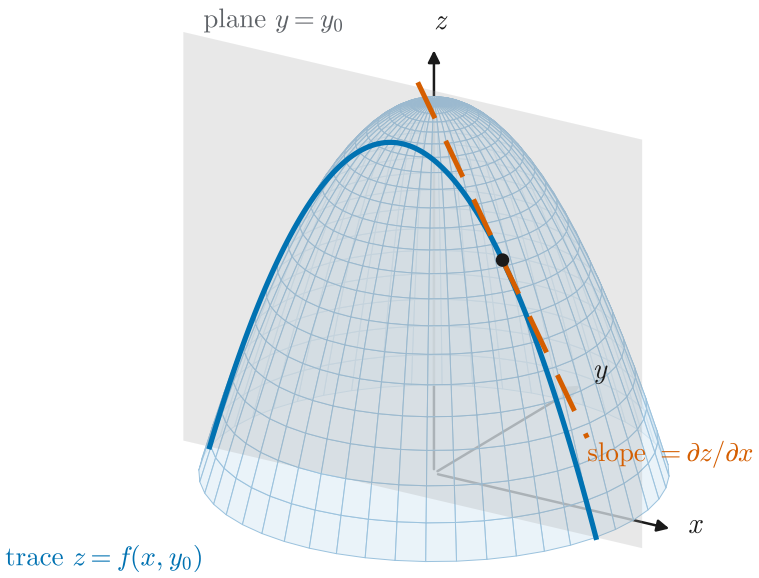


Figure G9.2. $\partial z/\partial x$: the tangent parallel to the x-axis. ESA ★★

S002·7 · V002 00:07:45 · T002 00:06:54

How to read it. The grey plane $y = y_0$ cuts the dome in the blue trace $z = f(x, y_0)$. The dashed tangent to that trace at P is parallel to the x-axis, and its slope is $\partial z/\partial x$ at (x_0, y_0).

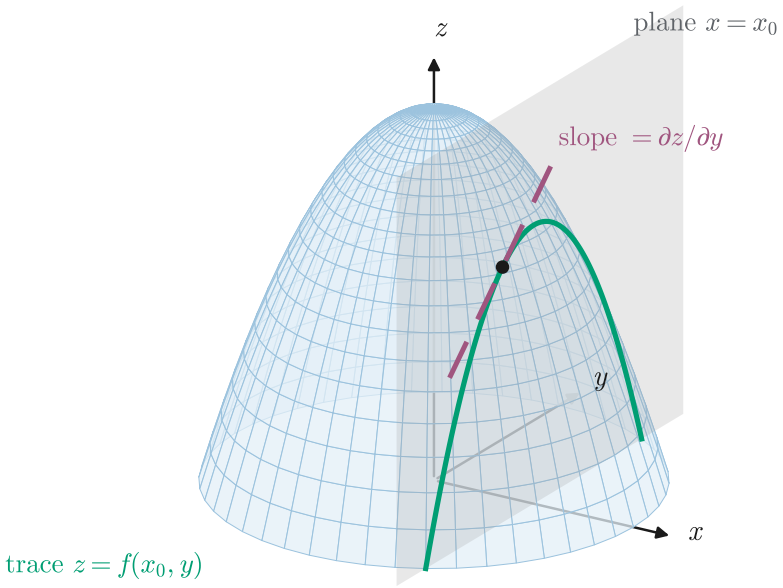


Figure G9.3. $\partial z/\partial y$: the tangent parallel to the y-axis. ESA ★★

S002·7 · V002 00:07:45 · T002 00:07:10

How to read it. The plane $x = x_0$ cuts the dome in the green trace $z = f(x_0, y)$, which "just depends on y". The dashed tangent at P is parallel to the y-axis, and its slope is $\partial z/\partial y$ at (x_0, y_0).

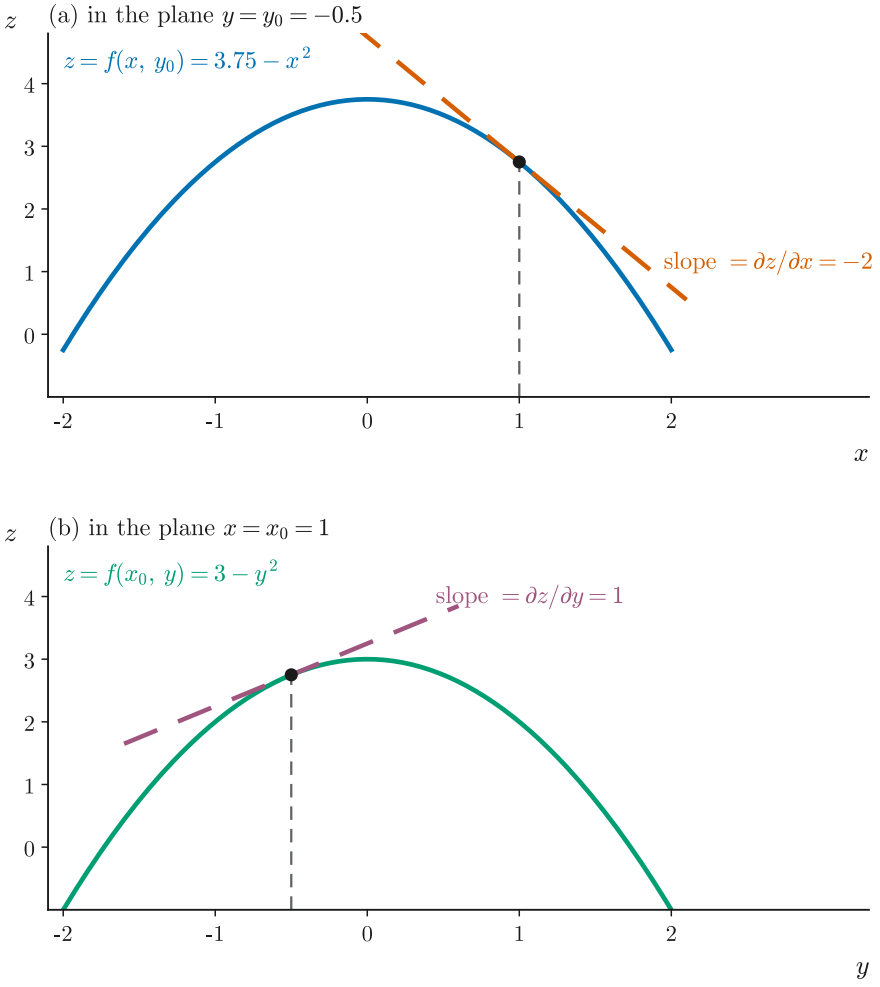
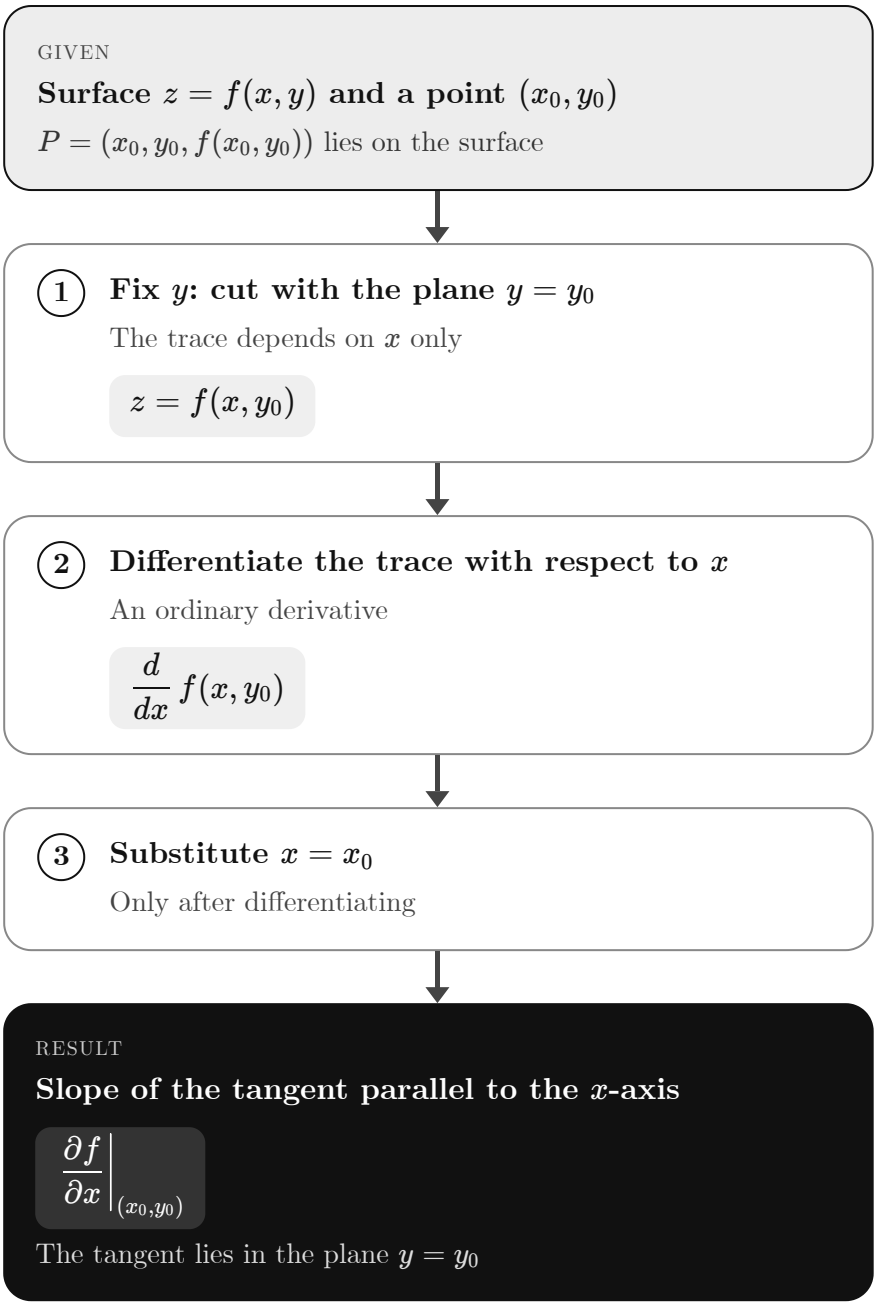


Figure G9.4. The two traces seen flat, with numerical slopes. ESA ★★

ADDED FOR CLARITY S002·7

How to read it. These are panels (a) and (b) from Figures G9.2 and G9.3, drawn in their own planes. At P, the tangent in (a) has slope $-2 = f_x(1, -0.5)$ and the tangent in (b) has slope $1 = f_y(1, -0.5)$. See worked example W9.1.

Flowcharts



Remark. For $\partial f/\partial y$, swap the roles: cut with $x = x_0$, differentiate $f(x_0, y)$ in y, then put $y = y_0$.

Figure F9.1. Partial derivative as the slope of a tangent, step by step. ESA ★★

S002·6 · S002·7 · V002 00:07:45 · T002 00:06:10

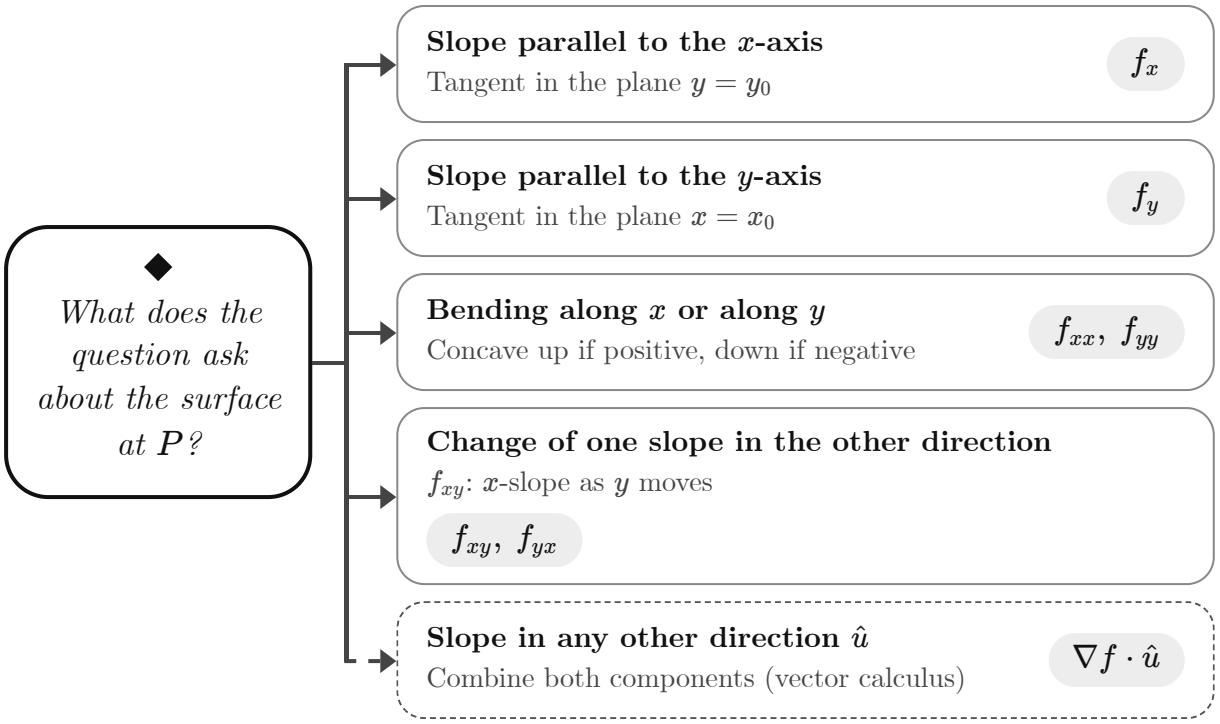


Figure F9.2. Which derivative answers which geometric question. ESA ★★

ADDED FOR CLARITY S002·6 · S002·8 · S002·9 · T002 00:07:54

Important points and why

1. A surface has infinitely many tangents at a given point, one for every direction. ESA ★★ T002 00:05:24 · T002 00:05:35
2. z_x at (x_0, y_0) is the slope of the tangent to the surface at that point which is parallel to the x-axis. ESA ★★ T002 00:06:10 · T002 00:06:27
3. z_y at (x_0, y_0) is the slope of the tangent parallel to the y-axis. ESA ★★ T002 00:07:10 · T002 00:07:54
4. To find $\partial f/\partial x$ at (x_0, y_0): put $y = y_0$, which gives a curve $f(x, y_0)$ in one variable, then differentiate it at $x = x_0$. ESA ★★ S002·7
5. The cutting plane $y = y_0$ slices the surface in the curve whose slope is f_x; the plane $x = x_0$ gives the curve whose slope is f_y. ESA ★★ S002·7 · V002 00:07:45

Why are there infinitely many tangents at one point of a surface?
From the point you can move in infinitely many directions in the xy-plane. Each direction traces a different curve on the surface, and each curve has its own tangent at P.
T002 00:04:36 · T002 00:05:24

Why does fixing $y = y_0$ turn $\partial f/\partial x$ into an ordinary derivative?
Once $y = y_0$, $f(x, y_0)$ depends on x alone. It is a plane curve in the plane $y = y_0$, and its ordinary derivative at x_0 is the slope of its tangent.

S002·7

Exam trap. "Parallel to the x-axis" describes the tangent's direction seen from above (it lies in the plane $y = y_0$). It does not mean the tangent is horizontal: its slope is $f_x(x_0, y_0)$, which is usually not zero. ADDED FOR CLARITY

Exam trap. Hold the other variable constant before differentiating. Do not substitute $x = x_0$ before differentiating with respect to x. ADDED FOR CLARITY

Worked examples

W9.1 · Slopes of the two tangents on a dome

ESA ★★

GIVEN $f(x, y) = 4 - x^2 - y^2$ (the dome in Figures G9.1–G9.4), point $(x_0, y_0) = (1, -0.5)$.

STEP 1 Height: $z_0 = 4 - 1 - 0.25 = 2.75$, so $P = (1, -0.5, 2.75)$.

STEP 2 Fix $y = -0.5$: $f(x, -0.5) = 3.75 - x^2$, so $\dfrac{d}{dx}(3.75 - x^2) = -2x = -2$ at $x = 1$.

STEP 3 Fix $x = 1$: $f(1, y) = 3 - y^2$, so $\dfrac{d}{dy}(3 - y^2) = -2y = 1$ at $y = -0.5$.

STEP 4 Read: moving parallel to the x-axis the surface falls 2 units per unit step; parallel to the y-axis it rises 1.

ANSWER $\boxed{f_x(1, -0.5) = -2, \quad f_y(1, -0.5) = 1}$

CHECK Directly: $f_x = -2x$ and $f_y = -2y$ give the same values.

S002·6 · S002·7

SHORT-NOTE RECAP

A surface has infinitely many tangents at a point $P(x_0, y_0, f(x_0, y_0))$. $\partial f/\partial x$ at (x_0, y_0) is the slope of the tangent parallel to the x-axis. It is found by cutting the surface with the plane $y = y_0$ and differentiating the curve $f(x, y_0)$ at x_0. Likewise $\partial f/\partial y$ is the slope of the tangent parallel to the y-axis, from the curve $f(x_0, y)$ in the plane $x = x_0$.

10 Slope in any other direction: the gradient

A tangent that is parallel to neither axis is handled by resolving the direction into x- and y-components. Treating f_x and f_y as vector components gives their resultant, the gradient ∇f. It is studied fully in vector calculus.

Key terms and definitions

Gradient. The vector $\nabla f = f_x\,\mathbf{i} + f_y\,\mathbf{j}$: the resultant of the two slope components along the x- and y-axes. S002·8 · T002 00:09:23

Directional derivative. The slope of the surface in the direction of a unit vector $\hat{u}$: $D_{\hat{u}} f = \nabla f \cdot \hat{u}$. It is studied in vector calculus. ADDED FOR CLARITY T002 00:09:54

> *"Slope of tangent along any arbitrary direction (Gradient of a function). What about other directions than parallel to x and y?"* S002·8

Formulas

E10.1 · Gradient ESA ★

$$\nabla f = f_x\,\mathbf{i} + f_y\,\mathbf{j}$$

GIVES	Vector whose components are the slopes along the x- and y-axes.
APPLIES	Taken at the point (x_0, y_0).
NOTE	Covered fully in vector calculus.

S002·8 · T002 00:09:23

E10.2 · Slope along any direction (directional derivative) ESA ★

$$D_{\hat{u}} f = \nabla f \cdot \hat{u} = a\,f_x + b\,f_y, \qquad \hat{u} = (a, b),\ a^2 + b^2 = 1$$

GIVES	Slope of the tangent in the direction $\hat{u}$.
APPLIES	$\hat{u}$ must be a unit vector.
NOTE	$\hat{u} = \mathbf{i}$ gives f_x; $\hat{u} = \mathbf{j}$ gives f_y. The largest slope is $\lvert\nabla f\rvert$.

ADDED FOR CLARITY S002·8

Graphs and diagrams

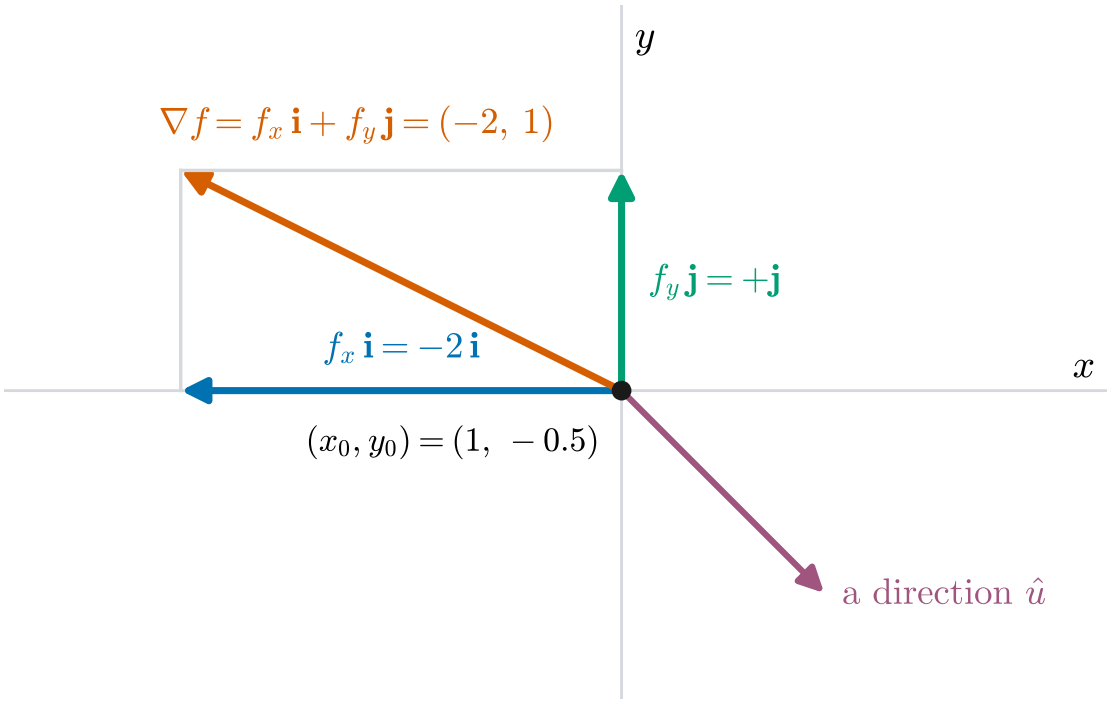
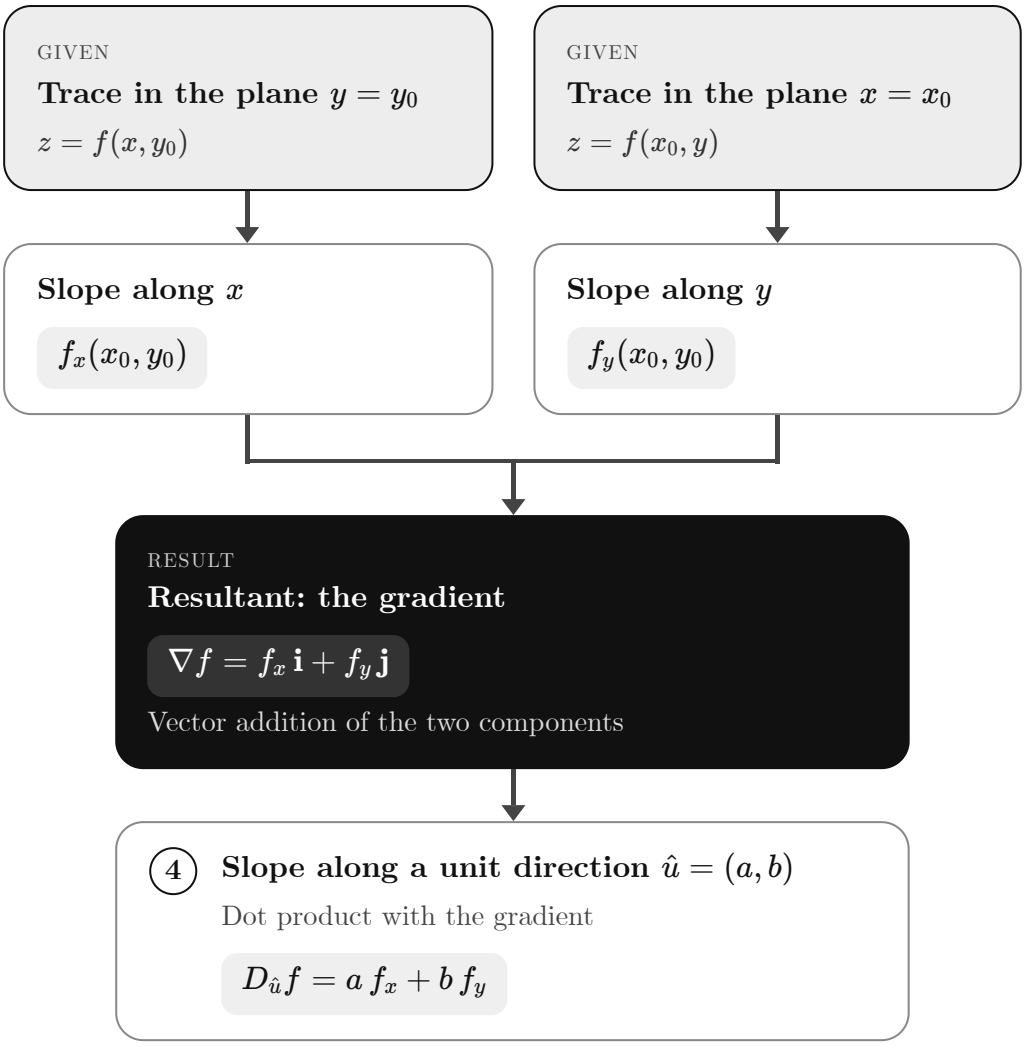


Figure G10.1. Resolving into components: the gradient. ESA ★

S002·8 · T002 00:09:06 · V002 00:09:05

How to read it. The slope components $f_x\,\mathbf{i}$ (blue) and $f_y\,\mathbf{j}$ (green) add by vector addition to give the resultant ∇f (vermilion), drawn here for the dome at $(1, -0.5)$. The purple arrow is another direction $\hat{u}$; the slope along it is $\nabla f \cdot \hat{u}$ (W10.1).

Flowcharts



Remark. Check $\hat{u} = (1, 0)$: you get back f_x. The steepest slope is $|\nabla f| = \sqrt{f_x^2 + f_y^2}$. Example: the dome at $(1, -0.5)$ gives $\nabla f = (-2, 1)$ (W10.1).

Figure F10.1. Two slopes combine into the gradient. ESA ★

ADDED FOR CLARITY S002·8 · T002 00:09:06 · T002 00:09:23

Important points and why

1. f_x and f_y only give slopes in two directions; most tangents are parallel to neither axis. ESA ★ S002·8 · T002 00:08:22
2. Any line in the plane can be resolved into components along the x- and y-axes; vector addition of the components gives the resultant. ESA ★ T002 00:09:06 · T002 00:09:23
3. The resultant of $f_x\,\mathbf{i}$ and $f_y\,\mathbf{j}$ is the gradient ∇f, studied in vector calculus. ESA ★ S002·8 · T002 00:09:44
4. The slope in a unit direction $\hat{u} = (a, b)$ is $af_x + bf_y$. It is greatest, equal to $|\nabla f|$, in the direction of ∇f. ESA ★ ADDED FOR CLARITY S002·8

Why can't f_x or f_y alone give the slope along a diagonal direction?
Each is the slope along one axis only. A diagonal direction has both an x- and a y-component, so both slopes contribute. Combining them gives the gradient, and its dot product with the unit direction gives the slope.

S002·8 · T002 00:09:06

Worked examples

W10.1 · Gradient and the slope along a diagonal

ESA ★

given Same dome and point as W9.1; direction $\hat{u} = \frac{1}{\sqrt{2}}(1, -1)$.

step 1 $\nabla f = f_x\,\mathbf{i} + f_y\,\mathbf{j} = -2\,\mathbf{i} + \mathbf{j}$.

step 2 $D_{\hat{u}} f = \frac{1}{\sqrt{2}}(-2) + \left(-\frac{1}{\sqrt{2}}\right)(1) = -\frac{3}{\sqrt{2}}$.

step 3 Steepest slope: $|\nabla f| = \sqrt{4+1} = \sqrt{5}$, in the direction of ∇f.

answer $\boxed{D_{\hat{u}} f = -3/\sqrt{2} \approx -2.121, \quad |\nabla f| = \sqrt{5} \approx 2.236}$

check $|D_{\hat{u}} f| \le |\nabla f|$, as it must be.

S002·8

short-note recap

Tangents in other directions are handled by resolving the direction into x- and y-components. The resultant of $f_x\,\mathbf{i}$ and $f_y\,\mathbf{j}$ is the gradient ∇f. The slope along a unit vector $\hat{u}$ is $\nabla f \cdot \hat{u}$, and the largest slope is $|\nabla f|$.

11 Second-order partials: concavity and mixed partials

As in one variable, the second derivative describes bending. $f_{xx} > 0$ means the surface is concave up in the x direction, and $f_{yy} > 0$ means concave up in the y direction. The mixed partial f_{xy} measures how the x-slope changes as you move in the y direction.

Key terms and definitions

Concave up in the x direction. The trace in the x direction bends upward; the x-slope increases as x increases. Indicated by $f_{xx} > 0$. S002·9

Mixed partial f_{xy}. Tells how the rate of change of $f(x, y)$ in the x direction is changing as we move in the y direction. S002·9

> *"If $f_{xx} > 0$ then f(x,y) is concave up in the x direction. If $f_{yy} > 0$ then f(x,y) is concave up in the y direction. Mixed partials tell us how a partial in one variable is changing in the direction of the other. f_{xy} tells us how the rate of change of f(x,y) in the x direction is changing as we move in the y direction."* S002·9

Formulas

E11.1 · Concavity from second partials ESA ★★

$$f_{xx} > 0 \Rightarrow \text{concave up in } x, \qquad f_{yy} > 0 \Rightarrow \text{concave up in } y$$

gives	Shape of the surface along each axis direction.
applies	$f_{xx} < 0$ (or $f_{yy} < 0$) gives concave down in that direction.
note	Same test as d^2y/dx^2 for one variable.

S002·9 · T002 00:11:19

E11.2 · Mixed partial as a change of slope ESA ★★

$$f_{xy} = \frac{\partial}{\partial y}(f_x)$$

gives	How the x-slope f_x changes as you move in the y direction.
applies	Likewise $f_{yx} = \partial(f_y)/\partial x$: the change of the y-slope along x.

S002·9 · T002 00:11:45

E11.3 · Names of higher derivatives (motion) ESA ★

$$\frac{dy}{dt} = \text{velocity}, \quad \frac{d^2y}{dt^2} = \text{acceleration}, \quad \frac{d^3y}{dt^3} = \text{jerk}$$

GIVES Physical meaning when the independent variable is time.

APPLIES One-variable analogy only.

NOTE Names exist up to about the 6th derivative.

TRANSCRIPT ONLY T002 00:10:23 · T002 00:10:57

Graphs and diagrams

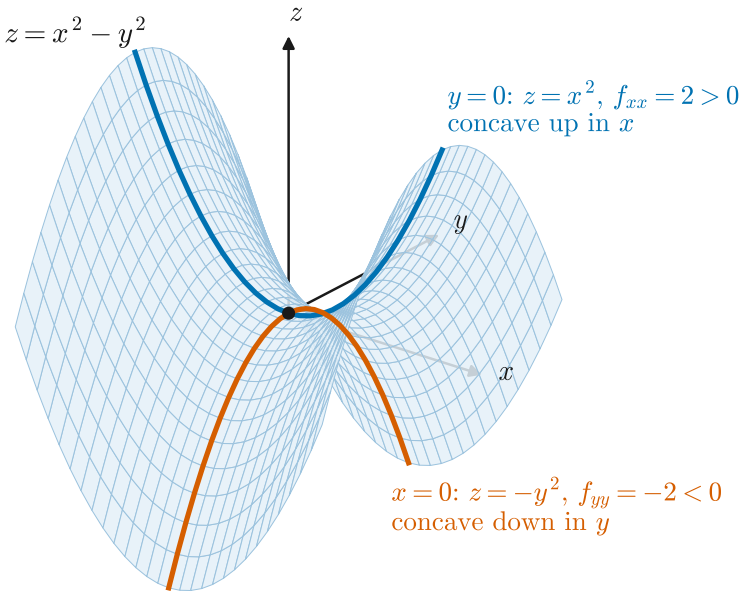


Figure G11.1. Concave up in x, concave down in y: the saddle z = x² − y². ESA ★★

ADDED FOR CLARITY S002·9 · S002·10 · T002 00:11:19

How to read it. Along the x direction (blue trace, $y = 0$) the surface bends up, so $f_{xx} = 2 > 0$. Along the y direction (vermilion trace, $x = 0$) it bends down, so $f_{yy} = -2 < 0$. One point can be concave up in one direction and concave down in another.

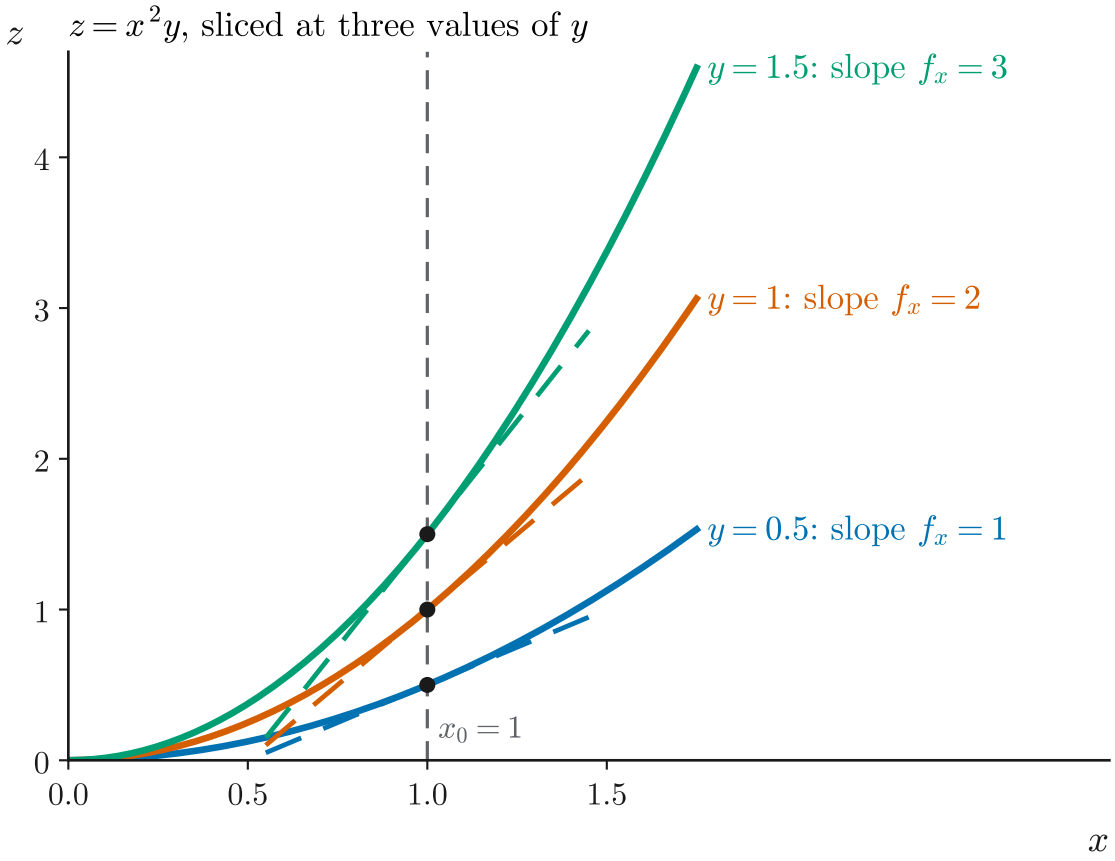
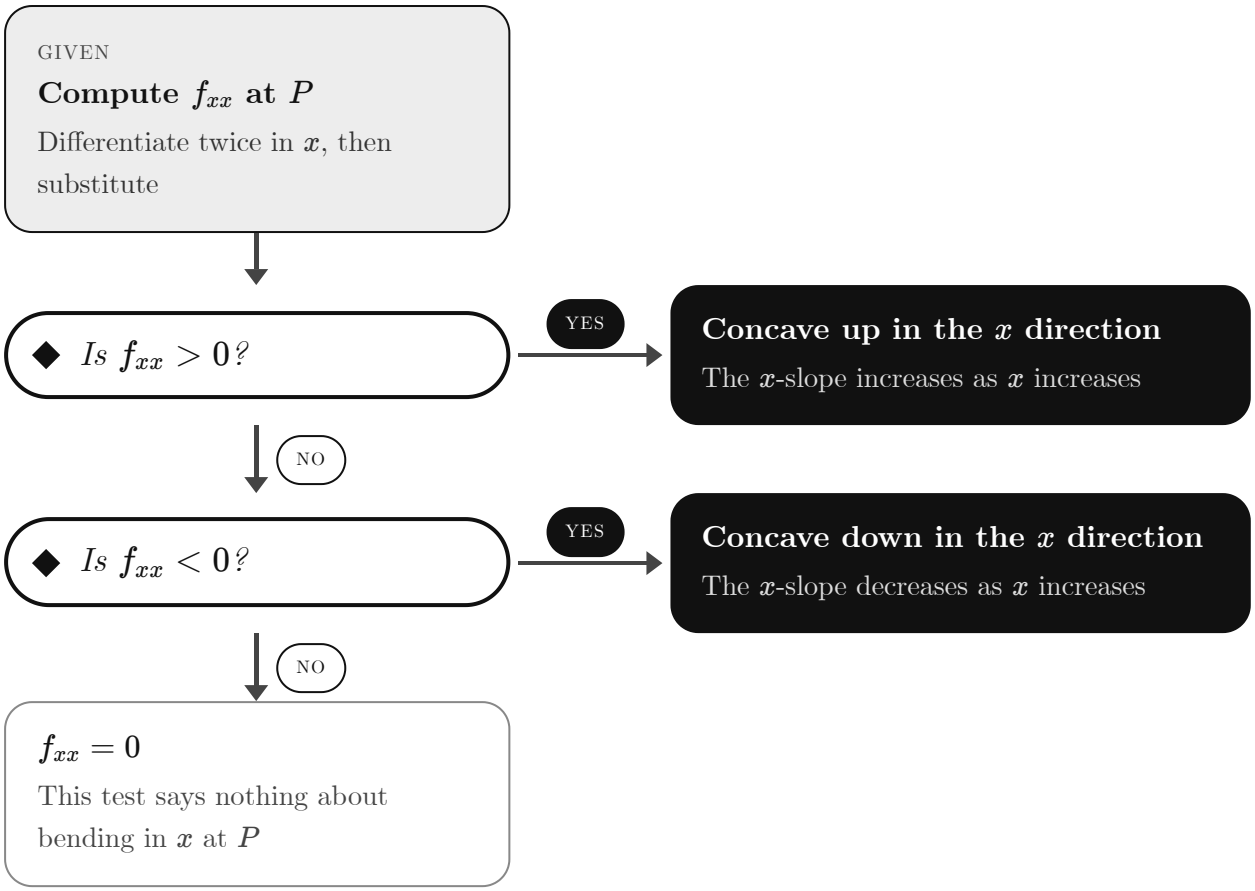


Figure G11.2. Mixed partial: how the x-slope changes as y changes. ESA ★★

ADDED FOR CLARITY S002·9 · T002 00:11:45

How to read it. Three traces of $z = x^2 y$, at $y = 0.5$, 1 and 1.5. At $x_0 = 1$ their tangents have slopes $f_x = 1, 2, 3$. The x-slope grows by 2 per unit of y, which is $f_{xy} = 2$.

Flowcharts



Remark. Run the same ladder with f_{yy} for the y direction. Read the sign of f_{xy} in the same way: positive means the x-slope increases as y increases (Figure G11.2).

Figure F11.1. Reading concavity from a second-order partial. ESA ★★

ADDED FOR CLARITY S002·9 · T002 00:11:19 · T002 00:11:38

One variable vs two variables

	One variable, $y = f(x)$	**Two variables, $z = f(x, y)$**
Graph	Curve in a plane	Surface in 3-D
Tangents at a point	One	Infinitely many, one per direction
First derivative	dy/dx: slope of the tangent	f_x, f_y: slopes parallel to the x- and y-axes; ∇f for other directions
Second derivative	d^2y/dx^2: concave up or down	f_{xx}, f_{yy}: concavity in x, y; f_{xy}, f_{yx}: change of one slope along the other direction
Number of order n	1	2^n (or m^n for m variables)

ADDED FOR CLARITY S002·4 · S002·6 · S002·9 · S002·3

Important points and why

1. As in one variable (concave up or down from d^2y/dx^2), the second-order partials describe how the surface bends. ESA ★★ T002 00:11:03 · T002 00:11:19
2. $f_{xx} > 0$ means concave up in the x direction; $f_{yy} > 0$ means concave up in the y direction. ESA ★★ S002·9 · T002 00:11:38
3. $f_{xx} < 0$ (or $f_{yy} < 0$) means concave down in that direction. One surface can be up in x and down in y, as in a saddle. ESA ★★ ADDED FOR CLARITY S002·9
4. f_{xy} is the change in f_x along the y direction; f_{yx} is the change in f_y along the x direction. ESA ★★ S002·9 · T002 00:11:45 · T002 00:12:04
5. The class demo, "Second-Order Partial Derivatives", lets you set the signs of f_{xx}, f_{yy} and f_{xy} and watch the red tangent line move in 3D and contour views. ESA ★ S002·10 · V002 00:12:40 · T002 00:12:40

Why does $f_{xx} > 0$ mean concave up in x?
f_{xx} is the rate of change of the x-slope as x increases. If it is positive, the tangent keeps tilting upward as you move along x, so the trace bends upward.
S002·9 · T002 00:11:19

Why do the 4 second-order partials usually give only 3 different values?
For the smooth functions used in this unit the mixed partials are equal, $f_{xy} = f_{yx}$ (Clairaut), leaving only f_{xx}, f_{yy} and f_{xy}.
ADDED FOR CLARITY T002 00:00:24

Worked examples

W11.1 · Concave up in x, concave down in y

ESA ★★

given $f(x, y) = x^2 - y^2$ at the origin (Figure G11.1).

step 1 $f_x = 2x$, $f_{xx} = 2 > 0$: concave up in the x direction.

step 2 $f_y = -2y$, $f_{yy} = -2 < 0$: concave down in the y direction.

step 3 $f_{xy} = \partial(2x)/\partial y = 0$: the x-slope does not change as y changes.

answer Saddle: up along x, down along y

check Traces $z = x^2$ (at $y = 0$) and $z = -y^2$ (at $x = 0$) agree.

S002·9

W11.2 · Reading a mixed partial

ESA ★★

given $f(x, y) = x^2 y$ at $x_0 = 1$ (Figure G11.2).

step 1 $f_x = 2xy$, so at $x = 1$ the x-slope is $2y$: 1, 2, 3 for $y = 0.5, 1, 1.5$.

step 2 $f_{xy} = \partial(2xy)/\partial y = 2x = 2$ at $x = 1$.

step 3 Each unit step in y raises the x-slope by 2, which matches the slopes $1 \to 2 \to 3$ for steps of 0.5.

answer $f_{xy}(1, y) = 2$

check $f_{yx} = \partial(x^2)/\partial x = 2x = 2$ as well.

S002·9

short-note recap

Second-order partials describe bending. If $f_{xx} > 0$ the surface is concave up in the x direction; if $f_{yy} > 0$ it is concave up in the y direction (negative means concave down). The mixed partial f_{xy} tells how the rate of change of f in the x direction changes as we move in the y direction.

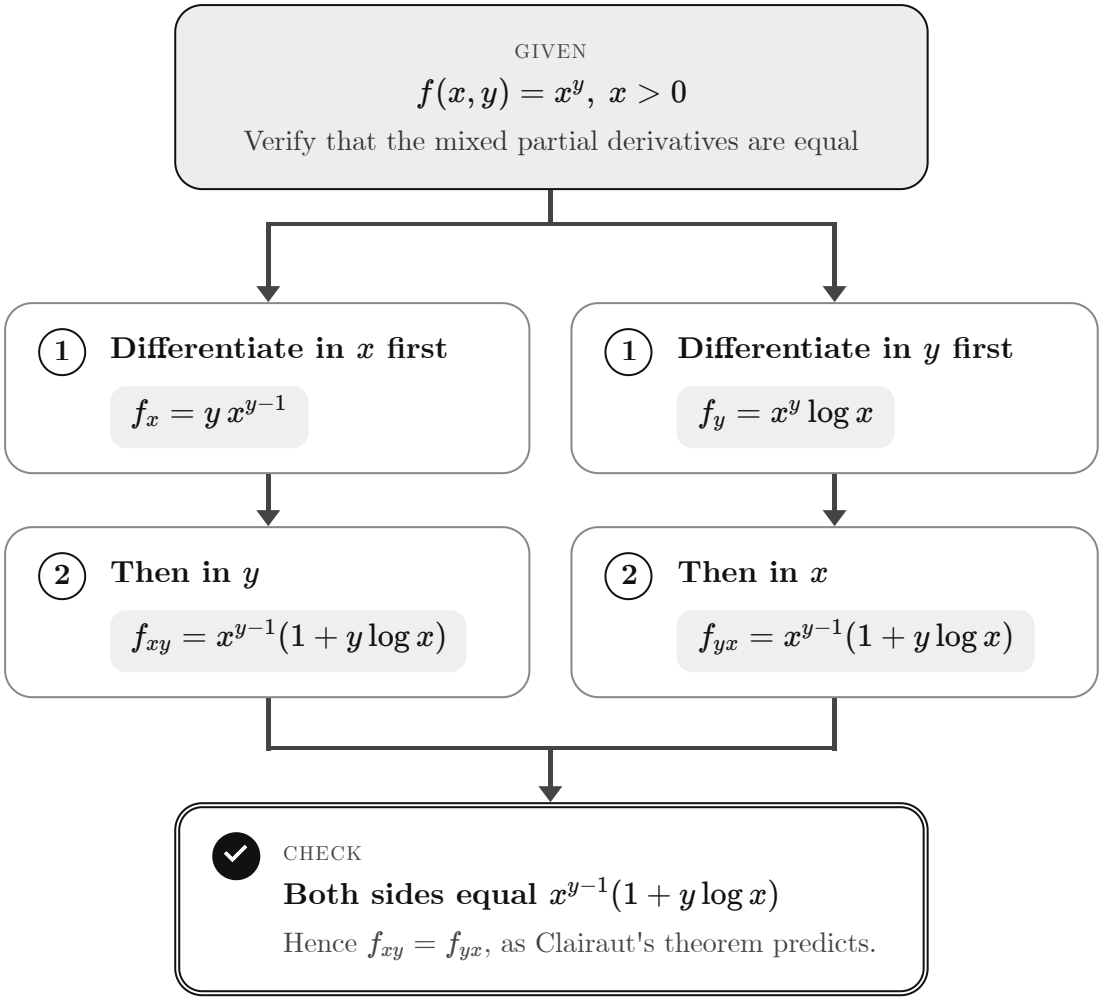
12 Problems on partial derivatives

The deck's title includes “Problems on partial derivatives”, but no problem slides are in it, and the transcript defers the problems to the next class. The problems here come from the Unit 1 class-work sheet and past ESA papers that test exactly this session's content: first-order partials at a point, higher-order partials, and equality of mixed partials.

Key terms and definitions

Clairaut's theorem. If f_{xy} and f_{yx} are continuous near a point, they are equal there. The class plans to verify it in the next session. TRANSCRIPT ONLY ADDED FOR CLARITY T002 00:14:59

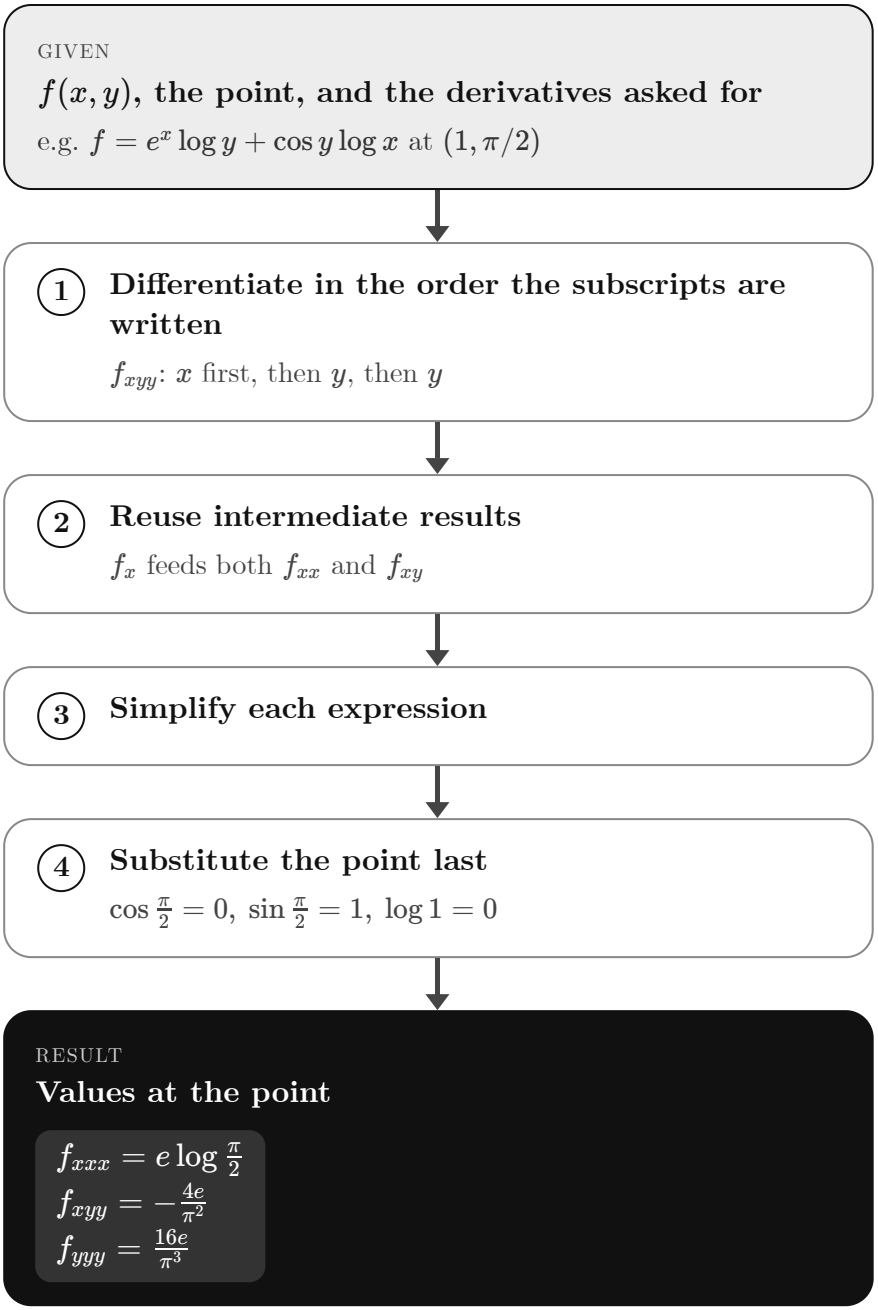
Flowcharts



Exam trap. In x, x^y is a power; in y, it is an exponential. Using the wrong rule is the usual slip.

Figure F12.1. Verifying equal mixed partial derivatives. ESA ★★

CW U1 P2 · T002 00:14:59



Remark. log is the natural logarithm, so $\frac{d}{dy}\log y = \frac{1}{y}$. Check one value against the CW answer key if you can.

Figure F12.2. Evaluating higher-order partials at a point (ESA Dec 2023 Q1.a). ESA ★★

ESA Dec-23 1a · CW U1 P5 · S002·3

Important points and why

1. Evaluate at the point only after all the differentiations are done. ESA ★★ ADDED FOR CLARITY CW U1 P1 · ESA Dec-23 1a
2. To verify $f_{xy} = f_{yx}$, work out each side separately and compare them. ESA ★★ CW U1 P2

Exam trap. In these problems $\log$ is the natural logarithm: $\frac{d}{dy}\log y = \frac{1}{y}$. ADDED FOR CLARITY

Exam trap. For x^y: with respect to x it is a power, $y\,x^{y-1}$; with respect to y it is an exponential, $x^y \log x$. ADDED FOR CLARITY

Worked examples

W12.1 · First-order partials at a point

ESA ★★

GIVEN $f(x,y) = x^4 - x^2y^2 + y^4$ at $(-1,1)$.

STEP 1 $f_x = 4x^3 - 2xy^2 = -4 + 2 = -2$.

STEP 2 $f_y = -2x^2y + 4y^3 = -2 + 4 = 2$.

STEP 3 Geometrically, the tangents parallel to the x- and y-axes at $(-1,1,1)$ have slopes -2 and 2.

ANSWER $\boxed{f_x(-1,1) = -2, \quad f_y(-1,1) = 2}$

CHECK Matches the answer printed on the CW sheet.

CW U1 P1(i)

W12.2 · First-order partials of a product

ESA ★★

GIVEN $f(x,y) = x^2 e^{-y/x}$ at $(4,2)$.

STEP 1 $f_x = 2x\,e^{-y/x} + x^2 e^{-y/x}\cdot\dfrac{y}{x^2} = (2x + y)\,e^{-y/x} = 10\,e^{-1/2}$.

STEP 2 $f_y = x^2 e^{-y/x}\cdot\left(-\dfrac{1}{x}\right) = -x\,e^{-y/x} = -4\,e^{-1/2}$.

ANSWER $\boxed{f_x = 10e^{-1/2}, \quad f_y = -4e^{-1/2}}$

CHECK Matches the CW answer key.

CW U1 P1(ii)

W12.3 · Verify that the mixed partials are equal ESA ★★

GIVEN $f(x,y) = x^{y}$, $x > 0$.

STEP 1 $f_x = y\,x^{y-1}$ (power rule); $f_y = x^{y}\log x$ (exponential rule).

STEP 2 $f_{xy} = \dfrac{\partial}{\partial y}\left(y\,x^{y-1}\right) = x^{y-1} + y\,x^{y-1}\log x = x^{y-1}(1 + y\log x)$.

STEP 3 $f_{yx} = \dfrac{\partial}{\partial x}\left(x^{y}\log x\right) = y\,x^{y-1}\log x + x^{y}\cdot\dfrac{1}{x} = x^{y-1}(1 + y\log x)$.

ANSWER $\boxed{f_{xy} = f_{yx} = x^{y-1}(1 + y\log x)}$

CHECK Both sides are identical, as Clairaut's theorem predicts.

CW U1 P2

W12.4 · Third-order partials at a point ESA ★★

GIVEN $f(x,y) = e^{x}\log y + \cos y\log x$ at $(1, \pi/2)$. Find f_{xxx}, f_{xyy}, f_{yyy}.

STEP 1 $f_x = e^{x}\log y + \dfrac{\cos y}{x}$, $f_{xx} = e^{x}\log y - \dfrac{\cos y}{x^{2}}$, $f_{xxx} = e^{x}\log y + \dfrac{2\cos y}{x^{3}}$.

STEP 2 $f_{xy} = \dfrac{e^{x}}{y} - \dfrac{\sin y}{x}$, $f_{xyy} = -\dfrac{e^{x}}{y^{2}} - \dfrac{\cos y}{x}$.

STEP 3 $f_y = \dfrac{e^{x}}{y} - \sin y\log x$, $f_{yy} = -\dfrac{e^{x}}{y^{2}} - \cos y\log x$, $f_{yyy} = \dfrac{2e^{x}}{y^{3}} + \sin y\log x$.

STEP 4 At $(1, \pi/2)$: $\cos y = 0$, $\sin y = 1$, $\log 1 = 0$.

ANSWER $\boxed{f_{xxx} = e\log\dfrac{\pi}{2}, \quad f_{xyy} = -\dfrac{4e}{\pi^{2}}, \quad f_{yyy} = \dfrac{16e}{\pi^{3}}}$

CHECK Matches the CW key; it also gives $f_{xxy} = 1 + 2e/\pi$.

ESA Dec-23 1a · CW U1 P5

W12.5 · Value of n from a second-order equation ESA ★★

GIVEN $\theta = t^{n}e^{-r^{2}/4t}$. Find n so that $\dfrac{1}{r^{2}}\dfrac{\partial}{\partial r}\left(r^{2}\dfrac{\partial\theta}{\partial r}\right) = \dfrac{\partial\theta}{\partial t}$.

STEP 1 $\dfrac{\partial\theta}{\partial r} = -\dfrac{r}{2}\,t^{n-1}e^{-r^{2}/4t}$, so $r^{2}\dfrac{\partial\theta}{\partial r} = -\dfrac{r^{3}}{2}\,t^{n-1}e^{-r^{2}/4t}$.

STEP 2 $\dfrac{1}{r^{2}}\dfrac{\partial}{\partial r}(\cdot) = \left(-\dfrac{3}{2}\,t^{n-1} + \dfrac{r^{2}}{4}\,t^{n-2}\right)e^{-r^{2}/4t}$.

STEP 3 $\dfrac{\partial\theta}{\partial t} = \left(n\,t^{n-1} + \dfrac{r^{2}}{4}\,t^{n-2}\right)e^{-r^{2}/4t}$.

STEP 4 Compare the t^{n-1} terms: $n = -\tfrac{3}{2}$.

ANSWER $\boxed{n = -\dfrac{3}{2}}$

CHECK The $r^{2}t^{n-2}$ terms agree on both sides for every n.

ESA Jan-24 s4 1c

Past-paper questions

ESA Dec 2023 · 6 marks · UE22MA141A Q1.a · solved in W12.4

If $f(x,y) = e^x \log y + \cos y \log x$, find f_{xxx}, f_{xyy} and f_{yyy} at $(1, \pi/2)$.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q1.c · solved in W12.5

If $\theta = t^n e^{-r^2/4t}$, what value of n will make $\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \theta}{\partial r}\right) = \frac{\partial \theta}{\partial t}$?

short-note recap

To find partials at a point, differentiate fully, holding the other variable constant, and substitute only at the end. To verify $f_{xy} = f_{yx}$, compute each side separately and compare. Watch the rule used: x^y is a power in x but an exponential in y.

M3

Classes 003–004 · Total Derivative, Chain Rule and Implicit Functions

13 Composite functions and the one-variable chain rule

A composite function is a function of a function, such as $y = e^{\sin(x^2)}$. You differentiate it from the outside in: the outer function first, then the next one, then the innermost, and multiply. Every rule in this pack is this same idea extended to several variables.

Key terms and definitions

Composite function. A function of a function: $f(g(h(x)))$, or a function of several variables whose variables are themselves functions of other variables. S003·4 · T003a 00:00:59

Formulas

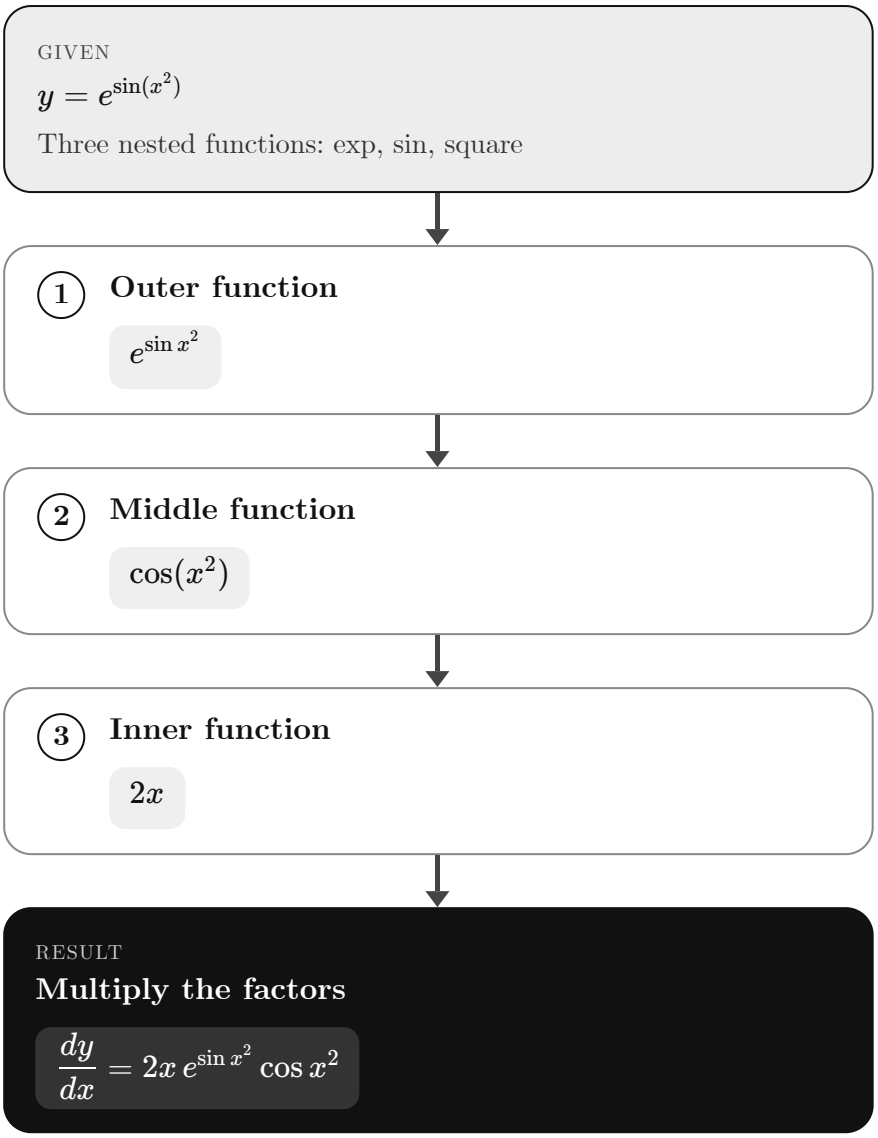
E13.1 · Chain rule, one variable ESA ★

$$\frac{d}{dx} f\big(g(h(x))\big) = f'\big(g(h(x))\big) \cdot g'\big(h(x)\big) \cdot h'(x)$$

gives	Derivative of a composition: outer to inner, multiplied.
applies	Each function differentiable.
note	Example: $\frac{d}{dx} e^{\sin x^2} = e^{\sin x^2} \cos x^2 \cdot 2x$.

S003·4 · T003a 00:00:33

Flowcharts



Remark. The multivariable rules in this pack work the same way, except that each intermediate variable adds its own term.

Figure F13.1. Differentiating a composite function from the outside in. ESA ★

S003·4 · T003a 00:00:33

Important points and why

1. For one variable you differentiate a composite function from the outside in: f first, then g, then h. ESA ★ T003a 00:01:19
2. For several variables the first question is always "differentiate with respect to what?", because there is more than one variable. ESA ★ T003a 00:01:45

Worked examples

W13.1 · Outside-in differentiation ESA ★

given $y = e^{\sin(x^2)}$.

step 1 Outer: $\dfrac{d}{d(\sin x^2)} e^{\sin x^2} = e^{\sin x^2}$.

step 2 Middle: $\dfrac{d}{d(x^2)} \sin(x^2) = \cos(x^2)$.

step 3 Inner: $\dfrac{d}{dx} x^2 = 2x$.

step 4 Multiply the three factors.

answer $\boxed{\dfrac{dy}{dx} = 2x\, e^{\sin x^2} \cos x^2}$

check sympy gives the same.

S003·4 · T003a 00:00:33

short-note recap

A composite function is a function of a function, e.g. $y = e^{\sin x^2}$. Its derivative is found from the outside in and the factors are multiplied: $e^{\sin x^2} \cdot \cos x^2 \cdot 2x$. For several variables the same idea gives the total derivative and the chain rule.

14 The total derivative rule

If $z = f(x, y)$ and both x and y are functions of one variable t, then z is ultimately a function of t alone. Its ordinary derivative dz/dt is the total derivative. It adds one term per path from z to t: $z_x\,dx/dt + z_y\,dy/dt$. A tree diagram shows the paths.

Key terms and definitions

Total derivative. If $z = f(x, y)$ where x and y are functions of one independent variable t, the ordinary derivative dz/dt is the total derivative of z with respect to t. S003·5 · T003b 00:00:52

Tree diagram. A diagram with z at the root, the intermediate variables in the middle and the final variable(s) at the leaves. Each branch is labelled with a derivative; multiply along each path and add over the paths. S003·5 · S003·14 · T003a 00:19:22 · T003a 00:25:02

> *"If $z = f(x, y)$ where x and y are functions of one independent variable 't' then we can find the derivative of z with respect t as* $\dfrac{dz}{dt} = \dfrac{\partial z}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial z}{\partial y}\dfrac{dy}{dt}$*."* S003·5

Formulas

E14.1 · Total derivative rule ESA ★★

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

gives	Rate of change of z with respect to the single variable t.
applies	$z = f(x, y)$, $x = x(t)$, $y = y(t)$.
note	Ordinary d on z, x, y with respect to t; partial ∂ only on z with respect to x, y.

S003·5 · T003a 00:05:54 · T003b 00:01:10

E14.2 · Total derivative, three variables ESA ★★

$$\frac{df}{dt} = f_x\frac{dx}{dt} + f_y\frac{dy}{dt} + f_z\frac{dz}{dt}$$

gives	One term per intermediate variable.
applies	$f = f(x, y, z)$, each of x, y, z a function of t.
note	Used in CW U1 P11 and in the triangle problem with x, y, θ.

added for clarity S003·9 · CW U1 P11

Graphs and diagrams

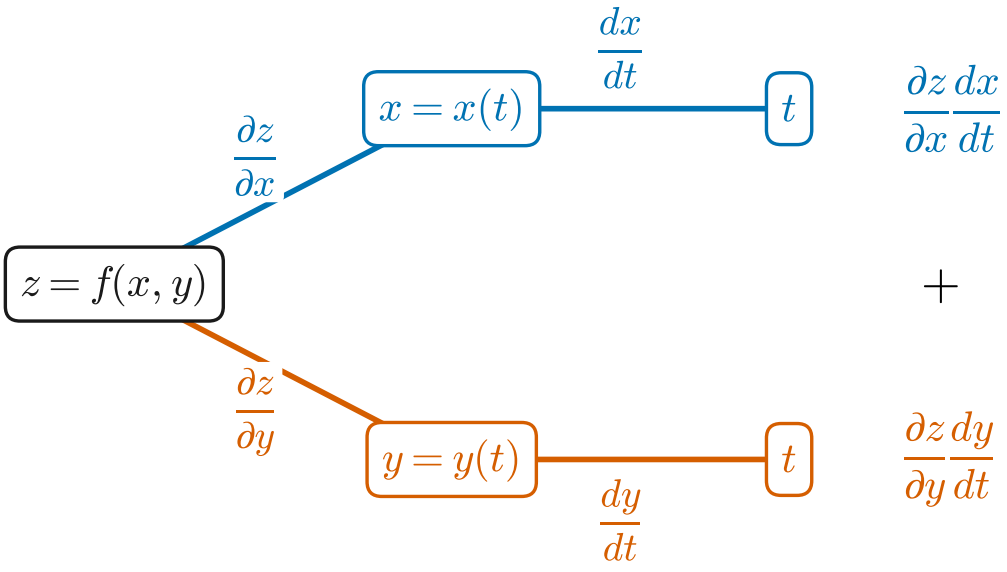


Figure G14.1. Tree diagram for the total derivative. ESA ★★

S003·5 · T003a 00:19:22

How to read it. Start at z. Each branch carries the derivative of the upper variable with respect to the lower one. Multiply along each path from z down to t, then add the two paths: $\dfrac{dz}{dt} = \dfrac{\partial z}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial z}{\partial y}\dfrac{dy}{dt}$. In an exam, draw it first so that no path is missed.

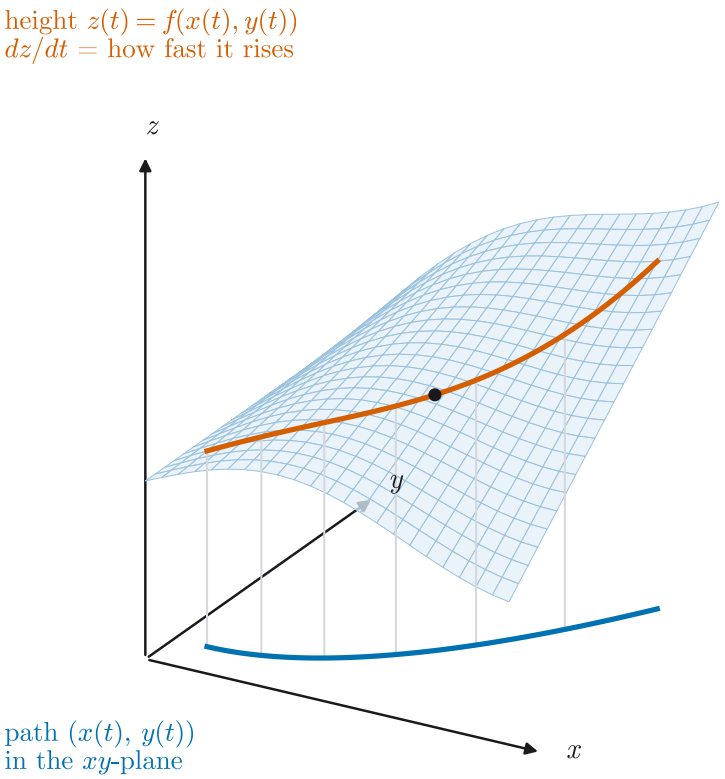
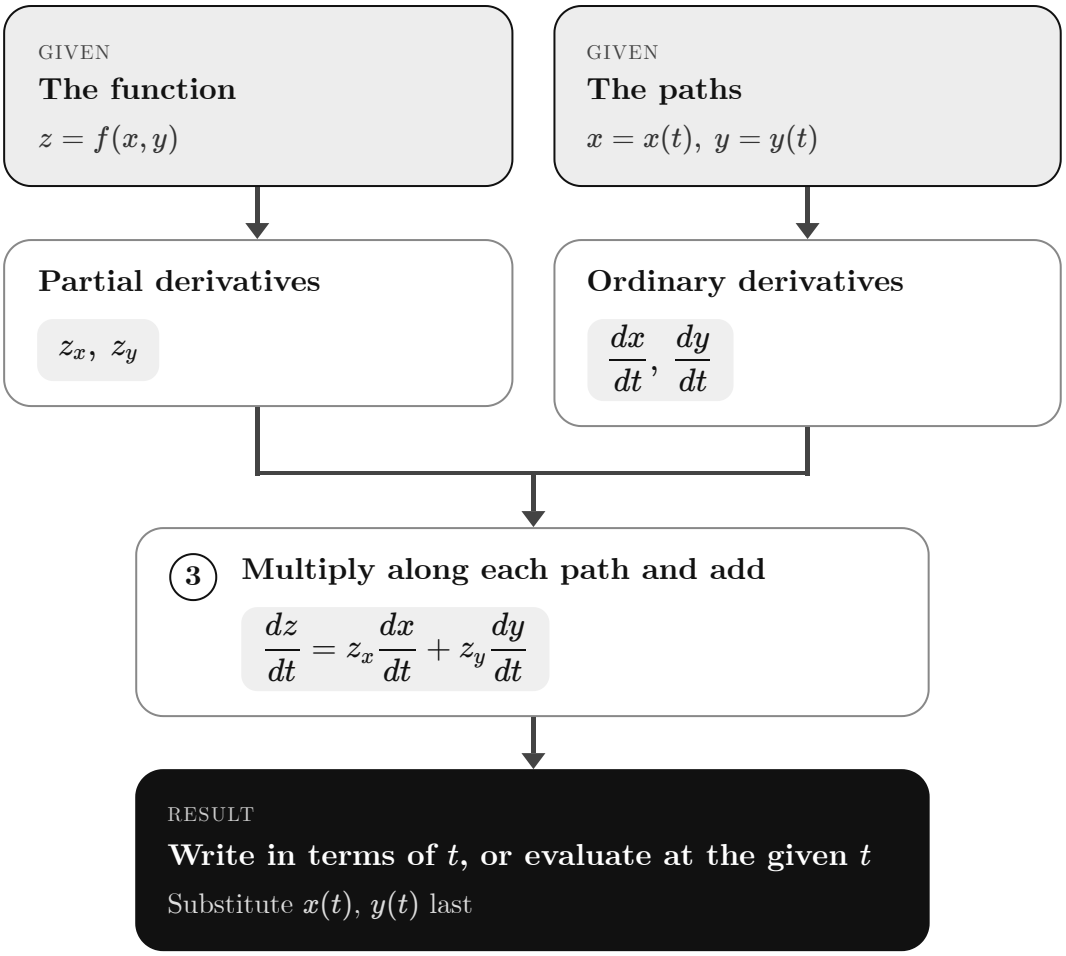


Figure G14.2. The total derivative as the slope along a path. ESA ★

ADDED FOR CLARITY S003·5 · T003a 00:05:20

How to read it. As t runs, the point $(x(t), y(t))$ moves along the blue path in the xy-plane. The vermilion curve is the surface height above it, $z(t) = f(x(t), y(t))$. dz/dt is how fast that height changes as t increases; it combines the slopes z_x and z_y weighted by how fast x and y move. The surface is an illustrative choice.

Flowcharts



Remark. Check on a small example by substituting first: both routes must agree. On slide 003·6 they do not, which exposes the slide's error (W14.1).

Figure F14.1. Total derivative in three moves. ESA ★★

S003·5 · S003·18 · T003b 00:01:10

Important points and why

1. You can always substitute $x(t)$ and $y(t)$ into z and differentiate directly, but this is often messy. The total derivative rule avoids substituting first. ESA ★★ T003a 00:02:39 · T003a 00:03:13
2. Both methods must give the same answer; substituting first is a good check on a short example. ESA ★★ (added for clarity) T003a 00:04:35
3. After using the rule, write the answer in terms of t alone (substitute $x(t)$, $y(t)$), or evaluate at the given t. ESA ★★ S003·19 · T003b 00:02:03
4. At a given t, a term vanishes if its dx/dt or dy/dt is 0 there (e.g. $dy/dt = \cos t = 0$ at $t = \pi/2$). ESA ★★ T003b 00:02:34

Why is it dz/dt (ordinary) and not $\partial z/\partial t$?
After substituting $x(t)$ and $y(t)$, z depends on the single variable t, so its derivative with respect to t is an ordinary derivative. The partial signs belong only to z with respect to x and y, where the other variable is held fixed.

S003·5 · T003a 00:02:23

Exam trap. Slide 003·6 has wrong numbers. For $z = x^2y^3$, $x = t^2$, $y = 2t^3$: $z_x = 2xy^3 = 16t^{11}$ (not $2t^{11}$), $z_y = 3x^2y^2 = 12t^{10}$ (not $9t^{10}$), and $dz/dt = 16t^{11}(2t) + 12t^{10}(6t^2) = 104t^{12}$ (not $60t^{12}$). Check: $z = 8t^{13}$. ADDED FOR CLARITY S003·6

Worked examples

W14.1 · Total derivative, both ways (slide 6, corrected) ESA ★★

GIVEN $z = x^2y^3$, $x = t^2$, $y = 2t^3$.

STEP 1 $z_x = 2xy^3 = 2t^2(8t^9) = 16t^{11}$, $z_y = 3x^2y^2 = 3t^4(4t^6) = 12t^{10}$.

STEP 2 $\dfrac{dx}{dt} = 2t$, $\dfrac{dy}{dt} = 6t^2$.

STEP 3 $\dfrac{dz}{dt} = 16t^{11}(2t) + 12t^{10}(6t^2) = 32t^{12} + 72t^{12}$.

ANSWER $\boxed{\dfrac{dz}{dt} = 104\,t^{12}}$

CHECK Substituting first: $z = t^4 \cdot 8t^9 = 8t^{13}$, so $dz/dt = 104t^{12}$. The slide's $60t^{12}$ is wrong.

ADDED FOR CLARITY S003·6

W14.2 · The lecture's version ESA ★

GIVEN $z = x^2y^3$, $x = t$, $y = 2t^2$.

STEP 1 Direct: $z = t^2(2t^2)^3 = 8t^8$, so $dz/dt = 64t^7$.

STEP 2 Rule: $z_x\,x' + z_y\,y' = 2xy^3(1) + 3x^2y^2(4t) = 16t^7 + 48t^7$.

ANSWER $\boxed{\dfrac{dz}{dt} = 64\,t^7}$

CHECK Both routes agree.

TRANSCRIPT ONLY T003a 00:02:05 · T003a 00:02:57

W14.3 · Total derivative at a point ESA ★★

GIVEN $u = x^2 y\,e^{xy}$, $x = t^2 + 1$, $y = \sin t$; find du/dt at $t = \pi/2$.

STEP 1 $u_x = y\,e^{xy}(2x + x^2 y)$, $u_y = x^2 e^{xy}(1 + xy)$; $\dfrac{dx}{dt} = 2t$, $\dfrac{dy}{dt} = \cos t$.

STEP 2 At $t = \pi/2$: $x = \frac{\pi^2}{4} + 1$, $y = 1$, $dx/dt = \pi$, $dy/dt = 0$, so the second term vanishes.

STEP 3 $\dfrac{du}{dt} = \pi\,e^{x}(2x + x^2)$ with $x = \frac{\pi^2}{4} + 1$.

ANSWER $\boxed{\pi\,e^{\frac{\pi^2}{4}+1}\left[2\left(\frac{\pi^2}{4}+1\right) + \left(\frac{\pi^2}{4}+1\right)^2\right]}$

CHECK Direct differentiation in sympy agrees.

S003·18 · S003·19 · T003b 00:00:23

W14.4 · Total derivative in terms of t ESA ★★

GIVEN $u = e^{xy} + \ln(y^2 + 1)$, $x = \tan t$, $y = \cos t$.

STEP 1 $u_x = y\,e^{xy}$, $u_y = x\,e^{xy} + \dfrac{2y}{y^2+1}$; $\dfrac{dx}{dt} = \sec^2 t$, $\dfrac{dy}{dt} = -\sin t$.

STEP 2 $\dfrac{du}{dt} = y\,e^{xy}\sec^2 t - \left(x\,e^{xy} + \dfrac{2y}{y^2+1}\right)\sin t$.

STEP 3 Substitute: $xy = \tan t \cos t = \sin t$ and $\cos t \sec^2 t - \tan t \sin t = \cos t$.

ANSWER $\boxed{\dfrac{du}{dt} = e^{\tan t \cos t}(\cos t \sec^2 t - \tan t \sin t) - \dfrac{2\cos t \sin t}{\cos^2 t + 1} = e^{\sin t}\cos t - \dfrac{\sin 2t}{1 + \cos^2 t}}$

CHECK The simplified form (added for clarity) equals the derivative of $e^{\sin t} + \ln(\cos^2 t + 1)$.

S003·20 · T003b 00:03:16

W14.5 · Total derivative at t = 0 ESA ★★

GIVEN $u = x^2 - y^2$, $x = e^t \cos t$, $y = e^t \sin t$; du/dt at $t = 0$.

STEP 1 $u_x = 2x$, $u_y = -2y$; $x' = e^t(\cos t - \sin t)$, $y' = e^t(\sin t + \cos t)$.

STEP 2 At $t = 0$: $x = 1$, $y = 0$, $x' = 1$, $y' = 1$.

STEP 3 $\dfrac{du}{dt} = 2(1)(1) - 2(0)(1)$.

ANSWER $\boxed{\left.\dfrac{du}{dt}\right|_{t=0} = 2}$

CHECK CW answer: 2.

CW U1 P10

W14.6 · Three intermediate variables ESA ★★

GIVEN $f = x^3 + xz^2 + y^3 + xyz$, $x = e^t$, $y = \cos t$, $z = t^3$; df/dt at $t = 0$.

STEP 1 At $t = 0$: $x = 1$, $y = 1$, $z = 0$; $x' = 1$, $y' = 0$, $z' = 0$.

STEP 2 $f_x = 3x^2 + z^2 + yz = 3$ (the y and z terms drop out because $y' = z' = 0$).

STEP 3 $\dfrac{df}{dt} = f_x\,x' + f_y\,y' + f_z\,z' = 3(1) + 0 + 0.$

ANSWER $\boxed{\left.\dfrac{df}{dt}\right|_{t=0} = 3}$

CHECK CW answer: 3.

CW U1 P11

SHORT-NOTE RECAP

If $z = f(x, y)$ and $x = x(t)$, $y = y(t)$, then z is a function of t alone and its total derivative is $\dfrac{dz}{dt} = \dfrac{\partial z}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial z}{\partial y}\dfrac{dy}{dt}$. A tree diagram with one branch per intermediate variable gives the terms. The answer is written in terms of t or evaluated at the given t.

15 The total differential: approximation and sensitivity

Without dividing by dt, the same rule gives the total differential $dz = f_x\,dx + f_y\,dy$. It approximates the change in z for small changes dx, dy, so it estimates function values near a known point. Comparing its coefficients shows which variable the output is most sensitive to. It also explains substitution in integrals.

Key terms and definitions

Total differential. For $z = f(x, y)$: $dz = f_x\,dx + f_y\,dy$. For $z = f(x_1, \ldots, x_n)$: $dz = f_{x_1}dx_1 + \cdots + f_{x_n}dx_n$. The total derivative is also called the differential. S004·3 · S004·4 · T004a 00:02:04

> *"The total derivative is also called the differential. The differential is used in integration by substitution and in approximation. It is also useful to calculate the Errors in approximation."* S004·4

Formulas

E15.1 · Total differential ESA ★★

$$dz = f_x(x, y)\,dx + f_y(x, y)\,dy$$

gives	Approximate change in z for small changes dx, dy.
applies	$z = f(x, y)$ differentiable.
note	n variables: $dz = f_{x_1}dx_1 + f_{x_2}dx_2 + \cdots + f_{x_n}dx_n$.

S004·3 · S004·4 · S003·8 · T003a 00:13:56

E15.2 · Approximating a nearby value ESA ★★

$$f(x_0 + dx,\, y_0 + dy) \approx f(x_0, y_0) + f_x(x_0, y_0)\,dx + f_y(x_0, y_0)\,dy$$

gives	Estimated function value near a known point.
applies	dx, dy small.
note	Slide 003·8: $f(2.1, -3.03) \approx 6 + 1.3(0.1) + (-0.6)(-0.03) = 6.148$.

S003·8 · T003a 00:13:09

Graphs and diagrams

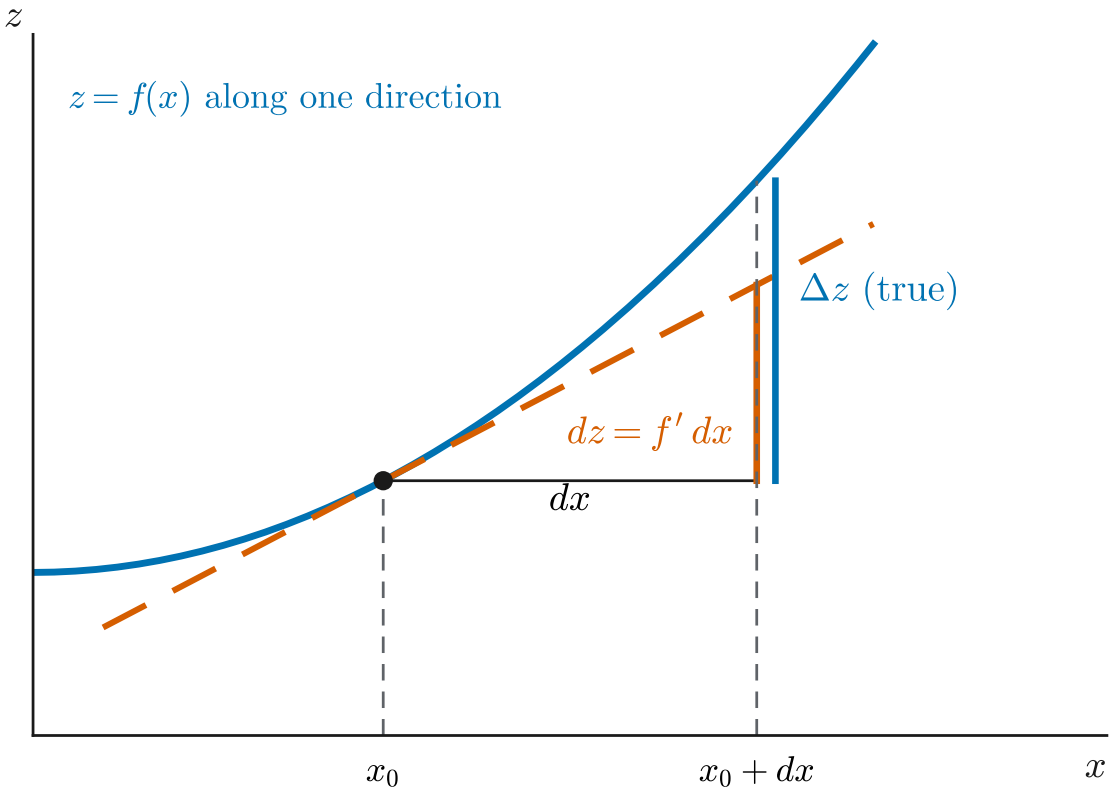
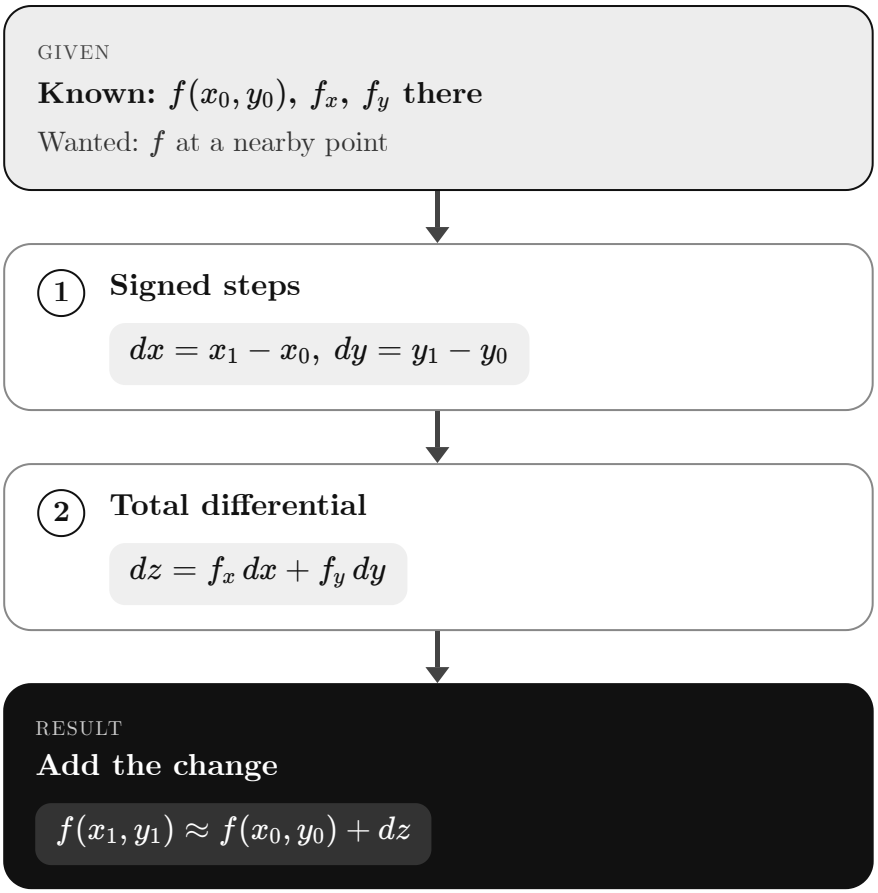


Figure G15.1. The differential approximates the true change. ESA ★★

ADDED FOR CLARITY S003·8 · S004·4

How to read it. Cut along one direction. The true change Δz follows the blue curve; the differential dz follows the dashed tangent line. For a small step dx the two are nearly equal, which is why $f(x_0 + dx) \approx f(x_0) + dz$. The gap grows as the step grows.

Flowcharts



Remark. Slide 003·8: $dz = 1.3(0.1) + (-0.6)(-0.03) = 0.148$, so $f(2.1, -3.03) \approx 6.148$.

Figure F15.1. Estimating a value near a known point. ESA ★★

S003·8 · T003a 00:13:09

Important points and why

1. The total differential estimates how much f changes from a known point to a nearby one. ESA ★★ S003·8
2. Sensitivity: in $dV = 40\pi\,dr + 4\pi\,dh$ the coefficient of dr is ten times that of dh, so the tank's volume is more sensitive to the radius than to the height. ESA ★★ S003·11 · S003·12 · T003a 00:18:30
3. Sometimes you need not finish the problem: the coefficients of the differential already tell you which variable contributes most. ESA ★ T003a 00:19:05
4. Integration by substitution uses the differential: with $t = x^2$, $dt = 2x\,dx$, so $\int e^{x^2} x\,dx = \frac{1}{2}\int e^t\,dt$. ESA ★ transcript only T004a 00:07:57 · T004a 00:08:29
5. Errors and approximations, another application of the differential, come at the end of the unit. ESA ★ transcript only T004a 00:08:47

Why does the tank's volume depend more on the radius?
$V = \pi r^2 h$ has $V_r = 2\pi r h = 40\pi$ and $V_h = \pi r^2 = 4\pi$ at $r = 2$, $h = 10$. So $dV = 4\pi(10\,dr + dh)$: a change in r is multiplied by ten times as much as the same change in h.

S003·11 · T003a 00:18:45

Exam trap. Use signed changes: from -3 to -3.03, $dy = -0.03$ (slide 003·8 writes 0.03 but correctly uses -0.03). S003·8

Worked examples

W15.1 · Approximate a function value

ESA ★★

given	$f(2,-3) = 6$, $f_x(2,-3) = 1.3$, $f_y(2,-3) = -0.6$; approximate $f(2.1,-3.03)$.
step 1	$dx = 2.1 - 2 = 0.1$, $dy = -3.03 - (-3) = -0.03$.
step 2	$dz = 1.3(0.1) + (-0.6)(-0.03) = 0.13 + 0.018 = 0.148$.
step 3	$f(2.1,-3.03) \approx 6 + 0.148$.
answer	$\boxed{f(2.1,-3.03) \approx 6.148}$
check	Signs: $dy < 0$ and $f_y < 0$, so that term adds.

S003·8 · T003a 00:13:09

W15.2 · Which dimension matters more?

ESA ★★

given	Cylindrical tank, $h = 10$ ft, diameter 4 ft ($r = 2$).
step 1	$V = \pi r^2 h$, $V_r = 2\pi r h$, $V_h = \pi r^2$.
step 2	$dV = 2\pi r h\,dr + \pi r^2\,dh = 40\pi\,dr + 4\pi\,dh$.
step 3	$40\pi \approx 125.7$ against $4\pi \approx 12.57$.
answer	$\boxed{\text{More sensitive to the radius (diameter) than to the height}}$
check	A diameter change of dD is $dr = dD/2$: still $20\pi\,dD$ against $4\pi\,dh$.

S003·11 · S003·12

W15.3 · Total differentials ESA ★★

GIVEN (i) $u = xy$, (ii) $u = x/y$.

STEP 1 (i) $u_x = y$, $u_y = x$.

STEP 2 (ii) $u_x = 1/y$, $u_y = -x/y^2$.

ANSWER (i) $du = y\,dx + x\,dy$; (ii) $du = \dfrac{y\,dx - x\,dy}{y^2}$

CHECK Matches the CW key; (ii) is the quotient rule.

CW U1 P9

SHORT-NOTE RECAP

The total differential $dz = f_x\,dx + f_y\,dy$ (or $\sum f_{x_i} dx_i$) gives the approximate change in z for small changes in the variables. It approximates nearby values, $f(x_0 + dx, y_0 + dy) \approx f + f_x\,dx + f_y\,dy$. It also measures sensitivity: the larger coefficient marks the variable that matters most. It is used in substitution and error estimates.

16 Rates of change problems

Many word problems ask how fast one quantity changes when others change at known rates. Write the quantity as a function of the changing variables, take the total derivative with respect to time, and substitute the given values. “Remains constant” means its derivative is 0. A constraint linking the variables is differentiated implicitly.

Key terms and definitions

Rate of change. The derivative of a quantity with respect to time t; positive means increasing, negative means decreasing. T003a 00:12:06

Formulas

E16.1 · Constant quantity ESA ★★

$$A = \text{constant} \implies \frac{dA}{dt} = 0$$

gives The equation that fixes the unknown rate.

applies “Remains constant” in the question.

S003·9 · T003a 00:16:32

Graphs and diagrams

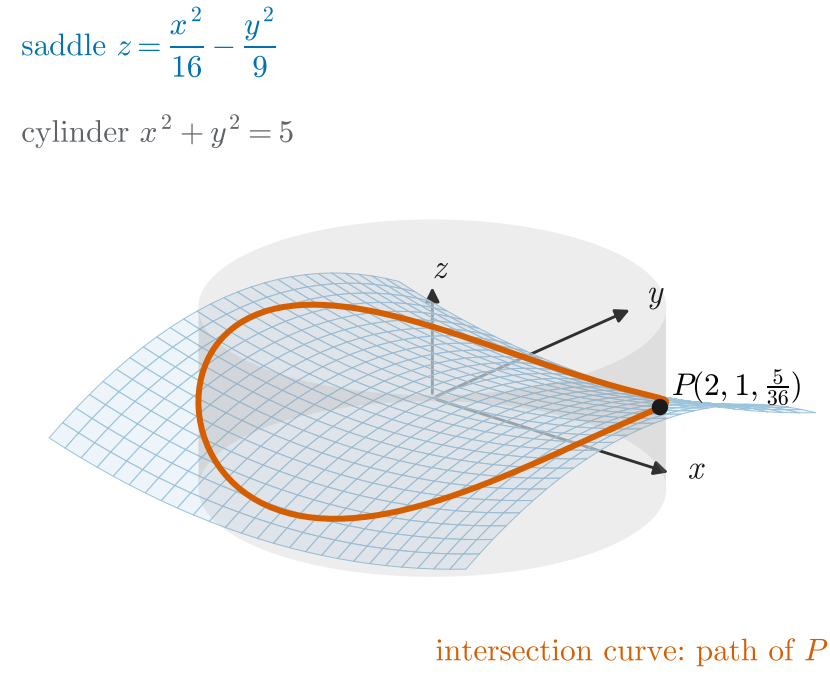


Figure G16.1. A point moving on a saddle and a cylinder. ESA ★★

S003·7 · T003a 00:06:27

How to read it. The saddle $z = x^2/16 - y^2/9$ (light blue) and the cylinder $x^2 + y^2 = 5$ (grey) meet in the vermilion curve. The point P moves along that curve; at $x = 2$, $y = 1$ its height rises at $dz/dt = 5/36$ cm/s when $dx/dt = 0.2$ cm/s (W16.1). The slide calls the surface a paraboloid, but it is a hyperbolic paraboloid (a saddle).

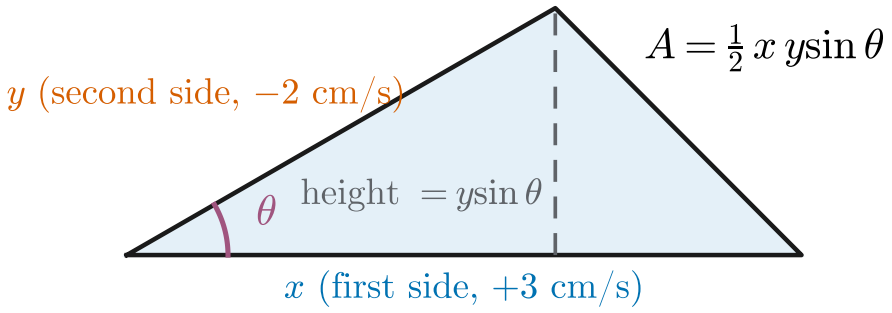
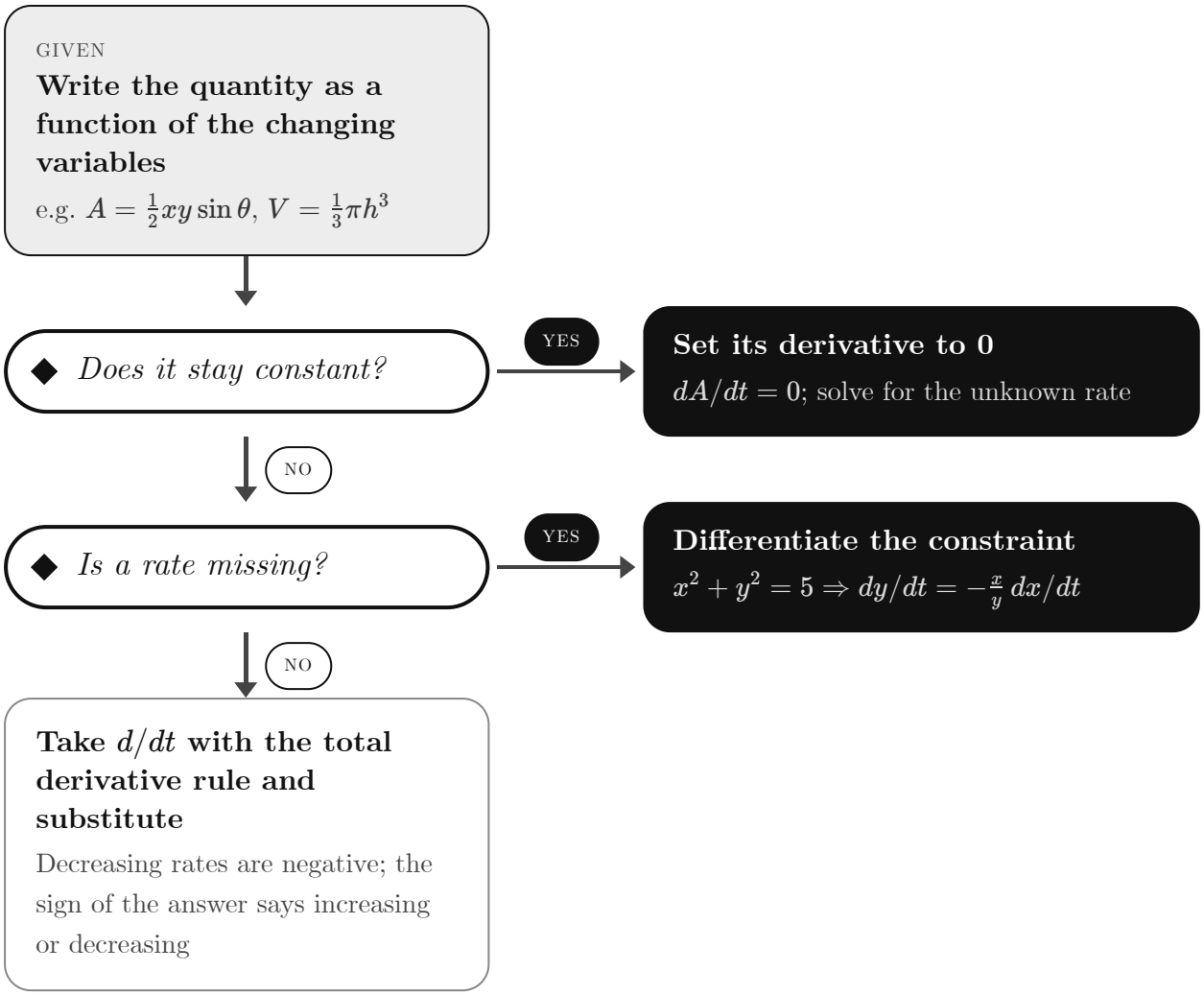


Figure G16.2. Triangle with two sides and the included angle. ESA ★★

S003·9 · T003a 00:15:25

How to read it. Sides x and y meet at the angle θ. The dashed perpendicular is the height $y \sin\theta$, so the area is $A = \frac{1}{2} x y \sin\theta$. Label the three changing quantities x, y, θ before differentiating.

Flowcharts



Remark. Both checks can apply in one problem; work down the ladder, then substitute the numbers last.

Figure F16.1. Solving a rates-of-change problem. ESA ★★

S003·7 · S003·9 · S003·21 · T003a 00:07:25

Important points and why

1. If $dz/dt > 0$ the quantity is increasing at that instant; if $dz/dt < 0$ it is decreasing. ESA ★★ T003a 00:12:06
2. When an unknown rate such as dy/dt is needed, find it from the constraint by implicit differentiation, e.g. $x^2 + y^2 = 5 \Rightarrow dy/dt = -\frac{x}{y}\,dx/dt$. ESA ★★ S003·7 · T003a 00:09:29
3. A decreasing side has a negative rate: "decreasing at 2 cm/s" means $dy/dt = -2$. ESA ★★ S003·10

Why does the angle have to decrease in the triangle problem?
$A = \frac{1}{2}xy\sin\theta$. One side grows ($+3$) and the other shrinks (-2), but at the given instant their combined effect $\frac{1}{2}\sin\theta(3y - 2x) = \frac{1}{2}\cdot\frac{1}{2}(90 - 40) > 0$ would increase the area. To keep A constant, θ must decrease.

ADDED FOR CLARITY S003·9 · S003·10

Exam trap. Slide 003·10 prints $d\theta/dt = 0.04831$. The value is $-25/(300\sqrt{3}) = -\sqrt{3}/36 \approx -0.0481$ rad/s: negative, and 481, not 4831. ADDED FOR CLARITY S003·10 · T003a 00:16:55

Exam trap. $x^2/16 - y^2/9 = z$ is a hyperbolic paraboloid (a saddle), not a cup-shaped paraboloid. The calculation does not change. ADDED FOR CLARITY S003·7 · T003a 00:06:27

Worked examples

W16.1 · Point moving on a saddle and a cylinder — ESA ★★

GIVEN $z = \dfrac{x^2}{16} - \dfrac{y^2}{9}$, $x^2 + y^2 = 5$, $\dfrac{dx}{dt} = 0.2$ cm/s; find dz/dt at $x = 2$ (so $y = 1$).

STEP 1 $\dfrac{dz}{dt} = \dfrac{x}{8}\dfrac{dx}{dt} - \dfrac{2y}{9}\dfrac{dy}{dt}$.

STEP 2 Constraint: $2x + 2y\,y' = 0$, so $\dfrac{dy}{dt} = -\dfrac{x}{y}\dfrac{dx}{dt} = -0.4$.

STEP 3 $\dfrac{dz}{dt} = \dfrac{x}{8}(0.2) + \dfrac{2x}{9}(0.2)$: the y cancels. At $x = 2$: $0.05 + 0.0889$.

ANSWER $\boxed{\dfrac{dz}{dt} = \dfrac{5}{36} \approx 0.139 \text{ cm/s (increasing)}}$

CHECK Positive, so z is increasing at $x = 2$.

S003·7 · T003a 00:06:27 · T003a 00:11:32

W16.2 · Angle of a triangle of constant area — ESA ★★

GIVEN $x = 20$, $y = 30$, $\theta = \pi/6$; $\dfrac{dx}{dt} = 3$, $\dfrac{dy}{dt} = -2$; area constant.

STEP 1 $A = \frac{1}{2}xy\sin\theta$ (height $= y\sin\theta$).

STEP 2 $0 = \dfrac{dA}{dt} = \frac{1}{2}\left(y\sin\theta(3) + x\sin\theta(-2) + xy\cos\theta\,\dfrac{d\theta}{dt}\right)$.

STEP 3 $= \frac{1}{2}\left(45 - 20 + 300\sqrt{3}\,\theta'\right)$, so $\theta' = -\dfrac{25}{300\sqrt{3}}$.

ANSWER $\boxed{\dfrac{d\theta}{dt} = -\dfrac{\sqrt{3}}{36} \approx -0.0481 \text{ rad/s (decreasing)}}$

CHECK The slide's "0.04831" is a misprint (see Exam trap).

S003·9 · S003·10 · T003a 00:14:31

W16.3 · Growing triangle ESA ★★

GIVEN $b = 10$, $h = 6$, $db/dt = 2$, $dh/dt = 1$ (cm, cm/s).

STEP 1 $A = \frac{1}{2}bh$, $\dfrac{dA}{dt} = \frac{1}{2}\left(b\dfrac{dh}{dt} + h\dfrac{db}{dt}\right)$.

STEP 2 $= \frac{1}{2}(10 \cdot 1 + 6 \cdot 2) = \frac{1}{2}(22)$.

ANSWER $\boxed{\dfrac{dA}{dt} = 11 \text{ cm}^2/\text{s}}$

CHECK Units: cm × cm/s = cm²/s.

S003·21 · S003·22 · T003b 00:04:57

W16.4 · Filling a cone ESA ★★

GIVEN $dV/dt = 9$ cm³/s, $r = h$ always, $h = 6$ cm.

STEP 1 $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi h^3$.

STEP 2 $\dfrac{dV}{dt} = \pi h^2 \dfrac{dh}{dt}$, so $9 = 36\pi\, \dfrac{dh}{dt}$.

ANSWER $\boxed{\dfrac{dh}{dt} = \dfrac{1}{4\pi} \approx 0.0796 \text{ cm/s}}$

CHECK Use $r = h$ before differentiating.

S003·23 · S003·24 · T003b 00:06:41

W16.5 · Growing rectangle ESA ★★

GIVEN Sides 5 ft and 4 ft, increasing at 1.5 ft/s and 0.5 ft/s.

STEP 1 $A = xy$, $\dfrac{dA}{dt} = y\dfrac{dx}{dt} + x\dfrac{dy}{dt} = 4(1.5) + 5(0.5)$.

ANSWER $\boxed{\dfrac{dA}{dt} = 8.5 \text{ ft}^2/\text{s}}$

CHECK CW answer: 8.5 sq. ft/s.

CW U1 P12

W16.6 · Changing cone

ESA ★★

GIVEN $h = 15$ cm, $dh/dt = 0.2$; $r = 10$ cm, $dr/dt = -0.3$.

STEP 1 $V = \frac{1}{3}\pi r^2 h$, $\dfrac{dV}{dt} = \frac{1}{3}\pi\left(2rh\dfrac{dr}{dt} + r^2\dfrac{dh}{dt}\right)$.

STEP 2 $= \frac{1}{3}\pi(2\cdot 10\cdot 15\cdot(-0.3) + 100\cdot 0.2) = \frac{1}{3}\pi(-90+20)$.

ANSWER $\boxed{\dfrac{dV}{dt} = -\dfrac{70\pi}{3}\ \text{cm}^3/\text{s (decreasing)}}$

CHECK CW answer: $-70\pi/3$.

CW U1 P13

SHORT-NOTE RECAP

For a rate problem, write the quantity as a function of the changing variables, take d/dt of it with the total derivative rule, and substitute the given values and rates (decreasing means negative). If something stays constant its derivative is 0; a constraint between variables gives a missing rate by implicit differentiation.

17 The chain rule for composite functions

If $z = f(x, y)$ and x, y are functions of two variables u, v, then z is a composite function of u and v. Its partial derivatives follow every path in the tree diagram: $z_u = z_x x_u + z_y y_u$ and $z_v = z_x x_v + z_y y_v$. When f is unknown, f_x, f_y stay as symbols. Substitution tricks ($r = 2x - 3y$, ...) and polar coordinates are the standard exam forms.

Key terms and definitions

Chain rule (several variables). If $z = f(x, y)$ is a bivariate function and x, y are functions of two or more independent variables u, v, then $\dfrac{\partial z}{\partial u} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial u} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial u}$ and $\dfrac{\partial z}{\partial v} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial v} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial v}$. S003·13 · T003a 00:21:07

The chain of a problem. The list of dependencies, e.g. "u depends on x, y; x, y depend on r, t; so u is a composite function of r, t". Knowing it fixes which chain-rule formula to write. T004b 00:02:31

> *"We can define chain rule for a function of two or more independent variable: If $z = f(x, y)$ is a bivariate function and x, y are functions of two or more independent variables say u, v then we can find the partial derivative of z with respect to u and v."* S003·13

Formulas

E17.1 · Chain rule, two intermediate variables ESA ★★★

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}, \qquad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

gives	Partial derivatives of a composite function.
applies	$z = f(x, y)$, $x = g(u, v)$, $y = h(u, v)$.
note	All derivatives are partial, because everything depends on two variables.

S003·13 · S003·14 · T003a 00:21:07

E17.2 · Chain rule, three intermediate variables ESA ★★★

$$\frac{\partial v}{\partial x} = \frac{\partial v}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial v}{\partial s}\frac{\partial s}{\partial x} + \frac{\partial v}{\partial t}\frac{\partial t}{\partial x}$$

gives	One term per intermediate variable.
applies	$v = f(r, s, t)$, each of r, s, t a function of x, y, z.
note	A term is 0 when that intermediate variable does not contain x.

S003·15 · T003a 00:23:10

E17.3 · Function of one intermediate variable ESA ★★

$$u = f(r),\ r = \sqrt{x^2+y^2+z^2} \implies \frac{\partial u}{\partial x} = f'(r)\,\frac{\partial r}{\partial x} = f'(r)\,\frac{x}{r}$$

GIVES $\partial u/\partial x$ in terms of $f'(r)$.

APPLIES u depends on x, y, z only through r.

S004·12 · T004b 00:05:07

E17.4 · Polar coordinates: derivatives you reuse ESA ★★

$$\begin{aligned} &x = r\cos\theta,\ y = r\sin\theta: && x_\theta = -r\sin\theta, \quad y_\theta = r\cos\theta \\ &r = \sqrt{x^2+y^2},\ \theta = \tan^{-1}\tfrac{y}{x}: && r_x = \frac{x}{r},\ r_y = \frac{y}{r},\ \theta_x = -\frac{y}{r^2},\ \theta_y = \frac{x}{r^2} \end{aligned}$$

GIVES The inner derivatives for polar problems.

APPLIES $r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1}(y/x)$.

NOTE So $u_x = f_r\frac{x}{r} - f_\theta\frac{y}{r^2}$ and $u_y = f_r\frac{y}{r} + f_\theta\frac{x}{r^2}$ for $u = f(r,\theta)$.

S004·13 · S004·15 · S004·17

E17.5 · Polar identity ESA ★★

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = \left(\frac{\partial u}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial u}{\partial \theta}\right)^2$$

GIVES The squared gradient in polar coordinates.

APPLIES $u = u(x,y)$, $x = r\cos\theta$, $y = r\sin\theta$.

NOTE Proved in W17.16.

CW U1 P15

Graphs and diagrams

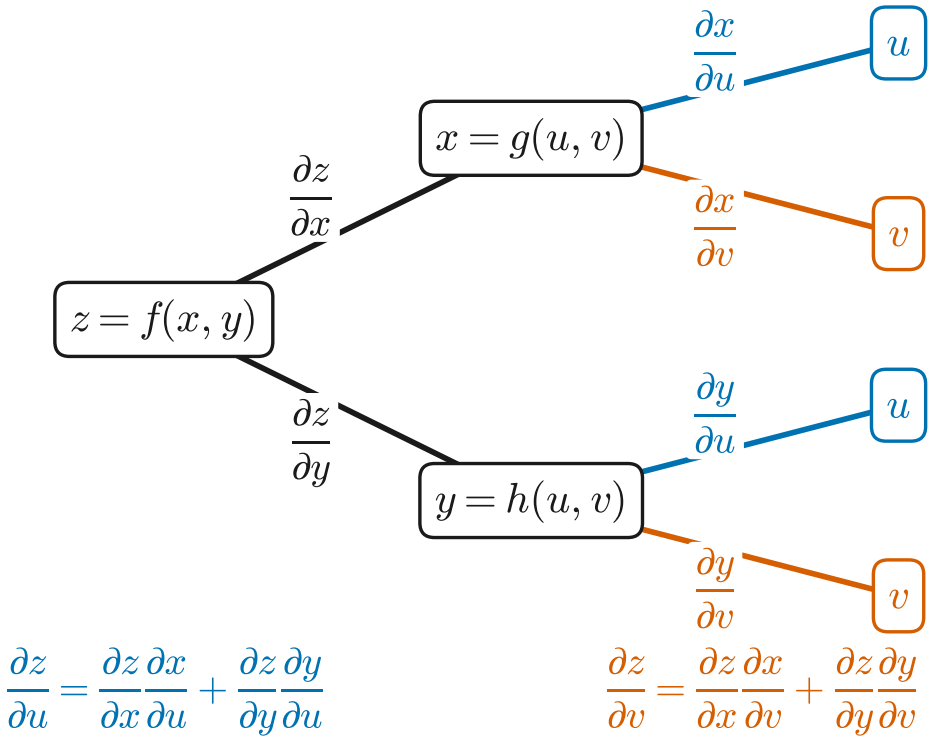
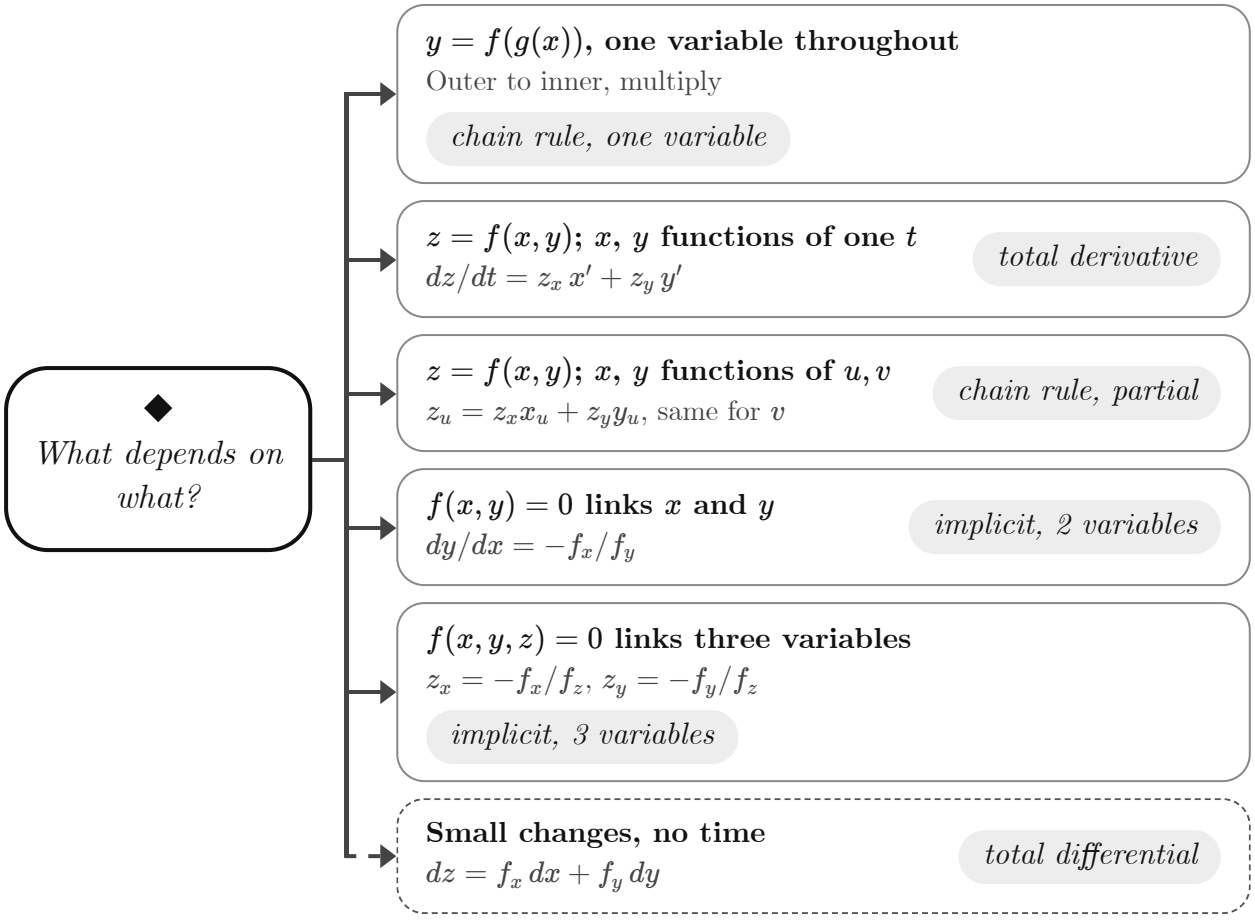


Figure G17.1. Tree diagram for the chain rule. ESA ★★★

S003·14 · T003a 00:19:42

How to read it. z branches to x and y, and each of those branches to both u and v, giving four leaves. For $\partial z/\partial u$ follow only the two paths ending at u (blue): $z_x x_u + z_y y_u$. For $\partial z/\partial v$ follow the two ending at v (vermilion). In an exam, draw the tree first and read the terms off it.

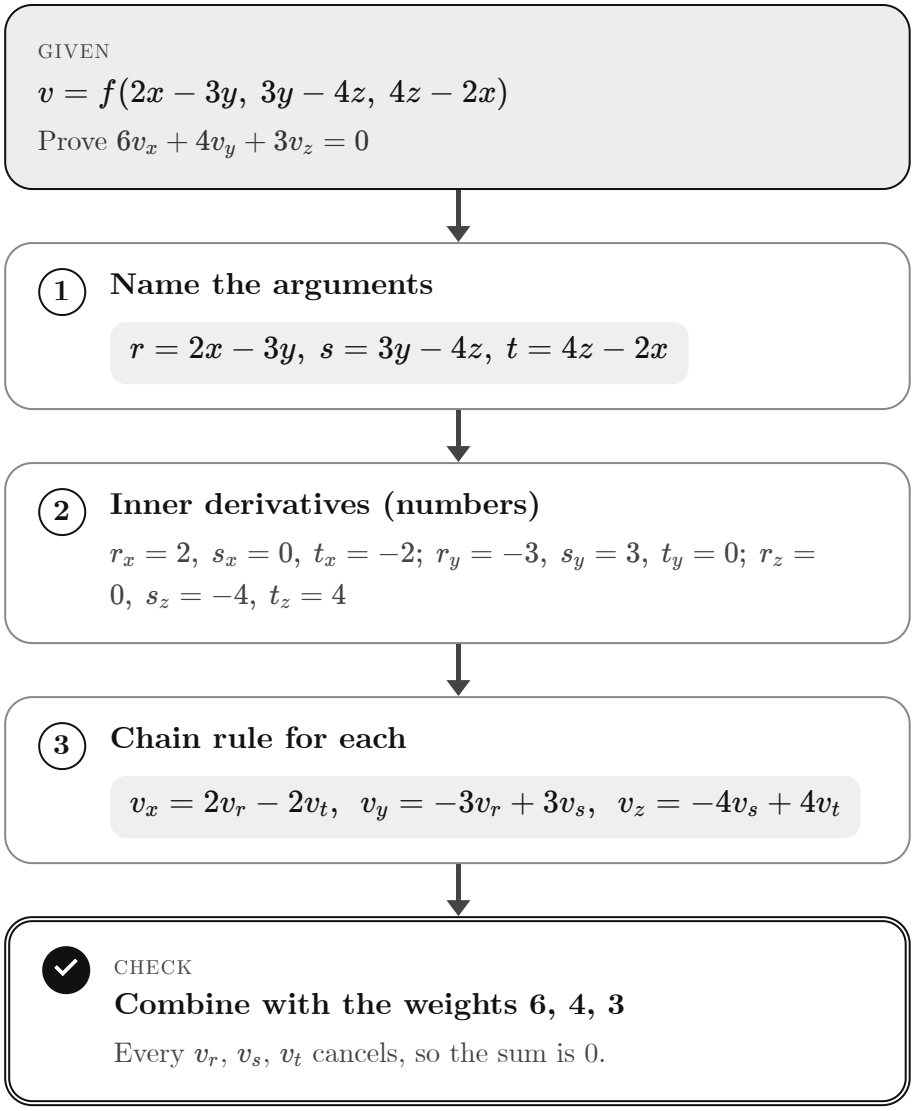
Flowcharts



Remark. The dashed route has no derivative with respect to a variable: it estimates a change. Everything else starts from the tree diagram.

Figure F17.1. Which rule? Read the chain first. ESA ★★★

S003·5 · S003·13 · S004·5 · S004·6 · T004b 00:02:31



Exam trap. Slide 003·16 drops the 3 in $v_y = -3v_r + 3v_s$; without it the terms do not cancel.

Figure F17.2. Proving an identity such as $6v_x + 4v_y + 3v_z = 0$. ESA ★★★

S003·15 · S003·16 · ESA Jan-24 s2 1d

Total derivative vs chain rule vs implicit differentiation

	Total derivative	Chain rule (partial)	Implicit function
Set-up	$z = f(x, y)$, $x(t)$, $y(t)$	$z = f(x, y)$, $x(u, v)$, $y(u, v)$	$f(x, y) = 0$ or $f(x, y, z) = 0$
Final variables	one: t	two: u, v	x (and y)
Formula	$z_x\, x' + z_y\, y'$	$z_x x_u + z_y y_u$	$-f_x/f_y$, $-f_x/f_z$
Result written	dz/dt	$\partial z/\partial u$, $\partial z/\partial v$	dy/dx, $\partial z/\partial x$
Typical question	rate of change at an instant	prove an identity	slope or tangent line

ADDED FOR CLARITY S003·5 · S003·13 · S004·4 · S004·6

Important points and why

1. When x and y depend on two variables u, v, z cannot be differentiated "totally": only z_u and z_v exist. ESA ★★ T003a 00:20:40
2. Write the tree diagram every time a function has more than one independent variable, so that no partial derivative is missed. ESA ★★★ T003a 00:25:02
3. It is essential to know the chain given in the problem: what depends on what. The formula follows from it. ESA ★★★ T004b 00:02:31
4. If f is not given ($u = f(x, y)$), keep f_x, f_y (or f_r, f_θ) as symbols in the answer. ESA ★★★ S003·17 · S003·25 · T003a 00:26:28
5. For $f(2x - 3y,\, 3y - 4z,\, 4z - 2x)$, name the arguments r, s, t and compute $r_x, s_x, t_x, \ldots$; most are constants or 0. ESA ★★★ S003·15 · T003a 00:22:33
6. When u is a composite function of r, θ (or s, t), eliminate x and y from the final answer: you must know why you substitute. ESA ★★★ T004b 00:07:52 · T004b 00:10:28
7. Functions of $x^2 + y^2$ alone (e.g. $\sin(x^2 + y^2)$, $e^{x^2+y^2}$) have $\partial u/\partial\theta = 0$ in polar coordinates, since $x^2 + y^2 = r^2$ does not involve θ. ESA ★★ ADDED FOR CLARITY S004·10 · S004·13 · S004·14
8. $\tan^{-1}(y/x)$ is exactly θ in polar coordinates, so $\partial u/\partial\theta = 1$. ESA ★★ ADDED FOR CLARITY S004·15
9. The chain rule extends to any number of variables: z of x, y, each of x, y of three variables, and so on. ESA ★ T003a 00:21:40

Why does $u = f(x - y,\, y - z,\, z - x)$ give $u_x + u_y + u_z = 0$?
Each argument is a difference whose derivatives sum to zero: for $r = x - y$, $r_x + r_y + r_z = 1 - 1 + 0 = 0$. Adding the three chain-rule expressions gives $f_r(0) + f_s(0) + f_t(0)$. In words: moving x, y, z by the same amount changes none of the arguments.
ADDED FOR CLARITY ESA Jan-24 s2 1d

Why do the chain-rule answers keep f_r and f_θ?
The outer function f is unknown, so its partial derivatives cannot be computed; only the inner derivatives ($r_s = t$, $\theta_s = s/\sqrt{s^2+t^2}$, ...) can.

S003·17 · T003a 00:26:28

Exam trap. Slide 003·16 shows $\dfrac{\partial v}{\partial y} = -3\dfrac{\partial v}{\partial r} + \dfrac{\partial v}{\partial s}$; it should be $-3\dfrac{\partial v}{\partial r} + 3\dfrac{\partial v}{\partial s}$ because $s = 3y - 4z$ gives $s_y = 3$. With the 3 the proof works. ADDED FOR CLARITY S003·16

Exam trap. Do not stop with x and y still in an answer that should be in r, θ or s, t. T004b 00:07:52

Exam trap. Write $\dfrac{\partial z}{\partial u}$, not $\dfrac{dz}{du}$: z depends on both u and v. ADDED FOR CLARITY S003·13

Worked examples

W17.1 · Prove $6v_x + 4v_y + 3v_z = 0$ — ESA ★★★

GIVEN $v = f(2x - 3y,\ 3y - 4z,\ 4z - 2x)$.

STEP 1 Let $r = 2x - 3y$, $s = 3y - 4z$, $t = 4z - 2x$: $r_x = 2,\ s_x = 0,\ t_x = -2$; $r_y = -3,\ s_y = 3,\ t_y = 0$; $r_z = 0,\ s_z = -4,\ t_z = 4$.

STEP 2 $v_x = 2v_r - 2v_t$, $v_y = -3v_r + 3v_s$, $v_z = -4v_s + 4v_t$.

STEP 3 $6v_x + 4v_y + 3v_z = (12 - 12)v_r + (12 - 12)v_s + (-12 + 12)v_t$.

ANSWER $\boxed{6v_x + 4v_y + 3v_z = 0}$

CHECK The weights 6, 4, 3 are chosen so that each of v_r, v_s, v_t cancels.

S003·15 · S003·16 · T003a 00:22:14

W17.2 · Unknown outer function — ESA ★★★

GIVEN $z = f(r, \theta)$, $r = ts - t^2$, $\theta = \sqrt{s^2 + t^2}$.

STEP 1 $r_s = t$, $\theta_s = \dfrac{s}{\sqrt{s^2+t^2}}$; $r_t = s - 2t$, $\theta_t = \dfrac{t}{\sqrt{s^2+t^2}}$.

STEP 2 $z_s = f_r\, r_s + f_\theta\, \theta_s$ and $z_t = f_r\, r_t + f_\theta\, \theta_t$.

ANSWER $\boxed{\dfrac{\partial z}{\partial s} = t\, f_r + \dfrac{s}{\sqrt{s^2+t^2}} f_\theta, \quad \dfrac{\partial z}{\partial t} = (s - 2t) f_r + \dfrac{t}{\sqrt{s^2+t^2}} f_\theta}$

CHECK sympy's chain rule on a symbolic f gives the same.

S003·17 · T003a 00:25:02

W17.3 · $u = f(x,y)$, $x = e^{st}$, $y = s^2 + t^2$ ESA ★★★

GIVEN $u = f(x,y)$, $x = e^{st}$, $y = s^2 + t^2$.

STEP 1 $x_s = te^{st}$, $x_t = se^{st}$, $y_s = 2s$, $y_t = 2t$.

STEP 2 $u_s = f_x x_s + f_y y_s$, $u_t = f_x x_t + f_y y_t$.

ANSWER $\boxed{\dfrac{\partial u}{\partial s} = te^{st} f_x + 2s\, f_y, \quad \dfrac{\partial u}{\partial t} = se^{st} f_x + 2t\, f_y}$

S003·25 · T003b 00:08:18

W17.4 · $\sin(x^2+y^2)$ in polar form ESA ★★

GIVEN $u = \sin(x^2+y^2)$, $x = r\cos t$, $y = r\sin t$; find u_t.

STEP 1 $u_x = 2x\cos(x^2+y^2)$, $u_y = 2y\cos(x^2+y^2)$; $x_t = -r\sin t$, $y_t = r\cos t$.

STEP 2 $u_t = 2\cos(r^2)\,[\,r\cos t(-r\sin t) + r\sin t(r\cos t)\,]$.

ANSWER $\boxed{\dfrac{\partial u}{\partial t} = 0}$

CHECK $u = \sin r^2$ does not involve t at all.

S004·10 · T004b 00:00:10

W17.5 · Three intermediate variables ESA ★★

GIVEN $u = x^2 y + y z^3$, $x = r\cos\theta$, $y = r\sin\theta$, $z = \theta$; find u_θ.

STEP 1 $u_x = 2xy$, $u_y = x^2 + z^3$, $u_z = 3yz^2$; $x_\theta = -r\sin\theta$, $y_\theta = r\cos\theta$, $z_\theta = 1$.

STEP 2 $u_\theta = 2xy(-r\sin\theta) + (x^2+z^3)(r\cos\theta) + 3yz^2$.

STEP 3 Substitute: $-2r^3\cos\theta\sin^2\theta + r^3\cos^3\theta = r^3\cos\theta(3\cos^2\theta - 2)$.

ANSWER $\boxed{\dfrac{\partial u}{\partial \theta} = r^3\cos\theta\,(3\cos^2\theta - 2) + r\,(3\theta^2\sin\theta + \theta^3\cos\theta)}$

CHECK sympy agrees.

S004·11 · T004b 00:03:09

W17.6 · $u = f(r)$ with r the distance ESA ★★

GIVEN $u = f(r)$, $r = \sqrt{x^2+y^2+z^2}$.

STEP 1 $u_x = \dfrac{du}{dr}\,r_x = f'(r)\,r_x$.

STEP 2 $r_x = \dfrac{x}{\sqrt{x^2+y^2+z^2}} = \dfrac{x}{r}$.

ANSWER $\boxed{\dfrac{\partial u}{\partial x} = f'(r)\,\dfrac{x}{r}}$

CHECK By symmetry $u_y = f'(r)\,y/r$, $u_z = f'(r)\,z/r$.

S004·12 · T004b 00:05:07

W17.7 · $e^{x^2+y^2}$ in polar form ESA ★★

GIVEN $u = e^{x^2+y^2}$, $x = r\cos\theta$, $y = r\sin\theta$.

STEP 1 $u_\theta = 2xe^{x^2+y^2}(-r\sin\theta) + 2ye^{x^2+y^2}(r\cos\theta)$.

STEP 2 $= 2re^{r^2}(-r\cos\theta\sin\theta + r\sin\theta\cos\theta)$.

ANSWER $\boxed{\dfrac{\partial u}{\partial \theta} = 0}$

CHECK $u = e^{r^2}$ has no θ.

S004·13 · S004·14 · T004b 00:06:38

W17.8 · Show $\partial u/\partial\theta = 1$ ESA ★★

GIVEN $u = \tan^{-1}(y/x)$, $x = r\cos\theta$, $y = r\sin\theta$.

STEP 1 $u_x = \dfrac{-y}{x^2+y^2}$, $u_y = \dfrac{x}{x^2+y^2}$; $x_\theta = -r\sin\theta$, $y_\theta = r\cos\theta$.

STEP 2 $u_\theta = \dfrac{r^2\sin^2\theta + r^2\cos^2\theta}{r^2}$.

ANSWER $\boxed{\dfrac{\partial u}{\partial \theta} = 1}$

CHECK $u = \tan^{-1}(\tan\theta) = \theta$.

S004·15 · T004b 00:08:40

W17.9 · $u = \sin(xy) + y^3,\ x = e^s,\ y = s + t$ ESA ★★

GIVEN $u = \sin(xy) + y^3$, $x = e^s$, $y = s + t$.

STEP 1 $u_x = y\cos(xy)$, $u_y = x\cos(xy) + 3y^2$; $x_s = e^s$, $y_s = 1$, $x_t = 0$, $y_t = 1$.

STEP 2 $u_s = y\cos(xy)e^s + (x\cos(xy) + 3y^2)$, $u_t = x\cos(xy) + 3y^2$, then substitute $x = e^s$, $y = s + t$.

ANSWER $\boxed{u_s = (s+t)e^s\cos\big(e^s(s+t)\big) + e^s\cos\big(e^s(s+t)\big) + 3(s+t)^2, \quad u_t = e^s\cos\big(e^s(s+t)\big) + 3(s+t)^2}$

CHECK sympy agrees.

S004·16 · T004b 00:09:32

W17.10 · $u = f(r,\theta)$ from Cartesian coordinates ESA ★★

GIVEN $u = f(r,\theta)$, $r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1}(y/x)$.

STEP 1 $r_x = \dfrac{x}{r}$, $r_y = \dfrac{y}{r}$, $\theta_x = -\dfrac{y}{r^2}$, $\theta_y = \dfrac{x}{r^2}$.

STEP 2 $u_x = f_r r_x + f_\theta \theta_x$, $u_y = f_r r_y + f_\theta \theta_y$.

ANSWER $\boxed{\dfrac{\partial u}{\partial x} = f_r\dfrac{x}{r} - f_\theta\dfrac{y}{r^2}, \quad \dfrac{\partial u}{\partial y} = f_r\dfrac{y}{r} + f_\theta\dfrac{x}{r^2}}$

S004·17 · T004b 00:10:47

W17.11 · $xu_x + yu_y + zu_z$ for $f(x/y,\ y/z,\ z/x)$ ESA ★★★

GIVEN $u = f\big(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\big)$; find $xu_x + yu_y + zu_z$.

STEP 1 Let $p = \frac{x}{y}$, $q = \frac{y}{z}$, $w = \frac{z}{x}$.

STEP 2 $xu_x = x\big(\frac{1}{y}f_p - \frac{z}{x^2}f_w\big) = p\,f_p - w\,f_w$.

STEP 3 $yu_y = y\big(-\frac{x}{y^2}f_p + \frac{1}{z}f_q\big) = -p\,f_p + q\,f_q$.

STEP 4 $zu_z = z\big(-\frac{y}{z^2}f_q + \frac{1}{x}f_w\big) = -q\,f_q + w\,f_w$.

STEP 5 Add the three lines.

ANSWER $\boxed{x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z} = 0}$

CHECK u is homogeneous of degree 0 (scaling x, y, z leaves every ratio unchanged), so Euler's theorem gives 0 too.

ADDED FOR CLARITY ESA Jan-24 s1 1d

W17.12 · $u_x + u_y + u_z$ for $f(x-y,\, y-z,\, z-x)$ ESA ★★★

GIVEN $u = f(x-y,\ y-z,\ z-x)$; find $u_x + u_y + u_z$.

STEP 1 Let $p = x-y$, $q = y-z$, $w = z-x$.

STEP 2 $u_x = f_p - f_w$, $u_y = -f_p + f_q$, $u_z = -f_q + f_w$.

STEP 3 Add.

ANSWER $\boxed{\dfrac{\partial u}{\partial x} + \dfrac{\partial u}{\partial y} + \dfrac{\partial u}{\partial z} = 0}$

CHECK Same pattern as W17.1 with equal weights.

ADDED FOR CLARITY ESA Jan-24 s2 1d

W17.13 · $z = f(u,v)$, $u = x^2 - y^2$, $v = 2xy$ ESA ★★★

GIVEN Show $z_x^2 + z_y^2 = 4(u^2+v^2)^{1/2}\,(z_u^2 + z_v^2)$.

STEP 1 $z_x = z_u(2x) + z_v(2y)$, $z_y = z_u(-2y) + z_v(2x)$.

STEP 2 $z_x^2 + z_y^2 = 4(x^2+y^2)(z_u^2+z_v^2)$: the cross terms $8xy\, z_u z_v$ cancel.

STEP 3 $u^2 + v^2 = (x^2-y^2)^2 + 4x^2y^2 = (x^2+y^2)^2$, so $x^2+y^2 = (u^2+v^2)^{1/2}$.

ANSWER $\boxed{z_x^2 + z_y^2 = 4\sqrt{u^2+v^2}\,(z_u^2+z_v^2)}$

CHECK Numerical check with a sample f in sympy: residual $\approx 10^{-14}$.

ADDED FOR CLARITY ESA Jan-24 s3 1b

W17.14 · $z = f(x,y)$, $x = e^u \sin v$, $y = e^u \cos v$ ESA ★★★

GIVEN Show $z_x^2 + z_y^2 = e^{-2u}(z_u^2 + z_v^2)$.

STEP 1 $z_u = z_x\, e^u \sin v + z_y\, e^u \cos v = x z_x + y z_y$.

STEP 2 $z_v = z_x\, e^u \cos v - z_y\, e^u \sin v = y z_x - x z_y$.

STEP 3 $z_u^2 + z_v^2 = (x^2+y^2)(z_x^2+z_y^2)$: the cross terms cancel.

STEP 4 $x^2 + y^2 = e^{2u}$.

ANSWER $\boxed{z_x^2 + z_y^2 = e^{-2u}\,(z_u^2 + z_v^2)}$

CHECK sympy with a sample f: exact 0.

ADDED FOR CLARITY ESA Jan-24 s4 1b

W17.15 · $z = \log(u^2 + v)$ ESA ★★★

GIVEN $z = \log(u^2+v)$, $u = e^{x+y^2}$, $v = x + y^2$; find $2y\,z_x - z_y$.

STEP 1 $z_u = \dfrac{2u}{u^2+v}$, $z_v = \dfrac{1}{u^2+v}$; $u_x = u$, $u_y = 2yu$, $v_x = 1$, $v_y = 2y$.

STEP 2 $z_x = \dfrac{2u^2+1}{u^2+v}$, $z_y = \dfrac{2y(2u^2+1)}{u^2+v}$.

ANSWER $\boxed{2y\,\dfrac{\partial z}{\partial x} - \dfrac{\partial z}{\partial y} = 0}$

CHECK CW answer: 0. Both depend on x, y only through $x + y^2$.

ESA Jun-24 1a · CW U1 P14

W17.16 · Polar identity ESA ★★

GIVEN $u = u(x,y)$, $x = r\cos\theta$, $y = r\sin\theta$.

STEP 1 $u_r = u_x\cos\theta + u_y\sin\theta$.

STEP 2 $u_\theta = -u_x\, r\sin\theta + u_y\, r\cos\theta$, so $\frac{1}{r}u_\theta = -u_x\sin\theta + u_y\cos\theta$.

STEP 3 Square and add: the cross terms cancel and $\sin^2\theta + \cos^2\theta = 1$.

ANSWER $\boxed{u_x^2 + u_y^2 = u_r^2 + \dfrac{1}{r^2}\,u_\theta^2}$

CHECK sympy with a sample u: residual $\approx 10^{-16}$.

CW U1 P15

Past-paper questions

ESA Jan 2024 (5 marks) UE23MA141A set 1, Q1.d · solved in W17.11

If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, find the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$.

ESA Jan 2024 (5 marks) UE23MA141A set 2, Q1.d · solved in W17.12

If $u = f(x-y,\, y-z,\, z-x)$, find the value of $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$.

ESA Jan 2024 (6 marks) UE23MA141A set 3, Q1.b · solved in W17.13

If $z = f(u,v)$ and $u = x^2 - y^2$, $v = 2xy$, show that $\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = 4(u^2+v^2)^{1/2}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right]$.

ESA Jan 2024 (6 marks) UE23MA141A set 4, Q1.b · solved in W17.14

If $z = f(x,y)$ and $x = e^u\sin v$, $y = e^u\cos v$, show that $\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = e^{-2u}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right]$.

ESA June 2024 | 6 marks | UE23MA141A Q1.a · solved in W17.15

If $z = \log(u^2 + v)$, where $u = e^{x+y^2}$, $v = x + y^2$, then find the value of $2y\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}$.

SHORT-NOTE RECAP

If $z = f(x, y)$ with $x = g(u, v)$, $y = h(u, v)$, then z is a composite function of u, v and $z_u = z_x x_u + z_y y_u$, $z_v = z_x x_v + z_y y_v$ (one term per path in the tree diagram). With three intermediate variables there are three terms. If f is unknown, keep f_x, f_y as symbols. Write the final answer in the new variables. Standard results: $f(x - y, y - z, z - x)$ gives $u_x + u_y + u_z = 0$; in polar form $u_x^2 + u_y^2 = u_r^2 + u_\theta^2/r^2$.

18 Implicit functions of two variables

A relation $f(x,y)=c$ defines y as a function of x without solving for it; geometrically it is a level curve. Its total differential is zero, $f_x\,dx+f_y\,dy=0$, which gives $dy/dx=-f_x/f_y$ with no product or quotient rule. The slope then gives tangent lines, and the same idea handles awkward relations such as $(\cos x)^y=(\sin y)^x$.

Key terms and definitions

Implicit function. A function given by a relation $f(x,y)=c$ (for example $x^3y+3xy^2=3$) rather than by a formula $y=g(x)$. S004·3 · T004a 00:00:18

Level surface (level curve). The set where $z=f(x,y)=c$ (or $f(x_1,\dots,x_n)=c$). Examples: equipotential surfaces, isotherms. The cylinders $x^2+y^2=c$ and the spheres $x^2+y^2+z^2=c$ are layers, one for each c. S004·4 · T004a 00:02:35 · T004a 00:03:29

> *"If z represents a level surface, i.e. if $z=f(x,y)=c$, then $dz=0$. The total derivative is also called the differential."* S004·4

> *"Setting $\dfrac{dz}{dx}=0$ leads to a simple computation formula for implicit differentiation: $f_x\cdot 1+f_y\dfrac{dy}{dx}=0$."* S004·5

Formulas

E18.1 · Implicit differentiation, two variables ESA ★★

$$f(x,y)=c \implies \frac{dy}{dx}=-\frac{f_x}{f_y}, \qquad f_y\neq 0$$

gives	Slope of the level curve at a point.
applies	$f_y\neq 0$ at the point.
note	From $df=f_x\,dx+f_y\,dy=0$. Slide 004·4 prints the fraction upside down.

S004·5 · S004·6 · T004a 00:06:36 · T003b 00:10:33

E18.2 · Tangent line at a point ESA ★★

$$y-y_0=m\,(x-x_0), \qquad m=\left.\frac{dy}{dx}\right|_{(x_0,y_0)}$$

gives	Equation of the tangent to the curve at (x_0,y_0).
applies	(x_0,y_0) on the curve.
note	Through the origin: $y=mx$.

S004·7 · T004a 00:14:25

E18.3 · Derivative of a constant power and a variable power ESA ★★

$$\frac{d}{dx}a^x = a^x \ln a, \qquad \frac{d}{dx}x^n = n\,x^{n-1}$$

GIVES	The two rules needed for $(\cos x)^y$ and $(\sin y)^x$.
APPLIES	In a partial derivative, the other variable is a constant.
NOTE	$\ln$ needs a positive argument: $\cos x > 0$, $\sin y > 0$.

S004·8 · T004a 00:16:43

Graphs and diagrams

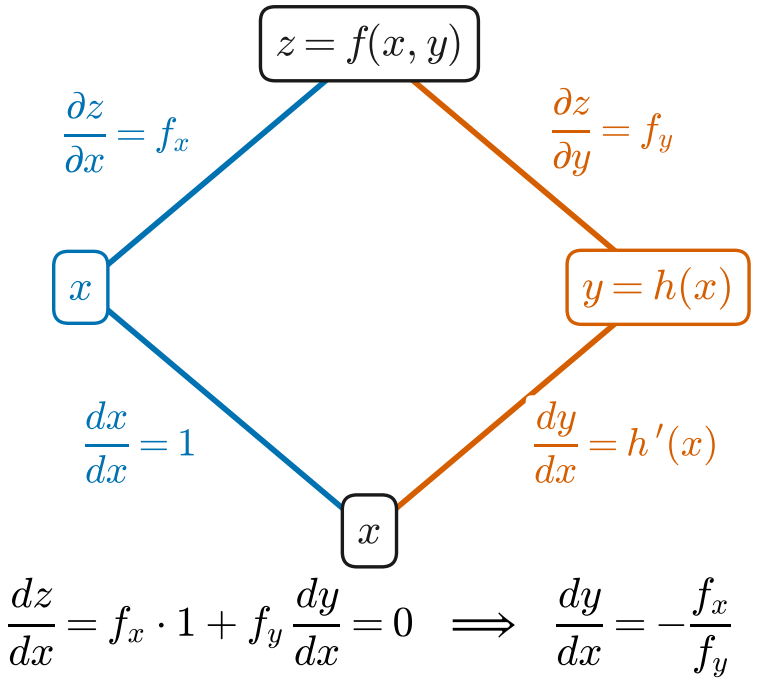


Figure G18.1. Branch diagram for implicit differentiation. ESA ★★

S004·5 · T004a 00:07:57

How to read it. Differentiate $z = f(x, y)$ with respect to x along both branches. The left branch is direct ($f_x \cdot dx/dx = f_x \cdot 1$). The right branch goes through $y = h(x)$ ($f_y \cdot dy/dx$). On a level curve the total is 0: $f_x + f_y\, dy/dx = 0$.

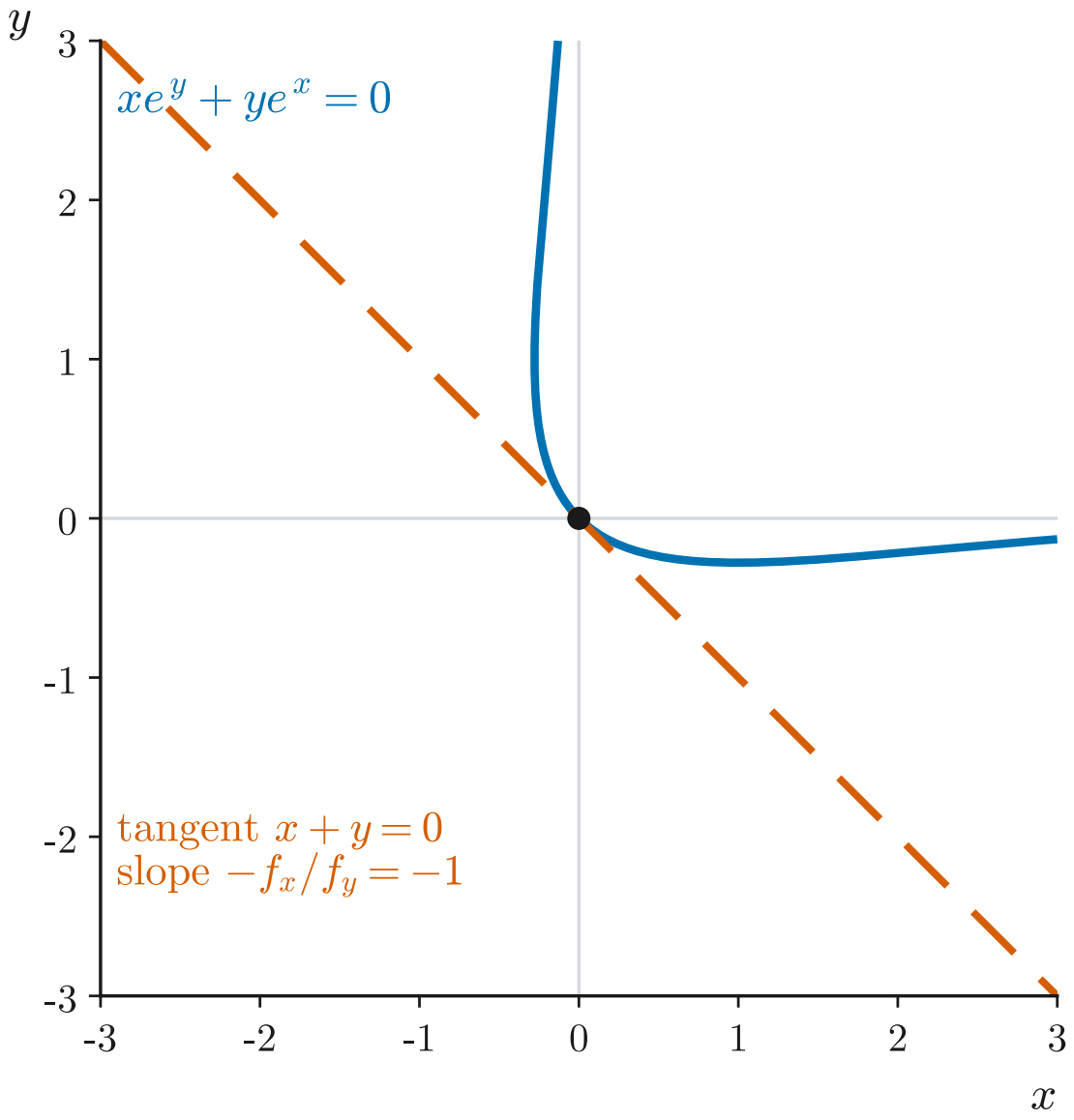


Figure G18.2. The curve $xe^y + ye^x = 0$ and its tangent at the origin. ESA ★★

S004·7 · T004a 00:13:48

How to read it. The blue curve is the set where $f(x, y) = xe^y + ye^x = 0$. At the origin $dy/dx = -f_x/f_y = -1$, so the dashed tangent is $y = -x$. The curve is plotted from its equation (the slide's picture is a 3-D grapher screenshot).

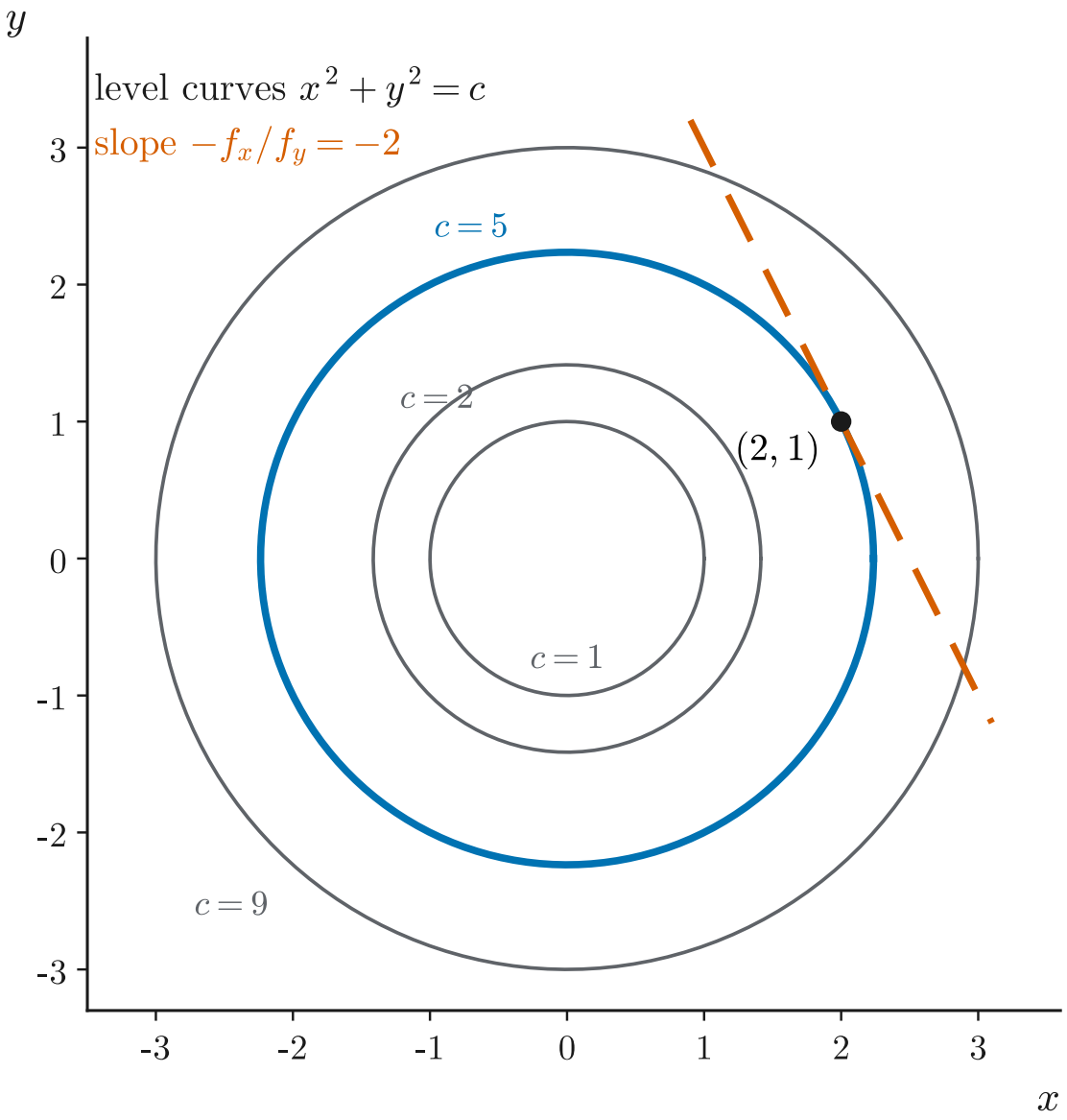
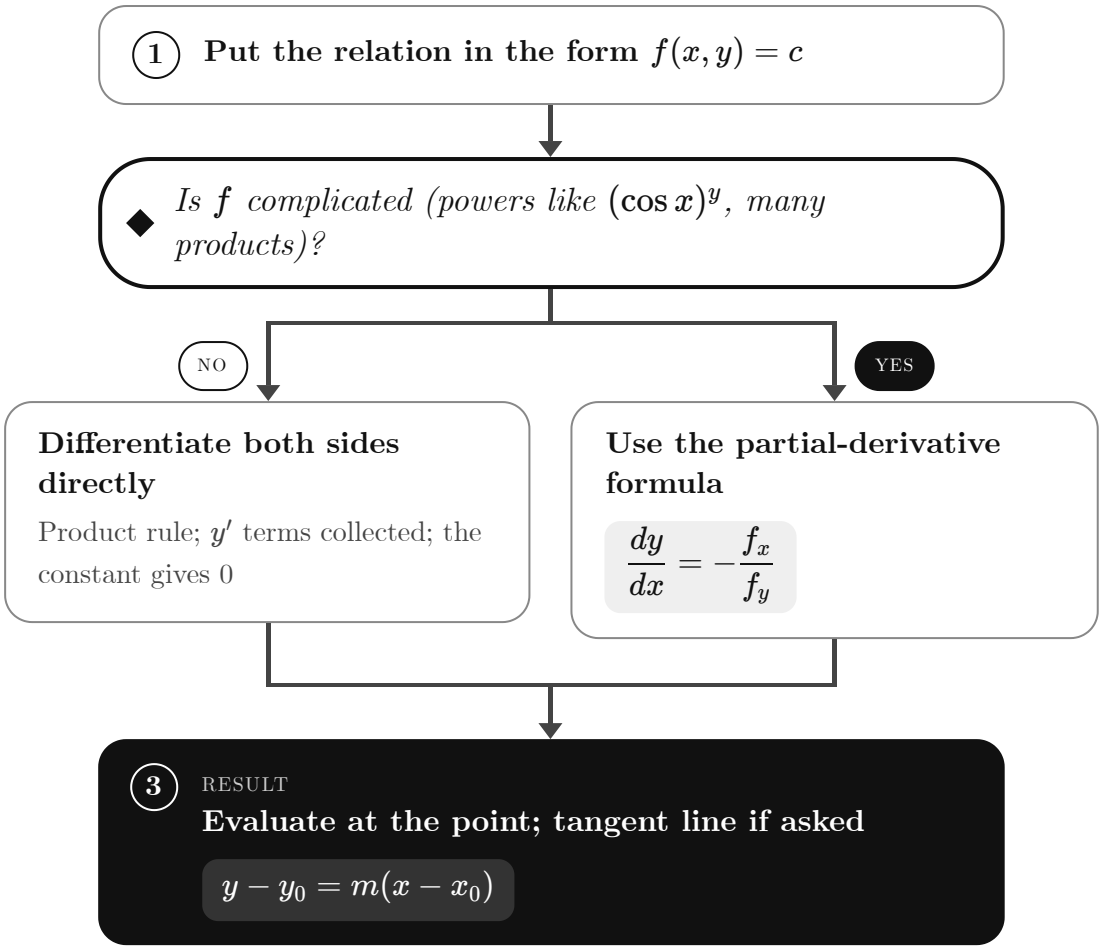


Figure G18.3. Level curves and the slope $-f_x/f_y$. ESA ★★

ADDED FOR CLARITY S004·4 · S003·7 · T004a 00:03:10

How to read it. Level curves of $f = x^2 + y^2$ for $c = 1, 2, 5, 9$: the cross-sections of the cylinders on slide 004·4. On $c = 5$ at $(2, 1)$ the dashed tangent has slope $-f_x/f_y = -2x/2y = -2$, the same $dy/dx = -x/y$ used in W16.1.

Flowcharts



Remark. Both routes give the same dy/dx (W18.1, W18.2). The formula needs $f_y \neq 0$ at the point.

Figure F18.1. Finding dy/dx for an implicit function: two routes. ESA ★★

S003·27 · S004·3 · S004·5 · T003b 00:10:33 · T004a 00:06:19

Important points and why

1. Differentiating an implicit relation directly needs the product rule on every term, and the right-hand constant gives 0. ESA ★★ S004·3 · T004a 00:00:48
2. The formula $dy/dx = -f_x/f_y$ gives the same answer and is far easier when f is complicated. ESA ★★ T004a 00:06:19 · T004a 00:06:36
3. Restriction to remember: the formula needs $z = f(x,y)$ to be a constant (a level curve). Move everything to one side first: $f(x,y)=c$ or $f(x,y)=0$. ESA ★★★ T004a 00:01:49 · T004a 00:02:04
4. There are two ways to solve "find dy/dx": differentiate both sides directly, or use $-f_x/f_y$. Both are accepted. ESA ★★ S003·27 · T003b 00:10:33
5. The slope dy/dx at a point gives the tangent line $y-y_0=m(x-x_0)$. ESA ★★ S004·7 · T004a 00:12:01
6. For $(\cos x)^y=(\sin y)^x$ the result needs $\cos x>0$ and $\sin y>0$, because of the logarithms. ESA ★ T004a 00:18:12

7. Implicit and composite together: if $u = \tan(x^2 + y^2)$ and $x^2 - y^2 = 2$, use the relation to eliminate y (or find dy/dx) before differentiating. ESA ★★ CW U1 P16 · CW U1 P18

Why is $dy/dx = -f_x/f_y$?
Along the curve $f(x, y) = c$ the value of f does not change, so $df = f_x\,dx + f_y\,dy = 0$. Divide by dx: $f_x + f_y\,\dfrac{dy}{dx} = 0$. The branch diagram (Figure G18.1) shows the same sum.
S004·4 · S004·5 · T004a 00:07:39

Why is the level-curve slope $-f_x/f_y$ and not $-f_y/f_x$ (as printed on slide 004·4)?
From $f_x + f_y\,y' = 0$ you get $y' = -f_x/f_y$. Test it on $x^2 + y^2 = 5$ at $(2, 1)$: $-f_x/f_y = -2x/(2y) = -2$, which matches differentiating directly; $-f_y/f_x$ would give $-1/2$.
ADDED FOR CLARITY S004·4 · S004·5

Exam trap. Slide 004·4 prints $dy/dx = -f_y(x, y)/f_x(x, y)$. The correct formula is $dy/dx = -f_x/f_y$. ADDED FOR CLARITY S004·4

Exam trap. In $(\cos x)^y$, with respect to x it is a power ($y(\cos x)^{y-1}(-\sin x)$); with respect to y it is an exponential ($(\cos x)^y \ln\cos x$). S004·8 · T004a 00:16:02

Worked examples

W18.1 · Direct differentiation vs the formula — ESA ★★

GIVEN $x^3y + 3xy^2 = 3$.

STEP 1 Direct: $x^3y' + 3x^2y + 3y^2 + 6xy\,y' = 0$.

STEP 2 Formula: $f_x = 3x^2y + 3y^2$, $f_y = x^3 + 6xy$.

STEP 3 $y' = -\dfrac{3x^2y + 3y^2}{x^3 + 6xy}$.

ANSWER $\boxed{\dfrac{dy}{dx} = -\dfrac{3y(x^2 + y)}{x(x^2 + 6y)}}$

CHECK Both methods agree; the slide's printed numerator is unclear.

S004·3 · T004a 00:00:48 · T004a 00:05:41

W18.2 · Two ways for $x^2 + xy + y^2 = 7$ ESA ★★

GIVEN $x^2 + xy + y^2 = 7$.

STEP 1 Direct: $2x + (xy' + y) + 2y\,y' = 0$, so $(x + 2y)y' = -(2x + y)$.

STEP 2 Formula: $f_x = 2x + y$, $f_y = x + 2y$, $y' = -f_x/f_y$.

ANSWER $\boxed{\dfrac{dy}{dx} = -\dfrac{2x + y}{x + 2y}}$

CHECK Same by both methods.

S003·27 · T003b 00:10:23

W18.3 · Tangent line at the origin ESA ★★

GIVEN $xe^y + ye^x = 0$; tangent at $(0, 0)$.

STEP 1 $f_x = e^y + ye^x$, $f_y = xe^y + e^x$.

STEP 2 $\dfrac{dy}{dx} = -\dfrac{e^y + ye^x}{xe^y + e^x}$; at $(0, 0)$: $-\dfrac{1 + 0}{0 + 1} = -1$.

STEP 3 $y - 0 = -1(x - 0)$.

ANSWER $\boxed{x + y = 0}$

CHECK $(0, 0)$ satisfies the equation.

S004·7 · T004a 00:11:48

W18.4 · $(\cos x)^y = (\sin y)^x$ ESA ★★

GIVEN $f(x, y) = (\cos x)^y - (\sin y)^x = 0$.

STEP 1 $f_x = y(\cos x)^{y-1}(-\sin x) - (\sin y)^x \ln \sin y$.

STEP 2 $f_y = (\cos x)^y \ln \cos x - x(\sin y)^{x-1} \cos y$.

STEP 3 $y' = -f_x/f_y$. Divide through by $(\cos x)^y = (\sin y)^x$, then multiply top and bottom by $\cos x \sin y$.

ANSWER $\boxed{\dfrac{dy}{dx} = \dfrac{\sin y\,(y \sin x + \cos x \ln \sin y)}{\cos x\,(\sin y \ln \cos x - x \cos y)}}$

CHECK Checked numerically at a point on the curve ($x = 0.5$, $y \approx 0.909$): both give -0.4980.

S004·8 · T004a 00:14:57

W18.5 · Slope at (1, 1) ESA ★★

GIVEN $e^y - e^x + xy = 1$ at $(1, 1)$.

STEP 1 Check: $e - e + 1 = 1$, so the point is on the curve.

STEP 2 $f_x = -e^x + y$, $f_y = e^y + x$.

STEP 3 $y' = -\dfrac{-e+1}{e+1}$.

ANSWER $\boxed{\dfrac{dy}{dx} = \dfrac{e-1}{e+1}}$

CHECK CW answer: $(e-1)/(e+1)$.

CW U1 P17

W18.6 · Composite with an implicit constraint ESA ★★

GIVEN $u = \tan(x^2 + y^2)$ with $x^2 - y^2 = 2$; find du/dx.

STEP 1 From the relation, $y^2 = x^2 - 2$, so $u = \tan(2x^2 - 2)$.

STEP 2 $\dfrac{du}{dx} = \sec^2(2x^2 - 2) \cdot 4x$.

ANSWER $\boxed{\dfrac{du}{dx} = 4x \sec^2(2x^2 - 2)}$

CHECK Via the chain rule: $u_x + u_y\, y'$ with $y' = x/y$ gives the same. CW answer agrees.

CW U1 P16

W18.7 · Total derivative along an implicit curve ESA ★★

GIVEN $u = x \log(xy)$ with $x^3 + y^3 - 3xy - 1 = 0$; find du/dx.

STEP 1 $\dfrac{du}{dx} = u_x + u_y \dfrac{dy}{dx}$, with $u_x = \log(xy) + 1$, $u_y = \dfrac{x}{y}$.

STEP 2 From the relation: $\dfrac{dy}{dx} = -\dfrac{3x^2 - 3y}{3y^2 - 3x} = \dfrac{y - x^2}{y^2 - x}$.

ANSWER $\boxed{\dfrac{du}{dx} = 1 + \log(xy) + \dfrac{x\,(y - x^2)}{y\,(y^2 - x)}}$

CHECK CW answer agrees; sympy residual 0.

CW U1 P18

SHORT-NOTE RECAP

An implicit function is given by a relation $f(x, y) = c$, a level curve. Along it $df = f_x\,dx + f_y\,dy = 0$, so $\dfrac{dy}{dx} = -\dfrac{f_x}{f_y}$ $(f_y \neq 0)$. Direct differentiation of both sides gives the same result. The slope at (x_0, y_0) gives the tangent $y - y_0 = m(x - x_0)$; e.g. $xe^y + ye^x = 0$ has tangent $x + y = 0$ at the origin.

19 Implicit functions of three variables

A relation $f(x,y,z)=0$ is a level surface and makes z a function of x and y. Holding y fixed gives $\partial z/\partial x = -f_x/f_z$. Holding z fixed gives $\partial y/\partial x = -f_x/f_y$. Each needs the denominator to be non-zero, and the point used must lie on the surface.

Key terms and definitions

Implicit function of three variables. $f(x,y,z)=0$ with z (or y) regarded as a function of the remaining variables; the other variables are intermediate variables. S004·6 · T004a 00:09:05

> *"Let $f(x,y,z)=0$ be the equation of an implicit function of three variables x,y,z. Suppose y and z are functions of x, then f is a function of one independent variable x and y, z are intermediate variables."* S004·6

Formulas

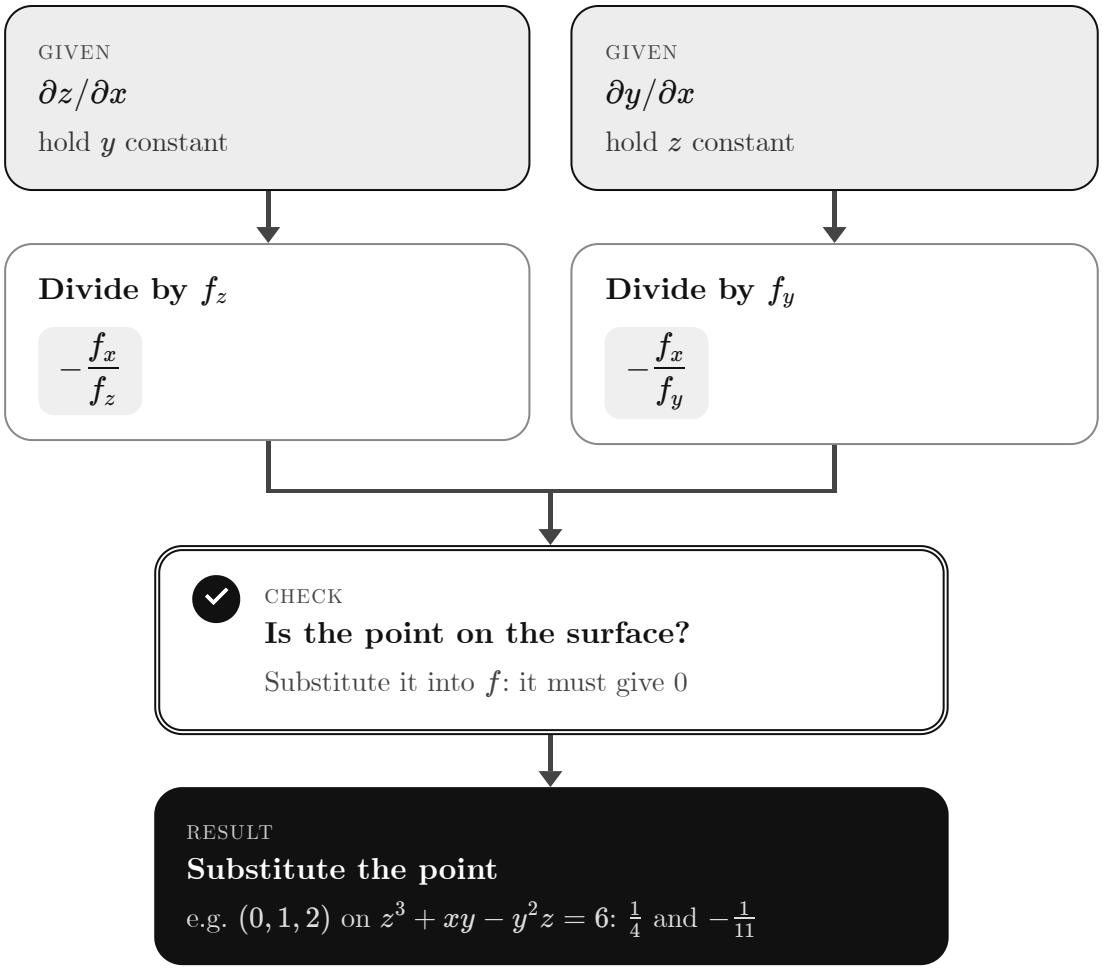
E19.1 · Implicit differentiation, three variables ESA ★★

$$f(x,y,z)=0: \quad \frac{\partial z}{\partial x}=-\frac{f_x}{f_z}, \quad \frac{\partial z}{\partial y}=-\frac{f_y}{f_z}\ (f_z\neq 0); \qquad \frac{\partial y}{\partial x}=-\frac{f_x}{f_y}\ (f_y\neq 0)$$

gives	Partial derivatives of the implicitly defined variable.
applies	Hold y constant for z_x; hold z constant for y_x.
note	Extends to any number of variables.

S004·6 · S003·26 · T004a 00:10:23 · T003b 00:09:29

Flowcharts



Exam trap. The lecture used $(1,0,2)$, which is not on the surface; always check the point first.

Figure F19.1. Partial derivatives from $f(x,y,z)=0$. ESA ★★

S004·6 · S004·9 · S003·26

Important points and why

1. For $f(x,y,z)=0$, x and y are independent, so dz alone has no meaning; find the partials z_x, z_y. ESA ★★ T004a 00:09:38
2. For $\partial z/\partial x$ treat y as constant; for $\partial y/\partial x$ treat z as constant. ESA ★★ S004·6 · S004·9 · T004a 00:19:00
3. The formula avoids separating z from a messy equation such as $x^3y^2+3xyz+\sin(e^z)=c$. ESA ★ T004a 00:10:55
4. Check that the given point satisfies $f=0$ before substituting. ESA ★★ (added for clarity) S004·9 · T004a 00:18:39

Exam trap. The lecture works problem 3 at $(1,0,2)$ and gets 0 for both derivatives, but $(1,0,2)$ is not on $z^3+xy-y^2z=6$ (it gives 8). Use the slide's point $(0,1,2)$: $\partial y/\partial x = 1/4$, $\partial z/\partial x = -1/11$. ADDED FOR CLARITY S004·9 · T004a 00:20:47

Exam trap. Do not differentiate $-f_x/f_z$ as if z were a constant when you need second derivatives: z is still a function of x and y (W19.3). ADDED FOR CLARITY ESA Jan-24 s3 1d

Worked examples

W19.1 · $\partial y/\partial x$ and $\partial z/\partial x$ at $(0,1,2)$ — ESA ★★

GIVEN $z^3+xy-y^2z=6$ at $(0,1,2)$.

STEP 1 Check the point: $8+0-2=6$.

STEP 2 $f=z^3+xy-y^2z-6$: $f_x=y$, $f_y=x-2yz$, $f_z=3z^2-y^2$.

STEP 3 $\dfrac{\partial y}{\partial x}=-\dfrac{y}{x-2yz}=-\dfrac{1}{0-4}$; $\dfrac{\partial z}{\partial x}=-\dfrac{y}{3z^2-y^2}=-\dfrac{1}{12-1}$.

ANSWER $\boxed{\dfrac{\partial y}{\partial x}=\dfrac{1}{4},\quad \dfrac{\partial z}{\partial x}=-\dfrac{1}{11}}$

CHECK The lecture's point $(1,0,2)$ is not on the surface (see Exam trap).

S004·9 · T004a 00:18:39

W19.2 · $\partial z/\partial x$ and $\partial z/\partial y$ at the origin — ESA ★★

GIVEN $x^3+z^2+ye^{xz}+z\cos y=0$ at $(0,0,0)$.

STEP 1 $F_x=3x^2+zye^{xz}$, $F_y=e^{xz}-z\sin y$, $F_z=2z+xye^{xz}+\cos y$.

STEP 2 At the origin: $F_x=0$, $F_y=1$, $F_z=1$.

ANSWER $\boxed{\dfrac{\partial z}{\partial x}=0,\quad \dfrac{\partial z}{\partial y}=-1}$

CHECK The origin satisfies the equation.

S003·26 · T003b 00:09:15

W19.3 · Second derivative of an implicit function ESA ★★

GIVEN $x^x y^y z^z = c$. Show that at $x = y = z$, $\dfrac{\partial^2 z}{\partial x\,\partial y} = -[x\log(ex)]^{-1}$.

STEP 1 Take logs: $f = x\log x + y\log y + z\log z - \log c = 0$, so $f_x = 1 + \log x$, $f_y = 1 + \log y$, $f_z = 1 + \log z$.

STEP 2 $z_x = -\dfrac{1+\log x}{1+\log z}$, $z_y = -\dfrac{1+\log y}{1+\log z}$.

STEP 3 $z_{xy} = \dfrac{\partial}{\partial y} z_x = \dfrac{1+\log x}{(1+\log z)^2}\cdot\dfrac{1}{z}\,z_y$ (z depends on y).

STEP 4 At $x = y = z$: $z_y = -1$, so $z_{xy} = -\dfrac{1}{x(1+\log x)}$, and $1 + \log x = \log(ex)$.

ANSWER $\boxed{\dfrac{\partial^2 z}{\partial x\,\partial y} = -\dfrac{1}{x\log(ex)}}$

CHECK sympy residual 0.

ADDED FOR CLARITY ESA Jan-24 s3 1d · CW U1 P6

Past-paper questions

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q1.d · solved in W19.3

For the point on the surface $x^x y^y z^z = c$, show that at $x = y = z$, $\frac{\partial^2 z}{\partial x \partial y} = -[x\log(ex)]^{-1}$.

SHORT-NOTE RECAP

For $f(x,y,z) = 0$, z is a function of x and y: $\dfrac{\partial z}{\partial x} = -\dfrac{f_x}{f_z}$ (hold y), $\dfrac{\partial z}{\partial y} = -\dfrac{f_y}{f_z}$, and $\dfrac{\partial y}{\partial x} = -\dfrac{f_x}{f_y}$ (hold z), provided the denominators are non-zero. Check that the point lies on the surface. For second derivatives, remember that z still depends on x, y.

M4

Class 005 · Homogeneous Functions and Euler's Theorem

20 Homogeneous functions

A function is homogeneous of degree n if it can be written as x^n times a function of the ratios (y/x, or y/x and z/x). Equivalently, scaling every variable by t scales the function by t^n. Polynomials are homogeneous when every term has the same degree. The degree can be positive, negative, fractional or zero.

Key terms and definitions

Homogeneous function (two variables). A function $u = f(x, y)$ is said to be a homogeneous function of degree n if it can be expressed in the form $x^n g\left(\frac{y}{x}\right)$ or $y^n g\left(\frac{x}{y}\right)$, g being an arbitrary function. S005·3 · T005 00:02:45

Homogeneous function (three variables). $u = f(x, y, z)$ is homogeneous of degree n if it can be expressed as $x^n g\left(\frac{y}{x}, \frac{z}{x}\right)$, or $y^n g\left(\frac{x}{y}, \frac{z}{y}\right)$, or $z^n g\left(\frac{x}{z}, \frac{y}{z}\right)$. S005·4 · T005 00:04:38

Scaling test. f is homogeneous of degree n exactly when $f(tx, ty) = t^n f(x, y)$ for all $t > 0$: scaling every variable by t scales the function by t^n. (added for clarity) S005·5

Degree. The power n in $x^n g(y/x)$; for a polynomial whose terms all have the same total degree, that common degree. T005 00:01:31

Formulas

E20.1 · Homogeneous of degree n ESA ★★

$$u = x^n g\left(\frac{y}{x}\right) = y^n h\left(\frac{x}{y}\right) \quad \Longleftrightarrow \quad u(tx, ty) = t^n\, u(x, y)$$

gives	The definition and the equivalent scaling test.
applies	Two variables; with three, use $x^n g(y/x, z/x)$.
note	The scaling test is added for clarity; the slides use the $x^n g(y/x)$ form.

S005·3 · S005·4 · S005·5

E20.2 · Degree of a quotient and of a product ESA ★★

$$\deg \frac{P}{Q} = \deg P - \deg Q, \qquad \deg(PQ) = \deg P + \deg Q$$

GIVES Degree of combinations of homogeneous functions.

APPLIES P, Q homogeneous.

NOTE $\frac{\sqrt{x}+\sqrt{y}}{x+y}$: $\frac{1}{2} - 1 = -\frac{1}{2}$. A ratio such as $\sin^{-1}(y/x)$ has degree 0.

ADDED FOR CLARITY T005 00:07:50

Graphs and diagrams

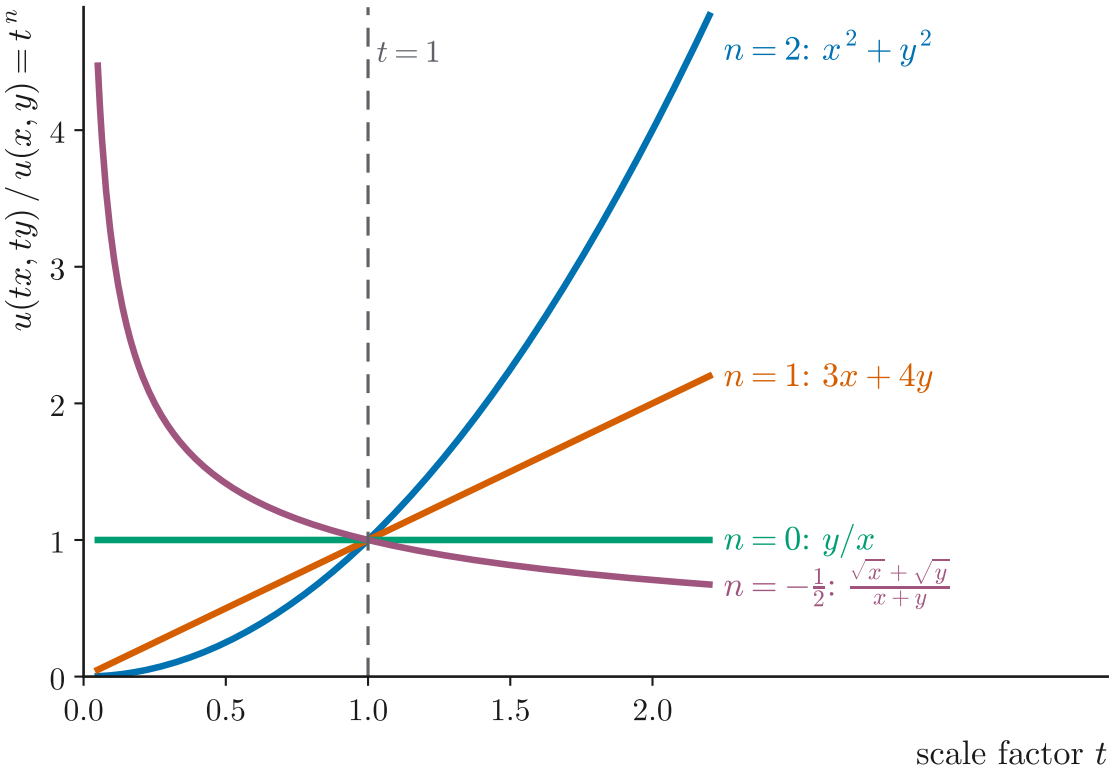
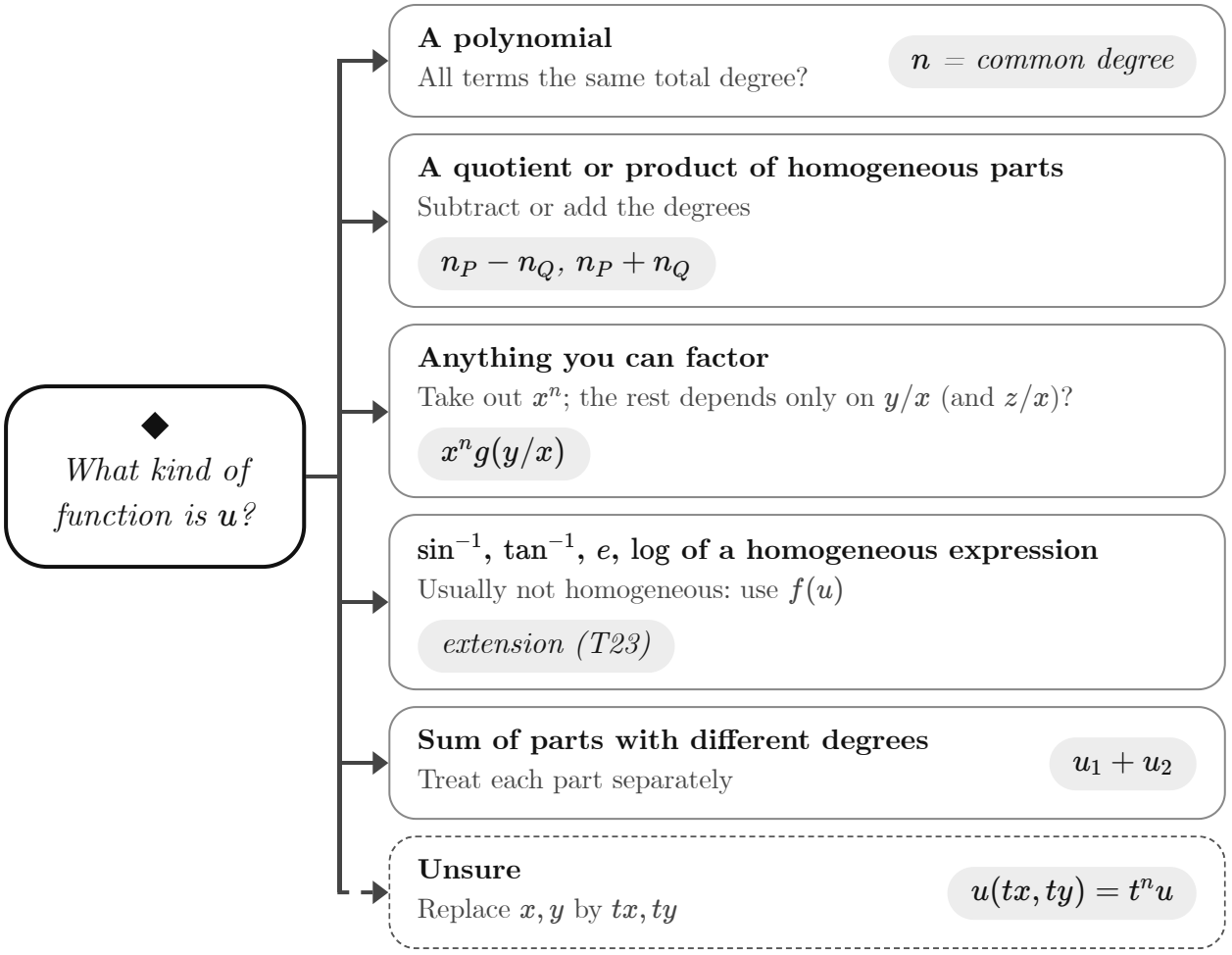


Figure G20.1. Scaling along a ray: $u(tx, ty) = t^n u(x, y)$. ESA ★★

ADDED FOR CLARITY S005·3 · S005·5

How to read it. Start at a point and move out along the ray through the origin by the factor t. A homogeneous function of degree n changes like t^n: degree 2 grows quadratically, degree 1 linearly, degree 0 stays constant along the ray (but can vary between rays), and degree $-\frac{1}{2}$ decays. Each curve is $u(t)/u(1)$.

Flowcharts



Remark. The dashed route (the scaling test) is added for clarity; it always works and confirms the degree found by factoring.

Figure F20.1. Is it homogeneous, and of what degree? ESA ★★

S005·3 · S005·4 · T005 00:02:36 · T005 00:05:42

Important points and why

1. For a polynomial, check that every term has the same total degree; that is the degree ($x^3 + 3x^2y + 3xy^2 + y^3$: degree 3). ESA ★★ T005 00:01:15 · T005 00:01:31
2. To test any function: take out x^n (or y^n) and check that what remains depends only on y/x (and z/x). ESA ★★ S005·3 · T005 00:05:42
3. Either form, $x^n g(y/x)$ or $y^n h(x/y)$, may be used — whichever is convenient — and both give the same degree. ESA ★★ T005 00:04:01
4. Inverse trigonometric or exponential functions of a homogeneous expression are usually not homogeneous; first find an f with $f(u)$ homogeneous (topic 4). ESA ★★★ T005 00:01:47 · T005 00:02:04
5. Homogeneous functions appeared earlier in homogeneous differential equations, solved by $y = vx$. ESA ★ (transcript only) T005 00:00:22

Exam trap. The lecture says the converse holds: "a function of degree 0 should be a constant function". It does not: y/x, $\sin^{-1}(y/x)$ and $\frac{x-y}{x+y}$ all have degree 0 and are not constant.

ADDED FOR CLARITY T005 00:07:11

Exam trap. "Homogeneous" does not mean "smooth everywhere" (as said at T005 00:11:06); it is a scaling property. $(\sqrt{x}+\sqrt{y})/(x+y)$ is homogeneous but not defined at the origin.

ADDED FOR CLARITY T005 00:11:06

Worked examples

W20.1 · Degree of a linear function — ESA ★

GIVEN $u = 3x + 4y.$

STEP 1 $u = x\left(3 + 4\frac{y}{x}\right) = x^1 g\left(\frac{y}{x}\right).$

ANSWER Homogeneous of degree 1

CHECK $u(tx, ty) = t\,u.$

S005·3

W20.2 · Three variables — ESA ★★

GIVEN $u = x^3 + y^3 + z^3 + 3xyz.$

STEP 1 $u = x^3\left(1 + \left(\frac{y}{x}\right)^3 + \left(\frac{z}{x}\right)^3 + 3\frac{y}{x}\frac{z}{x}\right) = x^3 g\left(\frac{y}{x}, \frac{z}{x}\right).$

ANSWER Homogeneous of degree 3

CHECK Every term has total degree 3. (The slide prints the second term's exponent as 2; it should be 3.)

S005·4 · T005 00:04:52

W20.3 · A negative, fractional degree — ESA ★★

GIVEN $u = \dfrac{\sqrt{x}+\sqrt{y}}{x+y}.$

STEP 1 Numerator: $\sqrt{x}\left(1+\sqrt{y/x}\right)$; denominator: $x(1+y/x)$.

STEP 2 $u = x^{1/2-1}\, g\left(\frac{y}{x}\right) = x^{-1/2} g\left(\frac{y}{x}\right).$

ANSWER Homogeneous of degree $-\frac{1}{2}$

CHECK Euler: $xu_x + yu_y = -\frac{1}{2}u$ (sympy).

TRANSCRIPT ONLY T005 00:07:50

SHORT-NOTE RECAP

$u = f(x, y)$ is homogeneous of degree n if $u = x^n g(y/x)$ or $y^n g(x/y)$; in three variables $x^n g(y/x, z/x)$. Equivalently $u(tx, ty) = t^n u$. A polynomial is homogeneous if all its terms have the same degree. The degree may be positive, negative, fractional or zero, e.g. $(\sqrt{x} + \sqrt{y})/(x + y)$ has degree $-\frac{1}{2}$.

21 Euler's theorem

For a homogeneous function of degree n, $x\,u_x + y\,u_y = n\,u$ (and $x\,u_x + y\,u_y + z\,u_z = n\,u$ in three variables). Once you know the function is homogeneous and its degree, the combination is known without differentiating. Geometrically, the changes along x and y, scaled by x and y, add up to the function scaled by its degree.

Key terms and definitions

> *"Statement 1: If $u = f(x, y)$ is a homogeneous function of degree n then $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = nu$."* S005·6

> *"Statement 2: If $u = f(x, y, z)$ is a homogeneous function of degree n then $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z} = nu$."* S005·6

Formulas

E21.1 · Euler's theorem, two variables ESA ★★★

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\,u$$

gives	The scaled sum of first partials.
applies	u homogeneous of degree n.
note	Compact form: $xu_x + yu_y = nu$.

S005·6 · T005 00:08:57

E21.2 · Euler's theorem, three variables ESA ★★

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = n\,u$$

gives	The same result with one term per variable.
applies	$u = f(x, y, z)$ homogeneous of degree n.
note	Extends to any number of variables.

S005·6 · T005 00:14:28

Graphs and diagrams

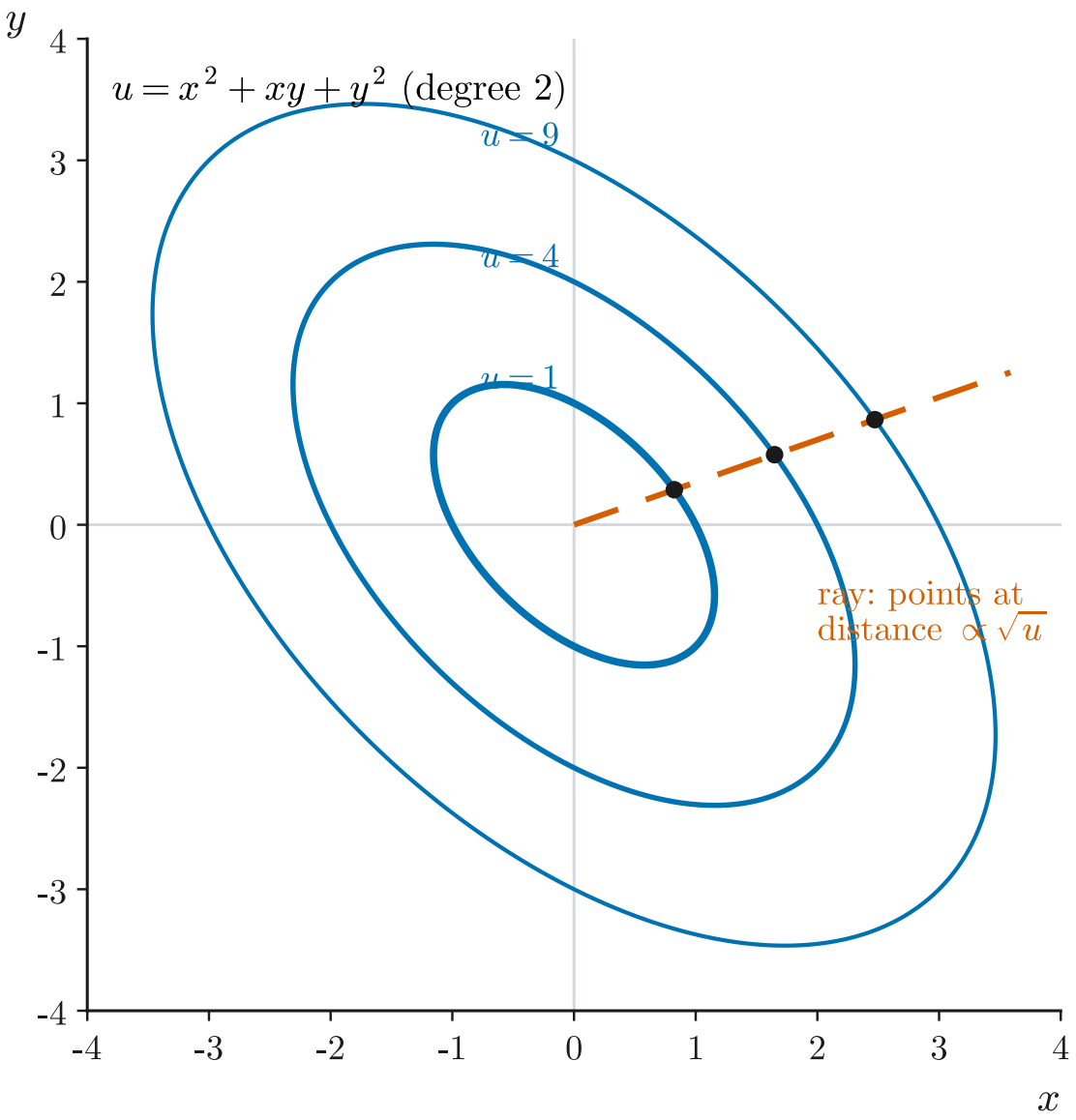


Figure G21.1. Level curves of a homogeneous function are scaled copies. ESA ★★

ADDED FOR CLARITY S005·5 · T005 00:10:18

How to read it. Level curves of $u = x^2 + xy + y^2$ (degree 2) for $u = 1, 4, 9$. Each is the first one scaled by $1, 2, 3 = \sqrt{u}$. Moving out along the dashed ray, the rate of change $xu_x + yu_y$ (the radial change scaled by distance) equals $2u$: this is Euler's theorem.

Euler's theorem and its relatives

	Applies when	Left side	Right side
Euler, first order	u homogeneous, degree n	$xu_x + yu_y$	nu
Corollary, second order	u homogeneous, degree n	$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$	$n(n-1)u$
Both added	u homogeneous, degree n	first + second order	n^2u
Extension, first order	$f(u)$ homogeneous, degree n	$xu_x + yu_y$	$g(u) = nf(u)/f'(u)$
Extension, second order	$f(u)$ homogeneous, degree n	$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$	$g(u)[g'(u) - 1]$
Three variables	$u(x, y, z)$ homogeneous	$xu_x + yu_y + zu_z$	nu

ADDED FOR CLARITY S005·6 · S005·7

Important points and why

1. You do not need to differentiate: check that u is homogeneous, find n, and write $xu_x + yu_y = nu$. ESA ★★★ T005 00:09:17 · T005 00:17:32
2. The lecturer calls Euler's theorem "the most important one" of the session. ESA ★★★ T005 00:08:57
3. Check: $u = x^2 + y^2$ gives $x(2x) + y(2y) = 2u$, degree 2. ESA ★★ T005 00:10:31
4. Meaning: the change along x scaled by x, plus the change along y scaled by y, equals the function scaled by its degree. ESA ★ T005 00:09:46 · T005 00:11:24

Why is $xu_x + yu_y = nu$? (A short proof.)
Differentiate $u(tx, ty) = t^n u(x, y)$ with respect to t using the chain rule: $x\,u_x(tx, ty) + y\,u_y(tx, ty) = nt^{n-1}u(x, y)$. Put $t = 1$.

ADDED FOR CLARITY S005·6

Worked examples

W21.1 · Euler's theorem checked directly ESA ★★

GIVEN $u = x^2 + y^2$.

STEP 1 $u_x = 2x$, $u_y = 2y$.

STEP 2 $xu_x + yu_y = 2x^2 + 2y^2$.

ANSWER $\boxed{xu_x + yu_y = 2u}$

CHECK Degree 2, as Euler predicts.

T005 00:10:31

W21.2 · Degree 1 with an inverse sine ESA ★★

GIVEN $u = \sqrt{y^2 - x^2}\,\sin^{-1}(x/y)$; find $xu_x + yu_y$.

STEP 1 $\sqrt{y^2 - x^2} = y\sqrt{1 - (x/y)^2}$ has degree 1; $\sin^{-1}(x/y)$ has degree 0.

STEP 2 So $u = y^1 h(x/y)$: degree 1.

ANSWER $\boxed{xu_x + yu_y = u}$

CHECK CW answer: u.

CW U1 P21

SHORT-NOTE RECAP

Euler's theorem: if $u = f(x, y)$ is homogeneous of degree n then $xu_x + yu_y = nu$; for $f(x, y, z)$, $xu_x + yu_y + zu_z = nu$. So only homogeneity and the degree are needed, not the derivatives. Proof: differentiate $u(tx, ty) = t^n u$ with respect to t and put $t = 1$.

22 Second and higher order: the corollary

Applying Euler's idea twice gives $x^2u_{xx}+2xy\,u_{xy}+y^2u_{yy}=n(n-1)\,u$, and three times gives $n(n-1)(n-2)\,u$. When u is a sum of homogeneous parts of different degrees, apply the results to each part separately. This is the commonest ESA question on the topic.

Key terms and definitions

> *"Corollary 1: If $u=f(x,y)$ is a homogeneous function of degree n then $x^2\dfrac{\partial^2u}{\partial x^2}+2xy\dfrac{\partial^2u}{\partial x\partial y}+y^2\dfrac{\partial^2u}{\partial y^2}=n(n-1)u$."* S005·6

Formulas

E22.1 · Corollary: second order

ESA ★★★

$$x^2\frac{\partial^2u}{\partial x^2}+2xy\frac{\partial^2u}{\partial x\,\partial y}+y^2\frac{\partial^2u}{\partial y^2}=n(n-1)\,u$$

GIVES	The scaled sum of second partials.
APPLIES	u homogeneous of degree n with continuous second partials.
NOTE	Coefficients $1,2,1$ as in $(x+y)^2$.

S005·6 · T005 00:12:21

E22.2 · Third order

ESA ★

$$x^3u_{xxx}+3x^2y\,u_{xxy}+3xy^2u_{xyy}+y^3u_{yyy}=n(n-1)(n-2)\,u$$

GIVES	The pattern continues with binomial coefficients.
APPLIES	Third partials continuous.
NOTE	The lecture states the right-hand side; the coefficients 1, 3, 3, 1 are added for clarity.

TRANSCRIPT ONLY · ADDED FOR CLARITY · T005 00:12:43 · T005 00:14:12

E22.3 · First plus second order

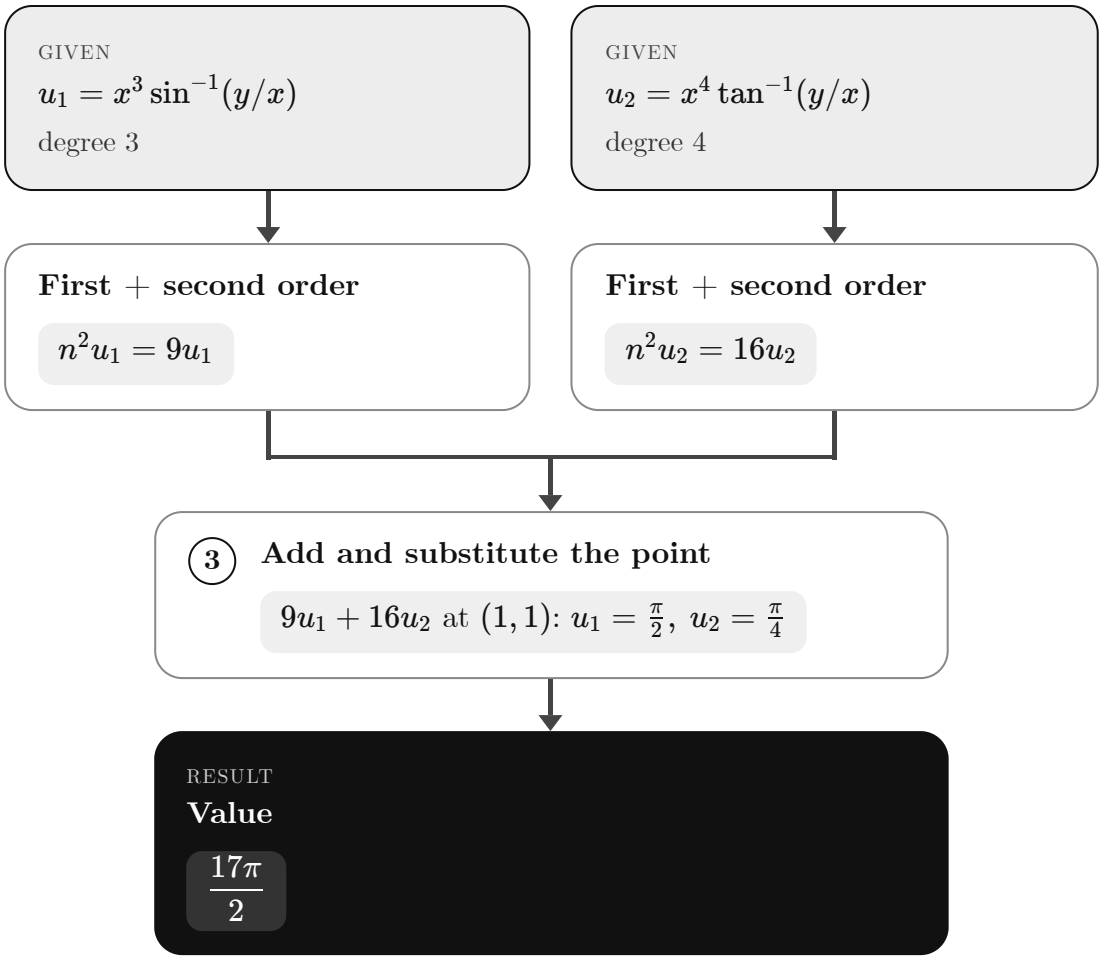
ESA ★★★

$$x^2u_{xx}+2xy\,u_{xy}+y^2u_{yy}+x\,u_x+y\,u_y=n(n-1)u+nu=n^2u$$

GIVES	The combination asked in ESA Jan 2024 sets 3 and 4.
APPLIES	u homogeneous of degree n; for $u=u_1+u_2$ apply to each part.
NOTE	Degrees 3 and -3 both give 9: so $9u$ for $x^3\sin^{-1}\frac{y}{x}+y^{-3}\tan^{-1}\frac{x}{y}$.

ADDED FOR CLARITY · ESA Jan-24 s3 1c · ESA Jan-24 s4 1d · CW U1 P23

Flowcharts



Remark. Only first order: n; only second order: $n(n-1)$; both: n^2. Read which combination the question asks for.

Figure F22.1. Evaluating Euler combinations for a sum of homogeneous parts. ESA ★★★

ESA Jan-24 s3 1c · ESA Jan-24 s4 1d

Important points and why

1. The mixed partials u_{xy} and u_{yx} are equal here (continuous derivatives), which is why the middle term is $2xy\,u_{xy}$. ESA ★★ T005 00:12:04
2. For $u = u_1 + u_2$ with different degrees n_1, n_2, apply the results to each part: $x^2 u_{xx} + \cdots = n_1(n_1 - 1)u_1 + n_2(n_2 - 1)u_2$. ESA ★★★ ADDED FOR CLARITY ESA Jan-24 s3 1c · CW U1 P22
3. Extend to three or more variables as an exercise; the lecture leaves it open and points to a research paper on higher-order extensions. ESA ★ S005·6 · T005 00:14:59

Why is the second-order right side $n(n-1)u$?
u_x and u_y are homogeneous of degree $n-1$, so Euler's theorem applied to them gives $xu_{xx}+yu_{xy}=(n-1)u_x$ and $xu_{yx}+yu_{yy}=(n-1)u_y$. Multiply by x and y, add, and use $xu_x+yu_y=nu$.

ADDED FOR CLARITY S005·6

Worked examples

W22.1 · Prove 5u and 20u — ESA ★★★

GIVEN $u=x^3y^2\sin^{-1}(y/x)$. Prove (i) $xu_x+yu_y=5u$, (ii) $x^2u_{xx}+2xy\,u_{xy}+y^2u_{yy}=20u$.

STEP 1 $x^3y^2=x^5\left(\frac{y}{x}\right)^2$ and $\sin^{-1}(y/x)$ depends only on y/x, so $u=x^5g(y/x)$: degree 5.

STEP 2 (i) Euler: $5u$.

STEP 3 (ii) Corollary: $n(n-1)u=5\cdot 4\,u$.

ANSWER (i) $5u$, (ii) $20u$

CHECK Direct differentiation in sympy gives the same.

ESA Jul-23 1a

W22.2 · Fractional degree, both orders — ESA ★★

GIVEN $u=\dfrac{y^{1/3}-x^{1/3}}{x+y}$.

STEP 1 Degree $\frac{1}{3}-1=-\frac{2}{3}$.

STEP 2 (i) $xu_x+yu_y=-\frac{2}{3}u$.

STEP 3 (ii) $n(n-1)=\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)=\frac{10}{9}$.

ANSWER (i) $-\frac{2}{3}u$, (ii) $\frac{10}{9}u$

CHECK CW answers agree.

CW U1 P19

W22.3 · Sum of two degree-3 parts — ESA ★★

GIVEN $u=x^3\sin^{-1}(y/x)+y^3\tan^{-1}(x/y)$; find $x^2u_{xx}+2xy\,u_{xy}+y^2u_{yy}$.

STEP 1 Both parts are homogeneous of degree 3, so u is.

STEP 2 $n(n-1)=3\cdot 2$.

ANSWER $6u$

CHECK CW answer: $6u$.

CW U1 P22

W22.4 · Degrees 3 and −3 together ESA ★★★

GIVEN $u = x^3 \sin^{-1}(y/x) + y^{-3}\tan^{-1}(x/y)$; evaluate $x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} + xu_x + yu_y$.

STEP 1 $u_1 = x^3\sin^{-1}(y/x)$: degree 3; $u_2 = y^{-3}\tan^{-1}(x/y)$: degree -3.

STEP 2 For each part the sum is $n(n-1)+n = n^2$: $9u_1$ and $9u_2$.

ANSWER $\boxed{9u}$

CHECK CW answer: $9u$; sympy agrees.

ESA Jan-24 s4 1d · CW U1 P23

W22.5 · Degrees 3 and 4, value at (1, 1) ESA ★★★

GIVEN $u = x^3\sin^{-1}(y/x) + x^4\tan^{-1}(y/x)$; find $x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} + xu_x + yu_y$ at $x=1, y=1$.

STEP 1 $u_1 = x^3\sin^{-1}(y/x)$ (degree 3) gives $9u_1$; $u_2 = x^4\tan^{-1}(y/x)$ (degree 4) gives $16u_2$.

STEP 2 At $(1,1)$: $u_1 = \sin^{-1}1 = \frac{\pi}{2}$, $u_2 = \tan^{-1}1 = \frac{\pi}{4}$.

STEP 3 $9\cdot\frac{\pi}{2} + 16\cdot\frac{\pi}{4}$.

ANSWER $\boxed{\dfrac{17\pi}{2}}$

CHECK sympy: the expression equals $9u_1 + 16u_2$ identically.

ESA Jan-24 s3 1c

Past-paper questions

ESA Jul 2023 (6 marks) UE22MA141A Q1.a · solved in W22.1

If $u(x,y) = x^3y^2\sin^{-1}\left(\frac{y}{x}\right)$, prove that (i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 5u$, (ii) $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 20u$.

ESA Jan 2024 (6 marks) UE23MA141A set 3, Q1.c · solved in W22.5

If $u = x^3\sin^{-1}\left(\frac{y}{x}\right) + x^4\tan^{-1}\left(\frac{y}{x}\right)$, find $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ at $x=1$, $y=1$ by using Euler's theorem and its extension.

ESA Jan 2024 (6 marks) UE23MA141A set 4, Q1.d · solved in W22.4

If $u = x^3\sin^{-1}\left(\frac{y}{x}\right) + y^{-3}\tan^{-1}\left(\frac{x}{y}\right)$, evaluate $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ by using Euler's theorem and its extension.

SHORT-NOTE RECAP

Corollary: $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$; adding the first-order result gives n^2u. For a sum of homogeneous parts of different degrees apply this to each part, e.g. degrees 3 and 4 give $9u_1 + 16u_2$. Third order: $n(n-1)(n-2)u$.

23 Extension: when f(u) is homogeneous

Functions such as $\tan^{-1}\frac{x^4+y^4}{xy}$ are not homogeneous, but $\tan u$ is. If $f(u)$ is homogeneous of degree n, then $x\,u_x + y\,u_y = g(u) = n\,\dfrac{f(u)}{f'(u)}$ and $x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = g(u)\,[g'(u) - 1]$. This extension is on the last content slide and is asked in two ESA papers.

Key terms and definitions

> *"If $f(u)$ be a homogeneous function of degree n in x and y, then $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = n\dfrac{f(u)}{f'(u)}$ and $x^2\dfrac{\partial^2 u}{\partial x^2} + 2xy\dfrac{\partial^2 u}{\partial x\partial y} + y^2\dfrac{\partial^2 u}{\partial y^2} = g(u)[g'(u) - 1]$ where $g(u) = n\dfrac{f(u)}{f'(u)}$."* S005·7

Formulas

E23.1 · Extension, first order

ESA ★★★

$$f(u) \text{ homogeneous of degree } n \implies x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\frac{f(u)}{f'(u)} = g(u)$$

gives	Euler's combination for a function that is not itself homogeneous.
applies	Some $f(u)$ is homogeneous (e.g. $\tan u$, $\sin u$, e^u).
note	$\tan u$: $g = n\tan u\cos^2 u = \frac{n}{2}\sin 2u$.

S005·7

E23.2 · Extension, second order

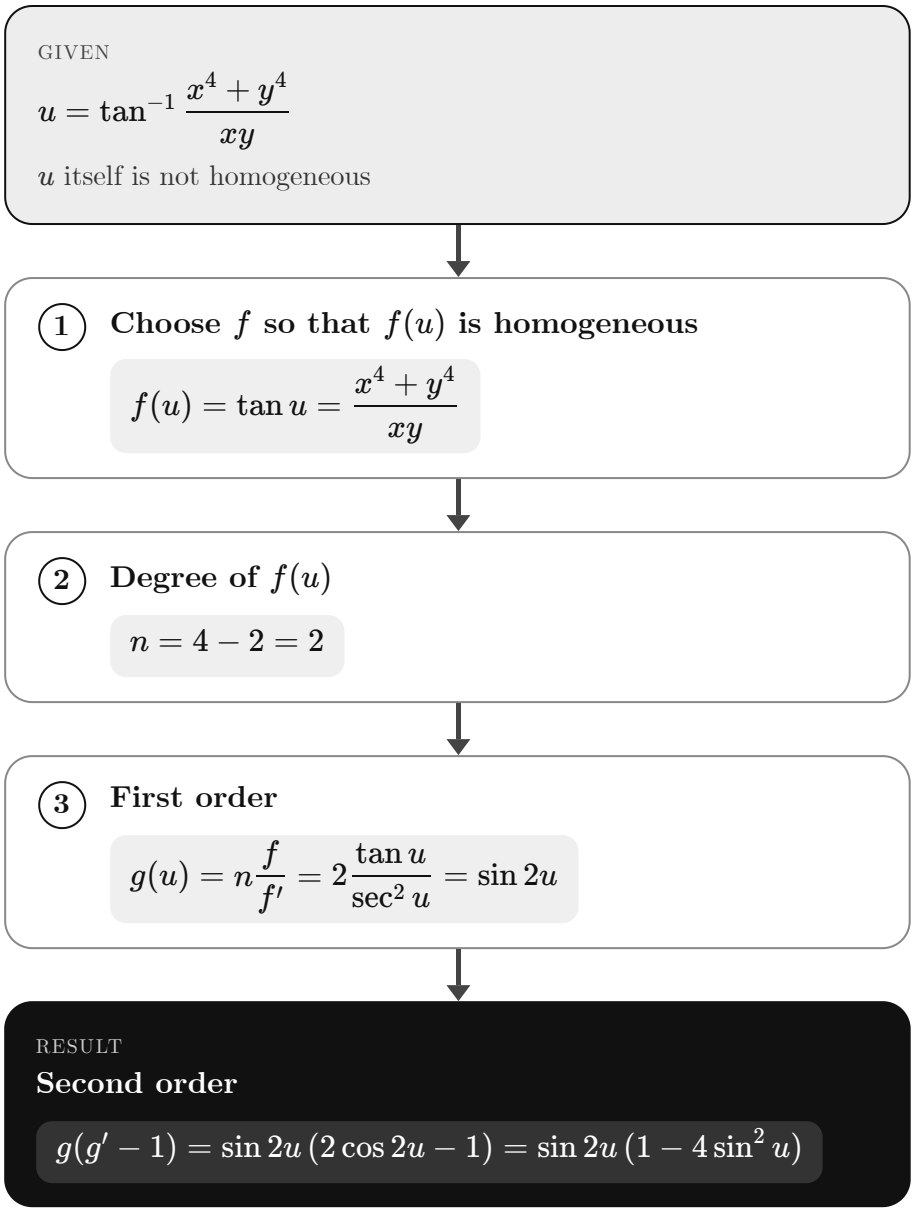
ESA ★★★

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = g(u)\left[g'(u) - 1\right]$$

gives	The second-order combination for the extension.
applies	$g(u) = n\,f(u)/f'(u)$ from E23.1.
note	Differentiate g with respect to u.

S005·7

Flowcharts



Remark. Common choices: $\sin^{-1}(\dots) \to f = \sin$ ($g = n\tan u$); $\tan^{-1} \to f = \tan$ ($g = \frac{n}{2}\sin 2u$); $\log(\dots) \to f = e^u$ ($g = n$); $e^{(\dots)} \to f = \log$ ($g = nu\log u$).

Figure F23.1. Using the extension of Euler's theorem. ESA ★★★

S005·7 · ESA May-24 1a

Important points and why

1. Spot the extension case: $u = \sin^{-1}(\dots)$, $\tan^{-1}(\dots)$, $\log(\dots)$ or $e^{(\dots)}$ of a homogeneous expression; then f is $\sin$, $\tan$, $\exp$ or $\log$ respectively. ESA ★★★ ADDED FOR CLARITY S005·7 · ESA May-24 1a
2. The degree n belongs to $f(u)$, not to u. ESA ★★★ S005·7

Where does $n\,f(u)/f'(u)$ come from?
Apply Euler to $F = f(u)$: $xF_x + yF_y = nF$. By the chain rule $F_x = f'(u)u_x$, $F_y = f'(u)u_y$, so $f'(u)(xu_x + yu_y) = nf(u)$.

ADDED FOR CLARITY S005·7

Exam trap. For $u = \tan^{-1}\frac{x^4+y^4}{xy}$ the degree is that of $\tan u = \frac{x^4+y^4}{xy}$, i.e. $4-2=2$, not of u.

ADDED FOR CLARITY ESA May-24 1a

Worked examples

W23.1 · Extension with tan u

ESA ★★★

GIVEN $u = \tan^{-1}\dfrac{x^4+y^4}{xy}$. Prove (i) $xu_x + yu_y = \sin 2u$, (ii) $x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = \sin 2u\,(1 - 4\sin^2 u)$.

STEP 1 $f(u) = \tan u = \dfrac{x^4+y^4}{xy}$ is homogeneous of degree $4-2=2$.

STEP 2 (i) $g(u) = 2\dfrac{\tan u}{\sec^2 u} = 2\sin u\cos u = \sin 2u$.

STEP 3 (ii) $g'(u) = 2\cos 2u$, so $g(g'-1) = \sin 2u\,(2\cos 2u - 1) = \sin 2u\,(1 - 4\sin^2 u)$, using $\cos 2u = 1 - 2\sin^2 u$.

ANSWER (i) $\sin 2u$, (ii) $\sin 2u\,(1 - 4\sin^2 u)$

CHECK sympy: residuals 0 and $\approx 10^{-16}$.

ESA May-24 1a · ESA Jun-24 1b · CW U1 P20

Past-paper questions

ESA May 2024 · 6 MARKS · UE22MA141A Q1.a · solved in W23.1

If $u = \tan^{-1}\left(\frac{x^4+y^4}{xy}\right)$, prove that (i) $xu_x + yu_y = \sin 2u$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \sin 2u(1 - 4\sin^2 u)$.

ESA June 2024 · 6 MARKS · UE23MA141A Q1.b · solved in W23.1

Same question as May 2024 Q1.a: $u = \tan^{-1}\left(\frac{x^4+y^4}{xy}\right)$, prove (i) and (ii).

SHORT-NOTE RECAP

If $f(u)$ (not u) is homogeneous of degree n: $xu_x + yu_y = g(u) = nf(u)/f'(u)$ and $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = g(u)[g'(u) - 1]$. For $u = \tan^{-1}\frac{x^4+y^4}{xy}$: $f = \tan$, $n = 2$, $g = \sin 2u$, second order $\sin 2u(1 - 4\sin^2 u)$.

24 Applications of homogeneous functions

Euler's theorem relates a homogeneous function to its partial derivatives and describes how the function scales. It is used in economics (production functions, marginal productivity, elasticity of demand) and in thermodynamics, where extensive variables are homogeneous of degree 1 and intensive variables of degree 0. A function of ratios only has degree 0, so its Euler combination is 0.

Key terms and definitions

Extensive / intensive variable. Extensive variables (mass, volume, energy) double when the system is doubled: homogeneous of degree 1. Intensive variables (pressure, temperature, density) do not change: degree 0. ADDED FOR CLARITY S005·5 · T005 00:16:26

> *"A constant function is homogeneous of degree 0. Linear functions are homogeneous of degree 1. The degree of the homogeneous function can be positive, negative or zero."* S005·5

Important points and why

1. A constant function is homogeneous of degree 0; linear functions are homogeneous of degree 1. ESA ★ S005·5 · T005 00:06:52
2. Euler's theorem also describes the scaling of the function: $n\,u$. ESA ★ S005·5 · T005 00:15:57
3. Applications: production functions, marginal productivities and elasticity of demand in economics; extensive (degree 1) and intensive (degree 0) variables in thermodynamics. ESA ★ S005·5 · T005 00:16:09
4. Any function of ratios only, such as $f(x/y,\, y/z,\, z/x)$, has degree 0, so $xu_x + yu_y + zu_z = 0$ (ESA Jan 2024 set 1). ESA ★★★ ADDED FOR CLARITY ESA Jan-24 s1 1d

Exam trap. The lecture swaps the examples: mass is extensive (degree 1); pressure and temperature are intensive (degree 0). ADDED FOR CLARITY T005 00:16:46

Worked examples

W24.1 · A function of ratios only ESA ★★★

GIVEN $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$; find $xu_x + yu_y + zu_z$.

STEP 1 Scaling x, y, z by t leaves every ratio unchanged, so $u(tx, ty, tz) = u$: degree 0.

STEP 2 Euler in three variables: $xu_x + yu_y + zu_z = 0 \cdot u$.

ANSWER $\boxed{xu_x + yu_y + zu_z = 0}$

CHECK Same answer as the chain-rule method (sympy).

ADDED FOR CLARITY ESA Jan-24 s1 1d

Past-paper questions

ESA Jan 2024 · 5 marks · UE23MA141A set 1, Q1.d · solved in W24.1

If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, find $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$ (degree 0; also solvable by the chain rule).

short-note recap

Constants are homogeneous of degree 0 and linear functions of degree 1. Euler's theorem gives the scaling of a function; it is used in economics (production functions, marginal productivity, elasticity) and in thermodynamics (extensive variables degree 1, intensive degree 0). A function of ratios has degree 0, so its Euler sum is 0.

M5

Class 006 · Problems on Euler's Theorem

25 Testing homogeneity: the scaling test

Before Euler's theorem can be used, you must show the function is homogeneous and find its degree. Class 006 does this with the scaling test: replace x by λx and y by λy; if the result is $\lambda^n u(x,y)$, then u is homogeneous of degree n. The degree can be positive, negative or zero; if no power of λ factors out, the function is not homogeneous.

Key terms and definitions

Homogeneous function (scaling form). $u(x,y)$ is homogeneous of degree n if $u(\lambda x, \lambda y) = \lambda^n\, u(x,y)$. Here n can be positive, negative or even zero. S006·5 · T006b 00:00:52 · T006b 00:01:11

Not homogeneous. u is not homogeneous when $u(\lambda x, \lambda y) \neq \lambda^n u(x,y)$ for every n: no single power of λ can be taken out. S006·8 · T006b 00:04:59

Formulas

E25.1 · Scaling test ESA ★★

$$u(\lambda x, \lambda y) = \lambda^n\, u(x,y) \implies u \text{ homogeneous of degree } n$$

gives	Whether u is homogeneous, and its degree.
applies	Any function; with three variables scale z too.
note	If no power λ^n factors out, u is not homogeneous.

S006·5 · T006b 00:00:52

E25.2 · Degree of a quotient ESA ★★

$$\deg \frac{P}{Q} = \deg P - \deg Q$$

gives	The degree without writing out λ.
applies	P, Q homogeneous.
note	$\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$: $1-4=-3$.

ADDED FOR CLARITY S006·4 · T006a 00:06:10

Graphs and diagrams

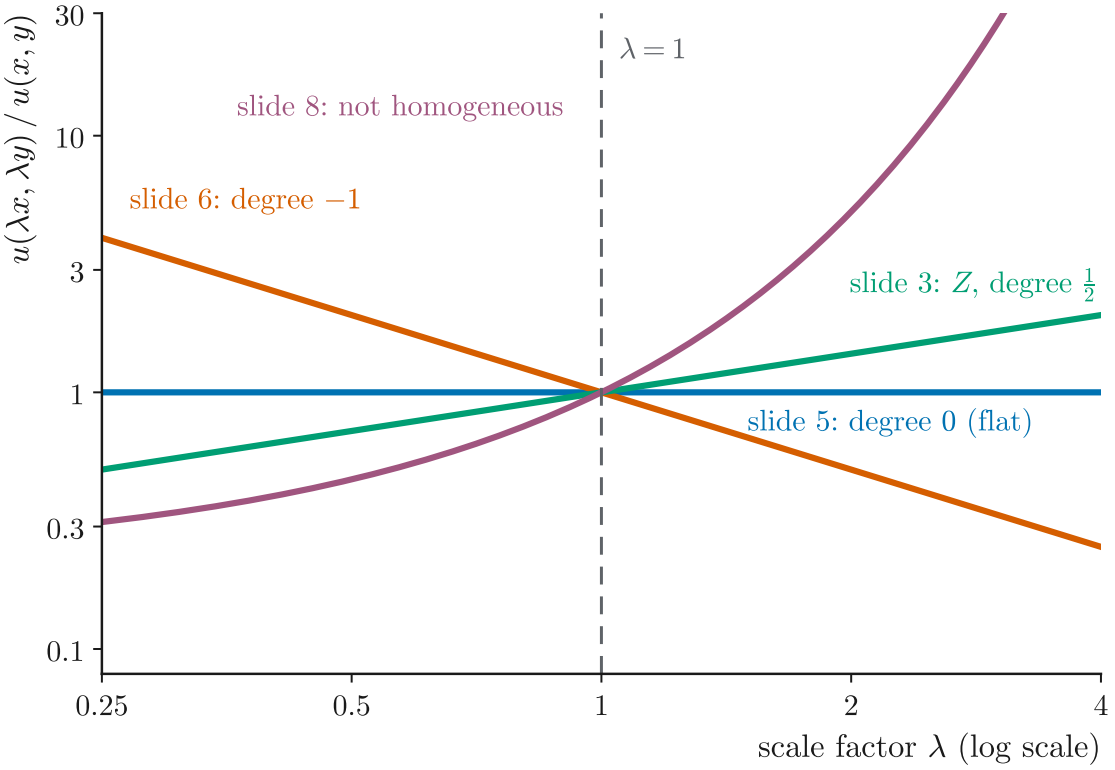
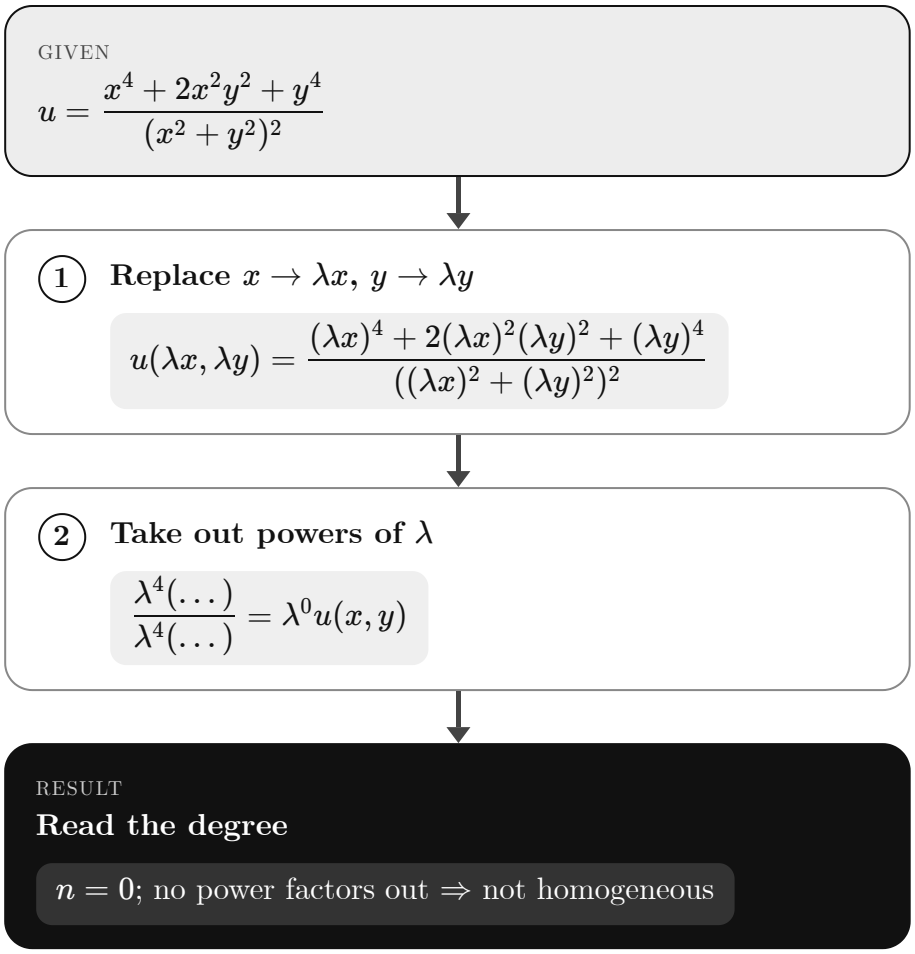


Figure G25.1. The scaling test on log-log axes. ESA ★★

ADDED FOR CLARITY · S006·3 · S006·5 · S006·6 · S006·8

How to read it. Each curve is $u(\lambda x, \lambda y)/u(x, y)$ for a deck function, plotted against λ on log-log axes. A homogeneous function gives λ^n: a straight line of slope n through $(1, 1)$. Slide 006·5 is flat (degree 0), slide 006·6 falls with slope -1, Z from slide 006·3 rises with slope $\frac{1}{2}$. Slide 006·8's function curves away: it is not homogeneous. Base point $(1, 2)$; $(2, 1)$ for slide 006·6, where $\sin^{-1}(y/x)$ needs $y \le x$.

Flowcharts



Remark. Use the test on u (topic 3) or on $f(u)$ (topic 4); the steps are the same.

Figure F25.1. The scaling test ESA ★★

S006·5 · T006b 00:01:41 · T006b 00:02:37

Important points and why

1. Write $u(\lambda x, \lambda y)$, take the highest power of λ out of the numerator and the denominator, and cancel. ESA ★★ S006·5 · T006b 00:02:17
2. A λ inside a ratio cancels: $\sin^{-1}\frac{\lambda y}{\lambda x} = \sin^{-1}\frac{y}{x}$, so functions of y/x add degree 0. ESA ★★ S006·6 · T006b 00:03:34
3. $e^{\lambda x}$ cannot be written as $\lambda^n e^x$, so exponentials of x or y alone break homogeneity (slide 006·8). ESA ★★ S006·8 · T006b 00:04:40
4. Simplify first when you can: slide 006·5 is $\frac{(x^2+y^2)^2}{(x^2+y^2)^2} = 1$ and slide 006·7 is $\frac{xy(x+y)}{(x+y)^3} = \frac{xy}{(x+y)^2}$; both have degree 0. ESA ★ (added for clarity) S006·5 · S006·7

Why does slide 006·8 fail the test?

$u(\lambda x, \lambda y) = \dfrac{e^{\lambda x} - e^{\lambda y}}{\lambda (x - y)}$. The denominator gives λ^{-1}, but $e^{\lambda x} - e^{\lambda y}$ is not a power of λ times $e^{x} - e^{y}$. Figure G25.1 shows the ratio is not a straight line on log-log axes.

ADDED FOR CLARITY S006·8

Worked examples

W25.1 · Degree 0 (slide 006·5) ESA ★

GIVEN $u = \dfrac{x^4 + 2x^2y^2 + y^4}{(x^2 + y^2)^2}$.

STEP 1 $u(\lambda x, \lambda y) = \dfrac{\lambda^4 (x^4 + 2x^2y^2 + y^4)}{\lambda^4 (x^2 + y^2)^2} = \lambda^0 u(x, y).$

ANSWER Homogeneous of degree 0

CHECK The numerator is $(x^2 + y^2)^2$, so $u \equiv 1$: a constant, degree 0.

S006·5 · T006b 00:00:24

W25.2 · Degree −1 (slide 006·6) ESA ★★

GIVEN $u = \dfrac{\sin^{-1}(y/x)}{\sqrt{x^2 + y^2}}$.

STEP 1 $\sin^{-1} \frac{\lambda y}{\lambda x} = \sin^{-1} \frac{y}{x}$: the λ cancels.

STEP 2 $\sqrt{(\lambda x)^2 + (\lambda y)^2} = \lambda \sqrt{x^2 + y^2}$.

STEP 3 $u(\lambda x, \lambda y) = \lambda^{-1} u(x, y)$.

ANSWER Homogeneous of degree -1

CHECK sympy: $u(\lambda x, \lambda y)/u = 1/\lambda$.

S006·6 · T006b 00:03:02

W25.3 · Degree 0 (slide 006·7) ESA ★

GIVEN $u = \dfrac{x^2 y + x y^2}{(x + y)^3}$.

STEP 1 $u(\lambda x, \lambda y) = \dfrac{\lambda^3 (x^2 y + x y^2)}{\lambda^3 (x + y)^3} = \lambda^0 u.$

ANSWER Homogeneous of degree 0

CHECK $u = \frac{xy}{(x+y)^2}$, a function of y/x only.

S006·7 · T006b 00:04:06

W25.4 · Not homogeneous (slide 006·8) ESA ★★

GIVEN $u = \dfrac{e^x - e^y}{x - y}.$

STEP 1 $u(\lambda x, \lambda y) = \dfrac{e^{\lambda x} - e^{\lambda y}}{\lambda (x - y)}.$

STEP 2 No factor λ^n can be taken out of $e^{\lambda x} - e^{\lambda y}$.

ANSWER u is not homogeneous

CHECK Figure G25.1: the ratio curves on log-log axes.

S006·8 · T006b 00:04:40

SHORT-NOTE RECAP

Scaling test: $u(\lambda x, \lambda y) = \lambda^n u(x, y)$ means degree n (positive, negative or zero). Slide examples: $\frac{x^4+2x^2y^2+y^4}{(x^2+y^2)^2}$ degree 0; $\frac{\sin^{-1}(y/x)}{\sqrt{x^2+y^2}}$ degree -1; $\frac{x^2y+xy^2}{(x+y)^3}$ degree 0; $\frac{e^x-e^y}{x-y}$ not homogeneous.

26 Verifying Euler's theorem directly

To verify the theorem, compute both sides separately: find the degree n, then differentiate, form $xu_x + yu_y$ and simplify until it equals nu. Slide 006·3 does this for $Z = \frac{x+y}{\sqrt{x}+\sqrt{y}}$ with the quotient rule. It is long; in every other problem the theorem is applied, not verified.

Key terms and definitions

> *"Euler's theorem: if $u = f(x, y)$ is a homogeneous function of degree n, then $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = nu$."*
>
> S006·10 · T006a 00:00:00

Formulas

E26.1 · Euler's theorem ESA ★★★

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\,u$$

gives	The first-order Euler sum.
applies	u homogeneous of degree n; add $z\,u_z$ for three variables.
note	To verify, compute the left side by differentiating and show it equals nu.

S006·3 · S006·10 · T006a 00:00:00

Important points and why

1. Verification is a check of the theorem, not a method: the lecture spends four minutes on slide 006·3; applying the theorem takes one line. ESA ★ T006a 00:04:57 · T006a 00:08:41
2. In slide 006·3, group the terms with the common denominator $(\sqrt{x} + \sqrt{y})^2$ and cancel one factor of $\sqrt{x} + \sqrt{y}$. ESA ★ T006a 00:02:59 · T006a 00:03:17

Exam trap. Slide 006·3 says "verify for $\cos^{-1}(\dots)$" but verifies for $Z = \cos u$. For u itself, $xu_x + yu_y = \frac{1}{2}\frac{\cos u}{-\sin u} = -\frac{1}{2}\cot u$, not $\frac{1}{2}u$. ADDED FOR CLARITY S006·3

Worked examples

W26.1 · Verify Euler's theorem (slide 006·3) ESA ★★

GIVEN $u = \cos^{-1}\dfrac{x+y}{\sqrt{x}+\sqrt{y}}$; verify Euler's theorem for $Z = \cos u = \dfrac{x+y}{\sqrt{x}+\sqrt{y}}$.

STEP 1 $Z = \dfrac{x(1+y/x)}{\sqrt{x}(1+\sqrt{y/x})} = \sqrt{x}\, f\left(\frac{y}{x}\right)$: degree $\frac{1}{2}$, so Euler predicts $xZ_x + yZ_y = \frac{1}{2}Z$.

STEP 2 Quotient rule: $Z_x = \dfrac{(\sqrt{x}+\sqrt{y}) - \frac{x+y}{2\sqrt{x}}}{(\sqrt{x}+\sqrt{y})^2}$, so $xZ_x = \dfrac{x\sqrt{x} + 2x\sqrt{y} - y\sqrt{x}}{2(\sqrt{x}+\sqrt{y})^2}$.

STEP 3 By symmetry $yZ_y = \dfrac{y\sqrt{y} + 2y\sqrt{x} - x\sqrt{y}}{2(\sqrt{x}+\sqrt{y})^2}$.

STEP 4 Add: numerator $x\sqrt{x} + x\sqrt{y} + y\sqrt{x} + y\sqrt{y} = (x+y)(\sqrt{x}+\sqrt{y})$, so the sum is $\dfrac{x+y}{2(\sqrt{x}+\sqrt{y})} = \frac{1}{2}Z$.

ANSWER $\boxed{xZ_x + yZ_y = \frac{1}{2}Z}$: verified

CHECK sympy: residual 0. For u itself: $xu_x + yu_y = -\frac{1}{2}\cot u$ (added for clarity).

S006·3 · T006a 00:00:18 · T006a 00:04:11

SHORT-NOTE RECAP

To verify: find n, differentiate, form $xu_x + yu_y$, simplify to nu. For $Z = \frac{x+y}{\sqrt{x}+\sqrt{y}}$ ($n = \frac{1}{2}$): $xZ_x + yZ_y = \frac{(x+y)(\sqrt{x}+\sqrt{y})}{2(\sqrt{x}+\sqrt{y})^2} = \frac{1}{2}Z$.

27 When u itself is homogeneous: first and second order

If u is homogeneous of degree n, write the answers down: $xu_x + yu_y = nu$ and $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$; their sum is n^2u. For a sum of parts with different degrees, apply the result to each part with its own degree. Most ESA questions on Euler's theorem are of this type.

Key terms and definitions

> *"By Euler's theorem for a homogeneous function of degree n: $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$."* S006·11

Formulas

E27.1 · Second order ESA ★★★

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = n(n-1)\,u$$

GIVES	The second-order Euler sum.
APPLIES	u homogeneous of degree n.
NOTE	$n=-1$ gives $2u$; $n=-\frac{2}{3}$ gives $\frac{10}{9}u$.

S006·11 · T006b 00:07:23

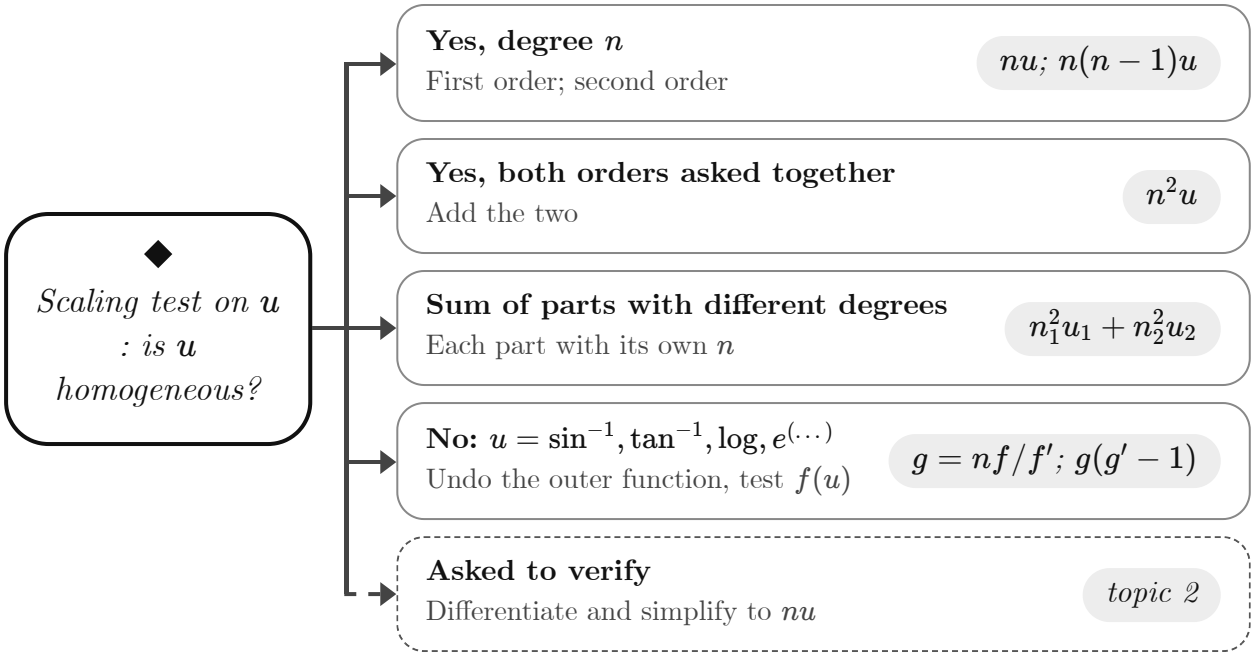
E27.2 · First plus second order ESA ★★★

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} + x\,u_x + y\,u_y = n^2u$$

GIVES	The combination asked in ESA Jan 2024 sets 3 and 4.
APPLIES	u homogeneous of degree n; for u_1+u_2 use $n_1^2u_1 + n_2^2u_2$.
NOTE	$n(n-1)+n = n^2$.

ADDED FOR CLARITY ESA Jan-24 s3 1c · ESA Jan-24 s4 1d · CW U1 P23

Flowcharts



Remark. Exam questions say "using Euler's theorem": always state the homogeneity and the degree first.

Figure F27.1. Which Euler formula? ESA ★★★

T006b 00:06:11 · T006b 00:09:08 · S006·13

Important points and why

1. First prove that u is homogeneous and find n; then write both answers without differentiating. ESA ★★★ T006b 00:06:11 · T006b 00:06:45
2. Give the answer in terms of x and y too: $-u = -\frac{x-y}{x^2+y^2}$ and $2u = \frac{2(x-y)}{x^2+y^2}$. ESA ★★ S006·10 · S006·11
3. For a sum of parts with different degrees, apply the corollary part by part: $n_1(n_1-1)u_1 + n_2(n_2-1)u_2$. ESA ★★★ ADDED FOR CLARITY ESA Jan-24 s3 1c · CW U1 P22

Exam trap. $n = -1$ gives $n(n-1) = (-1)(-2) = +2$, not -2: the second-order answer on slide 006·11 is $+2u$. ADDED FOR CLARITY S006·11

Worked examples

W27.1 · Degree −1, both orders (slides 006·9–11) ESA ★★★

GIVEN $u = \dfrac{x-y}{x^2+y^2}$; find (i) $xu_x + yu_y$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$.

STEP 1 $u(\lambda x, \lambda y) = \dfrac{\lambda(x-y)}{\lambda^2(x^2+y^2)} = \lambda^{-1}u$: $n = -1$.

STEP 2 (i) Euler: $-1 \cdot u$.

STEP 3 (ii) Corollary: $(-1)(-2)u = 2u$.

ANSWER $\boxed{\text{(i) } -\dfrac{x-y}{x^2+y^2},\ \text{(ii) } \dfrac{2(x-y)}{x^2+y^2}}$

CHECK sympy: both residuals 0.

S006·9 · S006·10 · S006·11 · T006b 00:05:27

W27.2 · Fractional degree (CW) ESA ★★

GIVEN $u = \dfrac{y^{1/3} - x^{1/3}}{x+y}$; find both Euler sums.

STEP 1 Degree $\frac{1}{3} - 1 = -\frac{2}{3}$.

STEP 2 (i) $-\frac{2}{3}u$.

STEP 3 (ii) $\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)u = \frac{10}{9}u$.

ANSWER $\boxed{\text{(i) } -\frac{2}{3}u,\ \text{(ii) } \frac{10}{9}u}$

CHECK CW answers agree.

CW U1 P19

W27.3 · Degree 1 with an inverse sine (CW) ESA ★★

GIVEN $u = \sqrt{y^2 - x^2}\,\sin^{-1}(x/y)$; find $xu_x + yu_y$.

STEP 1 $\sqrt{y^2-x^2} = y\sqrt{1-(x/y)^2}$: degree 1; $\sin^{-1}(x/y)$: degree 0.

STEP 2 Degree $1 + 0 = 1$.

ANSWER $\boxed{xu_x + yu_y = u}$

CHECK CW answer: u.

CW U1 P21

W27.4 · Prove 5u and 20u (ESA) ESA ★★★

GIVEN $u = x^3y^2\sin^{-1}(y/x)$. Prove (i) $xu_x + yu_y = 5u$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = 20u$.

STEP 1 $u(\lambda x, \lambda y) = \lambda^5 x^3y^2\sin^{-1}(y/x)$: degree 5.

STEP 2 (i) $5u$.

STEP 3 (ii) $5 \cdot 4\,u = 20u$.

ANSWER (i) $5u$, (ii) $20u$

CHECK Direct differentiation in sympy agrees.

ESA Jul-23 1a

W27.5 · Sum of two degree-3 parts (CW) ESA ★★

GIVEN $u = x^3\sin^{-1}(y/x) + y^3\tan^{-1}(x/y)$; find $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$.

STEP 1 Both parts have degree 3, so u does.

STEP 2 $3 \cdot 2\,u$.

ANSWER $6u$

CHECK CW answer: $6u$.

CW U1 P22

W27.6 · Degrees 3 and −3 together (ESA) ESA ★★★

GIVEN $u = x^3\sin^{-1}(y/x) + y^{-3}\tan^{-1}(x/y)$; evaluate $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$.

STEP 1 u_1: degree 3; $u_2 = y^{-3}\tan^{-1}(x/y)$: degree -3.

STEP 2 First + second order gives n^2: $9u_1 + 9u_2$.

ANSWER $9u$

CHECK CW answer: $9u$; sympy agrees.

ESA Jan-24 s4 1d · CW U1 P23

W27.7 · Degrees 3 and 4, value at (1, 1) (ESA) ESA ★★★

GIVEN $u = x^3 \sin^{-1}(y/x) + x^4 \tan^{-1}(y/x)$; find $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ at $(1,1)$.

STEP 1 $9u_1 + 16u_2$ with $u_1 = x^3 \sin^{-1}(y/x)$, $u_2 = x^4 \tan^{-1}(y/x)$.

STEP 2 At $(1,1)$: $u_1 = \frac{\pi}{2}$, $u_2 = \frac{\pi}{4}$.

STEP 3 $\frac{9\pi}{2} + 4\pi$.

ANSWER $\boxed{\dfrac{17\pi}{2}}$

CHECK sympy: the expression equals $9u_1 + 16u_2$ identically.

ESA Jan-24 s3 1c

W27.8 · A function of ratios only (ESA) ESA ★★

GIVEN $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$; find $xu_x + yu_y + zu_z$.

STEP 1 Scaling x, y, z by λ leaves every ratio unchanged: degree 0.

STEP 2 Euler in three variables: $0 \cdot u$.

ANSWER $\boxed{xu_x + yu_y + zu_z = 0}$

CHECK Same as the chain-rule method (sympy).

ESA Jan-24 s1 1d

Past-paper questions

ESA Jul 2023 (6 marks) UE22MA141A Q1.a · solved in W27.4

If $u(x,y) = x^3y^2 \sin^{-1}\left(\frac{y}{x}\right)$, prove that (i) $xu_x + yu_y = 5u$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = 20u$.

ESA Jan 2024 (6 marks) UE23MA141A set 3, Q1.c · solved in W27.7

If $u = x^3 \sin^{-1}\left(\frac{y}{x}\right) + x^4 \tan^{-1}\left(\frac{y}{x}\right)$, find $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ at $x = 1$, $y = 1$ by using Euler's theorem and its extension.

ESA Jan 2024 (6 marks) UE23MA141A set 4, Q1.d · solved in W27.6

If $u = x^3 \sin^{-1}\left(\frac{y}{x}\right) + y^{-3} \tan^{-1}\left(\frac{x}{y}\right)$, evaluate $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ by using Euler's theorem and its extension.

ESA Jan 2024 (5 marks) UE23MA141A set 1, Q1.d · solved in W27.8

If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, find $xu_x + yu_y + zu_z$ (degree 0; also solvable by the chain rule).

SHORT-NOTE RECAP

u homogeneous of degree n: $xu_x + yu_y = nu$, $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$, sum n^2u. $\frac{x-y}{x^2+y^2}$: $-u$, $2u$. ESA: $5u$ & $20u$; $9u$; $17\pi/2$; ratios only give 0.

28 When f(u) is homogeneous: the extension

For $u = \sin^{-1}(\dots)$, $\tan^{-1}(\dots)$, $\log(\dots)$ or $e^{(\cdots)}$, u is not homogeneous, but undoing the outer function gives a homogeneous $f(u)$: $\sin u$, $\tan u$, e^u or $\log u$. If $f(u)$ has degree n, then $xu_x + yu_y = g(u) = n\,f(u)/f'(u)$ and the second-order sum is $g(u)[g'(u) - 1]$. Class 006 works four such problems; ESA May and June 2024 ask the $\tan^{-1}$ case.

Key terms and definitions

Extension function g(u). $g(u) = n\dfrac{f(u)}{f'(u)}$, where $f(u)$ is homogeneous of degree n in x and y.

S006·13 · S006·15 · S006·17

> *"For $f(u)$ a homogeneous function of degree n: $xu_x + yu_y = n\left(\dfrac{f(u)}{f'(u)}\right) = g(u)$, and $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = g(u)\,[g'(u) - 1]$."* S006·13 · S006·15 · S006·17

Formulas

E28.1 · Extension, first order ESA ★★★

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\frac{f(u)}{f'(u)} = g(u)$$

gives	The Euler sum when u is not homogeneous but $f(u)$ is.
applies	$f(u)$ homogeneous of degree n; add $z\,u_z$ for three variables.
note	$\sin \to n\tan u$, $\tan \to \frac{n}{2}\sin 2u$, $e^u \to n$, $\log \to nu\log u$.

S006·13 · S006·15 · S006·17 · T006b 00:09:08

E28.2 · Extension, second order ESA ★★★

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = g(u)\,\big[g'(u) - 1\big]$$

gives	The second-order sum for the extension.
applies	g from E28.1; differentiate g with respect to u.
note	If g is a constant (the $\log$ case), the answer is $-g$.

S006·15 · S006·17 · T006b 00:10:59

E28.3 · Deriving the extension by the chain rule ESA ★★

$$F = f(u):\ xF_x + yF_y = nF, \quad F_x = f'(u)\,u_x \ \Rightarrow\ f'(u)\,(xu_x + yu_y) = nf(u)$$

gives	Where E28.1 comes from; T006a works slide 006·4 this way.
applies	When you prefer not to quote E28.1.
note	On slide 006·4: $\cos u\,(xu_x + yu_y + zu_z) = -3\sin u$; divide by $\cos u$.

S006·4 · T006a 00:07:33 · T006a 00:08:22

Graphs and diagrams

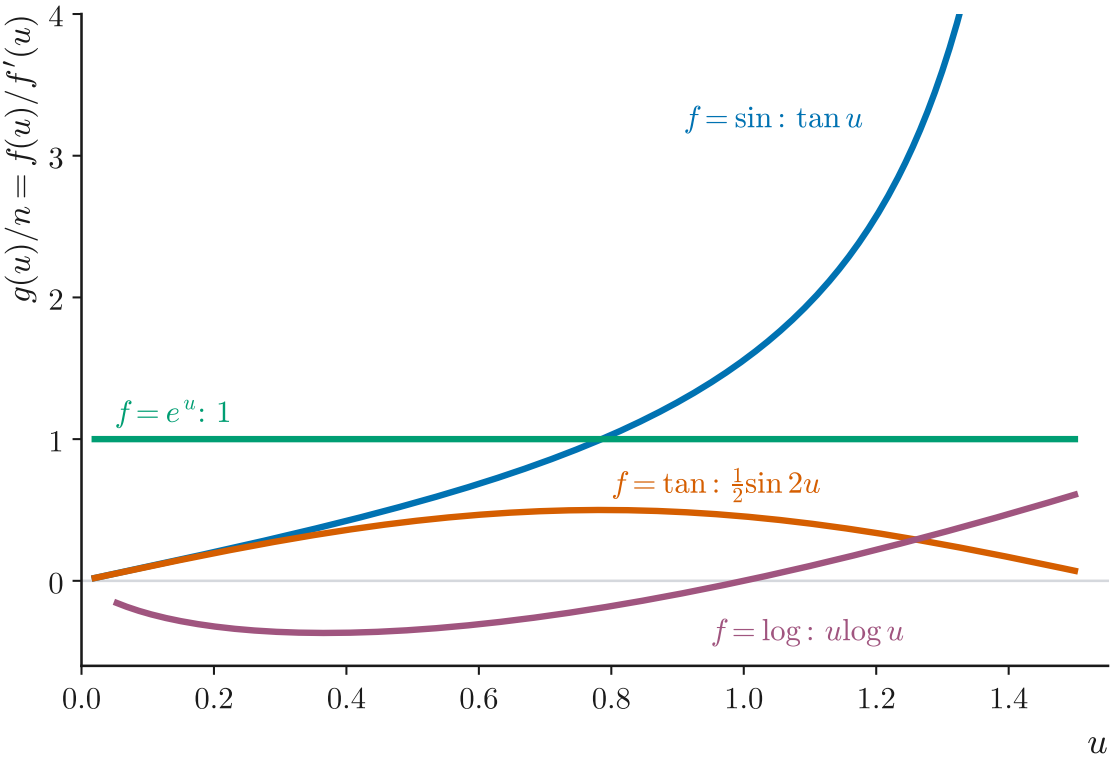
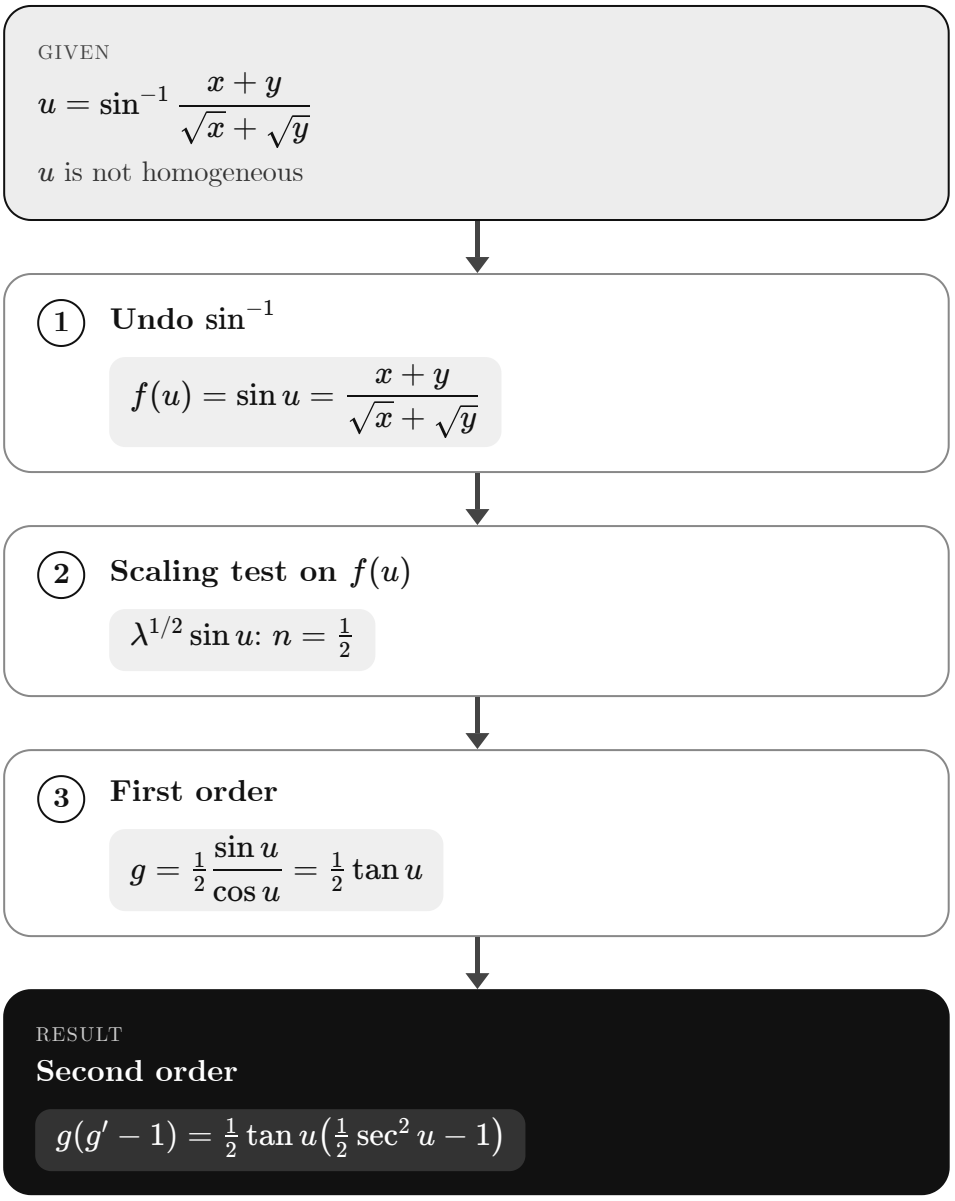


Figure G28.1. The extension function $f(u)/f'(u)$ for each choice of f. ESA ★★

ADDED FOR CLARITY S006·13 · S006·15 · S006·17 · ESA May-24 1a

How to read it. Multiply the curve by the degree n to get $g(u) = xu_x + yu_y$. For $f = \sin$ the sum grows without bound as $u \to \pi/2$; for $f = \tan$ it stays between $\pm\frac{n}{2}$; for $f = e^u$ it is the constant n, so the second order is $-n$; for $f = \log$ it is $nu \log u$, negative for $u < 1$ (on slide 006·12 $u = e^{(\cdots)} > 1$, so it is positive).

Flowcharts



Remark. The same four steps give $4u\log u$ for $e^{(\cdots)}$ (slide 006·12) and 1, -1 for $\log(\dots)$ (slide 006·14).

Figure F28.1. Using the extension (slides 006·16–17) ESA ★★★

S006·16 · S006·17 · T006b 00:11:40

The extension for each outer function

$u =$	$f(u)$	$g(u) = xu_x + yu_y$	$g(g'-1)$	**In class 006**
$\sin^{-1}(P)$	$\sin u$	$n\tan u$	$n\tan u\,(n\sec^2 u - 1)$	$n=-3$ (slide 4); $n=\frac{1}{2}$ (slide 16)
$\cos^{-1}(P)$	$\cos u$	$-n\cot u$	$-n\cot u\,(n\csc^2 u - 1)$	$n=\frac{1}{2}$ (slide 3, u itself)
$\tan^{-1}(P)$	$\tan u$	$\frac{n}{2}\sin 2u$	$\frac{n}{2}\sin 2u\,(n\cos 2u - 1)$	$n=2$ (ESA May/June 2024)
$\log(P)$	e^u	n	$-n$	$n=1$ (slide 14)
e^P	$\log u$	$nu\log u$	$nu\log u\,(n\log u + n - 1)$	$n=4$ (slide 12)

ADDED FOR CLARITY S006·4 · S006·13 · S006·15 · S006·17 · ESA May-24 1a

Important points and why

1. When u is $\sin^{-1}$, $\tan^{-1}$, $\log$ or e of something, apply the inverse function to both sides: that is $f(u)$. ESA ★★★ T006b 00:08:20 · T006b 00:09:55 · T006b 00:11:40
2. The degree belongs to $f(u)$, found by the scaling test on $f(u)$ (λ^4 on slide 006·12, λ on 006·14, $\lambda^{1/2}$ on 006·16). ESA ★★★ S006·12 · S006·14 · S006·16
3. "Whenever you have a complicated function, you can always use Euler's theorem if you feel that the function is homogeneous" — the lecture's closing advice for part I. ESA ★★ T006a 00:08:53
4. Three variables work the same way: on slide 006·4, $\sin u$ has degree -3, so $xu_x + yu_y + zu_z = -3\tan u$. ESA ★★ S006·4 · T006a 00:06:54

Why does the log case give -1 for the second order?
$f(u) = e^u$ gives $g = n\,e^u/e^u = n$, a constant. So $g' = 0$ and $g(g'-1) = -n$; slide 006·14 has $n = 1$.
S006·15

Why can't Euler's theorem be applied to u directly on slides 006·12–17?
u is not homogeneous: for $u = \log\frac{x^2+y^2}{x+y}$, $u(\lambda x, \lambda y) = \log\lambda + u$, which is a shift, not a factor λ^n.
ADDED FOR CLARITY T006b 00:09:55

Exam trap. For $u = e^{(\cdots)}$ the function is $f(u) = \log u$, so $f' = 1/u$ and $f/f' = u\log u$: the answer is $4u\log u$, not $4\log u$. ADDED FOR CLARITY S006·13

Exam trap. In the second order use $g'(u)$, the derivative of g with respect to u, not $f'(u)$: for $g = \frac{1}{2}\tan u$, $g' = \frac{1}{2}\sec^2 u$. S006·17 · T006b 00:12:42

Worked examples

W28.1 · Three variables, sin u (slide 006·4) ESA ★★★

GIVEN $u = \sin^{-1}\dfrac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$; show $xu_x + yu_y + zu_z + 3\tan u = 0$.

STEP 1 $Z = \sin u = \dfrac{x(1+2y/x+3z/x)}{x^4\sqrt{1+(y/x)^8+(z/x)^8}} = x^{-3}g\left(\frac{y}{x}, \frac{z}{x}\right)$: degree -3.

STEP 2 Euler for Z: $xZ_x + yZ_y + zZ_z = -3Z = -3\sin u$.

STEP 3 $Z_x = \cos u\, u_x$ (and so on): $\cos u\,(xu_x + yu_y + zu_z) = -3\sin u$.

STEP 4 Divide by $\cos u$.

ANSWER $\boxed{xu_x + yu_y + zu_z = -3\tan u}$

CHECK sympy at two points: residual $< 10^{-14}$. Same as E28.1 with $f = \sin$, $n = -3$.

S006·4 · T006a 00:04:40 · T006a 00:08:22

W28.2 · Exponential: f = log (slides 006·12–13) ESA ★★★

GIVEN $u = e^{\frac{x^3y^3}{x^2+y^2}}$; find $xu_x + yu_y$.

STEP 1 $f(u) = \log u = \dfrac{x^3y^3}{x^2+y^2}$; $f(\lambda x, \lambda y) = \dfrac{\lambda^6 x^3y^3}{\lambda^2(x^2+y^2)} = \lambda^4 f$: $n = 4$.

STEP 2 $g = 4\dfrac{\log u}{1/u} = 4u\log u$.

STEP 3 Second order (added for clarity): $g' = 4(\log u + 1)$, so $g(g'-1) = 4u\log u\,(4\log u + 3)$.

ANSWER $\boxed{xu_x + yu_y = 4u\log u}$

CHECK sympy: both orders give residual 0.

S006·12 · S006·13 · T006b 00:07:59 · T006b 00:09:27

W28.3 · Logarithm: $f = e^u$ (slides 006·14–15) ESA ★★★

GIVEN $u = \log\dfrac{x^2+y^2}{x+y}$; find (i) $xu_x + yu_y$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$.

STEP 1 $f(u) = e^u = \dfrac{x^2+y^2}{x+y}$; $f(\lambda x, \lambda y) = \lambda f$: $n = 1$.

STEP 2 (i) $g = 1 \cdot \dfrac{e^u}{e^u} = 1$.

STEP 3 (ii) $g' = 0$: $g(g'-1) = 1\,(0-1)$.

ANSWER $\boxed{\text{(i) } 1,\ \text{(ii) } -1}$

CHECK sympy: (i) 1, (ii) -1 exactly.

S006·14 · S006·15 · T006b 00:09:42 · T006b 00:10:59

W28.4 · Inverse sine, degree ½ (slides 006·16–17) ESA ★★★

GIVEN $u = \sin^{-1}\dfrac{x+y}{\sqrt{x}+\sqrt{y}}$; find (i) $xu_x + yu_y$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$.

STEP 1 $f(u) = \sin u$; $f(\lambda x, \lambda y) = \dfrac{\lambda(x+y)}{\sqrt{\lambda}(\sqrt{x}+\sqrt{y})} = \lambda^{1/2}\sin u$: $n = \frac{1}{2}$.

STEP 2 (i) $g = \frac{1}{2}\dfrac{\sin u}{\cos u} = \frac{1}{2}\tan u$.

STEP 3 (ii) $g' = \frac{1}{2}\sec^2 u$: $g(g'-1) = \frac{1}{2}\tan u\left(\frac{1}{2}\sec^2 u - 1\right)$.

ANSWER $\boxed{\text{(i) } \frac{1}{2}\tan u,\ \text{(ii) } \frac{1}{2}\tan u\left(\frac{1}{2}\sec^2 u - 1\right) = \frac{1}{4}\tan u\,(\tan^2 u - 1)}$

CHECK sympy at two points: residuals $< 10^{-16}$. Equivalent form: $-\dfrac{\sin u\cos 2u}{4\cos^3 u}$.

S006·16 · S006·17 · T006b 00:11:22 · T006b 00:12:42

W28.5 · Inverse tangent (ESA) ESA ★★★

GIVEN $u = \tan^{-1}\dfrac{x^4+y^4}{xy}$. Prove (i) $xu_x + yu_y = \sin 2u$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \sin 2u\,(1 - 4\sin^2 u)$.

STEP 1 $f(u) = \tan u = \dfrac{x^4+y^4}{xy}$: degree $4 - 2 = 2$.

STEP 2 (i) $g = 2\dfrac{\tan u}{\sec^2 u} = 2\sin u\cos u = \sin 2u$.

STEP 3 (ii) $g' = 2\cos 2u$: $g(g'-1) = \sin 2u\,(2\cos 2u - 1) = \sin 2u\,(1 - 4\sin^2 u)$, using $\cos 2u = 1 - 2\sin^2 u$.

ANSWER $\boxed{\text{(i) } \sin 2u,\ \text{(ii) } \sin 2u\,(1 - 4\sin^2 u)}$

CHECK sympy: residuals 0 and $\approx 10^{-16}$.

ESA May-24 1a · ESA Jun-24 1b · CW U1 P20

Past-paper questions

ESA May 2024 (6 marks) UE22MA141A Q1.a · solved in W28.5

If $u = \tan^{-1}\left(\frac{x^4+y^4}{xy}\right)$, prove that (i) $xu_x + yu_y = \sin 2u$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \sin 2u(1 - 4\sin^2 u)$.

ESA June 2024 (6 marks) UE23MA141A Q1.b · solved in W28.5

Same question as May 2024 Q1.a: $u = \tan^{-1}\left(\frac{x^4+y^4}{xy}\right)$, prove (i) and (ii).

SHORT-NOTE RECAP

If $f(u)$ has degree n: $g = nf/f'$, second order $g(g'-1)$. $e^{x^3y^3/(x^2+y^2)}$: $4u\log u$. $\log\frac{x^2+y^2}{x+y}$: 1, -1. $\sin^{-1}\frac{x+y}{\sqrt{x}+\sqrt{y}}$: $\frac{1}{2}\tan u$, $\frac{1}{2}\tan u(\frac{1}{2}\sec^2 u-1)$. $\sin^{-1}\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$: $-3\tan u$. ESA: $\tan^{-1}\frac{x^4+y^4}{xy}$ gives $\sin 2u$, $\sin 2u(1-4\sin^2 u)$.

M6

Class 007 · Taylor's and Maclaurin's Expansion

29 Why Taylor's and Maclaurin's series?

A complicated function is hard to evaluate, integrate or take limits of; a polynomial is easy. Taylor's series writes a function as a polynomial (of infinitely many terms) around a point, and a few terms are usually enough. This is how values such as $e^{\tan \sqrt[3]{3}}$, integrals such as $\int_0^1 \sin(x^2)\,dx$ and limits such as $\lim \frac{\sin x - x}{x^3}$ are handled, and it has many applications in physics and engineering.

Graphs and diagrams

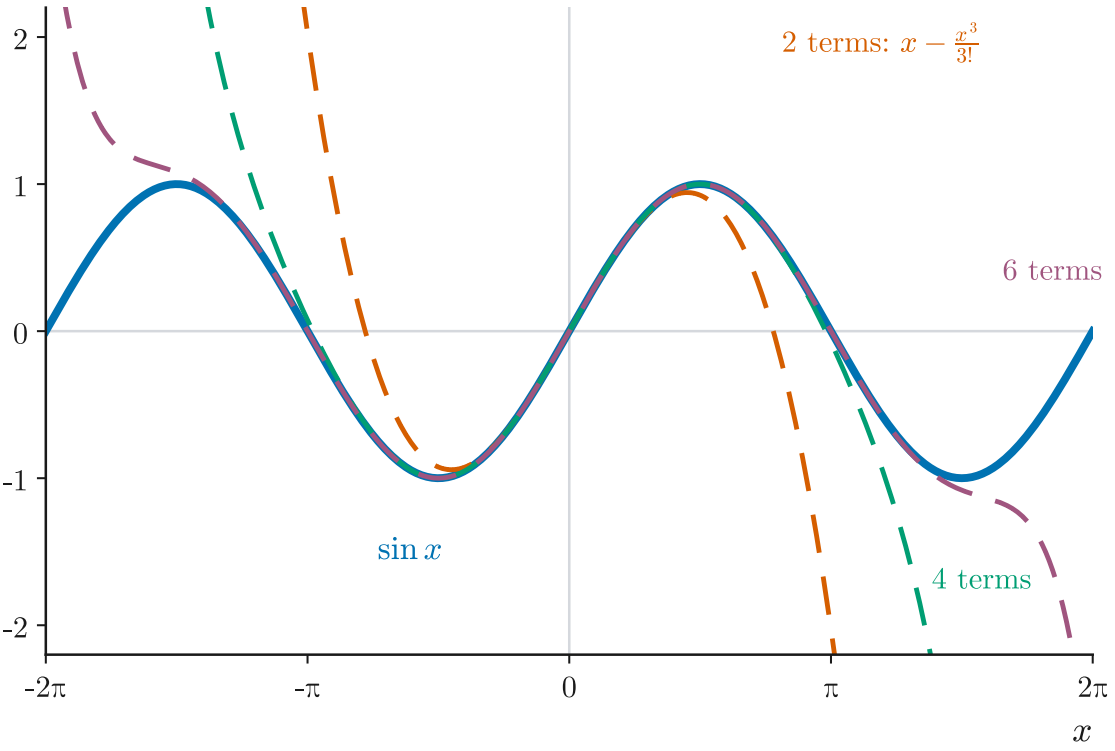


Figure G29.1. $\sin x$ and its Maclaurin polynomials. ESA ★

S007·8 · T007 00:14:45 · T007 00:15:58

How to read it. Blue is $\sin x$; the dashed curves keep 2, 4 and 6 terms of $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots$. Each extra term extends the range where the polynomial follows the function: near 0 all of them agree; far from 0 the short polynomials shoot off. The slide shows 4, 6 and 10 terms; with 10 terms the curves overlap on this range.

Important points and why

1. Once a function is a polynomial, you can differentiate, integrate, find limits and study how it changes. ESA ★ T007 00:06:50
2. Applications (one variable): polynomials for complicated functions, definite integrals with no closed form, asymptotic behaviour, growth of functions, differential equations, the Lorentz factor, $E = mc^2$ at low speed, $\sin\theta \approx \theta$ for a pendulum, power-flow analysis, generating functions. ESA ★ S007·6 · S007·7 · T007 00:11:31 · T007 00:13:44
3. Applications (two variables): optimization (a linear or quadratic model), accurate distance measurement with the sine law, variance of complicated configurations, visualizing complicated functions. ESA ★ S007·12 · T007 00:21:22
4. A few terms are often enough: four terms of the sine series already follow $\sin x$ closely near 0, and ten terms overlap it (Figure G29.1). ESA ★ S007·8 · T007 00:15:24 · T007 00:16:10
5. Taylor's theorem is the basis of L'Hôpital's rule and the mean value theorems of Rolle, Lagrange and Cauchy; the series is a consequence of the theorem. ESA ★ TRANSCRIPT ONLY T007 00:03:33

Why does a calculator need series at all?
It can only add and multiply. A polynomial needs nothing else, so a function such as $e^{\tan x}$ is evaluated through polynomial approximations.

T007 00:01:02 · T007 00:01:35

Exam trap. Slide 007·3's limit $\lim \frac{\sin x - x}{x^4}$ does not exist; with x^3 the answer is $-\frac{1}{6}$ (W29.2).

ADDED FOR CLARITY S007·3

Worked examples

W29.1 · A definite integral with no closed form — ESA ★

GIVEN Evaluate $\int_0^1 \sin(x^2)\,dx$.

STEP 1 $\sin(x^2) = x^2 - \frac{x^6}{3!} + \frac{x^{10}}{5!} - \dots$ (replace x by x^2 in the sine series).

STEP 2 Integrate term by term: $\frac{x^3}{3} - \frac{x^7}{7\cdot 3!} + \frac{x^{11}}{11\cdot 5!} - \dots$

STEP 3 At the limits: $\frac{1}{3} - \frac{1}{42} + \frac{1}{1320} - \dots$

ANSWER $\boxed{\approx 0.3103}$

CHECK sympy: $\int_0^1 \sin(x^2)\,dx = 0.310268$; three terms give 0.310281.

T007 00:11:51 · T007 00:12:20

W29.2 · A limit by series (slide 007·3, corrected) — ESA ★

GIVEN Find $\lim_{x\to 0} \dfrac{\sin x - x}{x^3}$.

STEP 1 $\sin x - x = -\frac{x^3}{3!} + \frac{x^5}{5!} - \dots$

STEP 2 Divide by x^3: $-\frac{1}{6} + \frac{x^2}{120} - \dots \to -\frac{1}{6}$.

ANSWER $\boxed{-\dfrac{1}{6}}$

CHECK sympy agrees. With x^4 (as printed on the slide) the quotient behaves like $-\frac{1}{6x}$ and has no limit.

ADDED FOR CLARITY S007·3

SHORT-NOTE RECAP

Series turn complicated functions into polynomials, which can be evaluated, integrated and used in limits: $\int_0^1 \sin(x^2)dx \approx \frac{1}{3} - \frac{1}{42} + \frac{1}{1320} = 0.3103$; $\lim \frac{\sin x - x}{x^3} = -\frac{1}{6}$. Applications: integrals, asymptotics, differential equations, Lorentz factor, $\sin\theta \approx \theta$, optimization, visualization.

30 Taylor's and Maclaurin's series in one variable

If f has continuous derivatives, $f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots$ about $x = a$; about the origin this is Maclaurin's series. The first two terms are the tangent line. The standard series for $(1+x)^n$, e^x, $\log(1+x)$, $\sin x$, $\cos x$, $\sinh x$, $\cosh x$, $\tan^{-1} x$, $\sin^{-1} x$ and $\cos^{-1} x$ are on slide 007·5. This one-variable part is a self-learning component of the syllabus.

Key terms and definitions

Taylor's series (one variable). If a function $f(x)$ has continuous derivatives of order $(n+1)$, then this function can be expressed as a polynomial at a point $x = a$ in the form $f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \cdots + \frac{(x-a)^n}{n!}f^{(n)}(a) + \dots$ S007·4 · T007 00:04:19

Maclaurin's series (one variable). If the function is expanded about the origin we get Maclaurin's series: $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(0) + \dots$; it is a particular case of Taylor's series. S007·4 · T007 00:06:17

Formulas

E30.1 · Taylor's series, one variable ESA ★

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \cdots + \frac{(x-a)^n}{n!}f^{(n)}(a) + \dots$$

gives	A polynomial form of f near $x = a$.
applies	f has derivatives of every order at a and the series converges to f.
note	Slide 007·4 writes $f^n(a)$ for the n-th derivative $f^{(n)}(a)$.

S007·4 · T007 00:04:56

E30.2 · Maclaurin's series, one variable ESA ★

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(0) + \dots$$

gives	Taylor's series about the origin.
applies	f and all its derivatives finite at 0 (not, e.g., $1/x$ or $\log x$).
note	"About a point $x = a$: Taylor; about the origin: Maclaurin."

S007·4 · T007 00:11:17

E30.3 · Exponential and logarithm ESA ★★

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \quad (x \in \mathbb{R})$$
$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \quad (|x| < 1)$$

GIVES Two standard series.

APPLIES As stated.

NOTE e^{-x}: replace x by $-x$.

S007·5 · T007 00:07:19 · T007 00:08:39

E30.4 · Binomial and geometric series ESA ★

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$
$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots \qquad (|x| < 1)$$

GIVES Series for powers.

APPLIES $|x| < 1$ (any n; for a positive integer n the series stops).

NOTE The geometric series is said in the lecture, not on the slide.

S007·5 · T007 00:07:37

E30.5 · Trigonometric and hyperbolic ESA ★★

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots, \qquad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$
$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots, \qquad \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$$

GIVES Odd and even standard series.

APPLIES All real x.

NOTE Slide 007·5 prints $\cos x = 1 - \frac{x^2}{2} + \dots$ ($2! = 2$).

S007·5

E30.6 · Inverse trigonometric

ESA ★

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

$$\sin^{-1} x = x + \frac{1}{2}\frac{x^3}{3} + \frac{1\cdot 3}{2\cdot 4}\frac{x^5}{5} + \dots, \qquad \cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$$

GIVES Series for the inverse functions.

APPLIES $-1 \le x \le 1$.

NOTE Slide 007·5 prints $\tan^{-1} x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$; the denominators are **3** and **5**, not **3!** and **5!**.

S007·5

Graphs and diagrams

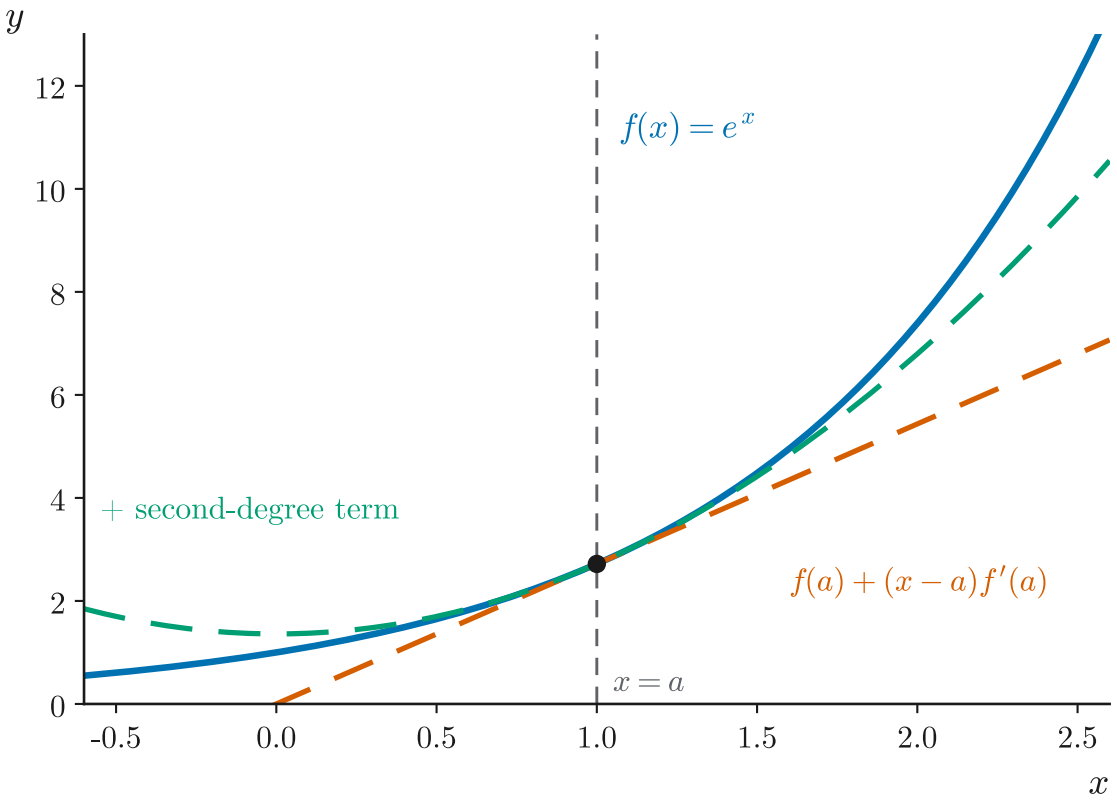


Figure G30.1. The first terms of Taylor's series: tangent line and parabola. ESA ★

ADDED FOR CLARITY T007 00:05:26 · T007 00:05:43

How to read it. $f(x) = e^x$ about $a = 1$. Two terms, $f(a) + (x-a)f'(a)$, give the tangent line (vermilion). Adding $\frac{(x-a)^2}{2!}f''(a)$ gives a parabola (green) that bends with the curve and stays close over a wider range.

Important points and why

1. Keep only the first two terms of Taylor's series and you get $f(x) \approx f(a) + (x-a)f'(a)$: the tangent line, i.e. the derivative as a slope (Figure G30.1). ESA ★ T007 00:05:26 · T007 00:05:43
2. To get a standard series, differentiate repeatedly and put $x = 0$: every derivative of e^x is e^x, which is 1 at the origin. ESA ★ T007 00:09:10 · T007 00:09:44
3. Substitute into a known series rather than differentiating: $\sin(x^2) = x^2 - \frac{x^6}{3!} + \frac{x^{10}}{5!} - \ldots$. ESA ★★ T007 00:11:51
4. No Maclaurin series exists if f or a derivative is infinite at 0: $1/x$, $\log x$, $\sqrt{x}$. ESA ★ T007 00:10:18

When is Maclaurin's series valid?
f must have derivatives of every order at 0 (so none is infinite there), and the series must converge to f. e^x, $\sin x$, $\cos x$ converge for every x; $\log(1+x)$ and $(1+x)^n$ only for $|x| < 1$.
ADDED FOR CLARITY T007 00:10:01 · S007·5

Exam trap. The lecture says $1 + 2 + 2^2 + \ldots$ "gives -1". It diverges: $\frac{1}{1-x} = 1 + x + x^2 + \ldots$ only for $|x| < 1$; $\frac{1}{1-2} = -1$ is outside that range. ADDED FOR CLARITY T007 00:07:52

Exam trap. $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \ldots$ has plain denominators; slide 007·5 shows $3!$ and $5!$, which belong to $\sin x$. ADDED FOR CLARITY S007·5

Worked examples

W30.1 · Maclaurin's series of e^x from the definition ESA ★

GIVEN $f(x) = e^x$.

STEP 1 $f = f' = f'' = \cdots = e^x$, so every value at 0 is 1.

STEP 2 Substitute into E30.2: $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$

ANSWER $\boxed{e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots}$

CHECK Replacing x by $-x$ gives $e^{-x} = 1 - x + \frac{x^2}{2!} - \ldots$

T007 00:09:10 · T007 00:09:44

SHORT-NOTE RECAP

Taylor about a: $f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \ldots$; Maclaurin about 0. Needs finite derivatives at the point. Standard: e^x, $\log(1+x)$, $(1+x)^n$, $\sin$, $\cos$, $\sinh$, $\cosh$, $\tan^{-1}$, $\sin^{-1}$, $\cos^{-1}$; $\log(1+x)$ and $(1+x)^n$ only for $|x| < 1$.

31 Taylor's expansion of a function of two variables

About a point (a,b): $f(x,y) = f(a,b) + \frac{1}{1!}\{(x-a)f_x + (y-b)f_y\} + \frac{1}{2!}\{(x-a)^2 f_{xx} + 2(x-a)(y-b)f_{xy} + (y-b)^2 f_{yy}\} + \ldots$, with every derivative evaluated at (a,b). The coefficients 1, 2, 1 (and 1, 3, 3, 1 at third order) are binomial, because the mixed partials are equal. ESA questions give the point and the degree.

Key terms and definitions

Taylor polynomial of degree n. The terms of Taylor's series up to total degree n in $(x-a)$ and $(y-b)$; "expand up to second (third) degree terms" means stop there. Degree 1 is the tangent plane.

added for clarity S007·11 · T007 00:20:13

> *"Taylor's series of $f(x,y)$ about (a,b) is given by* $f(x,y) = f(a,b) + \frac{1}{1!}\{(x-a)f_x(a,b) + (y-b)f_y(a,b)\} + \frac{1}{2!}\{(x-a)^2 f_{xx}(a,b) + 2(x-a)(y-b)f_{xy}(a,b) + (y-b)^2 f_{yy}(a,b)\} + \ldots$" S007·9 · T007 00:18:45

Formulas

E31.1 · Taylor's series, two variables (to second order) ESA ★★★

$$f(x,y) = f + \frac{1}{1!}\{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \ldots, \quad h = x-a,\ k = y-b$$

gives	A polynomial form of f near (a,b).
applies	All derivatives evaluated at (a,b), all finite there.
note	h, k are shorthand added for clarity; the slide writes $(x-a)$, $(y-b)$.

S007·9 · T007 00:18:07 · T007 00:18:45

E31.2 · Third-order term ESA ★★★

$$+\frac{1}{3!}\{h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$$

gives	The terms needed for "up to third degree".
applies	Add to E31.1 (or E32.1 with $h = x$, $k = y$).
note	Not on the slides; asked in four of six ESA Taylor questions.

added for clarity ESA Dec-23 1b · ESA Jan-24 s1 1b · ESA Jan-24 s2 1b · ESA Jun-24 1c

E31.3 · Operator form

ESA ★★

$$f(a+h,b+k)=\sum_{n=0}^{\infty}\frac{1}{n!}\Big(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\Big)^{n}f(a,b)$$

GIVES Why the coefficients are binomial: 1, 2, 1 and 1, 3, 3, 1.

APPLIES Mixed partials equal ($f_{xy}=f_{yx}$).

NOTE Expand $(h\partial_x+k\partial_y)^n$ like $(h+k)^n$.

ADDED FOR CLARITY T007 00:18:26

Graphs and diagrams

blue: $f(x,y)=xe^{y}+1$

grey: $L(x,y)=1+x+y$ (first degree)

point $(1,0,2)$

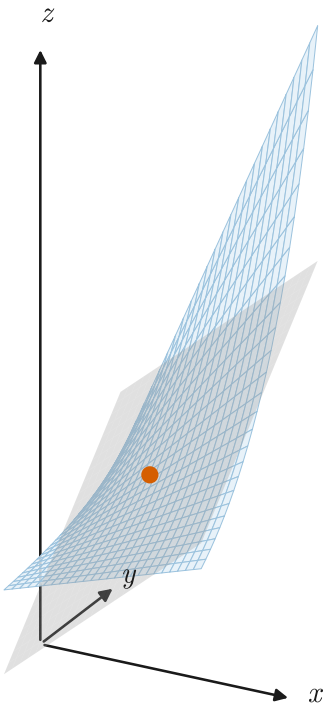


Figure G31.1. $xe^{y}+1$ and its first-degree Taylor polynomial. ESA ★

S007·11 · T007 00:20:13

How to read it. The surface $z=xe^{y}+1$ (blue mesh) and the plane $z=1+x+y$ (grey) touch at $(1,0,2)$. The first-degree Taylor polynomial is the tangent plane: it matches the surface's height and slopes at the point and separates from it away from the point.

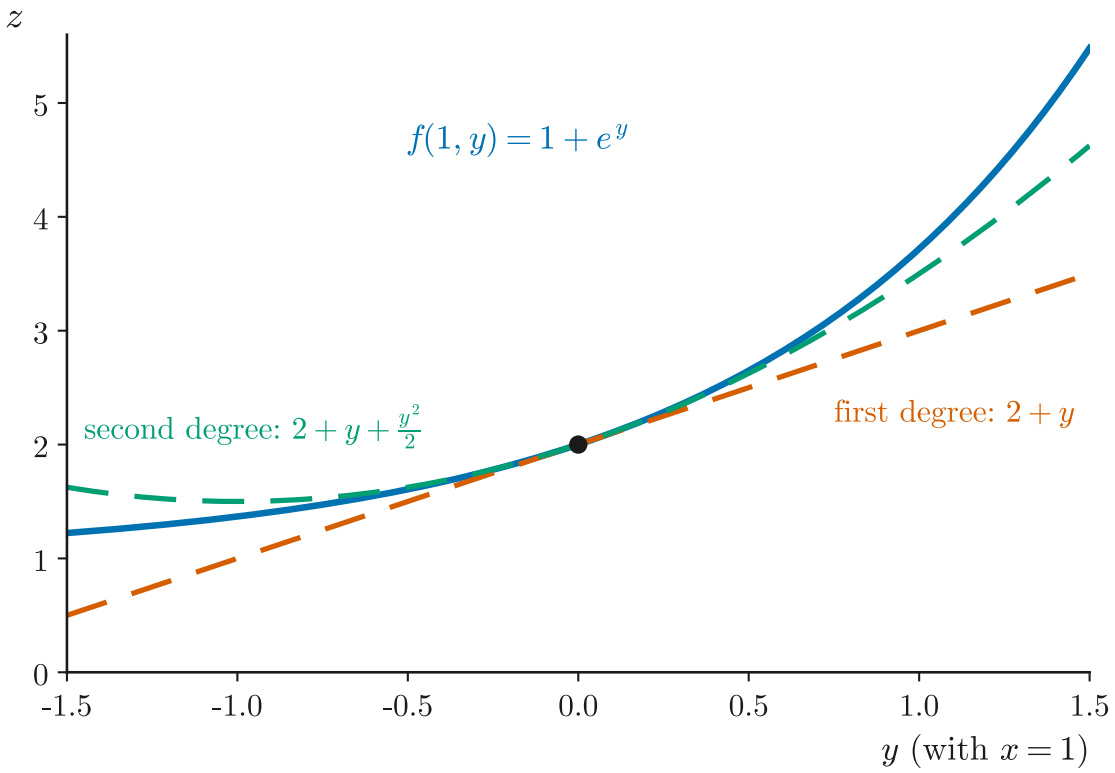
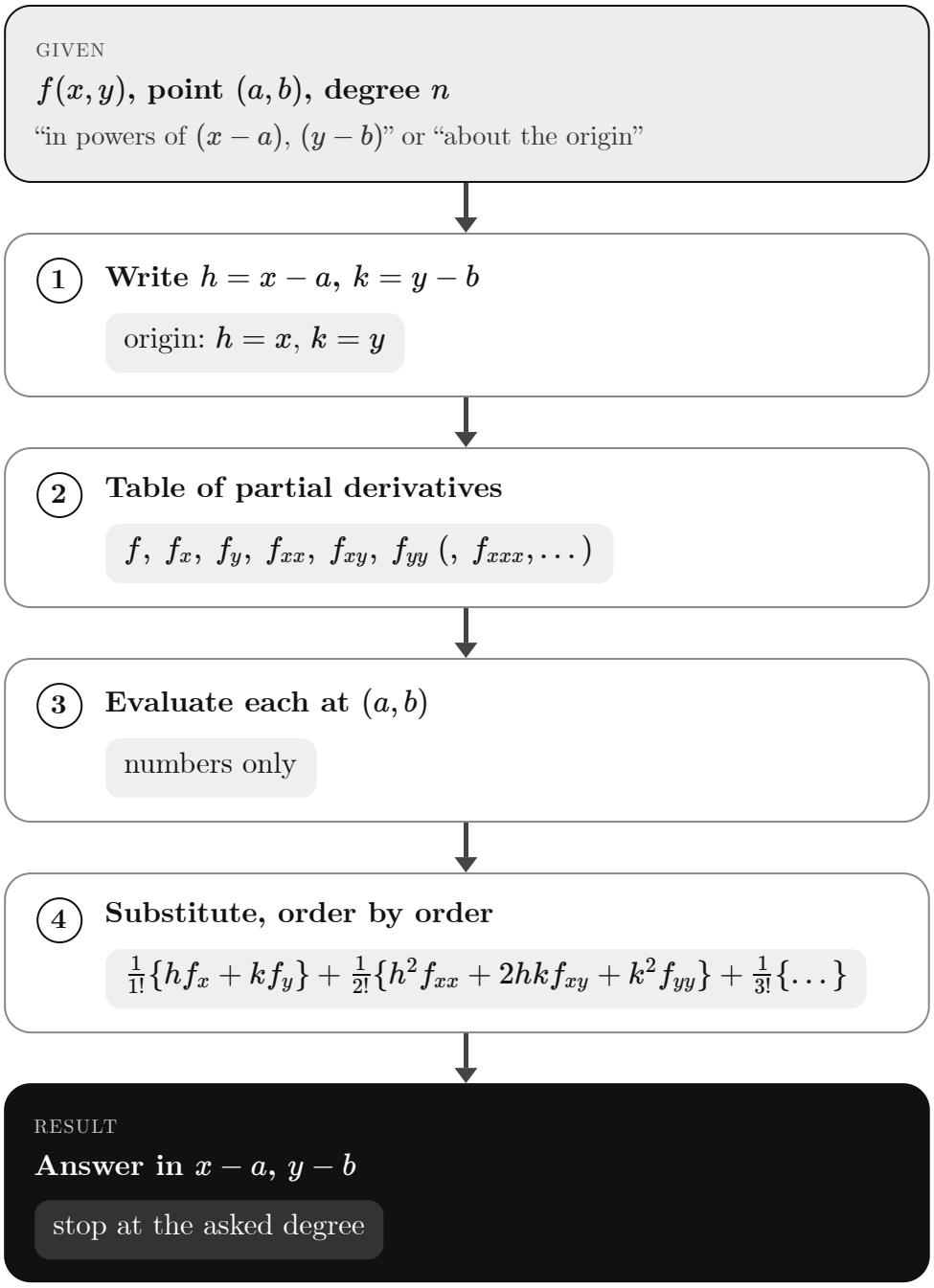


Figure G31.2. Along the line $x = 1$: first- vs second-degree polynomial. ESA ★

ADDED FOR CLARITY S007·11 · T007 00:20:51

How to read it. A slice of Figure G31.1 at $x = 1$: $f(1, y) = 1 + e^y$ (blue), the first-degree polynomial $2 + y$ (vermilion) and the second-degree $1 + x + xy + \frac{y^2}{2} = 2 + y + \frac{y^2}{2}$ (green). The second degree adds curvature and stays close much longer.

Flowcharts



Remark. Check: if f depends only on $x + 2y$, $x - y$, ..., the one-variable series gives the same answer.

Figure F31.1. Expanding f(x, y) in Taylor's series ESA ★★★

S007·9 · ESA Jul-23 1b · ESA Dec-23 1b

Important points and why

1. Lay out a table: f, f_x, f_y, f_{xx}, f_{xy}, f_{yy} (and third order if asked), each in general and then its value at (a, b). Then substitute into E31.1. ESA ★★★ ADDED FOR CLARITY CW U1 P24 · ESA Jul-23 1b

2. There are four second-order partials, but $f_{xy} = f_{yx}$, so they combine into $2(x-a)(y-b)f_{xy}$. ESA ★★ T007 00:18:26
3. "In powers of $(x-1)$ and $(y-\frac{\pi}{4})$" means about the point $(1, \frac{\pi}{4})$. ESA ★★ ADDED FOR CLARITY ESA May-24 1b · CW U1 P27
4. Taylor's series gives approximate values: $\tan^{-1}\frac{0.9}{1.1}$ from the expansion of $\tan^{-1}\frac{y}{x}$ about $(1,1)$ with $h=0.1$, $k=-0.1$. ESA ★★ CW U1 P26
5. The first-degree Taylor polynomial is the tangent plane; the second degree hugs the surface much better (slide 007·11, Figures G31.1–G31.2). ESA ★ S007·11 · T007 00:20:51

Why are the second-order coefficients 1, 2, 1?
Put $g(t) = f(a+th, b+tk)$ and use the one-variable Maclaurin series of g at $t=0$. By the chain rule $g'(0) = hf_x + kf_y$ and $g''(0) = h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}$ (the two equal mixed partials give the 2). Put $t=1$.
ADDED FOR CLARITY T007 00:18:26

Exam trap. Evaluate every derivative at (a,b) before substituting; never leave $f_{xx}(x,y)$ in the answer. ADDED FOR CLARITY S007·9

Exam trap. Don't forget $\frac{1}{2!}$ and $\frac{1}{3!}$, and the middle coefficients 2 and 3: the third-order bracket is $h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}$. ADDED FOR CLARITY ESA Dec-23 1b

Exam trap. Slide 007·11's polynomials ($1+x+y$, $1+x+xy+\frac{y^2}{2}$) are about $(1,0)$, not the origin; about the origin they are $1+x$ and $1+x+xy$. ADDED FOR CLARITY S007·11 · T007 00:20:32

Worked examples

W31.1 · sin x sin y about (π/4, π/4), second order — ESA ★★★

GIVEN Expand $f(x,y) = \sin x \sin y$ in Taylor's series up to second-order terms about $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$. Let $h = x - \frac{\pi}{4}$, $k = y - \frac{\pi}{4}$.

STEP 1 $f = \sin x \sin y = \frac{1}{2}$; $f_x = \cos x \sin y = \frac{1}{2}$; $f_y = \sin x \cos y = \frac{1}{2}$.

STEP 2 $f_{xx} = -\sin x \sin y = -\frac{1}{2}$; $f_{xy} = \cos x \cos y = \frac{1}{2}$; $f_{yy} = -\sin x \sin y = -\frac{1}{2}$.

STEP 3 E31.1: $\frac{1}{2} + \frac{1}{2}(h+k) + \frac{1}{2}\{-\frac{1}{2}h^2 + 2\cdot\frac{1}{2}hk - \frac{1}{2}k^2\}$.

ANSWER $\boxed{\sin x \sin y \approx \frac{1}{2} + \frac{1}{2}(h+k) - \frac{1}{4}\left(h^2 - 2hk + k^2\right)}$

CHECK CW answer agrees; sympy series agrees.

ESA Jul-23 1b · CW U1 P24

W31.2 · sin x sin y about (π/4, π/4), third degree ESA ★★★

GIVEN Expand $\sin x \sin y$ up to third-degree terms about $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$.

STEP 1 Second order as in W31.1.

STEP 2 $f_{xxx} = -\cos x \sin y = -\frac{1}{2}$, $f_{xxy} = -\sin x \cos y = -\frac{1}{2}$, $f_{xyy} = -\frac{1}{2}$, $f_{yyy} = -\frac{1}{2}$.

STEP 3 E31.2: $\frac{1}{6} \cdot \left(-\frac{1}{2}\right)\{h^3 + 3h^2k + 3hk^2 + k^3\} = -\frac{1}{12}(h+k)^3$.

ANSWER $\boxed{\frac{1}{2} + \frac{1}{2}(h+k) - \frac{1}{4}(h-k)^2 - \frac{1}{12}(h+k)^3}$

CHECK sympy series to third order agrees term by term.

ESA Dec-23 1b

W31.3 · $e^x \cos y$ about $(1, \frac{\pi}{4})$ ESA ★★★

GIVEN Expand $e^x \cos y$ in powers of $(x-1)$ and $\left(y - \frac{\pi}{4}\right)$ up to second-degree terms. Let $h = x - 1$, $k = y - \frac{\pi}{4}$.

STEP 1 At $(1, \frac{\pi}{4})$: $f = f_x = f_{xx} = \frac{e}{\sqrt{2}}$ $(e^x \cos y)$; $f_y = f_{xy} = -\frac{e}{\sqrt{2}}$ $(-e^x \sin y)$; $f_{yy} = -\frac{e}{\sqrt{2}}$ $(-e^x \cos y)$.

STEP 2 E31.1: $\frac{e}{\sqrt{2}}\left[1 + h - k + \frac{1}{2}(h^2 - 2hk - k^2)\right]$.

ANSWER $\boxed{e^x \cos y \approx \frac{e}{\sqrt{2}}\left[1 + (x-1) - \left(y - \frac{\pi}{4}\right) + \frac{1}{2}\left\{(x-1)^2 - 2(x-1)\left(y - \frac{\pi}{4}\right) - \left(y - \frac{\pi}{4}\right)^2\right\}\right]}$

CHECK sympy series agrees.

ESA May-24 1b

W31.4 · $\sqrt{1+x+y^2}$ in powers of (x−1) and y ESA ★★

GIVEN Expand $\sqrt{1 + x + y^2}$ about $(1, 0)$ up to second degree. Let $h = x - 1$, $k = y$.

STEP 1 $f = \sqrt{2}$; $f_x = \frac{1}{2\sqrt{1+x+y^2}} = \frac{\sqrt{2}}{4}$; $f_y = \frac{y}{\sqrt{\cdots}} = 0$.

STEP 2 $f_{xx} = -\frac{1}{4(1+x+y^2)^{3/2}} = -\frac{\sqrt{2}}{16}$; $f_{xy} = 0$; $f_{yy} = \frac{1+x}{(1+x+y^2)^{3/2}} = \frac{\sqrt{2}}{2}$.

STEP 3 E31.1: $\sqrt{2} + \frac{\sqrt{2}}{4}h + \frac{1}{2}\left(-\frac{\sqrt{2}}{16}h^2 + \frac{\sqrt{2}}{2}k^2\right)$.

ANSWER $\boxed{\sqrt{2}\left[1 + \frac{x-1}{4} - \frac{(x-1)^2}{32} + \frac{y^2}{4}\right]}$

CHECK CW answer agrees; sympy agrees.

CW U1 P27

W31.5 · $xy^2 + y\cos(x-y)$ about $(1,1)$ ESA ★★

GIVEN Expand up to second order about $(1,1)$. Let $h = x-1$, $k = y-1$.

STEP 1 $f = 2$, $f_x = y^2 - y\sin(x-y) = 1$, $f_y = 2xy + \cos(x-y) + y\sin(x-y) = 3$.

STEP 2 $f_{xx} = -y\cos(x-y) = -1$, $f_{xy} = 2y - \sin(x-y) + y\cos(x-y) = 3$, $f_{yy} = 2x + 2\sin(x-y) - y\cos(x-y) = 1$.

STEP 3 E31.1: $2 + (h+3k) + \frac{1}{2}(-h^2 + 6hk + k^2)$.

ANSWER $\boxed{2 + [(x-1) + 3(y-1)] + \frac{1}{2}[-(x-1)^2 + 6(x-1)(y-1) + (y-1)^2]}$

CHECK CW answer agrees; sympy agrees.

CW U1 P28

W31.6 · Approximate $\tan^{-1}(0.9/1.1)$ ESA ★★

GIVEN Using $f(x,y) = \tan^{-1}\frac{y}{x}$ about $(1,1)$, compute $\tan^{-1}\frac{0.9}{1.1}$.

STEP 1 At $(1,1)$: $f = \frac{\pi}{4}$, $f_x = \frac{-y}{x^2+y^2} = -\frac{1}{2}$, $f_y = \frac{x}{x^2+y^2} = \frac{1}{2}$, $f_{xx} = \frac{1}{2}$, $f_{xy} = 0$, $f_{yy} = -\frac{1}{2}$.

STEP 2 $\tan^{-1}\frac{y}{x} \approx \frac{\pi}{4} - \frac{1}{2}h + \frac{1}{2}k + \frac{1}{4}h^2 - \frac{1}{4}k^2$.

STEP 3 $h = 0.1$, $k = -0.1$: $0.7854 - 0.05 - 0.05 + 0.0025 - 0.0025$.

ANSWER $\boxed{\tan^{-1}\dfrac{0.9}{1.1} \approx 0.6854}$

CHECK Exact: 0.6857. The CW sheet prints 0.6904, an arithmetic slip.

CW U1 P26

W31.7 · The slide's $xe^y + 1$ polynomials ESA ★

GIVEN Find the first- and second-degree Taylor polynomials of $f = xe^y + 1$ about $(1,0)$.

STEP 1 $f = 2$, $f_x = e^y = 1$, $f_y = xe^y = 1$; $f_{xx} = 0$, $f_{xy} = e^y = 1$, $f_{yy} = xe^y = 1$.

STEP 2 First degree: $2 + (x-1) + y = 1 + x + y$.

STEP 3 Second degree: add $\frac{1}{2}\{2(x-1)y + y^2\}$, giving $1 + x + xy + \frac{y^2}{2}$.

ANSWER $\boxed{1 + x + y \text{ and } 1 + x + xy + \dfrac{y^2}{2}}$

CHECK These are slide 007·11's polynomials, so the slide expands about $(1,0)$.

ADDED FOR CLARITY S007·11 · T007 00:20:13

Past-paper questions

ESA Jul 2023 7 MARKS UE22MA141A Q1.b · solved in W31.1

Expand $f(x,y) = \sin x \sin y$ in Taylor's series up to second order terms about the point $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$.

ESA Dec 2023 (7 marks) UE22MA141A Q1.b · solved in W31.2

Expand $f(x,y) = \sin x \sin y$ in Taylor's series up to third degree term about $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$.

ESA May 2024 (7 marks) UE22MA141A Q1.b · solved in W31.3

Expand $f(x,y) = e^x \cos y$ in powers of $(x-1)$ and $\left(y - \frac{\pi}{4}\right)$ using Taylor's series up to second degree terms.

SHORT-NOTE RECAP

About (a,b) with $h = x-a$, $k = y-b$: $f + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \frac{1}{3!}\{h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$, all at (a,b). ESA: $\sin x \sin y$ about $(\frac{\pi}{4}, \frac{\pi}{4})$ gives $\frac{1}{2} + \frac{1}{2}(h+k) - \frac{1}{4}(h-k)^2 - \frac{1}{12}(h+k)^3$.

32 Maclaurin's expansion of a function of two variables

Put $(a,b)=(0,0)$: $f(x,y)=f(0,0)+\frac{1}{1!}\{xf_x+yf_y\}+\frac{1}{2!}\{x^2f_{xx}+2xyf_{xy}+y^2f_{yy}\}+\ldots$ at the origin. f_x goes with x, f_y with y, f_{xx} with x^2, f_{xy} with $2xy$, f_{yy} with y^2. When f is a function of $x+2y$ or $x-y$, the one-variable series gives the answer as a check.

Key terms and definitions

Maclaurin's series (two variables). Taylor's series of $f(x,y)$ about the origin $(0,0)$. S007·10 · T007 00:19:01

> *"Maclaurin's series for a function of two variables is given by* $f(x,y)=f(0,0)+\frac{1}{1!}\{xf_x(0,0)+yf_y(0,0)\}+\frac{1}{2!}\{x^2f_{xx}(0,0)+2xyf_{xy}(0,0)+y^2f_{yy}(0,0)\}+\ldots$*"* S007·10

Formulas

E32.1 · Maclaurin's series, two variables

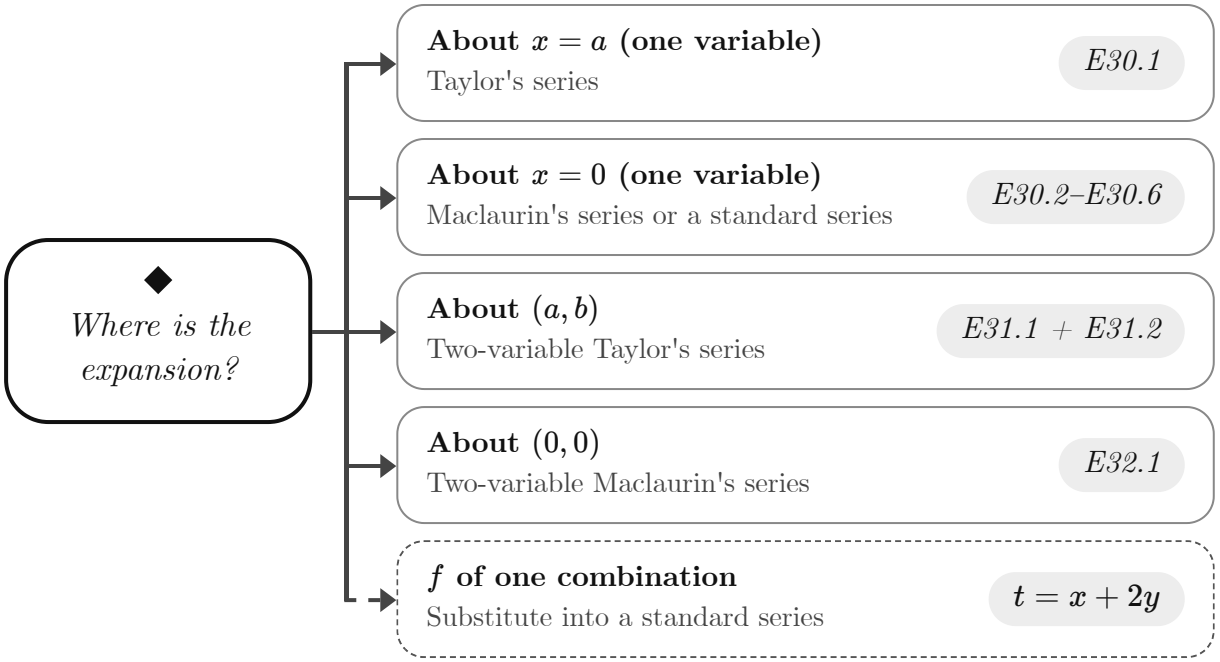
ESA ★★★

$$f(x,y)=f(0,0)+\frac{1}{1!}\{xf_x+yf_y\}+\frac{1}{2!}\{x^2f_{xx}+2xyf_{xy}+y^2f_{yy}\}+\ldots$$

gives	A polynomial form of f near the origin.
applies	Derivatives at $(0,0)$, all finite there.
note	Same as E31.1 with $a=b=0$; add E31.2 with $h=x$, $k=y$ for third degree.

S007·10 · T007 00:19:38

Flowcharts



Remark. The dashed route is a shortcut added for clarity; it is also the quickest way to check an answer.

Figure F32.1. Which series? ESA ★★

T007 00:11:17 · S007·9 · S007·10

Taylor's and Maclaurin's series side by side

	One variable	Two variables
Taylor (about a point)	$f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots$	$f(a,b) + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \dots$
Maclaurin (about the origin)	$f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots$	$f(0,0) + \{xf_x + yf_y\} + \frac{1}{2!}\{x^2 f_{xx} + 2xyf_{xy} + y^2 f_{yy}\} + \dots$
First degree	Tangent line	Tangent plane
Coefficients at order n	$\frac{1}{n!}$	$\frac{1}{n!}\times$ binomial (1,2,1; 1,3,3,1)
Condition	f and derivatives finite at the point	f and all partials finite at the point

ADDED FOR CLARITY S007·4 · S007·9 · S007·10

Important points and why

1. Memory aid from the lecture: f_x goes with x, f_y with y, f_{xx} with x^2, f_{xy} with $2xy$, f_{yy} with y^2. ESA ★★ T007 00:19:38
2. Shortcut and check: if f depends on one combination, use the one-variable series: $\sin(x+2y) = (x+2y) - \frac{(x+2y)^3}{6} + \ldots$, $\log(1+x-y) = (x-y) - \frac{(x-y)^2}{2} + \frac{(x-y)^3}{3} - \ldots$ ESA ★★★
   ADDED FOR CLARITY CW U1 P25 · ESA Jun-24 1c · ESA Jan-24 s1 1b
3. None of f, f_x, f_y, f_{xx}, f_{xy}, f_{yy} may be infinite at $(0,0)$. ESA ★ T007 00:19:26

Worked examples

W32.1 · log(1 + x − y) about the origin — ESA ★★★

GIVEN Expand $\log(1+x-y)$ about the origin up to third-degree terms.

STEP 1 At $(0,0)$: $f=0$; $f_x = \frac{1}{1+x-y} = 1$, $f_y = -1$.

STEP 2 $f_{xx} = -1$, $f_{xy} = 1$, $f_{yy} = -1$.

STEP 3 $f_{xxx} = 2$, $f_{xxy} = -2$, $f_{xyy} = 2$, $f_{yyy} = -2$.

STEP 4 E32.1 + E31.2: $(x-y) + \frac{1}{2}(-x^2+2xy-y^2) + \frac{1}{6}(2x^3 - 6x^2y + 6xy^2 - 2y^3)$.

ANSWER $\boxed{(x-y) - \dfrac{(x-y)^2}{2} + \dfrac{(x-y)^3}{3}}$

CHECK The same as $\log(1+t)$ with $t = x-y$; sympy agrees.

ESA Jan-24 s1 1b

W32.2 · $e^{ax}\sin by$ about the origin — ESA ★★★

GIVEN Expand $e^{ax}\sin by$ about the origin up to third-degree terms.

STEP 1 At $(0,0)$: $f=0$, $f_x = ae^{ax}\sin by = 0$, $f_y = be^{ax}\cos by = b$.

STEP 2 $f_{xx} = 0$, $f_{xy} = ab$, $f_{yy} = 0$.

STEP 3 $f_{xxx} = 0$, $f_{xxy} = a^2 b$, $f_{xyy} = 0$, $f_{yyy} = -b^3$.

STEP 4 E32.1 + E31.2: $by + \frac{1}{2}(2abxy) + \frac{1}{6}(3a^2bx^2y - b^3y^3)$.

ANSWER $\boxed{by + abxy + \dfrac{a^2 b\, x^2 y}{2} - \dfrac{b^3 y^3}{6}}$

CHECK Product of $1 + ax + \frac{a^2x^2}{2}$ and $by - \frac{b^3y^3}{6}$, keeping degree ≤ 3; sympy agrees.

ESA Jan-24 s2 1b

W32.3 · sin(x + 2y) about the origin ESA ★★★

GIVEN Expand $\sin(x+2y)$ in Taylor's series up to third-degree terms about $(0,0)$.

STEP 1 $f=0$, $f_x=\cos(x+2y)=1$, $f_y=2$.

STEP 2 $f_{xx}=-\sin(\cdot)=0$, $f_{xy}=0$, $f_{yy}=0$.

STEP 3 $f_{xxx}=-1$, $f_{xxy}=-2$, $f_{xyy}=-4$, $f_{yyy}=-8$.

STEP 4 $x+2y+\frac{1}{6}\{-x^3-6x^2y-12xy^2-8y^3\}$.

ANSWER $\boxed{(x+2y)-\dfrac{(x+2y)^3}{6}}$

CHECK CW answer agrees; the sine series with $t=x+2y$ gives the same.

ESA Jun-24 1c · CW U1 P25

Past-paper questions

ESA Jan 2024 · 7 marks · UE23MA141A set 1, Q1.b · solved in W32.1

Expand $\log(1+x-y)$ about origin up to third degree terms.

ESA Jan 2024 · 7 marks · UE23MA141A set 2, Q1.b · solved in W32.2

Expand $e^{ax}\sin by$ about origin up to third degree terms.

ESA June 2024 · 6 marks · UE23MA141A Q1.c · solved in W32.3

Expand $f(x,y)=\sin(x+2y)$ in Taylor's series up to third degree terms about the point $(0,0)$.

SHORT-NOTE RECAP

About $(0,0)$: $f+\{xf_x+yf_y\}+\frac{1}{2!}\{x^2f_{xx}+2xyf_{xy}+y^2f_{yy}\}+\ldots$ ESA: $\log(1+x-y)=(x-y)-\frac{(x-y)^2}{2}+\frac{(x-y)^3}{3}$; $e^{ax}\sin by=by+abxy+\frac{a^2b}{2}x^2y-\frac{b^3}{6}y^3$; $\sin(x+2y)=(x+2y)-\frac{(x+2y)^3}{6}$.

M7

Class 008 · Problems on Taylor's and Maclaurin's Series

33 Taylor expansion about a point (a, b)

Write $h = x - a$, $k = y - b$, tabulate f and its partial derivatives, evaluate them at (a, b) and substitute into $f + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \frac{1}{3!}\{\dots\}$. "In powers of $(x - 1)$ and $(y - \frac{\pi}{4})$" means about $(1, \frac{\pi}{4})$. Class 008 works $e^x \cos y$, $\sin(xy)$ and $xy^2 + y\cos(x - y)$; the ESA favourite is $\sin x \sin y$ about $(\frac{\pi}{4}, \frac{\pi}{4})$.

Key terms and definitions

Expansion "in powers of (x − a) and (y − b)". Taylor's series of $f(x, y)$ about the point (a, b).
S008·3 · T008 00:00:37

Up to n-th degree terms. Keep every term whose total power of $(x - a)$ and $(y - b)$ is at most n.
ADDED FOR CLARITY T008 00:06:10

> "$f(x,y) \approx f(a,b) + (x-a)f_x(a,b) + (y-b)f_y(a,b) + \frac{1}{2!}\big[(x-a)^2 f_{xx}(a,b) + 2(x-a)(y-b)f_{xy}(a,b) + (y-b)^2 f_{yy}(a,b)\big]$" S008·14 · S008·19

Formulas

E33.1 · Taylor's series about (a, b), to second order ESA ★★★

$$f(x,y) = f + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \dots, \quad h = x - a,\ k = y - b$$

GIVES	The polynomial to substitute into.
APPLIES	All derivatives evaluated at (a, b).
NOTE	h, k are shorthand added for clarity.

S008·3 · S008·14 · T008 00:00:37 · T008 00:01:25

E33.2 · Third-degree term

ESA ★★★

$$+\frac{1}{3!}\{h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$$

GIVES The terms for "up to third degree".

APPLIES Add to E33.1 or E34.1.

NOTE Coefficients 1, 3, 3, 1; asked in four of six ESA questions.

S008·10 · ESA Dec-23 1b · ESA Jun-24 1c

Graphs and diagrams

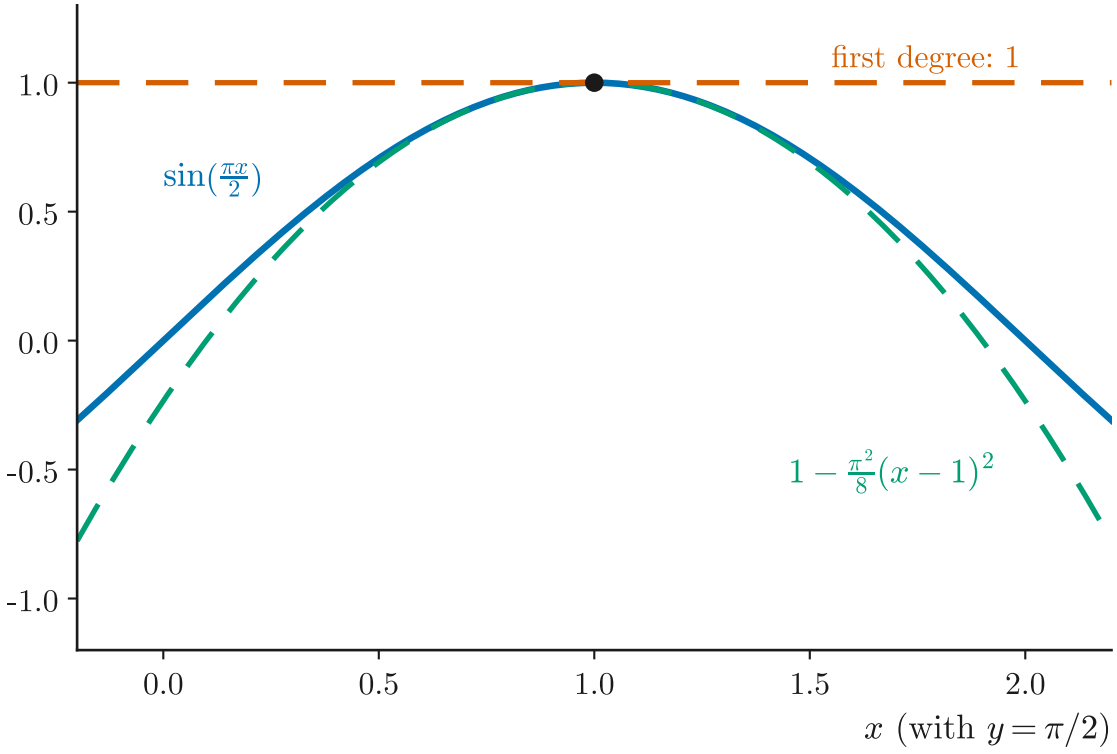
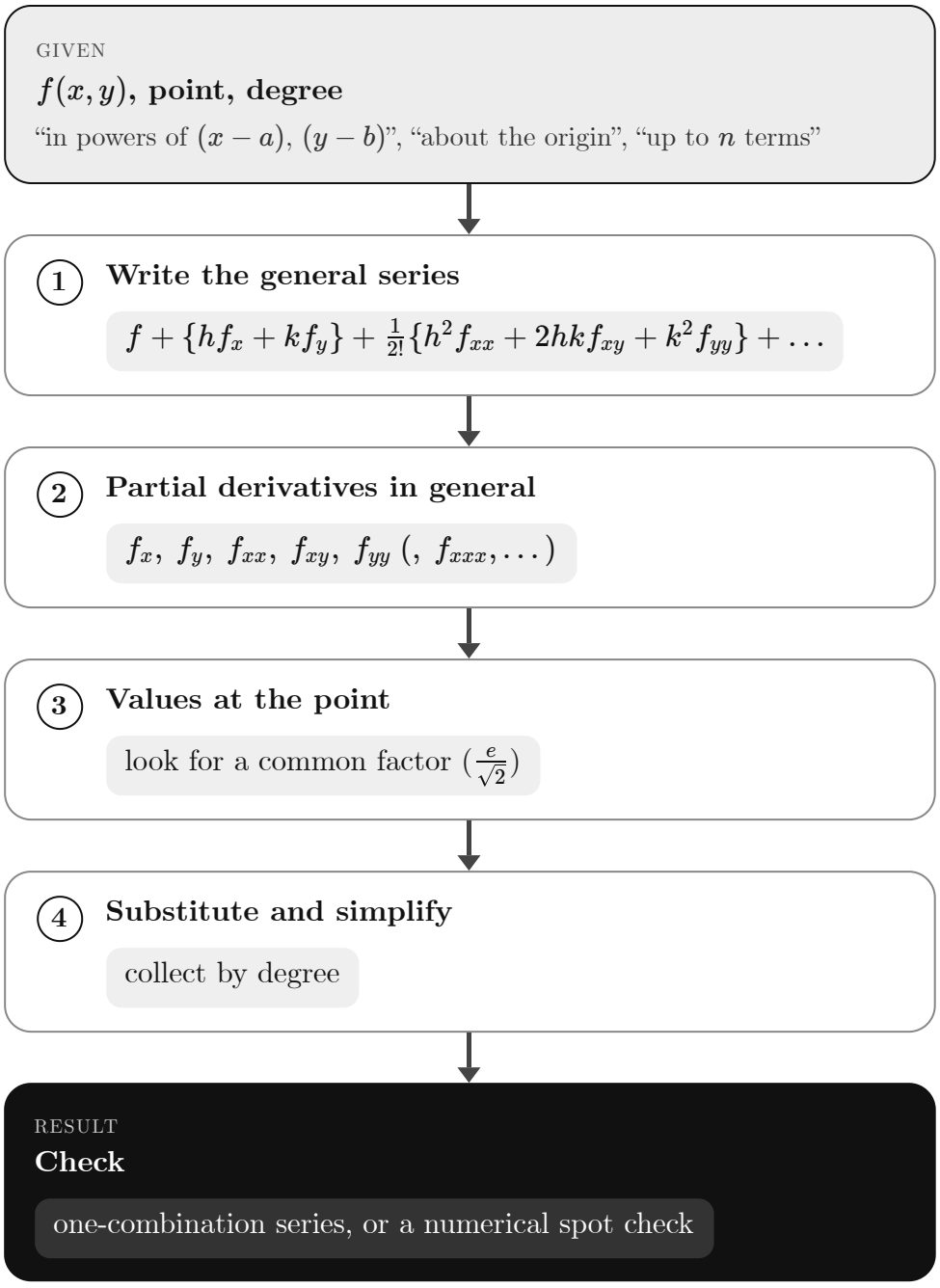


Figure G33.1. $\sin(xy)$ near $(1, \pi/2)$: a slice at $y = \pi/2$. ESA ★

ADDED FOR CLARITY S008·14 · S008·16

How to read it. Along $y = \frac{\pi}{2}$, $\sin(xy) = \sin\frac{\pi x}{2}$ (blue) has a maximum at $x = 1$, so both first partials vanish and the first-degree polynomial is the flat line 1 (vermilion). The second-degree polynomial $1 - \frac{\pi^2}{8}(x-1)^2$ (green) follows the curvature near $x = 1$.

Flowcharts



Remark. For "up to three terms", stop at three non-zero terms; for "third degree", include the $\frac{1}{3!}$ bracket.

Figure F33.1. Solving a Taylor or Maclaurin expansion problem ESA ★★★

T008 00:00:21 · T008 00:02:07 · T008 00:05:19

Important points and why

1. Write the general series first, then find the derivatives, then substitute. That is the order the lecture follows. ESA ★★ T008 00:00:21 · T008 00:02:07

2. For $e^x \cos y$ at $(1, \frac{\pi}{4})$ every derivative is $\pm\frac{e}{\sqrt{2}}$; only the signs differ, so take $\frac{e}{\sqrt{2}}$ out as a common factor. ESA ★★★ T008 00:04:54 · T008 00:05:19
3. If the question asks for third degree, you need the third-order partials $f_{xxx}, f_{xxy}, f_{xyy}, f_{yyy}$ too. ESA ★★★ T008 00:06:10
4. Derivatives that vanish at the point drop whole terms: for $\sin(xy)$ at $(1, \frac{\pi}{2})$ both first partials are 0. ESA ★★ S008·15 · S008·16
5. Differentiate carefully with the product and chain rules; the slide's own f_{yy} for $xy^2 + y\cos(x-y)$ has a sign slip (see the exam traps). ESA ★★ ADDED FOR CLARITY S008·17 · S008·18

Exam trap. For $xy^2 + y\cos(x-y)$: $f_{yy} = 2x + 2\sin(x-y) - y\cos(x-y)$, so $f_{yy}(1,1) = 1$, not 3 as slide 008·18 prints. ADDED FOR CLARITY S008·17 · S008·18

Exam trap. Every value on slide 008·3 is $\pm\frac{e}{\sqrt{2}}$, not $\pm\frac{e}{2}$ as said at T008 00:03:47.
ADDED FOR CLARITY S008·3 · T008 00:03:47

Worked examples

W33.1 · $e^x \cos y$ about $(1, \frac{\pi}{4})$ (slides 008·3–4) ESA ★★★

GIVEN Expand $e^x \cos y$ in powers of $(x-1)$ and $\left(y - \frac{\pi}{4}\right)$ up to second degree. Let $h = x - 1$, $k = y - \frac{\pi}{4}$.

STEP 1 $f = e^x \cos y = \frac{e}{\sqrt{2}}$; $f_x = e^x \cos y = \frac{e}{\sqrt{2}}$; $f_y = -e^x \sin y = -\frac{e}{\sqrt{2}}$.

STEP 2 $f_{xx} = e^x \cos y = \frac{e}{\sqrt{2}}$; $f_{xy} = -e^x \sin y = -\frac{e}{\sqrt{2}}$; $f_{yy} = -e^x \cos y = -\frac{e}{\sqrt{2}}$.

STEP 3 E33.1, taking $\frac{e}{\sqrt{2}}$ out: $\frac{e}{\sqrt{2}}\left[1 + h - k + \frac{1}{2!}(h^2 - 2hk - k^2)\right]$.

ANSWER
$$e^x \cos y \approx \frac{e}{\sqrt{2}}\left[1 + (x-1) - \left(y - \tfrac{\pi}{4}\right) + \tfrac{1}{2}\left\{(x-1)^2 - 2(x-1)\left(y - \tfrac{\pi}{4}\right) - \left(y - \tfrac{\pi}{4}\right)^2\right\}\right]$$

CHECK sympy series agrees. Same question as ESA May 2024 Q1.b.

S008·3 · S008·4 · T008 00:00:21 · T008 00:05:53 · ESA May-24 1b

W33.2 · $\sin(xy)$ about $(1, \frac{\pi}{2})$ (slides 008·14–16) ESA ★★

GIVEN Expand $\sin(xy)$ in powers of $(x-1)$ and $\left(y-\frac{\pi}{2}\right)$ up to second-degree terms. Let $h = x-1$, $k = y-\frac{\pi}{2}$.

STEP 1 $f = \sin\frac{\pi}{2} = 1$; $f_x = y\cos(xy) = 0$; $f_y = x\cos(xy) = 0$.

STEP 2 $f_{xx} = -y^2\sin(xy) = -\frac{\pi^2}{4}$; $f_{xy} = \cos(xy) - xy\sin(xy) = -\frac{\pi}{2}$; $f_{yy} = -x^2\sin(xy) = -1$.

STEP 3 E33.1: $1 + \frac{1}{2}\left[-\frac{\pi^2}{4}h^2 - \pi hk - k^2\right]$.

ANSWER $$\boxed{\sin(xy) \approx 1 - \frac{\pi^2}{8}(x-1)^2 - \frac{\pi}{2}(x-1)\left(y-\tfrac{\pi}{2}\right) - \frac{1}{2}\left(y-\tfrac{\pi}{2}\right)^2}$$

CHECK sympy series agrees; Figure G33.1.

S008·14 · S008·15 · S008·16

W33.3 · $xy^2 + y\cos(x-y)$ about $(1,1)$ (slides 008·17–19) ESA ★★

GIVEN Find the Taylor series of $f = xy^2 + y\cos(x-y)$ up to second order about $(1,1)$. Let $h = x-1$, $k = y-1$.

STEP 1 $f_x = y^2 - y\sin(x-y)$, $f_y = 2xy + y\sin(x-y) + \cos(x-y)$.

STEP 2 $f_{xx} = -y\cos(x-y)$, $f_{xy} = 2y + y\cos(x-y) - \sin(x-y)$, $f_{yy} = 2x + 2\sin(x-y) - y\cos(x-y)$.

STEP 3 At $(1,1)$: $f = 2$, $f_x = 1$, $f_y = 3$, $f_{xx} = -1$, $f_{xy} = 3$, $f_{yy} = 1$ (not 3, as slide 008·18 prints).

STEP 4 E33.1: $2 + (h+3k) + \frac{1}{2}(-h^2 + 6hk + k^2)$.

ANSWER $$\boxed{2 + (x-1) + 3(y-1) + \tfrac{1}{2}\left[-(x-1)^2 + 6(x-1)(y-1) + (y-1)^2\right]}$$

CHECK Slide 008·19 and the CW answer agree; sympy agrees.

S008·17 · S008·18 · S008·19 · CW U1 P28

W33.4 · sin x sin y about (π/4, π/4), second order (ESA) ESA ★★★

GIVEN Expand $\sin x \sin y$ up to second-order terms about $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$. Let $h = x - \frac{\pi}{4}$, $k = y - \frac{\pi}{4}$.

STEP 1 $f = f_x = f_y = \frac{1}{2}$.

STEP 2 $f_{xx} = f_{yy} = -\sin x \sin y = -\frac{1}{2}$; $f_{xy} = \cos x \cos y = \frac{1}{2}$.

STEP 3 $\frac{1}{2} + \frac{1}{2}(h+k) + \frac{1}{2}\{-\frac{1}{2}h^2 + hk - \frac{1}{2}k^2\}$.

ANSWER $$\boxed{\tfrac{1}{2} + \tfrac{1}{2}(h+k) - \tfrac{1}{4}(h-k)^2}$$

CHECK CW answer agrees; sympy agrees.

ESA Jul-23 1b · CW U1 P24

W33.5 · sin x sin y, third degree (ESA) ESA ★★★

GIVEN The same function up to third-degree terms.

STEP 1 $f_{xxx} = f_{xxy} = f_{xyy} = f_{yyy} = -\tfrac{1}{2}$ at $(\tfrac{\pi}{4}, \tfrac{\pi}{4})$.

STEP 2 E33.2: $\tfrac{1}{6} \cdot \big(-\tfrac{1}{2}\big)(h^3 + 3h^2k + 3hk^2 + k^3) = -\tfrac{1}{12}(h+k)^3$.

ANSWER $\boxed{\tfrac{1}{2} + \tfrac{1}{2}(h+k) - \tfrac{1}{4}(h-k)^2 - \tfrac{1}{12}(h+k)^3}$

CHECK sympy series to third order agrees.

ESA Dec-23 1b

W33.6 · $\sqrt{1+x+y^2}$ in powers of $(x-1)$ and y (CW) ESA ★★

GIVEN Expand $\sqrt{1+x+y^2}$ about $(1,0)$ up to second degree. Let $h = x-1$, $k = y$.

STEP 1 $f = \sqrt{2}$, $f_x = \tfrac{\sqrt{2}}{4}$, $f_y = 0$.

STEP 2 $f_{xx} = -\tfrac{\sqrt{2}}{16}$, $f_{xy} = 0$, $f_{yy} = \tfrac{\sqrt{2}}{2}$.

STEP 3 $\sqrt{2} + \tfrac{\sqrt{2}}{4}h + \tfrac{1}{2}\big(-\tfrac{\sqrt{2}}{16}h^2 + \tfrac{\sqrt{2}}{2}k^2\big)$.

ANSWER $\boxed{\sqrt{2}\Big[1 + \dfrac{x-1}{4} - \dfrac{(x-1)^2}{32} + \dfrac{y^2}{4}\Big]}$

CHECK CW answer agrees; sympy agrees.

CW U1 P27

Past-paper questions

ESA May 2024 (7 marks) UE22MA141A Q1.b · solved in W33.1

Expand $f(x,y) = e^x \cos y$ in powers of $(x-1)$ and $\big(y - \tfrac{\pi}{4}\big)$ using Taylor's series up to second degree terms.

ESA Jul 2023 (7 marks) UE22MA141A Q1.b · solved in W33.4

Expand $f(x,y) = \sin x \sin y$ in Taylor's series up to second order terms about the point $\big(\tfrac{\pi}{4}, \tfrac{\pi}{4}\big)$.

ESA Dec 2023 (7 marks) UE22MA141A Q1.b · solved in W33.5

Expand $f(x,y) = \sin x \sin y$ in Taylor's series up to third degree term about $\big(\tfrac{\pi}{4}, \tfrac{\pi}{4}\big)$.

SHORT-NOTE RECAP

About (a,b): $h = x-a$, $k = y-b$; table of $f, f_x, f_y, f_{xx}, f_{xy}, f_{yy}$ at the point; $f + \{hf_x + kf_y\} + \tfrac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\}$. $e^x \cos y$ at $(1, \tfrac{\pi}{4})$: $\tfrac{e}{\sqrt{2}}[1 + h - k + \tfrac{1}{2}(h^2 - 2hk - k^2)]$. $\sin(xy)$ at $(1, \tfrac{\pi}{2})$: $1 - \tfrac{\pi^2}{8}h^2 - \tfrac{\pi}{2}hk - \tfrac{1}{2}k^2$. $xy^2 + y\cos(x-y)$ at $(1,1)$: $2 + h + 3k + \tfrac{1}{2}(-h^2 + 6hk + k^2)$.

34 Maclaurin expansion about the origin

The same method with $(a,b)=(0,0)$, so $h=x$, $k=y$. Class 008 works $\log(2x+y+1)$, e^{x-y} and $e^{x}\log(1+y)$; the ESA papers ask $\log(1+x-y)$, $e^{ax}\sin by$ and $\sin(x+2y)$, all up to third degree. When f depends on one combination such as $2x+y$ or $x-y$, the one-variable series gives the answer at once and is the best check.

Key terms and definitions

Up to three terms. Keep going until there are three non-zero terms: "make sure that you will get three non-zero terms". S008·5 · T008 00:07:38

> *"Taylor's series about* $(0,0)$ *is given by* $f(x,y)\approx f(0,0)+f_x(0,0)\,x+f_y(0,0)\,y+\frac{1}{2!}\left[f_{xx}(0,0)x^2+2f_{xy}(0,0)xy+f_{yy}(0,0)y^2\right]+\frac{1}{3!}\left[f_{xxx}(0,0)x^3+3f_{xxy}(0,0)x^2y+3f_{xyy}(0,0)xy^2+f_{yyy}(0,0)y^3\right]+\ldots$" S008·10

Formulas

E34.1 · Maclaurin's series (about the origin) ESA ★★★

$$f(x,y)=f(0,0)+\{xf_x+yf_y\}+\frac{1}{2!}\{x^2f_{xx}+2xyf_{xy}+y^2f_{yy}\}+\ldots$$

gives	The polynomial about $(0,0)$.
applies	Derivatives at $(0,0)$.
note	Slide 008·10 calls this "Taylor's series about (0,0)"; it is the same thing.

S008·10 · S008·12 · T008 00:06:58

E34.2 · One-combination check ESA ★★

$$f=g(t),\ t=px+qy\ \Longrightarrow\ f(x,y)=g(0)+g'(0)\,t+\frac{g''(0)}{2!}t^2+\frac{g'''(0)}{3!}t^3+\ldots$$

gives	The same expansion in one line.
applies	f depends on x and y only through $px+qy$.
note	$\log(2x+y+1)=t-\frac{t^2}{2}+\frac{t^3}{3}$ with $t=2x+y$; $e^{x-y}=1+t+\frac{t^2}{2}+\frac{t^3}{6}$ with $t=x-y$.

ADDED FOR CLARITY S008·6 · S008·11

Graphs and diagrams

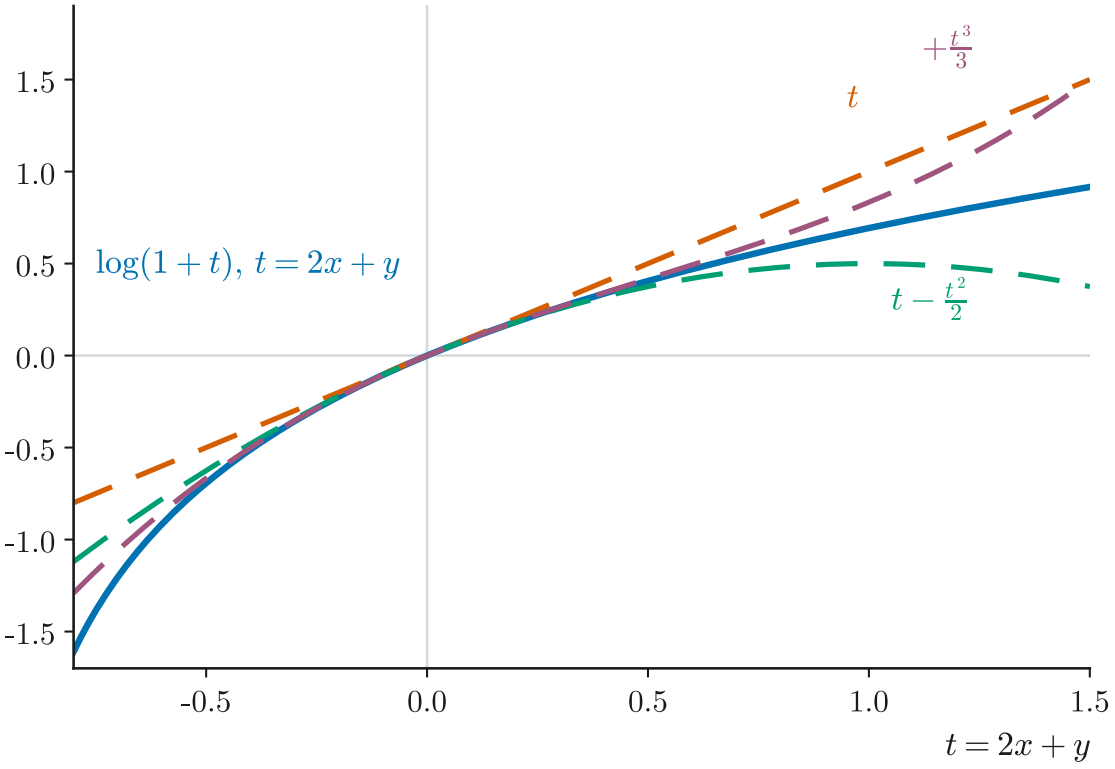


Figure G34.1. $\log(2x + y + 1)$ as a function of $t = 2x + y$. ESA ★★

ADDED FOR CLARITY S008·5 · S008·6

How to read it. The function depends only on $t = 2x + y$, so its Maclaurin polynomials are t, $t - \frac{t^2}{2}$ and $t - \frac{t^2}{2} + \frac{t^3}{3}$. Each extra degree follows $\log(1 + t)$ (blue) further from $t = 0$; all fail as $t \to -1$, where the logarithm blows up.

Quick checks with a one-variable series

Function	Combination t	One-variable series	Problem
$\log(2x + y + 1)$	$2x + y$	$t - \frac{t^2}{2} + \frac{t^3}{3} - \dots$	W34.1
e^{x-y}	$x - y$	$1 + t + \frac{t^2}{2} + \frac{t^3}{6} + \dots$	W34.2
$\log(1 + x - y)$	$x - y$	$t - \frac{t^2}{2} + \frac{t^3}{3} - \dots$	W34.4
$\sin(x + 2y)$	$x + 2y$	$t - \frac{t^3}{6} + \dots$	W34.6
$e^x \log(1 + y)$, $e^{ax} \sin by$	product	multiply the two series, keep degree ≤ 3	W34.3, W34.5

ADDED FOR CLARITY S008·6 · S008·11 · ESA Jan-24 s1 1b · ESA Jun-24 1c

Important points and why

1. About $(0,0)$ the constant term is often 0 ($\log 1 = 0$, $\sin 0 = 0$), so the series starts at first degree. ESA ★★ T008 00:07:57
2. "Up to three terms" means three non-zero terms; the lecture still writes all second-order terms. ESA ★★ T008 00:07:38 · T008 00:10:55
3. Collect and simplify at the end: $2x + y - 2x^2 - 2xy - \frac{y^2}{2}$. ESA ★ T008 00:11:11
4. Check a two-variable answer with the one-variable series of the combination (E34.2, Figure G34.1). ESA ★★★ ADDED FOR CLARITY S008·6 · S008·11

Why does $\log(2x+y+1)$ give $(2x+y) - \frac{(2x+y)^2}{2} + \ldots$?
It is $\log(1+t)$ with $t = 2x+y$, and $\log(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \ldots$; the two-variable Maclaurin series must agree with it.
ADDED FOR CLARITY S008·6

Exam trap. For $e^x \log(1+y)$: $f_{xy} = \frac{e^x}{1+y}$, so $f_{xy}(0,0) = 1$, not 0 as slide 008·12 lists.
ADDED FOR CLARITY S008·12

Exam trap. "First six terms" of $e^x \log(1+y)$ needs third order: $y + xy - \frac{y^2}{2} + \frac{x^2 y}{2} - \frac{xy^2}{2} + \frac{y^3}{3}$. The slide stops at three. ADDED FOR CLARITY S008·12 · S008·13

Worked examples

W34.1 · $\log(2x+y+1)$ about the origin (slides 008·5–6) — ESA ★★

GIVEN Expand $\log(2x+y+1)$ about $(0,0)$ up to three terms.

STEP 1 $f = \log 1 = 0$; $f_x = \frac{2}{2x+y+1} = 2$; $f_y = \frac{1}{2x+y+1} = 1$.

STEP 2 $f_{xx} = \frac{-4}{(2x+y+1)^2} = -4$; $f_{xy} = \frac{-2}{(\ldots)^2} = -2$; $f_{yy} = \frac{-1}{(\ldots)^2} = -1$.

STEP 3 E34.1: $2x + y + \frac{1}{2!}\{-4x^2 - 4xy - y^2\}$.

ANSWER
$$\log(2x+y+1) \approx 2x + y - 2x^2 - 2xy - \frac{y^2}{2} = (2x+y) - \frac{(2x+y)^2}{2}$$

CHECK E34.2 with $t = 2x+y$; sympy agrees. The lecture's spoken simplification "$-xy$" (T008 00:11:11) should be $-2xy$.

S008·5 · S008·6 · T008 00:06:34 · T008 00:11:11

W34.2 · e^{x-y} about the origin (slides 008·10–11) ESA ★★

GIVEN Find the Taylor expansion of e^{x-y} at $(0,0)$.

STEP 1 At $(0,0)$ every derivative is ± 1: $f_x = 1$, $f_y = -1$, $f_{xx} = 1$, $f_{xy} = -1$, $f_{yy} = 1$.

STEP 2 Third order: $f_{xxx} = 1$, $f_{xxy} = -1$, $f_{xyy} = 1$, $f_{yyy} = -1$.

STEP 3 E34.1 + E33.2: $1 + x - y + \frac{1}{2}(x^2 - 2xy + y^2) + \frac{1}{6}(x^3 - 3x^2y + 3xy^2 - y^3)$.

ANSWER $\boxed{e^{x-y} \approx 1 + (x-y) + \dfrac{(x-y)^2}{2} + \dfrac{(x-y)^3}{6}}$

CHECK The slide stops at second order; the third-degree term is added (sympy). E34.2 with $t = x - y$.

S008·10 · S008·11

W34.3 · $e^x \log(1+y)$, first six terms (slides 008·12–13) ESA ★★

GIVEN Find the first six terms of the expansion of $e^x \log(1+y)$ about $(0,0)$.

STEP 1 $f = 0$, $f_x = e^x \log(1+y) = 0$, $f_y = \frac{e^x}{1+y} = 1$.

STEP 2 $f_{xx} = 0$, $f_{xy} = \frac{e^x}{1+y} = 1$ (slide 008·12 prints 0), $f_{yy} = -\frac{e^x}{(1+y)^2} = -1$.

STEP 3 Third order: $f_{xxx} = 0$, $f_{xxy} = 1$, $f_{xyy} = -1$, $f_{yyy} = \frac{2e^x}{(1+y)^3} = 2$.

STEP 4 $y + \frac{1}{2}(2xy - y^2) + \frac{1}{6}(3x^2y - 3xy^2 + 2y^3)$.

ANSWER $\boxed{e^x \log(1+y) \approx y + xy - \dfrac{y^2}{2} + \dfrac{x^2y}{2} - \dfrac{xy^2}{2} + \dfrac{y^3}{3}}$

CHECK The slide gives the first three ($y + xy - \frac{y^2}{2}$); the other three are added. Product of $1 + x + \frac{x^2}{2}$ and $y - \frac{y^2}{2} + \frac{y^3}{3}$; sympy agrees.

S008·12 · S008·13

W34.4 · log(1 + x − y) to third degree (ESA) ESA ★★★

GIVEN Expand $\log(1 + x - y)$ about the origin up to third-degree terms.

STEP 1 $f_x = 1$, $f_y = -1$; $f_{xx} = -1$, $f_{xy} = 1$, $f_{yy} = -1$; $f_{xxx} = 2$, $f_{xxy} = -2$, $f_{xyy} = 2$, $f_{yyy} = -2$.

STEP 2 $(x-y) + \frac{1}{2}(-x^2 + 2xy - y^2) + \frac{1}{6}(2x^3 - 6x^2y + 6xy^2 - 2y^3)$.

ANSWER $\boxed{(x-y) - \dfrac{(x-y)^2}{2} + \dfrac{(x-y)^3}{3}}$

CHECK E34.2 with $t = x - y$; sympy agrees.

ESA Jan-24 s1 1b

W34.5 · $e^{ax}\sin by$ to third degree (ESA) ESA ★★★

GIVEN Expand $e^{ax}\sin by$ about the origin up to third-degree terms.

STEP 1 $f = f_x = 0$, $f_y = b$; $f_{xx} = 0$, $f_{xy} = ab$, $f_{yy} = 0$.

STEP 2 $f_{xxx} = 0$, $f_{xxy} = a^2 b$, $f_{xyy} = 0$, $f_{yyy} = -b^3$.

STEP 3 $by + abxy + \frac{1}{6}(3a^2bx^2y - b^3y^3)$.

ANSWER $\boxed{by + abxy + \dfrac{a^2 b\, x^2 y}{2} - \dfrac{b^3 y^3}{6}}$

CHECK Product of the series of e^{ax} and $\sin by$; sympy agrees.

ESA Jan-24 s2 1b

W34.6 · sin(x + 2y) to third degree (ESA) ESA ★★★

GIVEN Expand $\sin(x+2y)$ up to third-degree terms about $(0,0)$.

STEP 1 $f_x = 1$, $f_y = 2$; all second partials 0.

STEP 2 $f_{xxx} = -1$, $f_{xxy} = -2$, $f_{xyy} = -4$, $f_{yyy} = -8$.

STEP 3 $x + 2y + \frac{1}{6}(-x^3 - 6x^2y - 12xy^2 - 8y^3)$.

ANSWER $\boxed{(x+2y) - \dfrac{(x+2y)^3}{6}}$

CHECK CW answer agrees; E34.2 with $t = x + 2y$.

ESA Jun-24 1c · CW U1 P25

Past-paper questions

ESA Jan 2024 · 7 marks · UE23MA141A set 1, Q1.b · solved in W34.4

Expand $\log(1 + x - y)$ about origin up to third degree terms.

ESA Jan 2024 · 7 marks · UE23MA141A set 2, Q1.b · solved in W34.5

Expand $e^{ax}\sin by$ about origin up to third degree terms.

ESA June 2024 · 6 marks · UE23MA141A Q1.c · solved in W34.6

Expand $f(x,y) = \sin(x+2y)$ in Taylor's series up to third degree terms about the point $(0,0)$.

SHORT-NOTE RECAP

About $(0,0)$: $f + \{xf_x + yf_y\} + \frac{1}{2!}\{\dots\} + \frac{1}{3!}\{\dots\}$. $\log(2x+y+1) = (2x+y) - \frac{(2x+y)^2}{2}$; $e^{x-y} = 1 + (x-y) + \frac{(x-y)^2}{2} + \frac{(x-y)^3}{6}$; $e^x \log(1+y) = y + xy - \frac{y^2}{2} + \frac{x^2 y}{2} - \frac{xy^2}{2} + \frac{y^3}{3}$. Check one-combination functions with the one-variable series.

35 Approximate values from a Taylor polynomial

To estimate f at an awkward point, expand about a nearby point where everything is known, then substitute the small h and k. Class 008 estimates $\tan^{-1}(xy)$ at $(0.9, -1.2)$ from $(1, -1)$; the class-work sheet estimates $\tan^{-1}\frac{0.9}{1.1}$ from $(1, 1)$. More terms give a better value.

Key terms and definitions

Approximate value. The value of a truncated Taylor polynomial at a point near the expansion point. S008·7 · T008 00:11:47

Formulas

E35.1 · Approximation by a Taylor polynomial ESA ★★

$$f(a+h, b+k) \approx f(a,b) + hf_x + kf_y + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\}$$

gives	A numerical estimate.
applies	h, k small; (a, b) a point where f and its derivatives are easy.
note	Choose (a, b) with "nice" values: $(1, -1)$ for $(0.9, -1.2)$.

S008·7 · S008·9 · T008 00:12:10

Graphs and diagrams

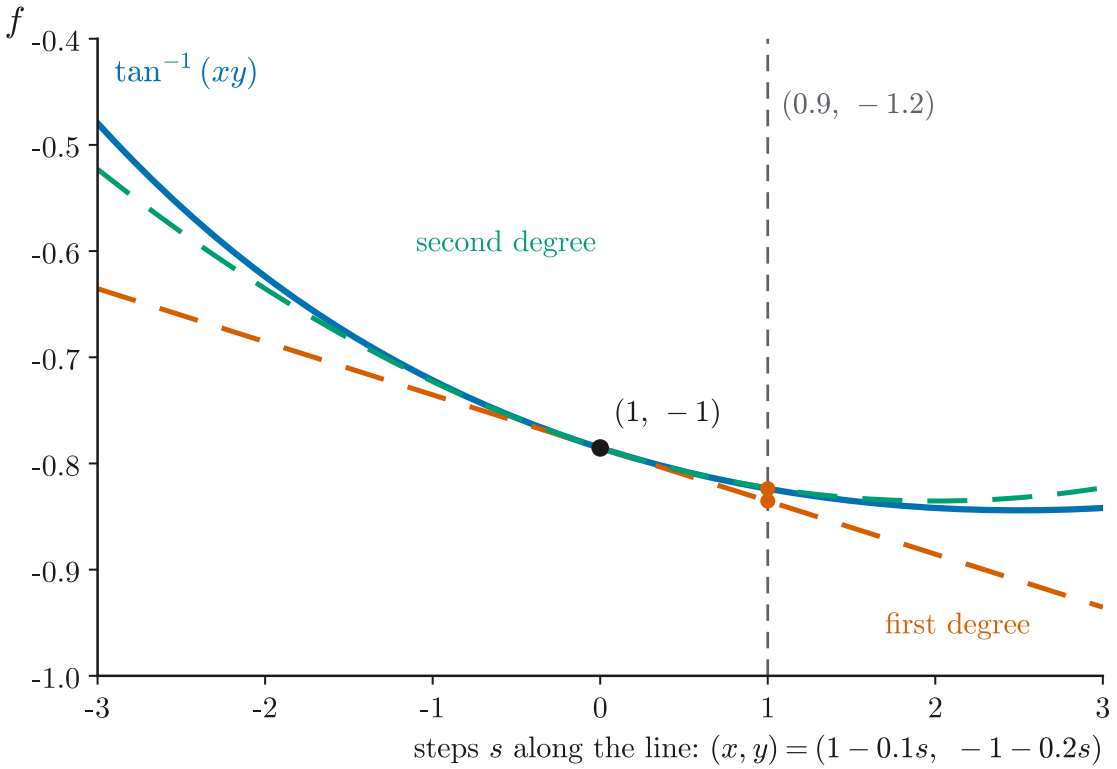
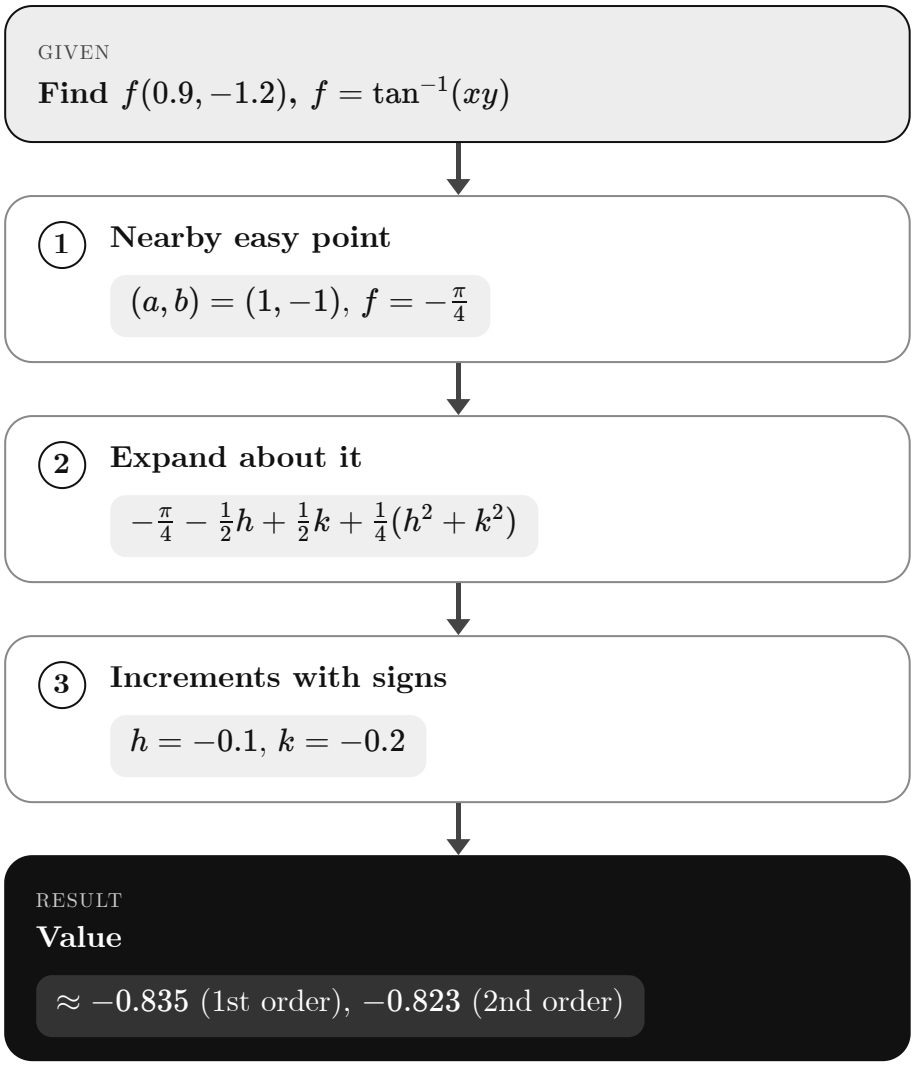


Figure G35.1. Approximating $\tan^{-1}(xy)$ at $(0.9, -1.2)$. ESA ★★

ADDED FOR CLARITY S008·7 · S008·9 · T008 00:14:18

How to read it. Values along the straight line from $(1, -1)$ (black dot, $s = 0$) through $(0.9, -1.2)$ ($s = 1$, grey dashed line). At $s = 1$ the first-degree polynomial (vermilion) gives -0.835, while the function (blue) is -0.824; the second degree (green) is almost on the curve there.

Flowcharts



Remark. Exact value -0.8238: the second-order term is worth keeping.

Figure F35.1. Approximate value from a Taylor polynomial ESA ★★

S008·7 · S008·9 · T008 00:12:10

Important points and why

1. Pick an expansion point near the target where f is easy: $(1, -1)$ for $(0.9, -1.2)$, $(1, 1)$ for $(1.1, 0.9)$. ESA ★★ T008 00:12:10 · CW U1 P26
2. "The more terms you take, the better the approximation": the first-order estimate of $\tan^{-1}(-1.08)$ is off by 0.012, the second-order one by 0.001. ESA ★★ T008 00:14:18
3. Substitute $h = x - a$ and $k = y - b$ with their signs: $x = 0.9$ gives $h = -0.1$, $y = -1.2$ gives $k = -0.2$. ESA ★★ T008 00:15:13 · T008 00:15:28

Why expand about (1, −1) to evaluate at (0.9, −1.2)?
At $(1,-1)$ every value is exact ($\tan^{-1}(-1)=-\frac{\pi}{4}$, simple derivatives), and $h=-0.1$, $k=-0.2$ are small, so the higher terms shrink quickly.

T008 00:12:10 · T008 00:12:48

Exam trap. With $x=0.9$, $h=x-1=-0.1$ (negative). A sign slip here changes the answer by 0.1. ADDED FOR CLARITY T008 00:15:13

Worked examples

W35.1 · tan⁻¹(xy) at (0.9, −1.2) (slides 008·7–9) — ESA ★★

GIVEN Find the Taylor series of $\tan^{-1}(xy)$ about $(1,-1)$ and hence approximate $f(0.9,-1.2)$. Let $h=x-1$, $k=y+1$.

STEP 1 $f=\tan^{-1}(-1)=-\frac{\pi}{4}$; $f_x=\frac{y}{1+x^2y^2}=-\frac{1}{2}$; $f_y=\frac{x}{1+x^2y^2}=\frac{1}{2}$.

STEP 2 $f_{xx}=\frac{-2xy^3}{(1+x^2y^2)^2}=\frac{1}{2}$; $f_{xy}=\frac{1-x^2y^2}{(1+x^2y^2)^2}=0$; $f_{yy}=\frac{-2x^3y}{(1+x^2y^2)^2}=\frac{1}{2}$.

STEP 3 Series: $-\frac{\pi}{4}-\frac{1}{2}h+\frac{1}{2}k+\frac{1}{4}(h^2+k^2)$.

STEP 4 $h=-0.1$, $k=-0.2$. First order: $-0.7854+0.05-0.1=-0.8354$. Adding $\frac{1}{4}(0.01+0.04)=0.0125$ gives -0.8229.

ANSWER $\boxed{\tan^{-1}(-1.08)\approx -0.835}$ (first order, as on the slide); ≈ -0.823 (second order)

CHECK Exact -0.82384 (sympy). The second-order value is ten times closer.

S008·7 · S008·8 · S008·9 · T008 00:11:53 · T008 00:15:42

W35.2 · tan⁻¹(0.9/1.1) (CW) — ESA ★★

GIVEN Using $\tan^{-1}\frac{y}{x}$ about $(1,1)$, compute $\tan^{-1}\frac{0.9}{1.1}$. Let $h=x-1$, $k=y-1$.

STEP 1 At $(1,1)$: $f=\frac{\pi}{4}$, $f_x=-\frac{1}{2}$, $f_y=\frac{1}{2}$, $f_{xx}=\frac{1}{2}$, $f_{xy}=0$, $f_{yy}=-\frac{1}{2}$.

STEP 2 $\frac{\pi}{4}-\frac{1}{2}h+\frac{1}{2}k+\frac{1}{4}h^2-\frac{1}{4}k^2$ with $h=0.1$, $k=-0.1$.

ANSWER $\boxed{\approx 0.6854}$

CHECK Exact 0.6857. The CW sheet prints 0.6904, an arithmetic slip.

CW U1 P26

SHORT-NOTE RECAP

Expand about a nearby easy point and substitute small h, k with signs. $\tan^{-1}(xy)$ at $(0.9,-1.2)$ from $(1,-1)$: -0.835 (first order), -0.823 (second); exact -0.824. $\tan^{-1}\frac{0.9}{1.1}\approx 0.6854$.

M8

Class 009 · Maxima and Minima of a Function of Two Variables

36 Maxima, minima and saddle points

A surface can have several peaks (local maxima) and several pits (local minima); at each the tangent plane is horizontal. A saddle point also has a horizontal tangent plane, but the surface rises in one direction and falls in another, like a horse saddle. The word "relative" (or local) means the comparison is only with nearby points.

Key terms and definitions

Saddle point. A stationary point at which $f(a,b)$ is neither a maximum nor a minimum: the surface rises in some directions and falls in others. Test: $rt - s^2 < 0$. S009·5 · S009·7 · T009 00:11:06

Graphs and diagrams

two local maxima (peaks)

vermilion: horizontal tangent plane

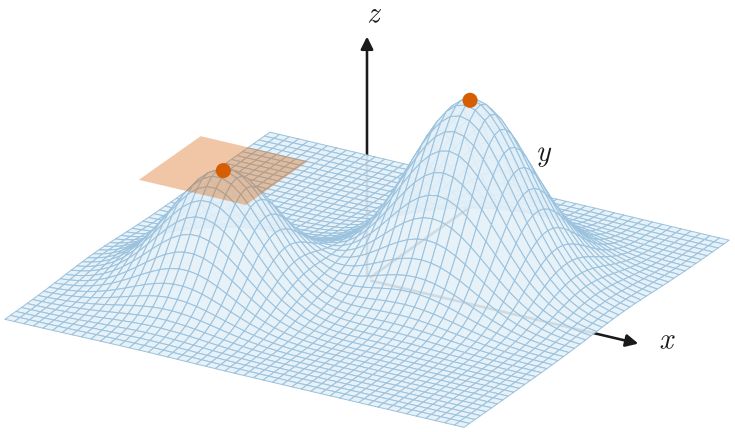


Figure G36.1. Local maxima: peaks with horizontal tangent planes. ESA ★

S009·3 · T009 00:00:16

How to read it. A surface with two peaks (illustrative: a sum of two bell shapes). Both peaks are local maxima; the taller one is also the largest value overall. The vermilion square is the horizontal tangent plane at the lower peak.

$z = x^2 + y^2$

minimum at $(0,0)$: $rt - s^2 = 4 > 0,\ r = 2 > 0$

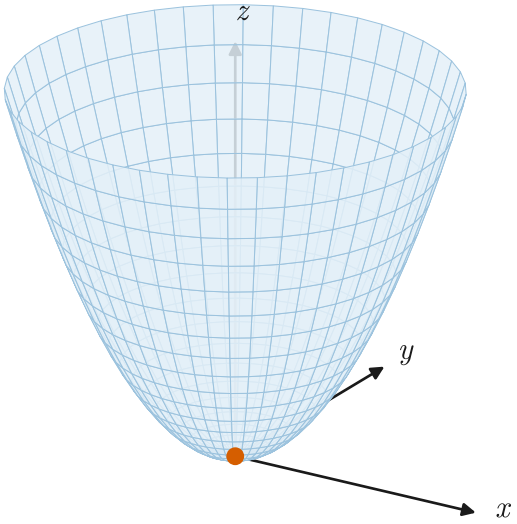


Figure G36.2. A local minimum: $z = x^2 + y^2$. ESA ★

S009·4 · T009 00:02:11

How to read it. The bowl $z = x^2 + y^2$ has its lowest point at the origin. Every nearby point is higher: $\Delta > 0$. Its test values are $r = t = 2$, $s = 0$, $rt - s^2 = 4 > 0$, $r > 0$ (W38.1).

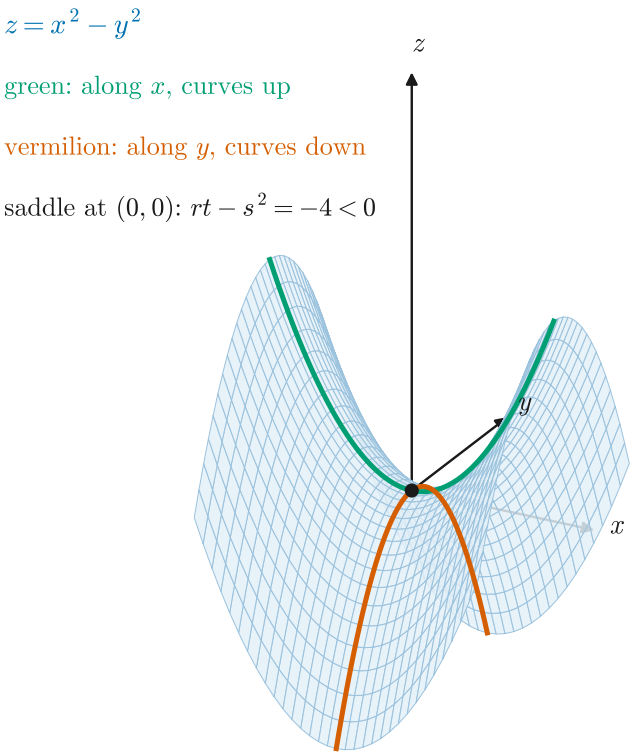


Figure G36.3. A saddle point: $z = x^2 - y^2$. ESA ★★

S009·5 · S009·7

How to read it. Along the x-direction (green) the surface curves up, along the y-direction (vermilion) it curves down. The origin is stationary but neither a maximum nor a minimum; $rt - s^2 = -4 < 0$ (W38.2).

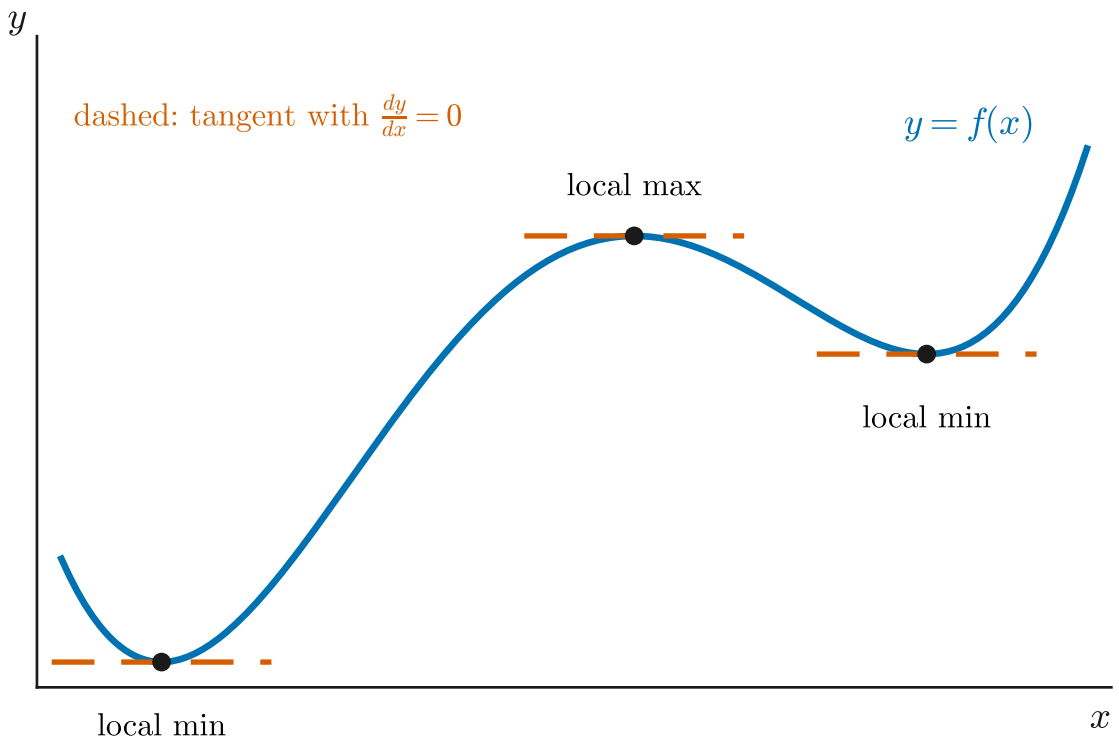


Figure G36.4. One-variable picture: maxima, minima and $\frac{dy}{dx} = 0$. ESA ★

T009 00:00:50 · T009 00:01:10

How to read it. On $y = f(x)$ the tangent is horizontal (slope 0) at every local maximum and minimum (short dashed segments). The two-variable test copies this: $f_x = f_y = 0$ replaces $\frac{dy}{dx} = 0$, and $rt - s^2$ with r replaces $\frac{d^2y}{dx^2}$.

Important points and why

1. One variable recap: solve $\frac{dy}{dx} = 0$ for critical points; $\frac{d^2y}{dx^2} < 0$ gives a maximum, > 0 a minimum; $= 0$ gives no conclusion. The tangent at a maximum or minimum is parallel to the x-axis (slope 0). ESA ★★ T009 00:00:50 · T009 00:01:40 · T009 00:02:57
2. In two variables the tangent plane at a maximum or minimum is horizontal (parallel to the xy-plane); slide 009·3 shows a horizontal tangent plane at a relative maximum (Figure G36.4). ESA ★ S009·3 · S009·4 · T009 00:01:10
3. A function can have several local maxima and several local minima (slides 009·3–4: "These are all local maxima / minima"). ESA ★ S009·3 · S009·4 · T009 00:00:16

Exam trap. In one variable $\frac{d^2y}{dx^2} = 0$ is not "a saddle point"; it is only inconclusive (x^4 has a minimum, x^3 an inflection). ADDED FOR CLARITY T009 00:03:14

SHORT-NOTE RECAP

Local maxima are peaks, local minima are pits; at both the tangent plane is horizontal. A saddle point also has a horizontal tangent plane but rises one way and falls another. A function can have several of each.

37 Definitions: relative maximum, minimum and extreme value

f has a relative maximum at (a, b) if $f(a, b) > f(a + h, b + k)$ for all small h, k, and a relative minimum if $f(a, b) < f(a + h, b + k)$. Equivalently, $\Delta = f(a + h, b + k) - f(a, b)$ keeps one sign: negative for a maximum, positive for a minimum. Either is an extreme value.

Key terms and definitions

Relative maximum. A function $f(x, y)$ is said to have a relative maximum at a point (a, b) if there exists a neighbourhood of the point (a, b), say $(a + h, b + k)$, h and k small, such that $f(a, b) > f(a + h, b + k)$. S009·6 · T009 00:03:46

Relative minimum. If $f(a, b) < f(a + h, b + k)$ then $f(x, y)$ is said to have a relative minimum at (a, b). S009·6 · T009 00:04:58

Extreme value. $f(a, b)$ is said to be an extreme value of f if it is either a maximum or a minimum. S009·6

Neighbourhood. The points very close to (a, b): $(a + h, b + k)$ with h, k small. "Relative" means largest or smallest only in such a neighbourhood, not over the whole plane. T009 00:04:12 · T009 00:04:40

> *"In other words if $\Delta = f(a + h, b + k) - f(a, b)$ is of the same sign for all small values of h, k and if this sign is negative then $f(a, b)$ is maximum & if this sign is positive then $f(a, b)$ is a minimum."* S009·6 · T009 00:05:23

Formulas

E37.1 · Sign of the increment

ESA ★

$$\Delta = f(a + h, b + k) - f(a, b) \qquad \begin{cases} < 0 \text{ for all small } h, k: & \text{maximum} \\ > 0 \text{ for all small } h, k: & \text{minimum} \end{cases}$$

gives	The definition of an extreme value.
applies	Used to settle the doubtful case $rt - s^2 = 0$.
note	Same sign in every direction.

S009·6 · T009 00:05:23

Important points and why

1. "Relative" because only points in a neighbourhood of (a, b) are compared, not the whole plane. ESA ★ T009 00:04:40

SHORT-NOTE RECAP

Relative maximum at (a, b): $f(a, b) > f(a + h, b + k)$ for small h, k; minimum: $<$. Equivalently $\Delta = f(a + h, b + k) - f(a, b)$ has one sign: negative $\Rightarrow$ maximum, positive $\Rightarrow$ minimum. Either is an extreme value.

38 Conditions and the working rule

As in one variable, first make the first derivatives zero: $f_x = 0$, $f_y = 0$ give the stationary (critical) points. Then use the second derivatives $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$ and the Hessian determinant $rt - s^2$: $rt - s^2 > 0$ with $r < 0$ is a maximum, with $r > 0$ a minimum; $rt - s^2 < 0$ is a saddle point; $rt - s^2 = 0$ is doubtful and needs further investigation.

Key terms and definitions

Stationary (critical) point. A point where $f_x = 0$ and $f_y = 0$: the tangent plane is horizontal, and the function may have an extreme value there. S009·8 · T009 00:02:47 · T009 00:06:19

Hessian. The determinant $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = rt - s^2$ of second partial derivatives. S009·7 · T009 00:07:22

> *"The necessary and sufficient condition for $f(x, y)$ to have a maximum value at (a, b) is $f_x(a, b) = 0$, $f_y(a, b) = 0$ and $rt - s^2 > 0$, where $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$. $rt - s^2$ is the determinant of the Hessian matrix $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix}$."* S009·7

> *"Note: If $rt - s^2 < 0$, $f(a, b)$ is not an extreme value, i.e. $f(x, y)$ is neither a maximum nor a minimum at the point (a, b) and the point is called a saddle point. If $rt - s^2 = 0$, then the test provides no information about the nature of the point."* S009·7

Formulas

E38.1 · Stationary points (necessary condition) ESA ★★★

$$f_x = \frac{\partial f}{\partial x} = 0, \qquad f_y = \frac{\partial f}{\partial y} = 0$$

gives	The candidates $(a, b), (c, d), \ldots$ for extreme values.
applies	f differentiable; solve the two equations simultaneously.
note	Two-variable version of $\frac{dy}{dx} = 0$.

S009·7 · S009·8 · T009 00:06:19

E38.2 · Second derivatives used in the test ESA ★★★

$$r = \frac{\partial^2 f}{\partial x^2}, \qquad s = \frac{\partial^2 f}{\partial x\, \partial y}, \qquad t = \frac{\partial^2 f}{\partial y^2}$$

gives	The three numbers at each stationary point.
applies	Evaluate at each stationary point separately.
note	$s = f_{xy} = f_{yx}$ (Clairaut).

S009·7 · S009·8 · T009 00:07:04

E38.3 · Hessian determinant

ESA ★★★

$$rt - s^2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = f_{xx} f_{yy} - f_{xy}^2$$

GIVES Whether the stationary point is an extremum.

APPLIES At a stationary point.

NOTE Positive: extremum (decide by r). Negative: saddle. Zero: doubtful.

S009·7 · T009 00:07:37

E38.4 · The test (working rule, steps 3–6)

ESA ★★★

$$rt - s^2 > 0,\ r < 0 \ \Rightarrow\ \text{maximum}; \qquad rt - s^2 > 0,\ r > 0 \ \Rightarrow\ \text{minimum}$$
$$rt - s^2 < 0 \ \Rightarrow\ \text{saddle point}; \qquad rt - s^2 = 0 \ \Rightarrow\ \text{doubtful}$$

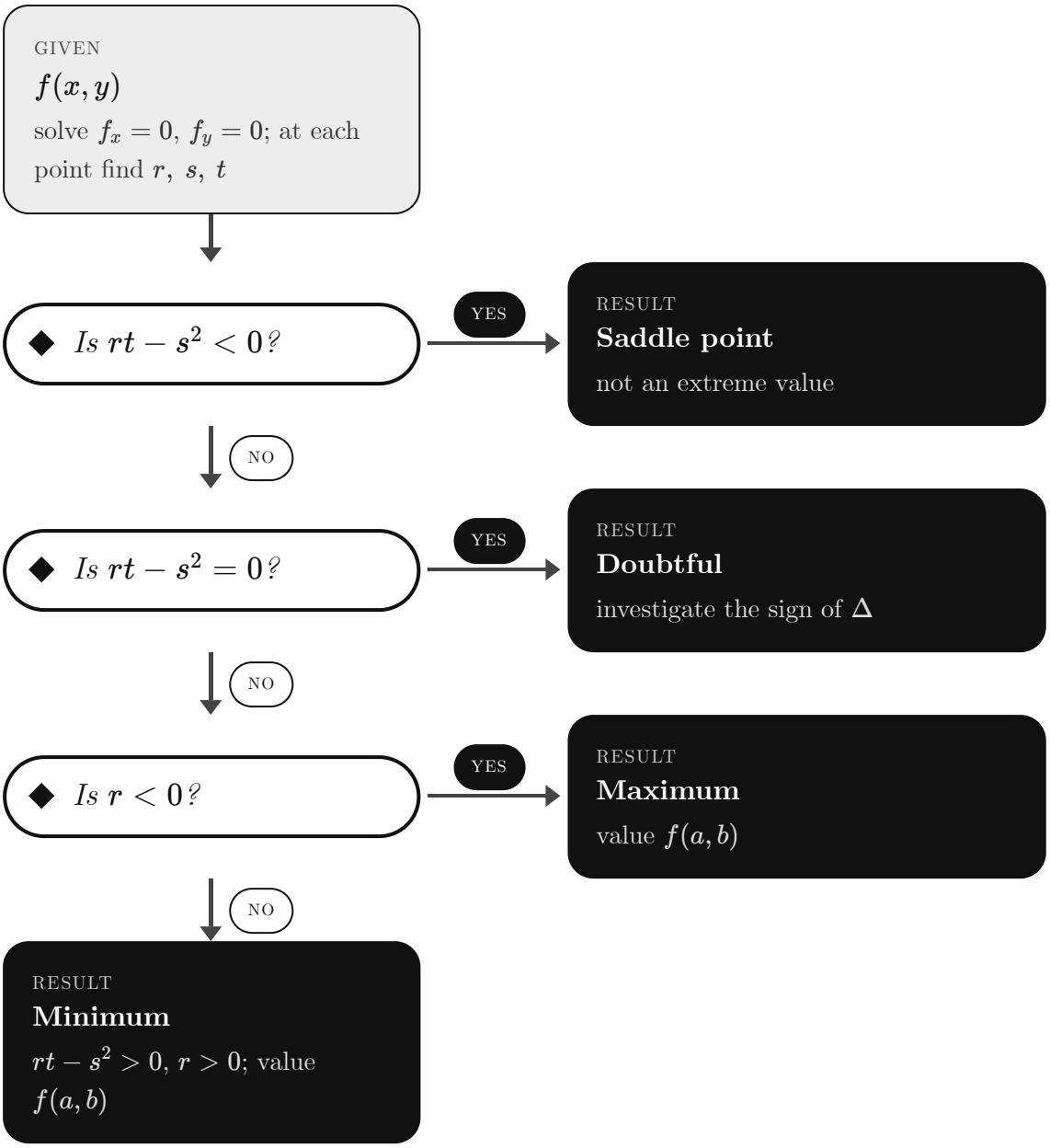
GIVES The nature of each stationary point.

APPLIES After E38.1–E38.3.

NOTE $r < 0$ plays the role of $\frac{d^2y}{dx^2} < 0$ in one variable. When $rt - s^2 > 0$, r and t have the same sign, so t can be used instead.

S009·8 · S009·9 · T009 00:11:32

Flowcharts



Remark. Repeat the ladder for every stationary point.

Exam trap. Never classify from r alone.

Figure F38.1. Working rule: nature of a stationary point ESA ★★★

S009·8 · S009·9 · T009 00:11:32

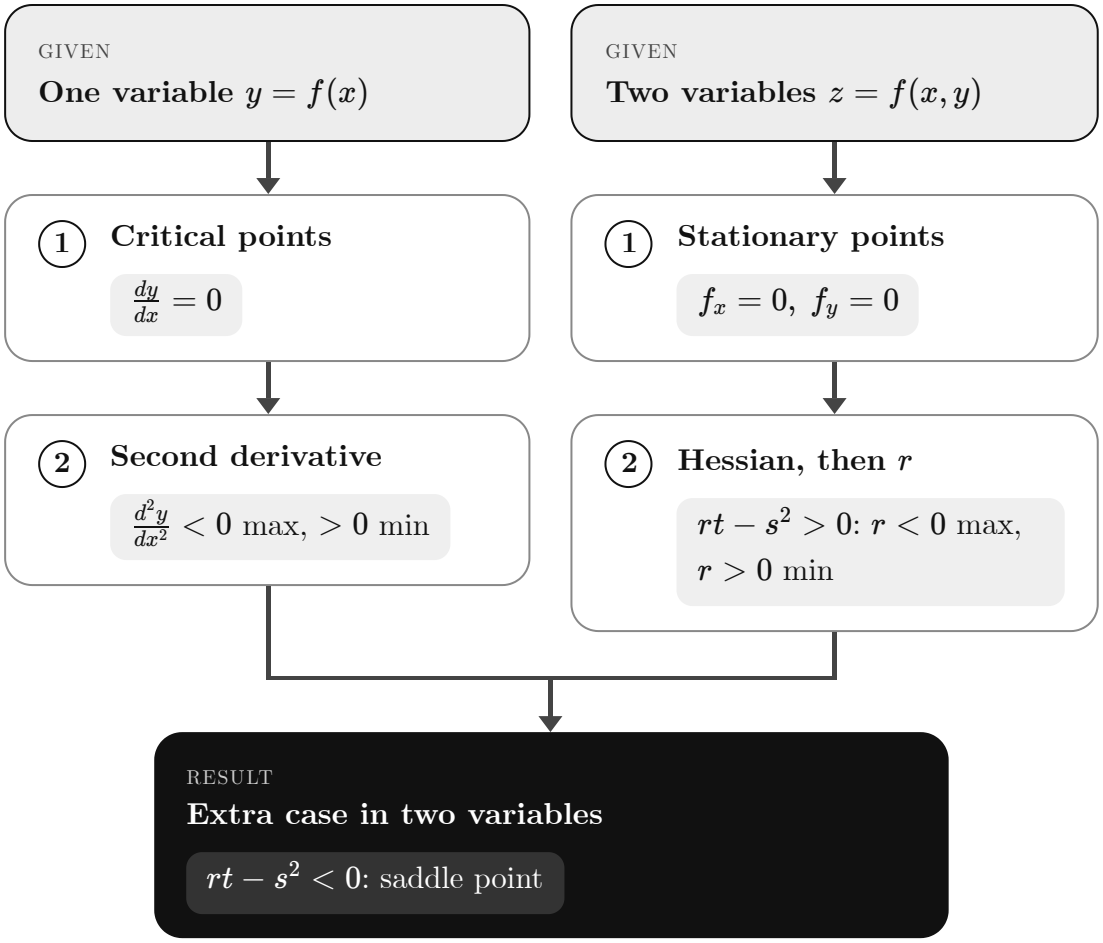


Figure F38.2. From one variable to two ESA ★★

T009 00:06:00 · T009 00:10:20

The four outcomes of the test

$rt - s^2$	r	Nature of (a, b)	Example at the origin
> 0	< 0	Maximum	$-x^2 - y^2$
> 0	> 0	Minimum	$x^2 + y^2$ (W38.1)
< 0	any	Saddle point	$x^2 - y^2$ (W38.2)
$= 0$	any	Doubtful: further investigation	$x^4 + y^4$ (min), $x^4 - y^4$ (neither)

S009·7 · S009·8 · S009·9

Important points and why

1. Working rule step 1: find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$, equate each to zero, solve simultaneously; let $(a, b), (c, d), \ldots$ be the pairs of values. ESA ★★★ S009·8 · T009 00:11:32
2. Step 2: calculate $r = \frac{\partial^2 f}{\partial x^2}$, $s = \frac{\partial^2 f}{\partial x \partial y}$, $t = \frac{\partial^2 f}{\partial y^2}$ for each pair of values. ESA ★★★ S009·8

3. Steps 3–6: $rt - s^2 > 0$ and $r < 0$: maximum; $rt - s^2 > 0$ and $r > 0$: minimum; $rt - s^2 < 0$: not an extreme value, (a, b) is a saddle point; $rt - s^2 = 0$: the case is doubtful and needs further investigation. ESA ★★★ S009·8 · S009·9 · T009 00:12:33
4. Two important conditions to remember (lecture): both first partials zero, and $rt - s^2$ from the second partials. ESA ★★★ TRANSCRIPT ONLY T009 00:08:50
5. Unlike one variable, the sign of r alone is not enough: first check $rt - s^2 > 0$, then look at r. ESA ★★★ T009 00:07:51 · T009 00:09:43
6. The value of the function at a maximum or minimum point is the maximum or minimum value: always substitute the point back into f. ESA ★★ ADDED FOR CLARITY T009 00:04:51
7. Each stationary point is tested separately: a function can have maxima, minima and saddle points together (class 010: four points for $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$). ESA ★★★ ESA Dec-23 1c · ESA Jan-24 s1 1a

Why must f_x and f_y both vanish at an extremum?
Along the line $y = b$, $f(x, b)$ is a function of x alone with an extremum at $x = a$, so its derivative $f_x(a, b) = 0$. The same along $x = a$ gives $f_y(a, b) = 0$. Geometrically, the tangent plane is horizontal.

ADDED FOR CLARITY T009 00:06:19

Why does $rt - s^2 < 0$ mean a saddle?
Near a stationary point $f(a + h, b + k) - f(a, b) \approx \frac{1}{2}(rh^2 + 2shk + tk^2)$ (Taylor, class 007). This quadratic keeps one sign for all h, k exactly when $rt - s^2 > 0$ (sign of r); when $rt - s^2 < 0$ it is positive in some directions and negative in others.

ADDED FOR CLARITY S009·6 · S009·7

Why is r enough to separate maximum from minimum when $rt - s^2 > 0$?
$rt > s^2 \geq 0$, so r and t have the same sign: the surface curves the same way along x and along y.

ADDED FOR CLARITY S009·8

Exam trap. $rt - s^2 < 0$ is a saddle point (a definite answer); only $rt - s^2 = 0$ is "no conclusion". The lecture mixes the two at one point. S009·7 · T009 00:08:23 · T009 00:11:06

Exam trap. Do not decide maximum or minimum from r before checking $rt - s^2 > 0$: $x^2 - 3xy + y^2$ has $r = 2 > 0$ at the origin but $rt - s^2 = -5$: a saddle. ADDED FOR CLARITY S009·8

Exam trap. $f_x = f_y = 0$ is necessary, $rt - s^2 > 0$ sufficient; slide 009·7 calls the pair "necessary and sufficient". ADDED FOR CLARITY S009·7

Exam trap. Evaluate r, s, t at each stationary point; a sign that holds at one point need not hold at another. S009·8

Worked examples

W38.1 · A bowl: minimum ESA ★

GIVEN $f(x,y) = x^2 + y^2$.

STEP 1 $f_x = 2x = 0$, $f_y = 2y = 0$: stationary point $(0,0)$.

STEP 2 $r = 2$, $s = 0$, $t = 2$; $rt - s^2 = 4 > 0$ and $r > 0$.

ANSWER Minimum at $(0,0)$; minimum value 0

CHECK $f \geq 0$ everywhere, so the answer agrees with the definition.

ADDED FOR CLARITY S009·4 · S009·8

W38.2 · A saddle ESA ★

GIVEN $f(x,y) = x^2 - y^2$.

STEP 1 $f_x = 2x = 0$, $f_y = -2y = 0$: $(0,0)$.

STEP 2 $r = 2$, $s = 0$, $t = -2$; $rt - s^2 = -4 < 0$.

ANSWER $(0,0)$ is a saddle point

CHECK Along the x-axis $f = x^2 \geq 0$, along the y-axis $f = -y^2 \leq 0$: Δ changes sign.

ADDED FOR CLARITY S009·5 · S009·9

W38.3 · Doubtful case resolved: a minimum ESA ★

GIVEN $f(x,y) = x^4 + y^4$.

STEP 1 $f_x = 4x^3$, $f_y = 4y^3$: stationary point $(0,0)$.

STEP 2 $r = 12x^2 = 0$, $s = 0$, $t = 12y^2 = 0$: $rt - s^2 = 0$, doubtful.

STEP 3 Definition: $\Delta = f(h,k) - f(0,0) = h^4 + k^4 > 0$ for all small $(h,k) \neq (0,0)$.

ANSWER Minimum at $(0,0)$ (value 0)

CHECK The test gave no information; the sign of Δ settles it.

ADDED FOR CLARITY S009·9 · S009·6

W38.4 · Doubtful case resolved: neither ESA ★

GIVEN $f(x,y) = x^4 - y^4$.

STEP 1 Stationary point $(0,0)$; again $r = s = t = 0$, so $rt - s^2 = 0$.

STEP 2 $\Delta = h^4 - k^4$: positive for $k = 0$, negative for $h = 0$.

ANSWER Neither a maximum nor a minimum at $(0,0)$

CHECK Compare with W38.3: $rt - s^2 = 0$ can hide either answer.

ADDED FOR CLARITY S009·9 · S009·6

W38.5 · The working rule on a quadratic ESA ★★

GIVEN Find the maximum and minimum values of $f(x,y) = 4x^2 + 2y^2 + 4xy - 10x - 2y - 3$.

STEP 1 $f_x = 8x + 4y - 10 = 0$, $f_y = 4y + 4x - 2 = 0$.

STEP 2 Subtract: $4x - 8 = 0$, so $x = 2$, $y = -\frac{3}{2}$: one stationary point $(2, -\frac{3}{2})$.

STEP 3 $r = 8$, $s = 4$, $t = 4$: $rt - s^2 = 32 - 16 = 16 > 0$, $r > 0$.

STEP 4 $f(2, -\frac{3}{2}) = 16 + \frac{9}{2} - 12 - 20 + 3 - 3 = -\frac{23}{2}$.

ANSWER Minimum value $-\dfrac{23}{2}$ at $\left(2, -\frac{3}{2}\right)$; no maximum

CHECK CW answer agrees; sympy agrees.

CW U1 P31

SHORT-NOTE RECAP

Solve $f_x = 0$, $f_y = 0$. At each point find $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$. $rt - s^2 > 0$: $r < 0$ max, $r > 0$ min; $rt - s^2 < 0$: saddle; $rt - s^2 = 0$: doubtful (use Δ). Substitute the point in f for the value.

M9

Class 010 · Problems on Maxima and Minima

39 The method, and the ESA cubic functions

Every problem follows one routine: find f_x and f_y, solve $f_x = f_y = 0$ for the stationary points (factorise one equation and split into cases), compute $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$, and classify each point by $rt - s^2$ and r. Both ESA functions are cubics with four stationary points: one maximum, one minimum and two saddle points. Always finish by substituting each extremum into f for the value.

Key terms and definitions

Stationary (critical) points. The solutions of $f_x = 0$, $f_y = 0$, solved simultaneously; the points at which the function may have an extreme value. S010·7 · T010a 00:04:13 · T010b 00:01:23

Maximum / minimum value. The value $f(a, b)$ at a point of maximum or minimum, obtained by substituting the point into f. S010·8 · T010b 00:04:02

> *"If $rt - s^2 > 0$ and $r < 0$ at (a, b), $f(a, b)$ is a maximum value; if $rt - s^2 > 0$ and $r > 0$, a minimum value; if $rt - s^2 < 0$, $f(a, b)$ is not an extreme value, i.e. (a, b) is a saddle point; if $rt - s^2 = 0$ the case is doubtful and needs further investigation (class 009, restated)."* S010·8 · T010b 00:05:00

Formulas

E39.1 · Stationary points ESA ★★★

$$f_x = 0, \qquad f_y = 0 \quad \text{(solve simultaneously)}$$

gives	All candidate points.
applies	Every problem, first step.
note	Factorise: $6xy - 30y = 6y(x - 5) = 0$ gives the cases $y = 0$ or $x = 5$.

S010·7 · T010b 00:01:23 · T010b 00:01:40

E39.2 · The test

ESA ★★★

$$r = f_{xx}, \quad s = f_{xy}, \quad t = f_{yy}$$

$$rt - s^2 > 0 :\ r < 0 \text{ max},\ r > 0 \text{ min}; \qquad rt - s^2 < 0 :\ \text{saddle}; \qquad rt - s^2 = 0 :\ \text{doubtful}$$

GIVES The nature of each stationary point.

APPLIES At each point separately.

NOTE Restated from class 009.

S010·8 · T010b 00:03:12 · T010b 00:04:02

E39.3 · Hessian of the ESA cubic

ESA ★★★

$$f = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x : \qquad rt - s^2 = 36\left[(x-5)^2 - y^2\right]$$

GIVES $rt - s^2$ at every point at once.

APPLIES Dec-23 1c, Jan-24 s1 1a.

NOTE $r = t = 6x - 30$, $s = 6y$. Slide 010·8 drops the 36 on y^2.

S010·8 · T010b 00:03:30

Graphs and diagrams

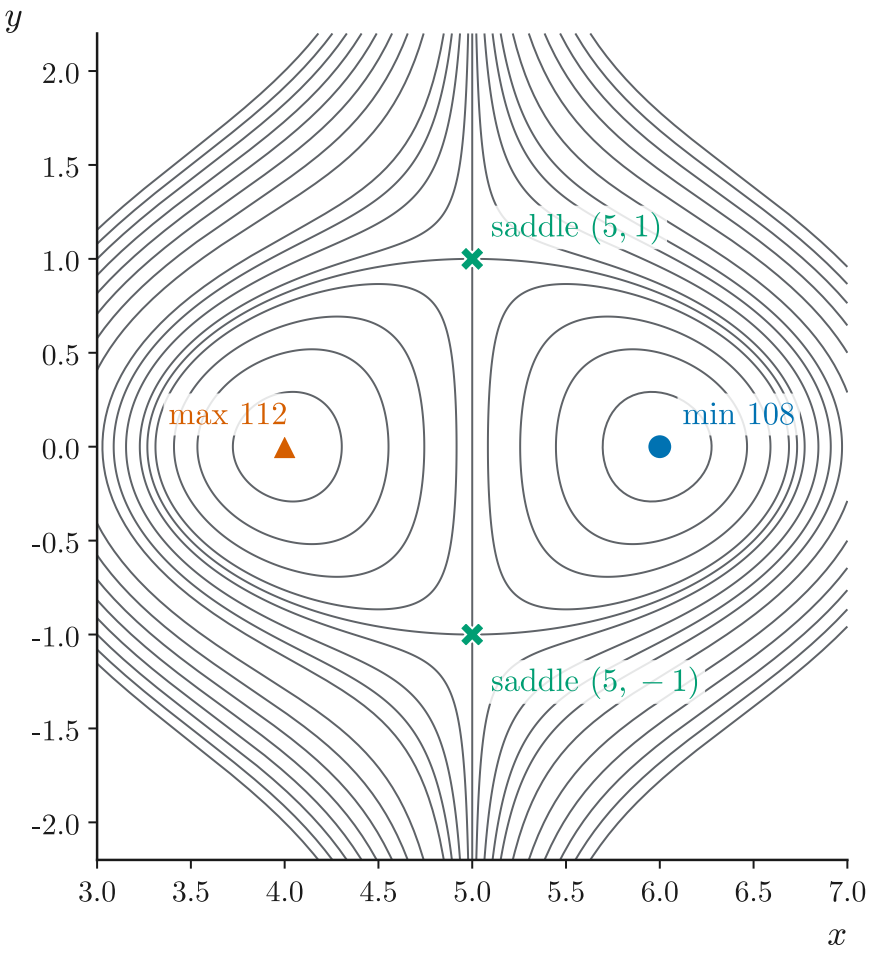


Figure G39.1. Level curves of $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$. ESA ★★★

ADDED FOR CLARITY S010·8 · ESA Dec-23 1c · ESA Jan-24 s1 1a

How to read it. Grey lines are level curves $f =$ constant. Around the maximum $(4, 0)$ and the minimum $(6, 0)$ they close into small loops; at the saddle points $(5, \pm 1)$ two level curves cross, the sign of the picture of a saddle.

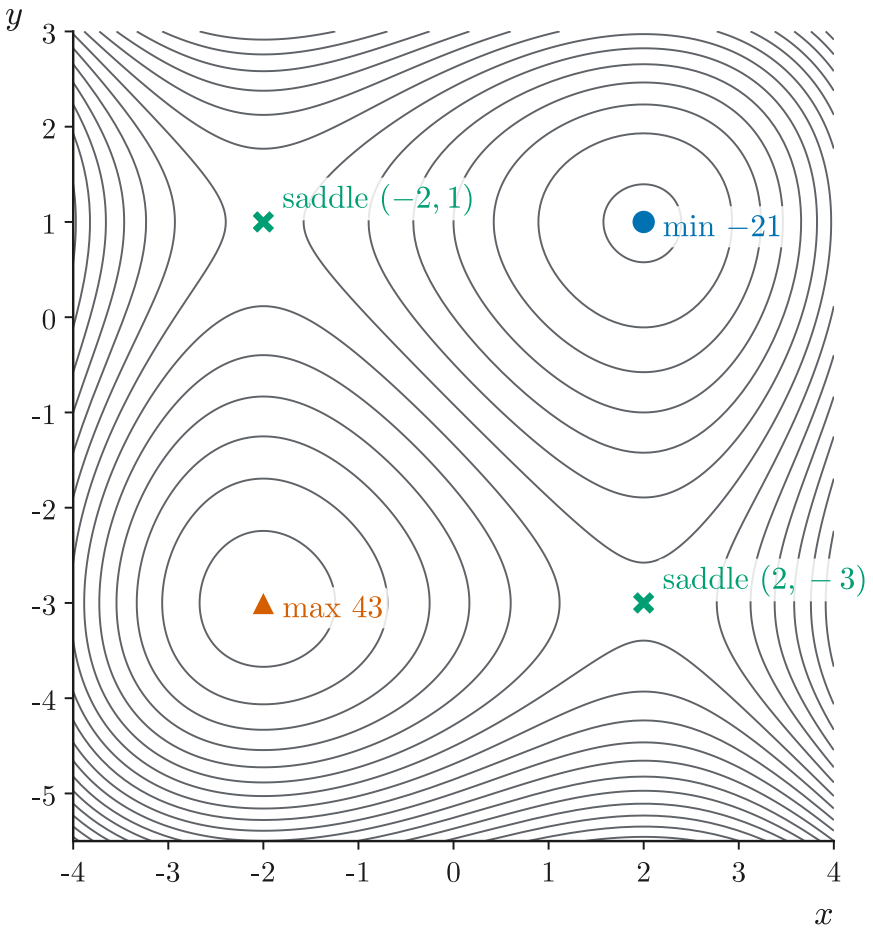
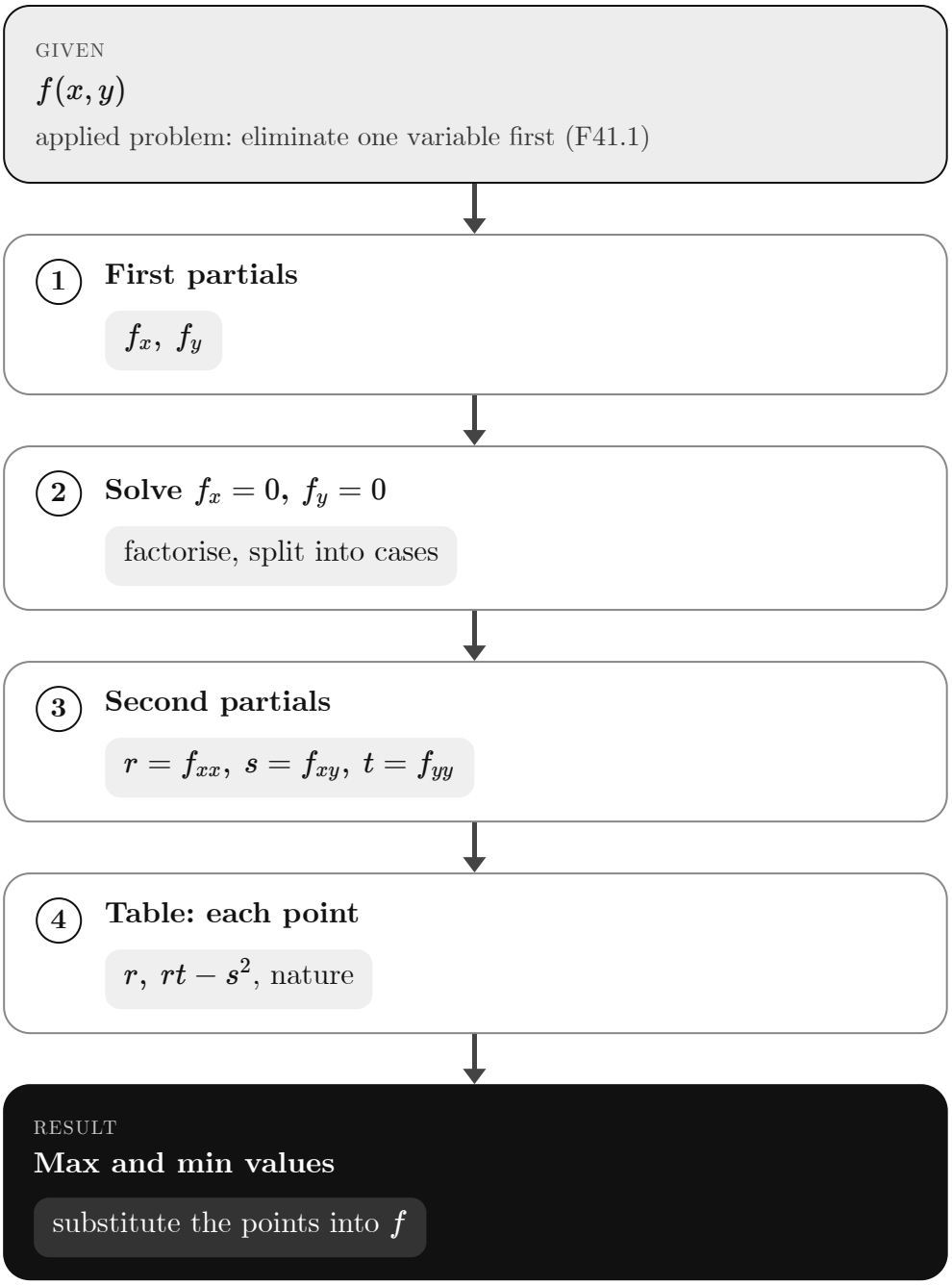


Figure G39.2. Level curves of $x^3 - 12x + y^3 + 3y^2 - 9y$. ESA ★★★

ADDED FOR CLARITY ESA Jan-24 s2 1a · CW U1 P29

How to read it. The four stationary points sit at the corners of a rectangle, $x = \pm 2$, $y = 1, -3$. Closed loops surround the maximum $(-2, -3)$ and the minimum $(2, 1)$; curves cross at the saddles $(2, -3)$ and $(-2, 1)$.

Flowcharts



Remark. Write a table with one row per stationary point: point, r, s, t, $rt - s^2$, nature, value.

Figure F39.1. Solving a maxima–minima problem ESA ★★★

S010·7 · S010·8 · T010b 00:05:00

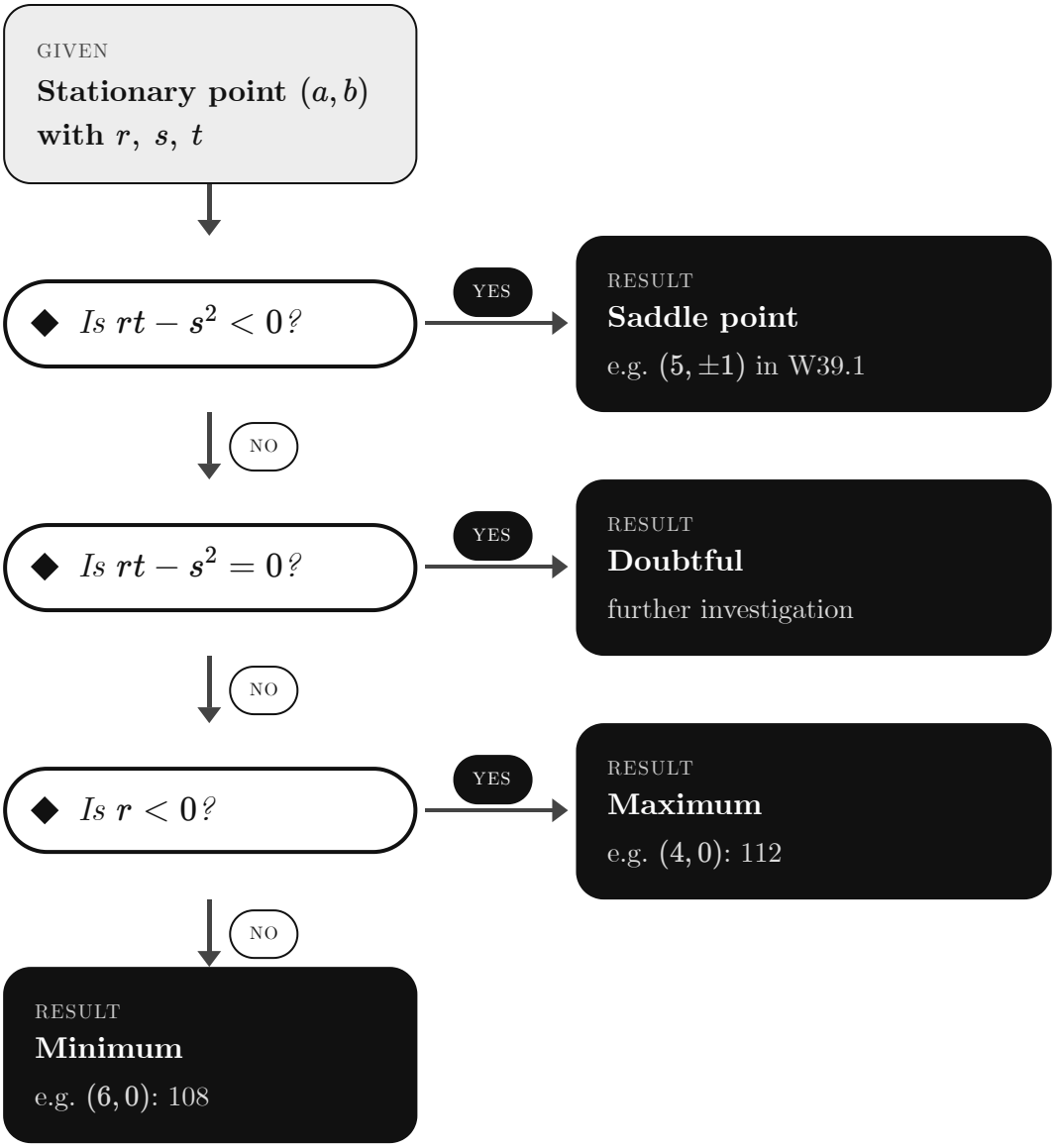


Figure F39.2. Classifying each stationary point ESA ★★★

S010·8 · T010b 00:04:02

Stationary points of the two ESA functions

Function	Point	r	$rt - s^2$	Nature	f
$x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$	$(4, 0)$	-6	36	Maximum	112
	$(6, 0)$	6	36	Minimum	108
	$(5, \pm 1)$	0	-36	Saddle	—
$x^3 - 12x + y^3 + 3y^2 - 9y$	$(-2, -3)$	-12	144	Maximum	43
	$(2, 1)$	12	144	Minimum	-21
	$(2, -3)$, $(-2, 1)$	± 12	-144	Saddle	—

S010·8 · ESA Dec-23 1c · ESA Jan-24 s2 1a

Important points and why

1. Procedure (lecture): find $p = f_x$, $q = f_y$; equate both to zero; find the critical points; compute r, s, t; evaluate $rt - s^2$ at each critical point; conclude. ESA ★★★ T010b 00:05:00 · T010b 00:05:16
2. Split $f_y = 0$ into cases and substitute each into $f_x = 0$: for the ESA cubic, $y = 0$ gives $x = 4, 6$ and $x = 5$ gives $y = \pm 1$. ESA ★★★ S010·7 · S010·8 · T010b 00:01:40
3. Count the stationary points before classifying; a cubic in x and y typically gives four, a quartic up to nine. ESA ★★ added for clarity S010·8 · S010·11
4. A saddle point has no value to report; give the maximum and minimum values only. ESA ★★ S010·8
5. When $rt - s^2 > 0$, r and t have the same sign, so either decides between maximum and minimum. ESA ★ added for clarity S010·8

Why does $f_y = 6y(x - 5) = 0$ give two separate cases?
A product is zero when either factor is zero: $y = 0$ or $x = 5$. Each case is substituted into $f_x = 0$ to find its own points; missing a case loses stationary points.
added for clarity S010·7 · T010b 00:01:40

Exam trap. Slide 010·8 says the maximum value is at $(6, 0)$; the maximum 112 is at $(4, 0)$, the minimum 108 at $(6, 0)$. S010·8 · T010b 00:04:42

Exam trap. $rt - s^2 = 36(x - 5)^2 - 36y^2$: the 36 multiplies y^2 too (slide 010·8 drops it).
added for clarity S010·8

Worked examples

W39.1 · Extreme values of $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$ ESA ★★★

given Find the extreme values (locate the stationary points and discuss their nature) of $f(x,y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.

step 1 $f_x = 3x^2 + 3y^2 - 30x + 72$, $f_y = 6xy - 30y = 6y(x-5)$.

step 2 $f_y = 0$: $y = 0$ or $x = 5$. If $y = 0$: $3x^2 - 30x + 72 = 0$, $x^2 - 10x + 24 = 0$, $x = 4$ or 6. If $x = 5$: $75 + 3y^2 - 150 + 72 = 0$, $y^2 = 1$, $y = \pm 1$.

step 3 Stationary points: $(4,0)$, $(6,0)$, $(5,1)$, $(5,-1)$.

step 4 $r = 6x - 30$, $s = 6y$, $t = 6x - 30$, so $rt - s^2 = 36[(x-5)^2 - y^2]$.

step 5 $(4,0)$: $r = -6$, $rt - s^2 = 36 > 0$: maximum, $f = 64 - 240 + 288 = 112$.

step 6 $(6,0)$: $r = 6$, $rt - s^2 = 36 > 0$: minimum, $f = 216 - 540 + 432 = 108$.

step 7 $(5, \pm 1)$: $r = 0$, $s = \pm 6$, $rt - s^2 = -36 < 0$: saddle points.

answer Maximum 112 at $(4,0)$; minimum 108 at $(6,0)$; $(5,1)$ and $(5,-1)$ are saddle points

check CW P32 answer agrees; sympy finds the same four points and values. Slide 010·8's "maximum ... at $(6,0)$" is a slip for $(4,0)$.

S010·7 · S010·8 · T010b 00:00:23 · ESA Dec-23 1c · ESA Jan-24 s1 1a · CW U1 P32

W39.2 · Stationary points of $x^3 - 12x + y^3 + 3y^2 - 9y$ ESA ★★★

given Locate the stationary points of $f(x,y) = x^3 - 12x + y^3 + 3y^2 - 9y$ and discuss their nature.

step 1 $f_x = 3x^2 - 12 = 0 \Rightarrow x = \pm 2$; $f_y = 3y^2 + 6y - 9 = 3(y+3)(y-1) = 0 \Rightarrow y = 1, -3$.

step 2 Stationary points: $(2,1)$, $(2,-3)$, $(-2,1)$, $(-2,-3)$.

step 3 $r = 6x$, $s = 0$, $t = 6y + 6$; $rt - s^2 = 36x(y+1)$.

step 4 $(2,1)$: $r = 12$, $rt - s^2 = 144 > 0$: minimum, $f = 8 - 24 + 1 + 3 - 9 = -21$.

step 5 $(-2,-3)$: $r = -12$, $rt - s^2 = 144 > 0$: maximum, $f = -8 + 24 - 27 + 27 + 27 = 43$.

step 6 $(2,-3)$ and $(-2,1)$: $rt - s^2 = -144 < 0$: saddle points.

answer Maximum 43 at $(-2,-3)$; minimum -21 at $(2,1)$; $(2,-3)$ and $(-2,1)$ are saddle points

check CW P29 answer agrees; sympy agrees.

ESA Jan-24 s2 1a · CW U1 P29

Past-paper questions

ESA Dec 2023 (7 marks) UE22MA141A Q1.c · solved in W39.1

Find the extreme values of the function $f(x,y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.

ESA Jan 2024 (7 marks) UE23MA141A set 1, Q1.a · solved in W39.1

Locate the stationary points of $f(x,y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$. Discuss their nature.

ESA Jan 2024 (7 marks) UE23MA141A set 2, Q1.a · solved in W39.2

Locate the stationary points of $f(x,y) = x^3 - 12x + y^3 + 3y^2 - 9y$. Discuss their nature.

SHORT-NOTE RECAP

$f_x = 0$, $f_y = 0$ → split into cases → stationary points → r, s, t → $rt - s^2$ and r → values. ESA: $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$: max 112 at $(4,0)$, min 108 at $(6,0)$, saddles $(5, \pm 1)$. $x^3 - 12x + y^3 + 3y^2 - 9y$: max 43 at $(-2,-3)$, min -21 at $(2,1)$, saddles $(2,-3)$, $(-2,1)$.

40 Functions with many stationary points

The slides test two more functions. $x^4 + y^4 - x^2 - y^2 + 1$ has nine stationary points (a maximum at the origin, four minima and four saddle points). $\sin x + \sin y + \sin(x + y)$ needs trigonometric equations: subtracting the two conditions gives $x = y$, then $\cos 2x = -\cos x$ gives $x = \frac{\pi}{3}$, a maximum $\frac{3\sqrt{3}}{2}$.

Graphs and diagrams

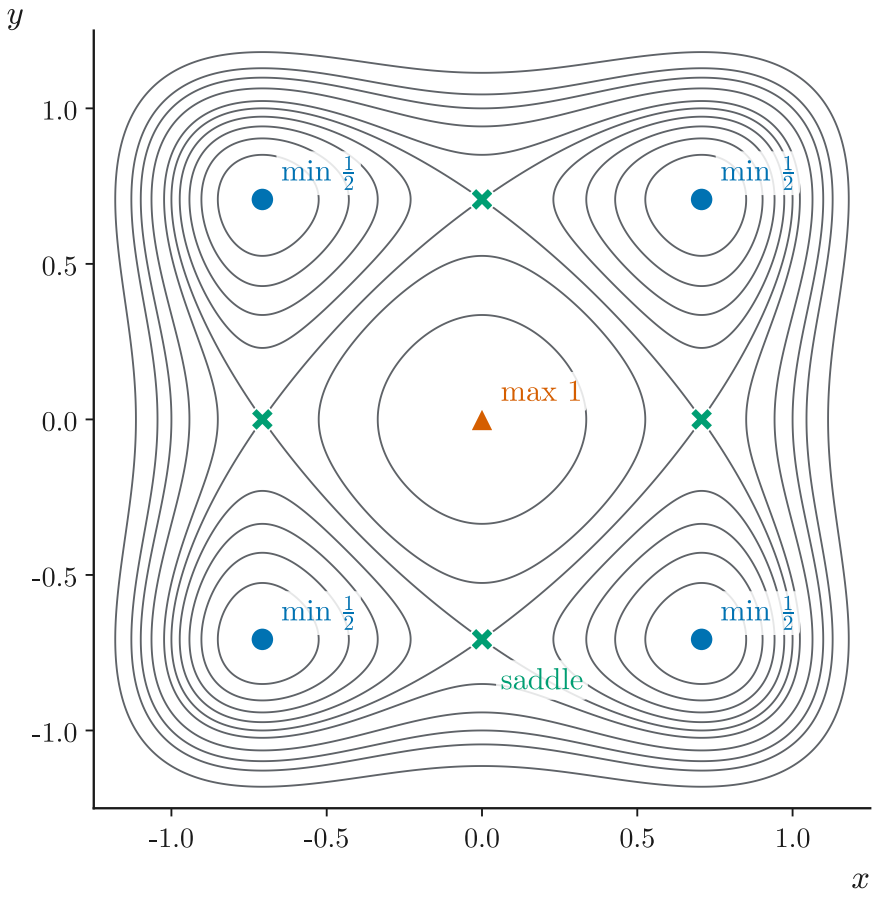


Figure G40.1. Level curves of $x^4 + y^4 - x^2 - y^2 + 1$. ESA ★★

ADDED FOR CLARITY S010·11 · S010·12

How to read it. Nine stationary points on a 3×3 grid: a maximum at the centre, minima at the four corners $(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}})$ and saddles at the four edge midpoints on the axes.

Important points and why

1. Separable functions such as $x^4 + y^4 - x^2 - y^2 + 1$ give f_x in x only and f_y in y only: combine every root of one with every root of the other. ESA ★★ S010·11 · T010b 00:08:29
2. For $\sin x + \sin y + \sin(x + y)$, subtract the two conditions to get $\cos x = \cos y$, i.e. $x = y$ in the working range; then $\cos 2x = -\cos x = \cos(\pi - x)$ gives $x = \frac{\pi}{3}$. ESA ★★ S010·13 · T010b 00:10:23

Exam trap. When $rt - s^2 < 0$ the sign of r is irrelevant; at $(\pm\frac{1}{\sqrt{2}}, 0)$, $r = 4$ (not -2 as slide 010·12 says), and the point is still a saddle. ADDED FOR CLARITY S010·12 · T010b 00:09:26

Exam trap. At $(\frac{\pi}{3}, \frac{\pi}{3})$, $s = -\frac{\sqrt{3}}{2}$, so $rt - s^2 = 3 - \frac{3}{4} = \frac{9}{4}$ (slide 010·14 prints $\frac{3}{4}$).
ADDED FOR CLARITY S010·14

Worked examples

W40.1 · $x^4 + y^4 - x^2 - y^2 + 1$: nine stationary points — ESA ★★

GIVEN Test $f(x, y) = x^4 + y^4 - x^2 - y^2 + 1$ for maxima, minima and saddle points.

STEP 1 $f_x = 4x^3 - 2x = 2x(2x^2 - 1) = 0 \Rightarrow x = 0, \pm\frac{1}{\sqrt{2}}$; likewise $y = 0, \pm\frac{1}{\sqrt{2}}$: nine points.

STEP 2 $r = 12x^2 - 2$, $s = 0$, $t = 12y^2 - 2$.

STEP 3 $(0, 0)$: $r = t = -2$, $rt - s^2 = 4 > 0$, $r < 0$: maximum $f = 1$.

STEP 4 $(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}})$: $r = t = 4$, $rt - s^2 = 16 > 0$: minima, $f = \frac{1}{4} + \frac{1}{4} - \frac{1}{2} - \frac{1}{2} + 1 = \frac{1}{2}$.

STEP 5 $(\pm\frac{1}{\sqrt{2}}, 0)$: $r = 4$, $t = -2$; $(0, \pm\frac{1}{\sqrt{2}})$: $r = -2$, $t = 4$; both $rt - s^2 = -8 < 0$: saddle points.

ANSWER Maximum 1 at $(0, 0)$; minimum $\frac{1}{2}$ at the four points $(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}})$; four saddle points on the axes

CHECK sympy agrees (saddle values $\frac{3}{4}$).

S010·11 · S010·12 · T010b 00:08:12

W40.2 · $\sin x + \sin y + \sin(x + y)$ — ESA ★★

GIVEN Test $f(x, y) = \sin x + \sin y + \sin(x + y)$ for maxima, minima and saddle points ($0 < x, y < \pi$).

STEP 1 $f_x = \cos x + \cos(x + y) = 0$ (1), $f_y = \cos y + \cos(x + y) = 0$ (2).

STEP 2 (1) − (2): $\cos x = \cos y$, so $x = y$. Then (1): $\cos 2x = -\cos x = \cos(\pi - x)$, $2x = \pi - x$, $x = \frac{\pi}{3}$.

STEP 3 $r = -\sin x - \sin(x + y)$, $s = -\sin(x + y)$, $t = -\sin y - \sin(x + y)$.

STEP 4 At $(\frac{\pi}{3}, \frac{\pi}{3})$: $r = t = -\sqrt{3}$, $s = -\frac{\sqrt{3}}{2}$, $rt - s^2 = 3 - \frac{3}{4} = \frac{9}{4} > 0$, $r < 0$: maximum.

STEP 5 $f = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$.

ANSWER Maximum $\dfrac{3\sqrt{3}}{2}$ at $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$

CHECK sympy: $rt - s^2 = \frac{9}{4}$ (slide 010·14 prints $\frac{3}{4}$); the conclusion is the same.

S010·13 · S010·14 · T010b 00:09:50

SHORT-NOTE RECAP

$x^4+y^4-x^2-y^2+1$: max 1 at $(0,0)$, min $\frac{1}{2}$ at $(\pm\frac{1}{\sqrt{2}},\pm\frac{1}{\sqrt{2}})$, four saddles on the axes. $\sin x+\sin y+\sin(x+y)$: $x=y=\frac{\pi}{3}$, max $\frac{3\sqrt{3}}{2}$.

41 Applied problems: eliminate one variable first

When the quantity depends on three variables tied by a condition (a sphere, a fixed volume), use the condition to eliminate one variable, then apply the two-variable method. The temperature $400xyz^2$ on the unit sphere is highest, 50, at $(\pm\frac{1}{2},\pm\frac{1}{2})$ with equal signs. An open box of volume 10 m³ uses least metal when the base is $\sqrt[3]{20}$ square and the height half of that. Class 011 does such problems without eliminating.

Key terms and definitions

Elimination. Using the given condition (constraint) to express one variable in terms of the others, so that the quantity becomes a function of two independent variables. T010a 00:00:40 · T010a 00:08:27

Formulas

E41.1 · Eliminating with a sphere ESA ★★

$$x^2+y^2+z^2=a^2 \;\Rightarrow\; z^2=a^2-x^2-y^2, \qquad T=kxyz^2=kxy(a^2-x^2-y^2)$$

gives	A function of x, y only.
applies	Quantity contains z^2 and the point lies on a sphere.
note	Unit sphere: $a=1$.

S010·3 · S010·9 · T010a 00:01:00

E41.2 · Open box of volume V ESA ★★

$$A=xy+2yz+2zx, \quad xyz=V \;\Rightarrow\; A=xy+\frac{2V}{x}+\frac{2V}{y}$$

gives	Surface area (metal used) of a box with no top, base $x\times y$.
applies	Least material for a given volume.
note	Minimum at $x=y=\sqrt[3]{2V}$, $z=\frac{x}{2}$ (CW P34). The slide's $A=2xy+2yz+zx$ is not the area of an open box.

added for clarity S010·5 · CW U1 P34

Graphs and diagrams

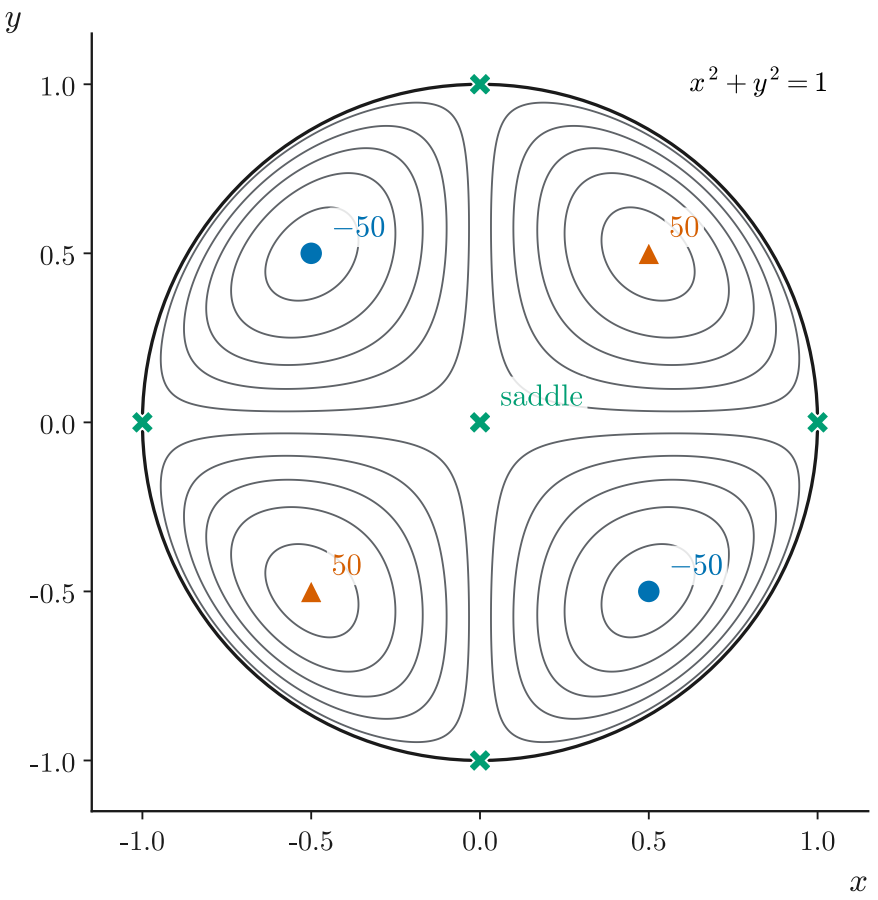


Figure G41.1. Temperature $400xy(1 - x^2 - y^2)$ over the unit disk. ESA ★★

ADDED FOR CLARITY S010·3 · S010·4

How to read it. Inside the unit circle (the projection of the sphere) the temperature is positive in the first and third quadrants and negative in the others. It peaks at 50 at $(\tfrac{1}{2}, \tfrac{1}{2})$ and $(-\tfrac{1}{2}, -\tfrac{1}{2})$ and is lowest, -50, at the other two. The origin and $(\pm 1, 0)$, $(0, \pm 1)$, where $T = 0$, are saddles.

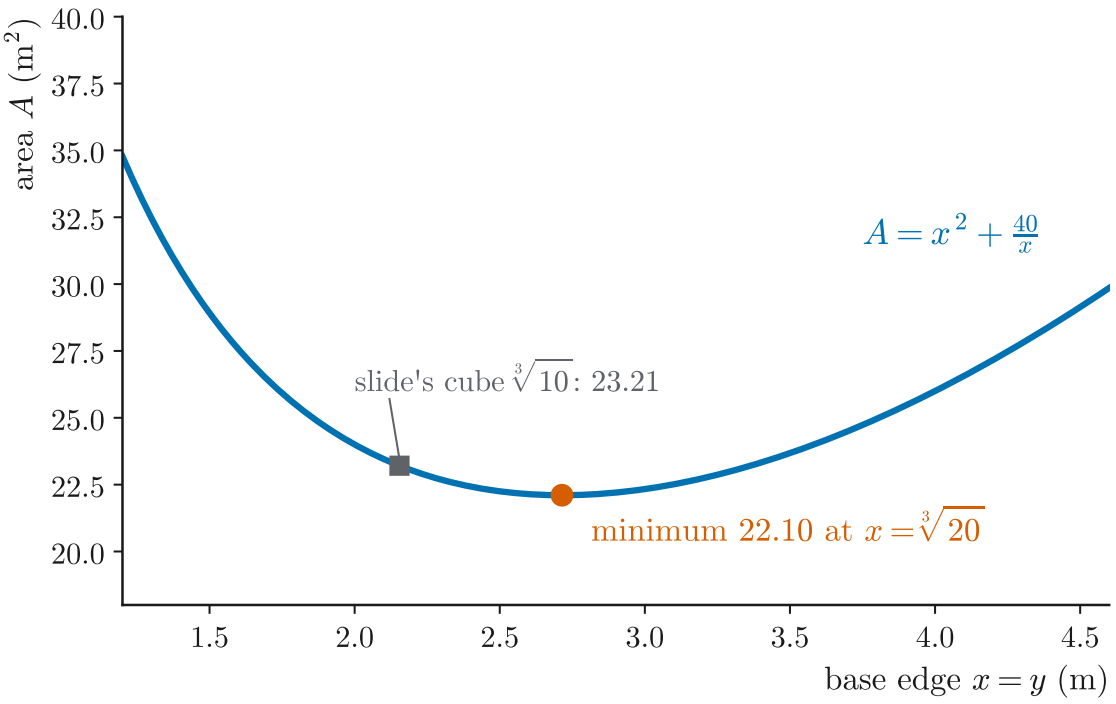
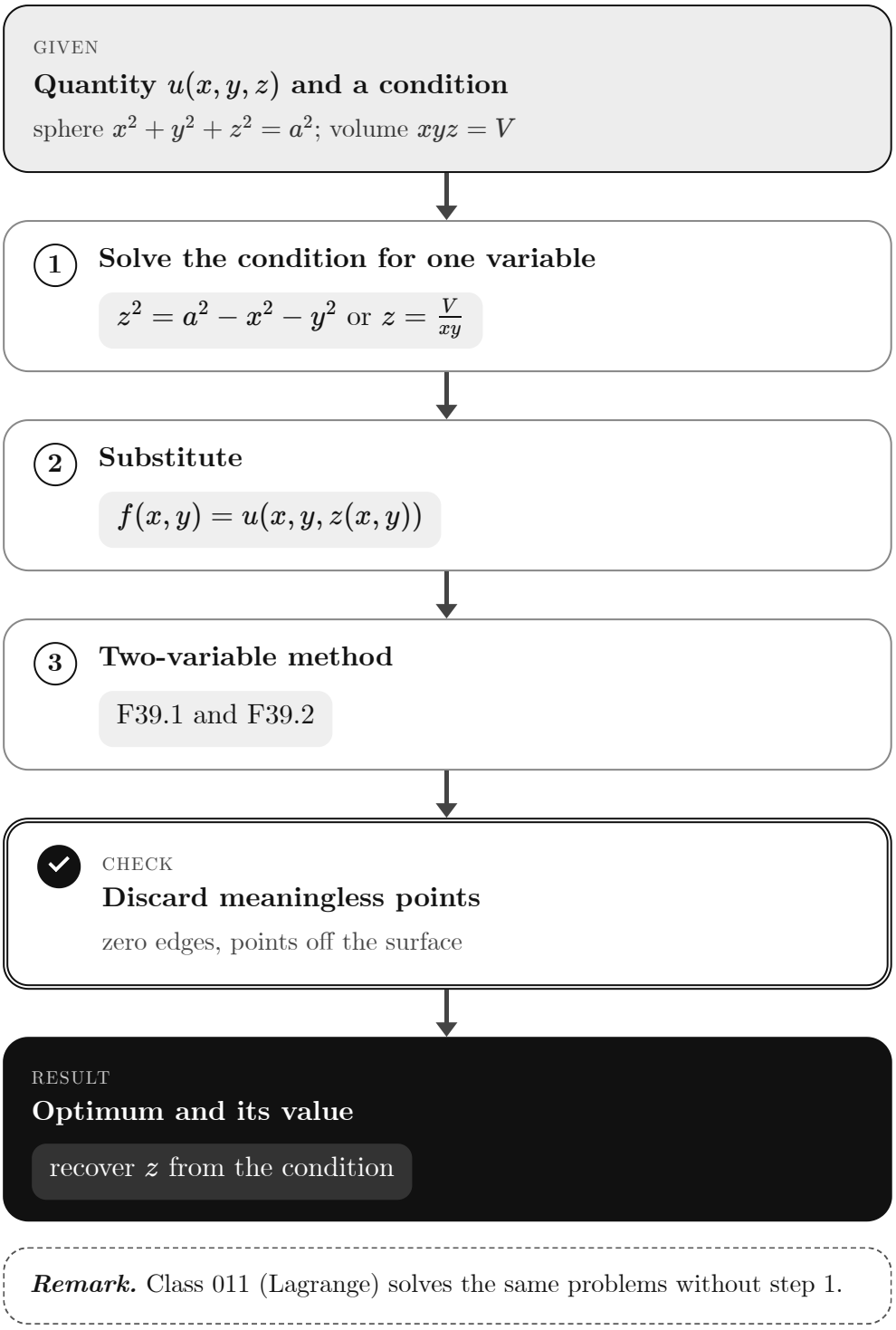


Figure G41.2. Metal used by an open box of 10 m³ with a square base. ESA ★

ADDED FOR CLARITY S010·6 · CW U1 P34

How to read it. Along $x = y$ the area is $A = x^2 + \frac{40}{x}$ (blue). Its minimum, **22.10** m², is at $x = \sqrt[3]{20} \approx 2.71$ m. The slide's cube, $x = \sqrt[3]{10} \approx 2.15$ m, sits higher on the curve (**23.21** m²).

Flowcharts



Remark. Class 011 (Lagrange) solves the same problems without step 1.

Figure F41.1. Applied problem by elimination ESA ★★

S010·3 · S010·5 · T010a 00:00:40 · T010a 00:08:27

Important points and why

1. We cannot yet find extrema of a function of three variables, so use the constraint to eliminate one (here z^2 or z). ESA ★★ T010a 00:00:40 · T010a 00:08:27

2. A product such as $kxyz^2$ is largest when x and y have the same sign and $k > 0$; points with x, y of opposite signs give the lowest temperature. ESA ★★ S010·10 · T010b 00:05:40
3. Reject stationary points that make no physical sense (an edge $y = 0$ in the box problem makes $x \to \infty$). ESA ★ T010a 00:13:07 · T010a 00:13:48
4. Elimination is not always easy; class 011's Lagrange multipliers avoid it (T010a 00:16:49). ESA ★ T010a 00:16:49

Why is a cube the wrong answer for an open box?
Without a top the base is used once and each side face twice in the area, so the box should be lower and wider: $x = y = \sqrt[3]{20}$, $z = \frac{1}{2}\sqrt[3]{20}$. The cube $x = y = z = \sqrt[3]{V}$ minimises the area of a closed box $2(xy + yz + zx)$.
ADDED FOR CLARITY S010·6 · CW U1 P34

Exam trap. $400xy(1 - x^2 - y^2)$ also has stationary points $(\pm 1, 0)$, $(0, \pm 1)$ (saddles) that the slides omit; and slide 010·4's $rt - s^2$ values are -160000 and 320000, not -1600 and 356000.
ADDED FOR CLARITY S010·3 · S010·4

Exam trap. Open box: area $xy + 2yz + 2zx$ (one base, four sides). The slide's $2xy + 2yz + zx$ and its derivative slip lead to a cube, which is wrong. ADDED FOR CLARITY S010·5 · S010·6 · T010a 00:07:38

Worked examples

W41.1 · Highest temperature $400xyz^2$ on the unit sphere — ESA ★★★

GIVEN The temperature at (x, y, z) is $T = 400xyz^2$. Find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$.

STEP 1 Eliminate $z^2 = 1 - x^2 - y^2$: $f(x, y) = 400xy(1 - x^2 - y^2) = 400(xy - x^3y - xy^3)$.

STEP 2 $f_x = 400y(1 - 3x^2 - y^2) = 0$, $f_y = 400x(1 - x^2 - 3y^2) = 0$.

STEP 3 Both brackets zero: subtracting gives $x^2 = y^2$, then $4x^2 = 1$: $(\pm\frac{1}{2}, \pm\frac{1}{2})$. Also $(0, 0)$, $(\pm 1, 0)$, $(0, \pm 1)$.

STEP 4 $r = -2400xy$, $s = 400(1 - 3x^2 - 3y^2)$, $t = -2400xy$.

STEP 5 $(\frac{1}{2}, \frac{1}{2})$, $(-\frac{1}{2}, -\frac{1}{2})$: $r = -600$, $s = -200$, $rt - s^2 = 320000 > 0$, $r < 0$: maxima, $T = 400 \cdot \frac{1}{4} \cdot \frac{1}{2} = 50$.

STEP 6 $(0, 0)$: $rt - s^2 = -160000 < 0$; $(\pm 1, 0)$, $(0, \pm 1)$: $-640000 < 0$: saddles. Opposite signs: minima, $T = -50$.

ANSWER Highest temperature 50, at $x = y = \pm\frac{1}{2}$, $z = \pm\frac{1}{\sqrt{2}}$

CHECK sympy agrees. ESA June 2024 Q1.d asks this by Lagrange's method (class 011 pack) and gets the same 50.

S010·3 · S010·4 · T010a 00:00:00 · T010a 00:06:33 · ESA Jun-24 1d

W41.2 · Highest temperature $kxyz^2$ on a sphere of radius a ESA ★★

GIVEN $T = kxyz^2$ ($k > 0$). Find the highest temperature on $x^2 + y^2 + z^2 = a^2$.

STEP 1 $T = kxy(a^2 - x^2 - y^2)$; $T_x = ky(a^2 - 3x^2 - y^2)$, $T_y = kx(a^2 - x^2 - 3y^2)$.

STEP 2 Stationary points: $(0,0)$ and $(\pm\frac{a}{2}, \pm\frac{a}{2})$ (also $(\pm a, 0)$, $(0, \pm a)$, saddles).

STEP 3 $r = -6kxy$, $s = k(a^2 - 3x^2 - 3y^2)$, $t = -6kxy$.

STEP 4 $(0,0)$: $r = 0$, $rt - s^2 = -k^2a^4 < 0$: saddle.

STEP 5 $(\frac{a}{2}, \frac{a}{2})$, $(-\frac{a}{2}, -\frac{a}{2})$: $r = -\frac{6ka^2}{4} < 0$, $s = -\frac{ka^2}{2}$, $rt - s^2 = \frac{9k^2a^4}{4} - \frac{k^2a^4}{4} = 2k^2a^4 > 0$: maximum.

ANSWER $$\boxed{T_{\max} = k \cdot \frac{a}{2} \cdot \frac{a}{2} \cdot \frac{a^2}{2} = \frac{ka^4}{8}}$$

CHECK With $k = 400$, $a = 1$: 50, as in W41.1; sympy agrees.

S010·9 · S010·10 · T010b 00:05:25

W41.3 · Open box of 10 m³ with least metal ESA ★★

GIVEN A container with an open top is to have 10 m³ capacity and be made of thin sheet metal. Find the dimensions that use the least metal.

STEP 1 Base $x \times y$, height z: $A = xy + 2yz + 2zx$, $xyz = 10$, so $A = xy + \frac{20}{x} + \frac{20}{y}$.

STEP 2 $A_x = y - \frac{20}{x^2} = 0$, $A_y = x - \frac{20}{y^2} = 0$; then $x^2y = xy^2 = 20$, so $x = y$ and $x^3 = 20$: $x = y = \sqrt[3]{20}$.

STEP 3 $z = \frac{10}{xy} = \frac{10}{20^{2/3}} = \frac{1}{2}\sqrt[3]{20}$.

STEP 4 $A_{xx} = \frac{40}{x^3} = 2$, $A_{xy} = 1$, $A_{yy} = \frac{40}{y^3} = 2$: $rt - s^2 = 3 > 0$, $r > 0$: minimum.

ANSWER $x = y = \sqrt[3]{20} \approx 2.71$ m, height $z = \frac{1}{2}\sqrt[3]{20} \approx 1.36$ m; least area $3 \cdot 20^{2/3} \approx 22.10$ m²

CHECK Matches CW P34 ($x = y = (2V)^{1/3}$, $z = x/2$); sympy agrees. The slide's cube of side $\sqrt[3]{10}$ needs 23.21 m², more metal.

ADDED FOR CLARITY S010·5 · S010·6 · T010a 00:07:06 · CW U1 P34

Past-paper questions

ESA June 2024 7 MARKS UE23MA141A Q1.d · solved in W41.1

The temperature T at any point (x, y, z) in space is $400xyz^2$. Using Lagrange's method of multipliers, find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$. (Asked by Lagrange's method; W41.1 checks the answer by elimination.)

SHORT-NOTE RECAP

Eliminate with the condition, then use the two-variable test. $400xyz^2$ on the unit sphere: max 50 at $(\pm\frac{1}{2},\pm\frac{1}{2})$ (same signs); $kxyz^2$ on radius a: $\frac{ka^4}{8}$. Open box, volume V: $A = xy + \frac{2V}{x} + \frac{2V}{y}$, $x = y = \sqrt[3]{2V}$, $z = \frac{x}{2}$.

M10

Class 011 · Lagrange's Method of Undetermined Multipliers

42 Lagrange's method: the auxiliary function and the procedure

To find the extreme values of $f(x, y, z)$ subject to a constraint $\phi(x, y, z) = 0$, introduce an unknown constant λ (the Lagrange multiplier) and form $F = f + \lambda\phi$. Solve the four equations $F_x = 0$, $F_y = 0$, $F_z = 0$, $\phi = 0$ for x, y, z, λ. No variable has to be eliminated first, and the method extends to more variables and constraints; but it gives only stationary points, not their nature.

Key terms and definitions

Constraint. A condition $\phi(x, y, z) = 0$ that the variables must satisfy (walk only on the surface of a sphere, a fixed surface area, a fixed sum). S011·3 · T011a 00:00:36

Lagrange multiplier. Suppose we need to minimize or maximize a function $f(x, y, z)$ subject to the constraint $\phi(x, y, z)$; then we can introduce an additional unknown constant λ, known as Lagrange's multiplier, to ease the process of finding the extrema. S011·3 · T011a 00:00:53

Auxiliary function. $F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z)$: the given function plus λ times its constraint (also called Lagrange's function). S011·3 · T011a 00:01:08 · T011b 00:02:06

> *"Working procedure. Step 1: Form the auxiliary equation $F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z)$ … (1). Step 2: Partially differentiate F in (1) w.r.t. x, y, z respectively. Step 3: Solve the four equations $F_x = 0$, $F_y = 0$, $F_z = 0$ and the constraint for the Lagrange multiplier λ and stationary values x, y, z."* S011·3

> *"Advantages: 1. The stationary points of $f(x, y, z)$ can be determined without determining x, y, z explicitly. 2. This method can be extended to function of several variables subject to many constraints. Disadvantages: 1. Nature of the stationary points can not be determined. 2. Equations $F_x = 0$, $F_y = 0$, $F_z = 0$ are only necessary conditions."* S011·4 · T011a 00:03:04

Formulas

E42.1 · Auxiliary (Lagrange) function ESA ★★★

$$F(x,y,z) = f(x,y,z) + \lambda\,\phi(x,y,z)$$

GIVES One function whose stationary points are the constrained stationary points.

APPLIES Optimise f subject to $\phi(x,y,z) = 0$.

NOTE Write the constraint with everything on one side: $x^2 + y^2 + z^2 - r^2$, $xy + 2yz + 2zx - 432$.

S011·3 · T011a 00:01:08

E42.2 · The four equations ESA ★★★

$$F_x = f_x + \lambda\phi_x = 0, \quad F_y = f_y + \lambda\phi_y = 0, \quad F_z = f_z + \lambda\phi_z = 0, \quad \phi = 0$$

GIVES x, y, z and λ (four unknowns, four equations).

APPLIES Always; with two variables drop F_z (three equations).

NOTE Equivalently $\nabla f = -\lambda\nabla\phi$: the gradients are parallel (added for clarity).

S011·3 · T011a 00:02:17 · T011a 00:02:30

E42.3 · Solving trick: make λ the subject ESA ★★

$$\lambda = -\frac{f_x}{\phi_x} = -\frac{f_y}{\phi_y} = -\frac{f_z}{\phi_z}$$

GIVES Relations between x, y, z without λ.

APPLIES When $\phi_x, \phi_y, \phi_z \neq 0$.

NOTE Or multiply the equations by x, y, z and subtract (slides 011·6, 011·11).

S011·13 · S011·14 · T011b 00:07:24

Graphs and diagrams

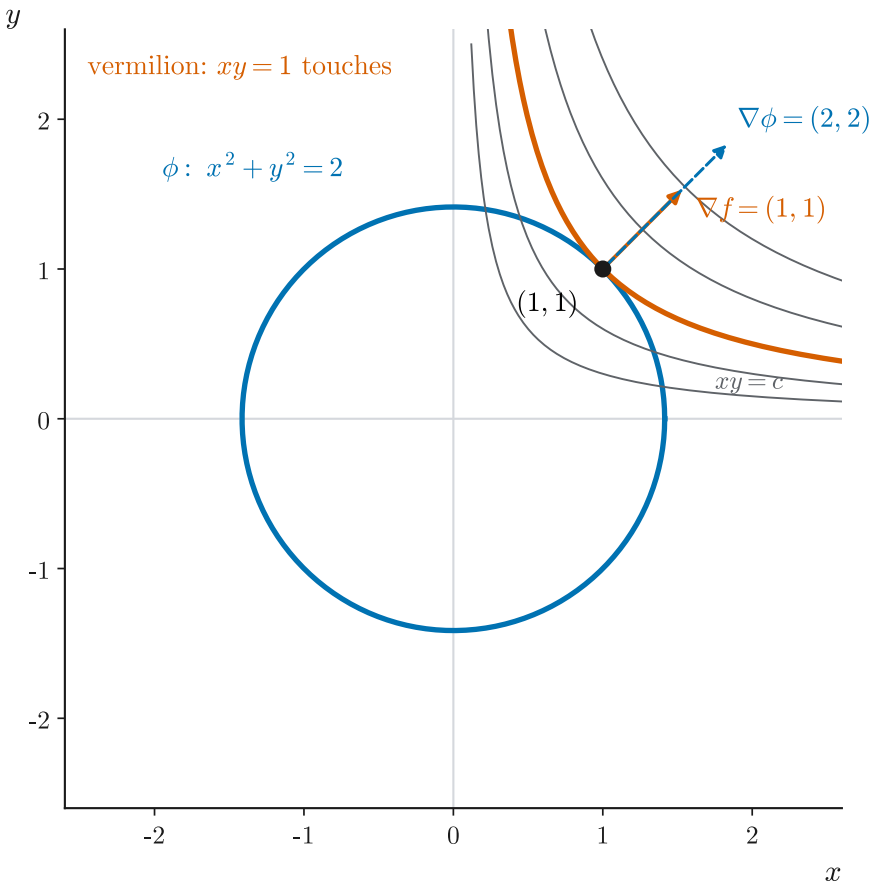
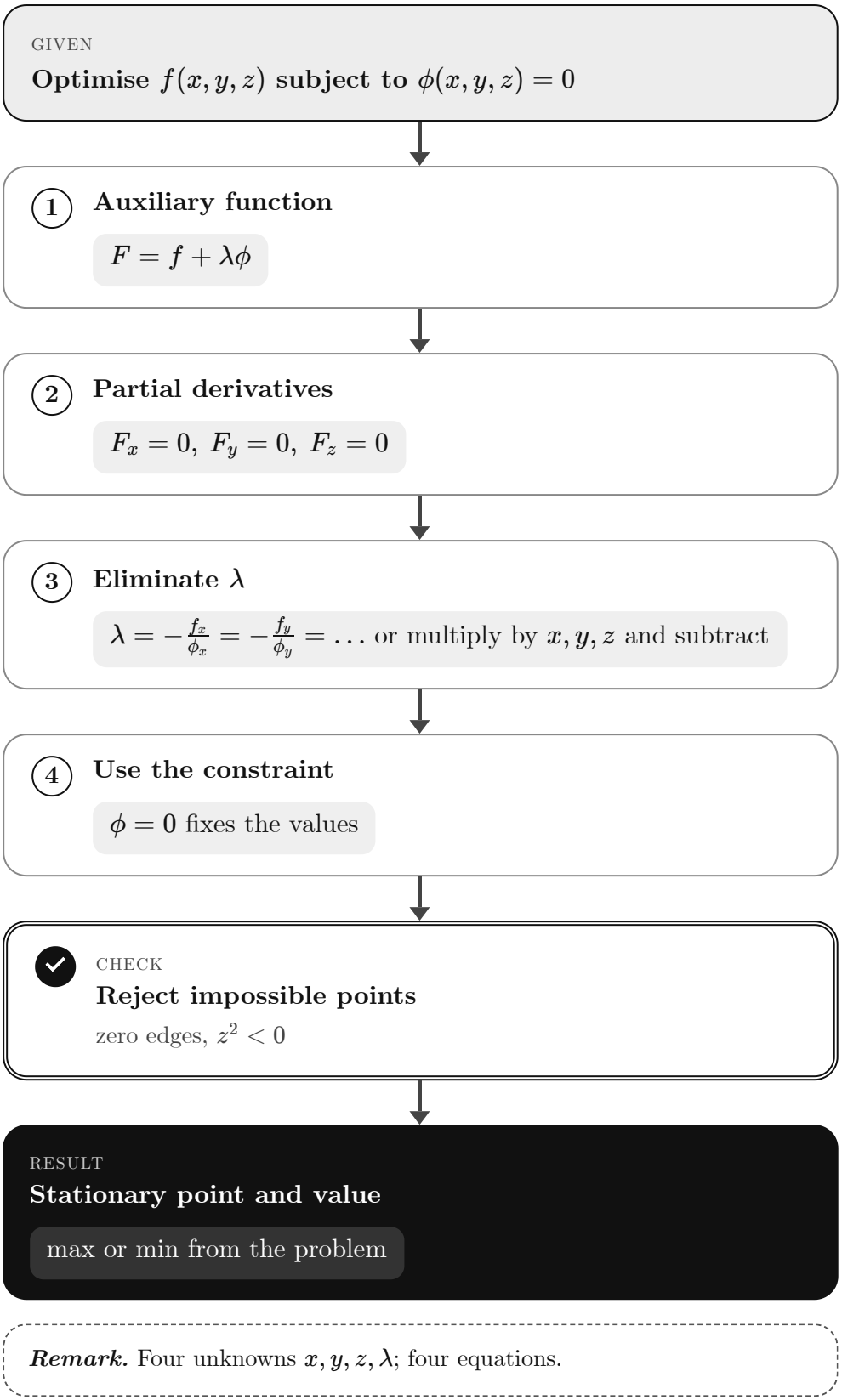


Figure G42.1. Why the method works: a level curve touches the constraint. ESA ★

ADDED FOR CLARITY S011·3 · S011·7

How to read it. Maximise $f = xy$ on the circle $x^2 + y^2 = 2$ (blue). The level curves $xy = c$ (grey hyperbolas) cross the circle until $c = 1$, where $xy = 1$ (vermilion) just touches it at $(1, 1)$. There ∇f and $\nabla\phi$ point the same way (arrows): $\nabla f + \lambda\nabla\phi = 0$ with $\lambda = -\frac{1}{2}$.

Flowcharts



Remark. Four unknowns x, y, z, λ; four equations.

Figure F42.1. Lagrange's working procedure ESA ★★★

S011·3 · T011a 00:01:18

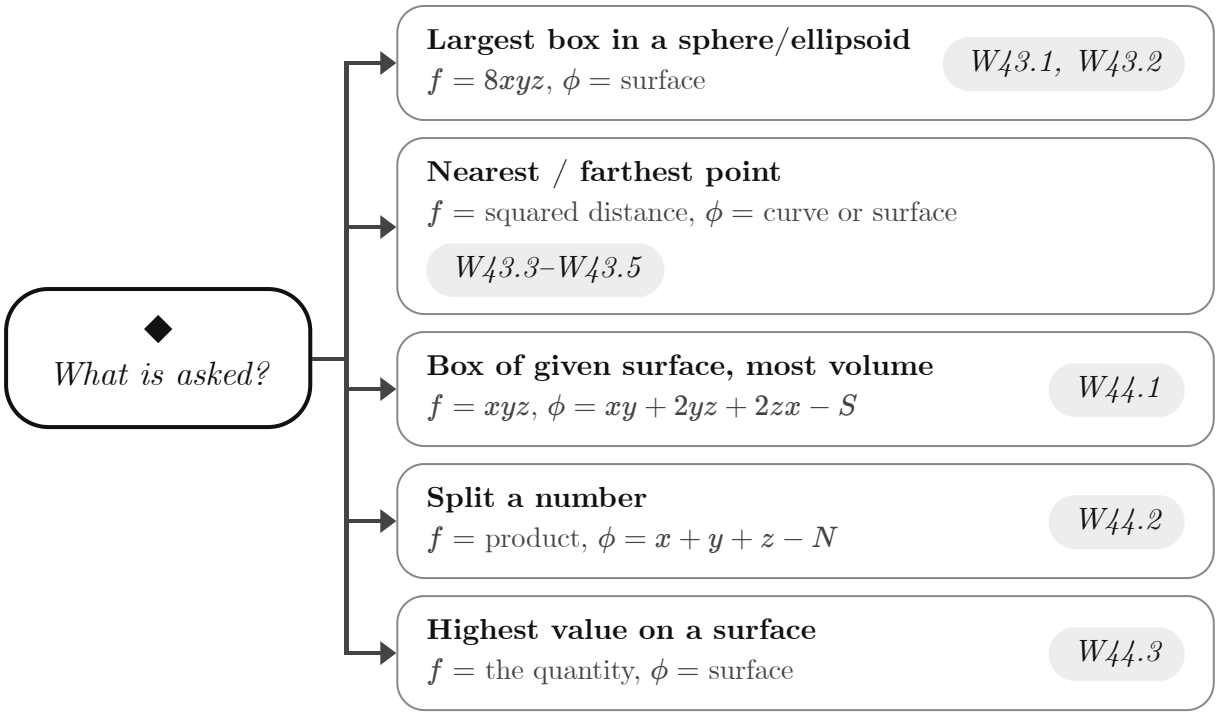


Figure F42.2. Choosing f and ϕ ESA ★★★

T011a 00:09:15 · T011b 00:01:46 · S011·10 · S011·12

Elimination (class 010) and Lagrange's method

	Elimination	Lagrange's multipliers
Constraint used	Solved for one variable and substituted	Added as $\lambda\phi$
Equations	$f_x = f_y = 0$ in two variables	$F_x = F_y = F_z = 0$, $\phi = 0$
Nature of the point	$rt - s^2$ and r decide	Not decided; from the problem
More variables / constraints	Hard	Extends directly
Temperature $400xyz^2$	50 (class 010)	50 (W44.3)

S011·4 · T011a 00:01:36 · T011a 00:04:23

Important points and why

1. The method avoids elimination: in class 010 we had to substitute $z^2 = 1 - x^2 - y^2$ for the temperature problem; with Lagrange we differentiate F directly. ESA ★★ T011a 00:01:18 · T011a 00:03:17
2. Four unknowns x, y, z, λ need four equations: $F_x = 0$, $F_y = 0$, $F_z = 0$ and the constraint $\phi = 0$. ESA ★★★ T011a 00:02:30
3. λ is "undetermined": we may find it, but it does not appear in the answer. ESA ★ T011a 00:02:04

4. The method gives only stationary points; to verify maximum or minimum by $rt - s^2$ we would have to go back to the previous method. In problems the question itself says which (maximum volume, closest point). ESA ★★ S011·4 · T011a 00:04:05 · T011a 00:14:01
5. The most important step is to identify the function f and the constraint ϕ correctly (Figure F42.2). ESA ★★★ T011a 00:09:15 · T011b 00:01:46
6. Geometrically, at a constrained extremum a level curve (surface) of f just touches the constraint, so ∇f and $\nabla \phi$ are parallel: $\nabla f = -\lambda \nabla \phi$ (Figure G42.1). ESA ★ ADDED FOR CLARITY S011·7

Why does $F_x = F_y = F_z = 0$ find the constrained extrema?
At a constrained extremum, f cannot increase along any direction that stays on the surface $\phi = 0$, so ∇f is perpendicular to the surface, i.e. parallel to $\nabla \phi$: $\nabla f + \lambda \nabla \phi = 0$, which is exactly $F_x = F_y = F_z = 0$.
ADDED FOR CLARITY S011·3 · S011·7

Exam trap. Do not forget the constraint as the fourth equation; three equations cannot fix four unknowns. T011a 00:02:30

Exam trap. Dividing by x, y, z or λ assumes it is not zero; check those cases separately (they usually give zero volume or an impossible point). ADDED FOR CLARITY S011·9 · T011b 00:03:40

SHORT-NOTE RECAP

Optimise f subject to $\phi = 0$: $F = f + \lambda \phi$; solve $F_x = F_y = F_z = 0$ with $\phi = 0$ for x, y, z, λ. Advantages: no elimination; extends to many variables and constraints. Disadvantages: nature not decided; the equations are only necessary.

43 Geometric problems: inscribed solids, nearest and farthest points

The largest rectangular solid in a sphere is a cube; in the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ its half-edges are $\frac{a}{\sqrt3},\frac{b}{\sqrt3},\frac{c}{\sqrt3}$ and its volume $\frac{8abc}{3\sqrt3}$. For nearest and farthest points, optimise the square of the distance: the hyperbolic cylinder $x^2-z^2=1$ is nearest the origin at $(\pm1,0,0)$; the curve $5x^2+6xy+5y^2=8$ is between distances 1 and 2.

Key terms and definitions

Continued product. The product obtained by multiplying together in succession: for parts x,y,z, the product of the first, the square of the second and the cube of the third is xy^2z^3. S011·12 · T011b 00:05:23

Formulas

E43.1 · Largest box in a sphere and in an ellipsoid ESA ★★★

$$x^2+y^2+z^2=r^2:\ x=y=z=\tfrac{r}{\sqrt3},\ V_{\max}=\tfrac{8r^3}{3\sqrt3}$$
$$\tfrac{x^2}{a^2}+\tfrac{y^2}{b^2}+\tfrac{z^2}{c^2}=1:\ x=\tfrac{a}{\sqrt3},\ y=\tfrac{b}{\sqrt3},\ z=\tfrac{c}{\sqrt3},\ V_{\max}=\tfrac{8abc}{3\sqrt3}$$

GIVES Half-edges and the largest volume $V=8xyz$.

APPLIES Box centred at the origin, edges $2x,2y,2z$ parallel to the axes.

NOTE The sphere is the ellipsoid with $a=b=c=r$: the box is a cube.

S011·5 · ESA May-24 1c · ESA Jan-24 s3 1a · ESA Jan-24 s4 1a

E43.2 · Distance problems: use the square ESA ★★★

$$f=x^2+y^2+z^2\ \text{(origin)},\qquad f=(x-x_0)^2+(y-y_0)^2+(z-z_0)^2$$

GIVES The squared distance, which has its extrema at the same points as the distance.

APPLIES Nearest or farthest point on a curve or surface.

NOTE Take the square root at the end.

ADDED FOR CLARITY S011·8 · ESA Jul-23 1c · CW U1 P36

Graphs and diagrams

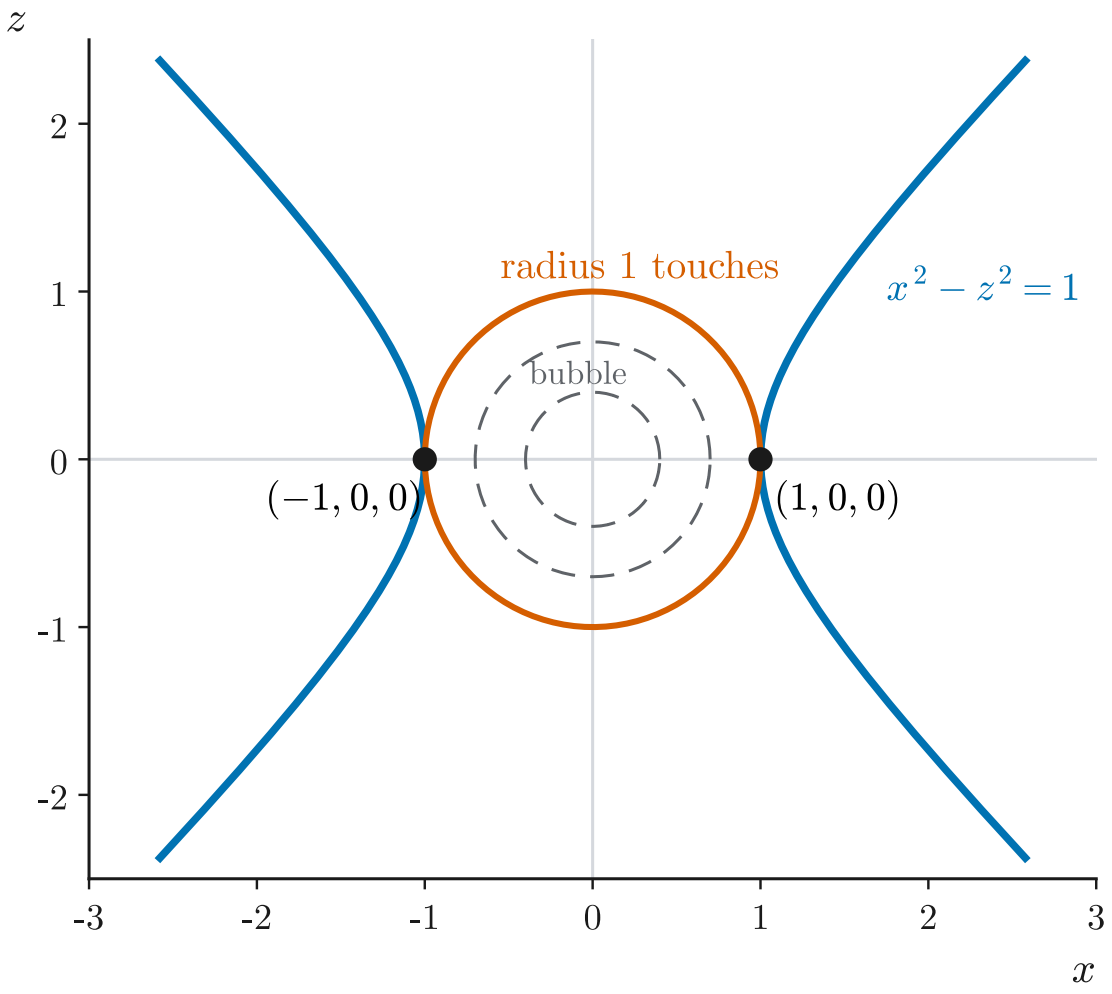


Figure G43.1. The soap bubble: $x^2 - z^2 = 1$ nearest the origin. ESA ★★

S011·7 · S011·9

How to read it. Cross-section $y = 0$. The hyperbolic cylinder appears as the hyperbola $x^2 - z^2 = 1$ (blue). Circles centred at the origin (grey, growing like a soap bubble) first touch it at $(\pm 1, 0)$ with radius 1 (vermilion): the nearest points $(\pm 1, 0, 0)$.

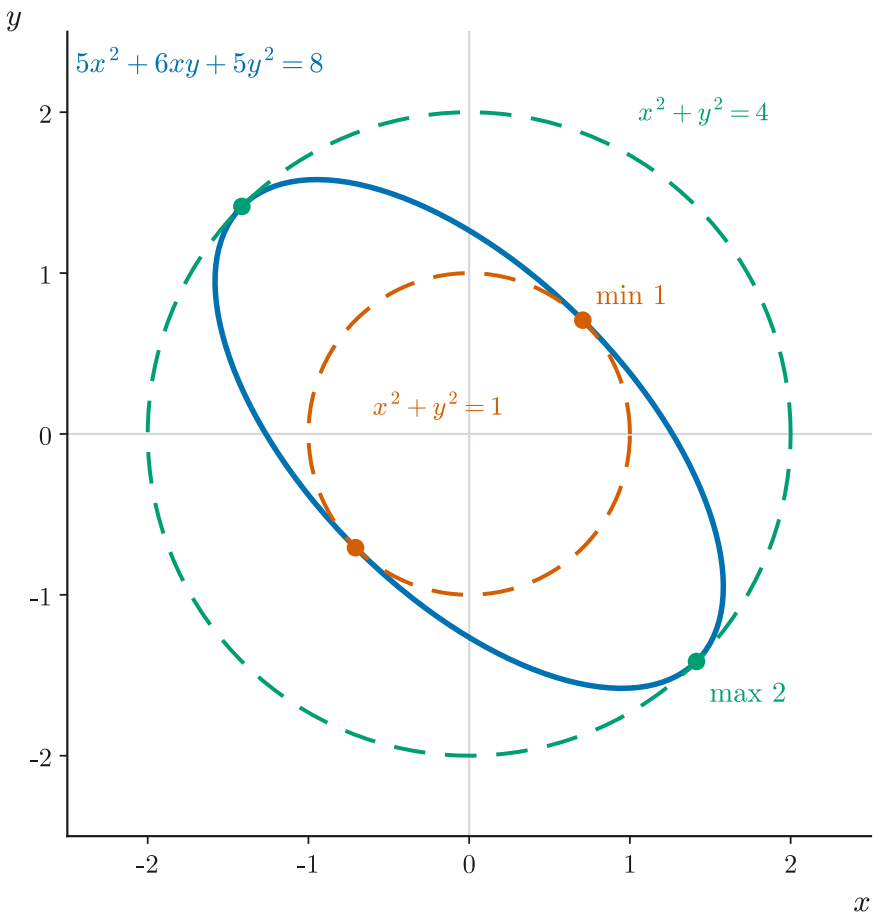


Figure G43.2. $5x^2 + 6xy + 5y^2 = 8$ between the circles of radius 1 and 2. ESA ★★★

ADDED FOR CLARITY ESA Jul-23 1c

How to read it. The curve is an ellipse (blue) tilted at $45°$. The circle $x^2 + y^2 = 1$ (vermilion) touches it from inside at $\pm(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$: minimum distance 1. The circle $x^2 + y^2 = 4$ (green) touches it from outside at $\pm(\sqrt{2}, -\sqrt{2})$: maximum distance 2.

Important points and why

1. For a box inscribed in a sphere or ellipsoid centred at the origin, take the corner (x, y, z) in the first octant: edges $2x, 2y, 2z$, volume $8xyz$. Using xyz instead gives the same point. ESA ★★★ S011·5 · T011a 00:06:08 · T011a 00:06:53
2. To find the nearest point, imagine a small sphere centred at the origin expanding like a soap bubble until it just touches the surface (Figure G43.1). ESA ★ S011·7
3. Reject solutions that are impossible: on $x^2 - z^2 = 1$, $x = 0$ would need $z^2 = -1$. ESA ★★ S011·9 · T011a 00:12:26
4. When there are two stationary values (W43.4: distances 1 and 2), the larger is the maximum and the smaller the minimum on a closed curve. ESA ★★★ ADDED FOR CLARITY ESA Jul-23 1c

Why minimise $x^2+y^2+z^2$ instead of the distance?
$\sqrt{u}$ increases with u, so both have their extrema at the same points, and the square has no square root to differentiate.

ADDED FOR CLARITY S011·8 · T011a 00:10:08

Exam trap. The box in the sphere has edges $2x, 2y, 2z$, so $V = 8xyz$: its largest volume is $\frac{8r^3}{3\sqrt{3}}$, not $\frac{r^3}{3\sqrt{3}}$. ADDED FOR CLARITY S011·5

Exam trap. Slide 011·6's third line should read $8xyz + 2z^2\lambda = 0$ (printed with y^2).

ADDED FOR CLARITY S011·6

Worked examples

W43.1 · The largest rectangular solid in a sphere is a cube — ESA ★★★

GIVEN Show that the rectangular solid of maximum volume that can be inscribed in a given sphere $x^2+y^2+z^2=r^2$ is a cube.

STEP 1 Edges $2x, 2y, 2z$: $f = 8xyz$; $\phi = x^2+y^2+z^2-r^2$; $F = 8xyz + \lambda(x^2+y^2+z^2-r^2)$.

STEP 2 $F_x = 8yz + 2x\lambda = 0$, $F_y = 8xz + 2y\lambda = 0$, $F_z = 8xy + 2z\lambda = 0$.

STEP 3 Multiply by x, y, z: $8xyz + 2x^2\lambda = 0$, $8xyz + 2y^2\lambda = 0$, $8xyz + 2z^2\lambda = 0$.

STEP 4 So $2x^2\lambda = 2y^2\lambda = 2z^2\lambda = -8xyz \neq 0$: $x^2 = y^2 = z^2$, and with $x, y, z > 0$, $x = y = z$.

STEP 5 From the constraint $3x^2 = r^2$: $x = y = z = \frac{r}{\sqrt{3}}$.

ANSWER $\boxed{x = y = z\text{: the solid is a cube of edge } \dfrac{2r}{\sqrt{3}}\text{, volume } \dfrac{8r^3}{3\sqrt{3}}}$

CHECK sympy solves the system to $x = y = z = \frac{r}{\sqrt{3}}$ and $V = \frac{8\sqrt{3}r^3}{9} = \frac{8r^3}{3\sqrt{3}}$.

S011·5 · S011·6 · S011·14 · T011a 00:04:49 · T011b 00:08:50 · ESA May-24 1c · Extra Q59

W43.2 · Largest parallelepiped in an ellipsoid ESA ★★★

GIVEN Use the method of Lagrange's multipliers to find the volume of the largest rectangular parallelepiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$.

STEP 1 $f=8xyz$, $F=8xyz+\lambda\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}-1\right)$.

STEP 2 $F_x=8yz+\frac{2\lambda x}{a^2}=0$, $F_y=8xz+\frac{2\lambda y}{b^2}=0$, $F_z=8xy+\frac{2\lambda z}{c^2}=0$.

STEP 3 Multiply by x, y, z: $\frac{2\lambda x^2}{a^2}=\frac{2\lambda y^2}{b^2}=\frac{2\lambda z^2}{c^2}=-8xyz$, so $\frac{x^2}{a^2}=\frac{y^2}{b^2}=\frac{z^2}{c^2}$.

STEP 4 Constraint: $3\frac{x^2}{a^2}=1$, so $x=\frac{a}{\sqrt{3}}$, $y=\frac{b}{\sqrt{3}}$, $z=\frac{c}{\sqrt{3}}$.

STEP 5 $V=8\cdot\frac{abc}{3\sqrt{3}}$.

ANSWER $\boxed{V_{\max}=\dfrac{8abc}{3\sqrt{3}}\ \text{(edges } \tfrac{2a}{\sqrt{3}},\tfrac{2b}{\sqrt{3}},\tfrac{2c}{\sqrt{3}}\text{)}}$

CHECK sympy: $\frac{8\sqrt{3}abc}{9}$, the same; with $a=b=c=r$ it reduces to W43.1.

ESA Jan-24 s3 1a · ESA Jan-24 s4 1a · Extra Q61

W43.3 · Points of $x^2-z^2=1$ closest to the origin ESA ★★

GIVEN Find the points closest to the origin on the hyperbolic cylinder $x^2-z^2=1$.

STEP 1 $f=x^2+y^2+z^2$ (squared distance), $\phi=x^2-z^2-1$; $F=x^2+y^2+z^2+\lambda(x^2-z^2-1)$.

STEP 2 $F_x=2x(1+\lambda)=0$ (1), $F_y=2y=0$ (2), $F_z=2z(1-\lambda)=0$ (3), $x^2-z^2=1$ (4).

STEP 3 (2): $y=0$. (1): $x=0$ or $\lambda=-1$; $x=0$ would need $z^2=-1$, so $\lambda=-1$.

STEP 4 (3) with $\lambda=-1$: $4z=0$, $z=0$. (4): $x^2=1$, $x=\pm1$.

ANSWER $\boxed{(\pm1,0,0)\text{, at distance 1 from the origin}}$

CHECK sympy finds only these two points ($\lambda=-1$).

S011·7 · S011·8 · S011·9 · T011a 00:09:35

W43.4 · Maximum and minimum distances from the origin to $5x^2 + 6xy + 5y^2 = 8$ ESA ★★★

GIVEN Find the maximum and minimum distances from the origin to the curve $5x^2 + 6xy + 5y^2 - 8 = 0$.

STEP 1 $f = x^2 + y^2$, $F = x^2 + y^2 + \lambda(5x^2 + 6xy + 5y^2 - 8)$.

STEP 2 $F_x = 2x + \lambda(10x + 6y) = 0$, $F_y = 2y + \lambda(6x + 10y) = 0$.

STEP 3 Eliminate λ: $\frac{x}{5x+3y} = \frac{y}{3x+5y}$, so $3x^2 + 5xy = 5xy + 3y^2$, $x^2 = y^2$, $y = \pm x$.

STEP 4 $y = x$: $16x^2 = 8$, $x^2 = \frac{1}{2}$; $f = x^2 + y^2 = 1$: distance 1, at $\pm(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$.

STEP 5 $y = -x$: $4x^2 = 8$, $x^2 = 2$; $f = 4$: distance 2, at $\pm(\sqrt{2}, -\sqrt{2})$.

ANSWER Maximum distance 2, minimum distance 1

CHECK sympy agrees. In rotated axes the curve is the ellipse $8u^2 + 2v^2 = 8$ with semi-axes 1 and 2 (Figure G43.2).

ESA Jul-23 1c

W43.5 · Distances of $(3, 4, 12)$ from the unit sphere ESA ★★

GIVEN Find the maximum and minimum distances of the point $(3, 4, 12)$ from the unit sphere $x^2 + y^2 + z^2 = 1$.

STEP 1 $f = (x-3)^2 + (y-4)^2 + (z-12)^2$, $F = f + \lambda(x^2 + y^2 + z^2 - 1)$.

STEP 2 $2(x-3) + 2\lambda x = 0 \Rightarrow x = \frac{3}{1+\lambda}$; likewise $y = \frac{4}{1+\lambda}$, $z = \frac{12}{1+\lambda}$.

STEP 3 Constraint: $\frac{9+16+144}{(1+\lambda)^2} = 1$, $1 + \lambda = \pm 13$: points $\pm\frac{1}{13}(3, 4, 12)$.

STEP 4 Distances: $13 - 1 = 12$ and $13 + 1 = 14$.

ANSWER Maximum 14, minimum 12

CHECK CW answer agrees; sympy agrees.

CW U1 P36 · Extra Q45

Past-paper questions

ESA Jul 2023 (7 marks) UE22MA141A Q1.c · solved in W43.4

Find the maximum and minimum distances from the origin to the curve $5x^2 + 6xy + 5y^2 - 8 = 0$.

ESA May 2024 (7 marks) UE22MA141A Q1.c · solved in W43.1

Show that the rectangular solid of maximum volume that can be inscribed in a given sphere is a cube by using Lagrange's method of multipliers.

ESA Jan 2024 7 marks UE23MA141A set 3, Q1.a · solved in W43.2

Use the method of Lagrange's multipliers to find the volume of the largest rectangular parallelepiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$.

ESA Jan 2024 7 marks UE23MA141A set 4, Q1.a · solved in W43.2

The same question as set 3 Q1.a (largest parallelepiped in the ellipsoid).

SHORT-NOTE RECAP

Box in sphere: $f=8xyz \Rightarrow x=y=z=\frac{r}{\sqrt{3}}$, cube. Ellipsoid: $V_{\max}=\frac{8abc}{3\sqrt{3}}$. Distances: optimise the square. $x^2-z^2=1$: nearest $(\pm1,0,0)$. $5x^2+6xy+5y^2=8$: distances 1 (min) and 2 (max). $(3,4,12)$ to unit sphere: 12 and 14.

44 Applied problems: boxes, splitting a number, temperature

The same procedure handles every applied problem once f and ϕ are identified: an open box of surface 432 cm² holds most as $12 \times 12 \times 6$; 24 split as $4 + 8 + 12$ maximises xy^2z^3; the temperature $400xyz^2$ on the unit sphere is at most 50. The trick in solving is to make λ the subject of each equation and equate.

Important points and why

1. Open box (no top): surface $xy + 2yz + 2zx$; closed box: $2(xy + yz + zx)$. ESA ★★ S011·10 · T011b 00:01:09
2. Dividing a number N into parts x, y, z: constraint $x + y + z = N$; the function is the product asked for. ESA ★ S011·12 · T011b 00:05:57
3. The temperature problem of class 010 gives the same 50 by Lagrange's method with no elimination (W44.3). ESA ★★★ T011a 00:14:33 · ESA Jun-24 1d

Exam trap. Open-top box: $x = y = 2z$ (height half the base edge), not a cube. S011·11 · CW U1 P34

Worked examples

W44.1 · Open box of surface 432 cm² with maximum capacity ESA ★★

given Find the dimensions of the rectangular box, open at the top, of maximum capacity whose surface is 432 sq cm.

step 1 $f = xyz$, $\phi = xy + 2yz + 2zx - 432$; $F = xyz + \lambda(xy + 2yz + 2zx - 432)$.

step 2 $yz + \lambda(y + 2z) = 0$ (1), $xz + \lambda(x + 2z) = 0$ (2), $xy + \lambda(2y + 2x) = 0$ (3).

step 3 $x\cdot(1) - y\cdot(2)$: $2z\lambda(x - y) = 0 \Rightarrow x = y$. $y\cdot(2) - z\cdot(3)$: $x\lambda(y - 2z) = 0 \Rightarrow y = 2z$.

step 4 Constraint with $x = y = 2z$: $x^2 + x^2 + x^2 = 3x^2 = 432$, $x = 12$.

answer 12 cm $\times$ 12 cm $\times$ 6 cm (capacity 864 cm³)

check sympy agrees.

S011·10 · S011·11 · T011b 00:00:19

W44.2 · Divide 24 to maximise xy^2z^3 ESA ★★

GIVEN Divide the number 24 into three parts such that the continued product of the first, square of the second and cube of the third may be maximum.

STEP 1 $f = xy^2z^3$, $\phi = x + y + z - 24$; $F = xy^2z^3 + \lambda(x + y + z - 24)$.

STEP 2 $\lambda = -y^2z^3 = -2xyz^3 = -3xy^2z^2$.

STEP 3 $y^2z^3 = 2xyz^3 \Rightarrow y = 2x$; $2xyz^3 = 3xy^2z^2 \Rightarrow z = \frac{3}{2}y = 3x$.

STEP 4 $x + 2x + 3x = 24$: $x = 4$.

ANSWER Parts 4, 8, 12 (product $4 \cdot 8^2 \cdot 12^3 = 442368$)

CHECK sympy agrees.

S011·12 · S011·13 · T011b 00:05:03 · Extra Q53

W44.3 · Highest temperature $400xyz^2$ on the unit sphere ESA ★★★

GIVEN The temperature T at any point (x, y, z) in space is $400xyz^2$. Using Lagrange's method of multipliers, find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$.

STEP 1 $F = 400xyz^2 + \lambda(x^2 + y^2 + z^2 - 1)$.

STEP 2 $400yz^2 + 2\lambda x = 0$, $400xz^2 + 2\lambda y = 0$, $800xyz + 2\lambda z = 0$.

STEP 3 Multiply by x, y, z: $400xyz^2 = -2\lambda x^2 = -2\lambda y^2$ and $800xyz^2 = -2\lambda z^2$.

STEP 4 So $x^2 = y^2$ and $z^2 = 2x^2$. Constraint: $x^2 + x^2 + 2x^2 = 1$, $x^2 = \frac{1}{4}$, $z^2 = \frac{1}{2}$.

STEP 5 With x, y of the same sign: $T = 400 \cdot \frac{1}{4} \cdot \frac{1}{2} = 50$ (opposite signs give -50).

ANSWER Highest temperature 50, at $x = y = \pm\frac{1}{2}$, $z = \pm\frac{1}{\sqrt{2}}$

CHECK sympy: stationary values $\{0, 50, -50\}$. Same as class 010's elimination answer.

ESA Jun-24 1d · Extra Q57 · T011a 00:00:17

W44.4 · Extreme value of $a^3x^2+b^3y^2+c^3z^2$ ESA ★★

GIVEN Find the extreme value of $a^3x^2+b^3y^2+c^3z^2$ subject to $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=1$, where $a,b,c>0$.

STEP 1 $F=a^3x^2+b^3y^2+c^3z^2+\lambda\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}-1\right)$.

STEP 2 $2a^3x-\frac{\lambda}{x^2}=0\Rightarrow a^3x^3=\frac{\lambda}{2}$; likewise $b^3y^3=c^3z^3=\frac{\lambda}{2}$. So $ax=by=cz=t$.

STEP 3 Constraint: $\frac{a}{t}+\frac{b}{t}+\frac{c}{t}=1$, $t=a+b+c$.

STEP 4 Value: $a^3\frac{t^2}{a^2}+b^3\frac{t^2}{b^2}+c^3\frac{t^2}{c^2}=t^2(a+b+c)=t^3$.

ANSWER $\boxed{(a+b+c)^3 \text{ at } \left(\frac{t}{a},\frac{t}{b},\frac{t}{c}\right),\ t=a+b+c}$

CHECK CW answer agrees; numerically with $a,b,c=1,2,3$: value $216=6^3$.

CW U1 P33

W44.5 · Extreme value of $x^3+8y^3+64z^3$ when $xyz=1$ ESA ★

GIVEN Find the extreme value of $x^3+8y^3+64z^3$ subject to $xyz=1$.

STEP 1 $F=x^3+8y^3+64z^3+\lambda(xyz-1)$: $3x^2+\lambda yz=0$, $24y^2+\lambda xz=0$, $192z^2+\lambda xy=0$.

STEP 2 Multiply by x, y, z: $3x^3=24y^3=192z^3=-\lambda$, so $x=2y=4z$.

STEP 3 $xyz=x\cdot\frac{x}{2}\cdot\frac{x}{4}=1$: $x^3=8$, $x=2$, $y=1$, $z=\frac{1}{2}$.

ANSWER $\boxed{24 \text{ at } \left(2,1,\tfrac{1}{2}\right) \text{ (a minimum for positive } x,y,z)}$

CHECK CW answer agrees; sympy agrees.

CW U1 P35

Past-paper questions

ESA June 2024 · 7 marks · UE23MA141A Q1.d · solved in W44.3

The temperature T at any point (x,y,z) in space is $400xyz^2$. Using Lagrange's method of multipliers, find the highest temperature on the surface of the unit sphere $x^2+y^2+z^2=1$.

SHORT-NOTE RECAP

Open box, surface 432: $12\times12\times6$. $24=4+8+12$ for xy^2z^3. $400xyz^2$ on the unit sphere: 50. CW: $a^3x^2+b^3y^2+c^3z^2$ with $\sum\frac{1}{x}=1$: $(a+b+c)^3$; $x^3+8y^3+64z^3$ with $xyz=1$: 24.

A Symbol glossary

Symbol	Name	Meaning	Appears in
$z = f(x, y)$	Function of two variables	x, y independent, z dependent	Class 001: T1
D	Domain	Set of ordered pairs where f is defined	Class 001: T1
L	Limit	Common value along every path	Class 001: E2.1
(a, b)	Point of approach	Where the limit or continuity is tested	Class 001: E2.1, E3.1
m	Slope of a path	Path $y = mx$ in the two-path test	Class 001: E2.2, W2.2
∂	Partial ("dou")	Partial derivative sign; not δ, not d	Class 001: E4.1–E5.3
h	Increment	Small change in one variable in the definition	Class 001: E4.1
$z_x,\ f_x$	First partial in x	$\frac{\partial z}{\partial x}$ with y constant	Class 001: E4.2
$z_{xx},\ z_{yy}$	Second partials	$\frac{\partial^2 z}{\partial x^2},\ \frac{\partial^2 z}{\partial y^2}$	Class 001: E5.1
z_{xy}	Mixed partial	x first, then y; $= \frac{\partial^2 z}{\partial y \partial x}$	Class 001: E5.2, E5.3
$f,\ z$	Function / dependent variable	$z = f(x, y)$; its graph is a surface	Class 002: E6.1–E11.2
$x,\ y$	Independent variables	Coordinates in the xy-plane	Class 002: all
(x_0, y_0)	Base point	Point where the derivatives are evaluated	Class 002: E9.1–E10.2
$P,\ z_0$	Point on the surface	$P = (x_0, y_0, z_0)$ with $z_0 = f(x_0, y_0)$	Class 002: E9.3
∂	"Partial", read "dou"	Differentiate one variable, hold the others constant	Class 002: E6.1–E6.4
$f_x,\ f_y$	First-order partials	Slopes parallel to the x- and y-axes	Class 002: E6.1, E9.1–E10.2
$f_{xx},\ f_{yy}$	Pure second-order partials	Concavity along x and along y	Class 002: E6.2, E11.1

Symbol	Name	Meaning	Appears in
$f_{xy},\ f_{yx}$	Mixed second-order partials	Change of one slope along the other direction	Class 002: E6.2, E6.4, E11.2
$m,\ n$	Counts	m = number of variables, n = order	Class 002: E6.3
$f'(x_0),\ dy/dx$	Ordinary derivative	Slope of a tangent to a plane curve	Class 002: E7.1, E7.2
∇f	Gradient ("del f")	$f_x\,\mathbf{i} + f_y\,\mathbf{j}$	Class 002: E10.1, E10.2
$\mathbf{i},\ \mathbf{j}$	Unit vectors	Along the x- and y-axes	Class 002: E10.1
$\hat{u} = (a, b)$	Unit direction vector	$a^2 + b^2 = 1$	Class 002: E10.2
$D_{\hat{u}} f$	Directional derivative	Slope of the surface in direction $\hat{u}$	Class 002: E10.2
t	Time (in F14)	Independent variable in the motion analogy	Class 002: E11.3
$z = f(x, y)$	Function of two variables	The quantity being differentiated	Classes 003–004: E14.1, E15.1, E17.1
x, y	Intermediate variables	Depend on t (total derivative) or on u, v (chain rule)	Classes 003–004: E14.1, E17.1
t	Parameter / time	Single final variable in the total derivative; also a new variable in some chain-rule problems	Classes 003–004: E14.1, E14.2
u, v / s, t	New independent variables	Final variables of a composite function	Classes 003–004: E17.1
r, s, t	Named arguments	$r = 2x - 3y$, ... in slide 003·15 (not a radius there)	Classes 003–004: E17.2
r, θ	Polar coordinates	$x = r\cos\theta$, $y = r\sin\theta$	Classes 003–004: E17.4, E17.5
$\dfrac{dz}{dt}$	Total derivative	Ordinary derivative when z ends up depending on t alone	Classes 003–004: E14.1
$\dfrac{\partial z}{\partial u}$	Partial derivative of a composite	z depends on u and v	Classes 003–004: E17.1
$dx,\ dy,\ dz$	Differentials	Small (signed) changes; $dz = f_x\,dx + f_y\,dy$	Classes 003–004: E15.1, E15.2

Symbol	Name	Meaning	Appears in
f_x, f_y, f_z	Partial derivatives of f	Kept as symbols when f is unknown	Classes 003–004: E17.1, E18.1, E19.1
$f(x, y) = c$	Level curve / implicit relation	Defines y as a function of x	Classes 003–004: E18.1
$f(x, y, z) = 0$	Level surface	Defines z as a function of x, y	Classes 003–004: E19.1
$f'(r)$	Ordinary derivative of $f(r)$	u depends on x, y, z only through r	Classes 003–004: E17.3
m	Slope	dy/dx at the point; tangent $y - y_0 = m(x - x_0)$	Classes 003–004: E18.2
A, V, h, b, θ	Geometric quantities	Area, volume, height, base, included angle in rate problems	Classes 003–004: E16.1
$u = f(x, y)$	Function	The function tested for homogeneity	Class 005: E20.1, E21.1
n	Degree	Power in $x^n g(y/x)$; any real number	Class 005: E20.1–E23.2
$g(\cdot)$ (topic 1)	Arbitrary function of ratios	$u = x^n g(y/x)$	Class 005: E20.1
$g(u)$ (topic 4)	Extension function	$nf(u)/f'(u)$ (a different g from topic 1)	Class 005: E23.1, E23.2
$f(u)$	Homogeneous function of u	e.g. $\tan u$ when $u = \tan^{-1}(\dots)$	Class 005: E23.1
t	Scale factor	$u(tx, ty) = t^n u$	Class 005: E20.1
u_x, u_y, u_z	First partials	Homogeneous of degree $n - 1$	Class 005: E21.1, E21.2
u_{xx}, u_{xy}, u_{yy}	Second partials	Degree $n - 2$; $u_{xy} = u_{yx}$	Class 005: E22.1
u_1, u_2	Homogeneous parts	Parts of different degrees in a sum	Class 005: E22.3
P, Q	Homogeneous factors	Numerator and denominator of a quotient	Class 005: E20.2
λ	Scale factor	Replaces x, y by $\lambda x, \lambda y$ in the scaling test	Class 006: E25.1

Symbol	Name	Meaning	Appears in
n	Degree	Of u (topic 3) or of $f(u)$ (topic 4)	Class 006: E25.1–E28.2
$f(u)$	Homogeneous function of u	$\sin u$, $\tan u$, e^u, $\log u$	Class 006: E28.1
$f(y/x)$ (slide 3)	Arbitrary function of the ratio	$Z = \sqrt{x}\, f(y/x)$; not the $f(u)$ of topic 4	Class 006: W26.1
$g(u)$	Extension function	$nf(u)/f'(u)$; equals $xu_x + yu_y$	Class 006: E28.1, E28.2
$g'(u)$	Derivative of g	With respect to u	Class 006: E28.2
Z	Auxiliary function	$\cos u$ (slide 3) or $\sin u$ (slide 4, transcript)	Class 006: W26.1, W28.1
z	Third variable	Lowercase, on slide 4 only	Class 006: W28.1
$u_x,\ u_{xy}$	Partial derivatives	$\partial u/\partial x$, $\partial^2 u/\partial x \partial y$	Class 006: E26.1, E27.1
a, (a, b)	Expansion point	Point about which the series is written	Class 007: E30.1, E31.1
$h,\ k$	Increments	$h = x - a$, $k = y - b$ (shorthand added for clarity)	Class 007: E31.1–E31.3
$f^{(n)}(a)$	n-th derivative	Printed $f^n(a)$ on slide 007·4; not a power	Class 007: E30.1
$n!$	Factorial	$1 \cdot 2 \cdots n$; divides the n-th order term	Class 007: E30.1–E32.1
$f_x,\ f_y$	First partials	At the expansion point	Class 007: E31.1, E32.1
$f_{xx},\ f_{xy},\ f_{yy}$	Second partials	$f_{xy} = f_{yx}$, hence the 2	Class 007: E31.1, E32.1
$f_{xxx}, \ldots, f_{yyy}$	Third partials	Coefficients 1, 3, 3, 1	Class 007: E31.2
n (in $(1+x)^n$)	Exponent	Any real number; not the order of a derivative	Class 007: E30.4
$L(x, y)$	Taylor polynomial	Slide 007·11's name for the truncated series	Class 007: W31.7
(a, b)	Expansion point	$(1, \frac{\pi}{4})$, $(1, \frac{\pi}{2})$, $(1, 1)$, $(1, -1)$ or $(0, 0)$ in class 008	Class 008: E33.1

Symbol	Name	Meaning	Appears in
$f_x, \ldots, f_{yy}$	Partial derivatives	Evaluated at the expansion point	Class 008: E33.1, E34.1
t	Combination	$2x + y$, $x - y$, $x + 2y$ in the one-variable check	Class 008: E34.2
s	Step along a line	Figure G35.1's parameter only	Class 008: G35.1
$a,\ b$ (in $e^{ax}\sin by$)	Constants	Not the expansion point here (it is the origin)	Class 008: W34.5
(a, b)	Stationary point	Solution of $f_x = f_y = 0$	Class 009: E38.1
$h,\ k$	Small increments	Neighbouring point $(a + h, b + k)$	Class 009: E37.1
Δ	Increment of f	$f(a + h, b + k) - f(a, b)$	Class 009: E37.1
r	f_{xx}	Second partial in x at the point	Class 009: E38.2–E38.4
s	f_{xy}	Mixed partial at the point (not arc length)	Class 009: E38.2–E38.4
t	f_{yy}	Second partial in y (not time)	Class 009: E38.2–E38.4
$rt - s^2$	Hessian determinant	$f_{xx}f_{yy} - f_{xy}^2$	Class 009: E38.3
$p,\ q$	First partials	$p = f_x$, $q = f_y$ (lecture T010b)	Class 010: E39.1
$r,\ s,\ t$	Second partials	f_{xx}, f_{xy}, f_{yy} at a point	Class 010: E39.2, E39.3
$rt - s^2$	Hessian	Decides extremum or saddle	Class 010: E39.2, E39.3
T	Temperature	$400xyz^2$ or $kxyz^2$; not the second partial t	Class 010: E41.1
k	Constant	Positive constant in $T = kxyz^2$	Class 010: E41.1
a	Radius	Sphere $x^2 + y^2 + z^2 = a^2$	Class 010: E41.1
A	Surface area	Metal used by the open box	Class 010: E41.2
V	Volume	Fixed capacity $xyz = V$ (10 m³)	Class 010: E41.2
$f(x, y, z)$	Objective	Function to maximise or minimise	Class 011: E42.1
$\phi(x, y, z)$	Constraint	$\phi = 0$; printed $\varnothing$ on the slides	Class 011: E42.1, E42.2

Symbol	Name	Meaning	Appears in
λ	Lagrange multiplier	Undetermined constant; not in the answer	Class 011: E42.1–E42.3
F	Auxiliary function	$f + \lambda\phi$	Class 011: E42.1
r	Radius	Of the sphere $x^2 + y^2 + z^2 = r^2$; not f_{xx}	Class 011: E43.1
$a,\ b,\ c$	Semi-axes	Of the ellipsoid (in W44.4: positive constants)	Class 011: E43.1, W44.4
$2x,\ 2y,\ 2z$	Edges	Of the inscribed box; volume $8xyz$	Class 011: E43.1

B Sources, coverage and gaps

Source	Path	Use
Class 001 · Deck 001 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.*	18 slides; content on 3–17
Class 001 · Transcript 001a (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Introduction_to_Partial_Differentiation__7d6d5767ec830adb.*	19.7 min: motivation, definition, limits
Class 001 · Transcript 001b (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Continuity_and_differentiability_of_a_function_of_two_independent_variable__543f4d65e9b6ed52.*	21.1 min: continuity, history, definition, mixed partials
Class 001 · Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 1–8 (with answers)
Class 001 · ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	No direct question; two partial-derivative computations (class 002)
Class 001 · Videos 001a, 001b (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_*.mp4	Not opened: every slide is readable and the board work is read out in the transcripts
Class 002 · Deck 002 (.pptx, rendered to images; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.*	11 slides; content on slides 3–10
Class 002 · Transcript 002 (.transcript.md / .json / .srt / .txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/002_Geometrical_interpretation_of_Partial_derivatives__374c7bb76e445af3.*	15.7 min lecture
Class 002 · Video 002 (.mp4): 3 single frames at 00:07:45, 00:09:05, 00:12:40	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/002_...374c7bb76e445af3.mp4	Remote seek; the full video was not downloaded
Class 002 · Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 1, 2, 5 used
Class 002 · ESA papers, UE22MA141A: Dec 2023, Jul 2023, May 2024	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE22MA141A/*.pdf	Q1 read for Unit 1

Source	Path	Use
Class 002 · ESA papers, UE23MA141A: Jan sets 1–4, June 2024	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE23MA141A/*.pdf	Q1 read for Unit 1
Classes 003–004 · Deck 003 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.*	28 slides; content on 4–27
Classes 003–004 · Deck 004 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.*	18 slides; content on 3–17
Classes 003–004 · Transcript 003 part I (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Total_derivative_Chain_rule_and_Implicit_functions__d194dd0f27d571b2.*	26.5 min lecture
Classes 003–004 · Transcript 003 part II (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Problems_on_Total_derivatives_Chain_rule__4d669ee31cdd4a30.*	11.6 min problems
Classes 003–004 · Transcript 004 part I (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Implicit_functions_and_problems__28ebf6864cd576d3.*	22.4 min lecture
Classes 003–004 · Transcript 004 part II (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Problems_on_Composite_and_Implicit_functions__d1ea652fce3206d2.*	11.7 min problems
Classes 003–004 · Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 9–18 (with answers)
Classes 003–004 · ESA papers, UE22MA141A: Dec 2023, Jul 2023, May 2024	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE22MA141A/*.pdf	Q1 read; none in scope
Classes 003–004 · ESA papers, UE23MA141A: Jan 2024 sets 1–4, June 2024	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE23MA141A/*.pdf	6 in-scope Q1 parts
Classes 003–004 · Videos 003 and 004 (4 .mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/00[34]_*.mp4	Not opened: every gap was settled from the slides, transcripts and algebra

Source	Path	Use
Class 005 · Deck 005 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.*	8 slides; content on 3–7
Class 005 · Transcript 005 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/005_Homogeneous_functions_and_Euler_s_theorem__a0c2c43d28f6ec69.*	19.4 min lecture
Class 005 · Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 19–23 (with answers)
Class 005 · ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	6 in-scope Q1 parts
Class 005 · Video 005 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/005_*.mp4	Not opened: slide 7 is readable in the rendered image
Class 006 · Deck 006 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.*	18 slides; content on 3–17
Class 006 · Transcript 006 part I (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_Theorem__6c4797f9ed9a5b61.*	9.1 min: slides 3–4
Class 006 · Transcript 006 part II (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_theorem__0d726228252093d7.*	12.9 min: slides 5–17
Class 006 · Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 19–23 (with answers)
Class 006 · ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	6 in-scope Q1 parts
Class 006 · Videos 006 (2 .mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_*.mp4	Not opened: every slide is readable and every gap was settled by algebra

Source	Path	Use
Class 007 · Deck 007 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.*	13 slides; content on 3–12
Class 007 · Transcript 007 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/007_Taylor_s_and_Maclaurin_s_expansion__ff2506f0c8105451.*	22.2 min lecture
Class 007 · Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 24–28 (with answers)
Class 007 · ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	6 in-scope Q1 parts
Class 007 · Video 007 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/007_*.mp4	Not opened: every slide is readable, and the one garbled formula is read out in full in the transcript
Class 008 · Deck 008 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.*	20 slides; content on 3–19
Class 008 · Transcript 008 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/008_Problems_on_Taylor_s_and_Maclaurin_s_series__4b73a1bc5fc83388.*	16.2 min: slides 3–9
Class 008 · Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 24–28 (with answers)
Class 008 · ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	6 in-scope Q1 parts
Class 008 · Video 008 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/008_*.mp4	Not opened: every garbled slide was settled from the transcript and by recomputing

Source	Path	Use
Class 009 · Deck 009 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.*	10 slides; content on 3–9
Class 009 · Transcript 009 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/009_Extremas_of_functions__5ebc2dd46f817380.*	13.2 min lecture
Class 009 · Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problem 31 (with answer)
Class 009 · ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	3 questions use this test (worked in class 010)
Class 009 · Video 009 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/009_*.mp4	Not opened: the slides are readable and the board work is read out
Class 010 · Deck 010 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.*	15 slides; content on 3–14
Class 010 · Transcript T010a (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Extremas_of_functions_continued__5cf502cc80940c87.*	17.3 min: slides 3–6
Class 010 · Transcript T010b (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__e276ddb59439e6ec.*	11.1 min: slides 7–14
Class 010 · Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 29, 31, 32 (with answers); 30 is blank; 34 confirms the open-box answer
Class 010 · ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	3 in-scope Q1 parts (2 distinct functions)
Class 010 · Videos 010 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_*.mp4	Not opened: every slide is readable and every disputed number was settled by recomputing

Source	Path	Use
Class 011 · Deck 011 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.*	15 slides; content on 3–14
Class 011 · Transcript T011a (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Lagrange_s_method_of_undetermined_coefficient__6812c905c391978e.*	14.8 min: method, cube in a sphere, hyperbolic cylinder
Class 011 · Transcript T011b (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Problems_on_Lagrange_s_method_of_undetermined_multipliers__94e58d0103d5b799.*	10.2 min: open box, dividing 24, cube in a sphere
Class 011 · Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 33–36 (with answers)
Class 011 · Extra questions “lagrange-ques” (third-party list)	pesuc/First_Year_8enterprise/Files/Math 1/Extra/Extra Questions/lagrange-ques__1smrsCddi3.pdf	Confirms ESA problems (Q57, Q59, Q61); no solutions used
Class 011 · ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	5 in-scope Q1 parts (4 distinct problems)
Class 011 · Videos 011 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_*.mp4	Not opened: every slide is readable and every result was recomputed

Class 001. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. No question asks for a limit, continuity or the definition of a partial derivative. Computing partial derivatives is asked directly twice: Dec 2023 Q1.a (third-order partials of $e^x \log y + \cos y \log x$) and Jan 2024 set 4 Q1.c (find n for $\theta = t^n e^{-r^2/4t}$). Both are worked in the class 002 pack; this pack works the class-work problems of the same type (CW P4, P7, P8). Every other Unit 1 question (chain rule, Euler, Taylor, maxima and minima, Lagrange) uses the rules of this session.

Class 002. Eight past ESA papers were checked: UE22MA141A Dec 2023, May 2024 and Jul 2023, and UE23MA141A Jan sets 1–4 and June 2024. None asks for the geometrical interpretation directly. Two 6-mark Q1 parts test higher-order partials, the computational half of this session. Other Unit 1 questions (Euler, Taylor, maxima, Lagrange, chain rule) belong to later sessions and are left out.

Classes 003–004. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. In-scope Q1 parts: five chain-rule questions (Jan 2024 set 1 Q1.d and set 2 Q1.d, 5 marks each; set 3 Q1.b, set 4 Q1.b and June 2024 Q1.a, 6 marks each) and one implicit-function question (Jan 2024 set 3 Q1.d, 6 marks). The total derivative and rates of change were not asked in these papers; they are computed in class-work problems 9–13. No paper repeats another. Questions on Euler's theorem, Taylor series, maxima and Lagrange belong to later sessions and are left out.

Class 005. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. Euler's theorem appears in six of them: Jul 2023 Q1.a, May 2024 Q1.a, June 2024

Q1.b (the same question as May 2024), Jan 2024 set 3 Q1.c, set 4 Q1.d, and set 1 Q1.d (a degree-0 function, also solvable by the chain rule). Five are 6-mark questions, so this is one of the most examined topics in Unit 1. The definition, applications and third-order result were not asked.

Class 006. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. Euler problems appear in six of them: Jul 2023 Q1.a (degree 5: $5u$, $20u$), May 2024 Q1.a and June 2024 Q1.b (the same $\tan^{-1}$ extension question), Jan 2024 set 3 Q1.c ($17\pi/2$), set 4 Q1.d ($9u$) and set 1 Q1.d (degree 0). Five are 6-mark questions. Testing homogeneity alone and verifying Euler's theorem by direct differentiation (topics 1–2) were not asked on their own; they appear only as the first step of a longer question.

Class 007. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. A Taylor or Maclaurin expansion in two variables is asked in six of them, always as a 6–7 mark Q1 part. About a point (a, b): Dec 2023 Q1.b ($\sin x \sin y$, third degree), Jul 2023 Q1.b (the same function, second order) and May 2024 Q1.b ($e^x \cos y$ about $(1, \pi/4)$). About the origin: Jan 2024 set 1 Q1.b ($\log(1 + x - y)$), set 2 Q1.b ($e^{ax} \sin by$) and June 2024 Q1.c ($\sin(x + 2y)$). Four of the six ask for third-degree terms, which the slides do not write out. Jan 2024 sets 3 and 4 have no Taylor question. One-variable series and the applications were never asked.

Class 008. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. A two-variable Taylor or Maclaurin expansion is asked in six of them, always a 6–7 mark Q1 part: Jul 2023 Q1.b and Dec 2023 Q1.b ($\sin x \sin y$ about $(\frac{\pi}{4}, \frac{\pi}{4})$, second order and third degree), May 2024 Q1.b ($e^x \cos y$ about $(1, \frac{\pi}{4})$, the same as slide 008·3), Jan 2024 set 1 Q1.b ($\log(1 + x - y)$), set 2 Q1.b ($e^{ax} \sin by$) and June 2024 Q1.c ($\sin(x + 2y)$). Four ask for third-degree terms. Jan 2024 sets 3 and 4 have no Taylor question. Approximate values (topic 3) were never asked in these papers; they appear in class and in CW problem 26.

Class 009. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. The definitions are never asked by themselves, but the test of this session is the whole of three 7-mark Q1 parts: Dec 2023 Q1.c and Jan 2024 set 1 Q1.a (the same function $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$) and Jan 2024 set 2 Q1.a ($x^3 - 12x + y^3 + 3y^2 - 9y$). They are worked in the class 010 pack; the method is starred ★★★ here.

Class 010. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. Free extrema of $f(x, y)$ are asked in three of them, always a 7-mark Q1 part: Dec 2023 Q1.c and Jan 2024 set 1 Q1.a ask the same function $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$ (also slide 010·7 and CW problem 32), and Jan 2024 set 2 Q1.a asks $x^3 - 12x + y^3 + 3y^2 - 9y$ (CW problem 29). Both are cubic polynomials with four stationary points: one maximum, one minimum and two saddle points. The other five papers ask Lagrange's method (class 011) instead. The temperature-on-a-sphere problem of slide 010·3 is asked in June 2024 Q1.d, but by Lagrange's method; the elimination method of topic 3 gives the same answer. Trigonometric and quartic functions (topic 2) were never asked.

Class 011. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. Lagrange's method is asked in five of them, always as a 7-mark Q1 part, with four distinct problems: Jul 2023 Q1.c (maximum and minimum distances from the origin to $5x^2 + 6xy + 5y^2 - 8 = 0$), May 2024 Q1.c (the rectangular solid of maximum volume in a sphere is a cube; also slides 011·5–6), Jan 2024 sets 3 and 4 Q1.a (largest parallelepiped in an ellipsoid) and June 2024 Q1.d (highest temperature $400xyz^2$ on the unit sphere; also class 010 by elimination). Three of the four also appear in the extra question list (Q57, Q59, Q61). Dec 2023 and Jan 2024 sets 1–2 ask free extrema (class 010) instead.

Known gaps, conflicts and corrections

1. Class 001: Slide 001·9 asks for $\lim_{(x,y)\to(2,1)}(x^2 - 2xy + 3y^2 - 4x + 3y - 6)$, which is -8. The lecture (T001a 00:12:59) works the point $(2, -1)$ instead and gets -6. Both values are correct for their points (sympy); W2.1 gives both.
2. Class 001: Slide 001·12's text is about $f = \frac{3x^2y}{x^2+y^2}$ (whose limit at the origin is 0), but the Gnuplot caption on the slide's picture reads $\frac{3xy}{y^2+x^2}$, a different function whose limit at the origin does not exist (along $y = mx$ it is $\frac{3m}{1+m^2}$). Figure G3.1 redraws the function in the text.

3. Class 001: Slide 001·12 says the function is not continuous at $(0,0)$ "as $f(0,0)$ is not defined". Added for clarity: the limit is 0, so defining $f(0,0)=0$ removes the discontinuity (a removable discontinuity); with $2xy/(3x^2+y^2)$ (slide 001·13) no value of $f(0,0)$ helps.
4. Class 001: The lecture says a function is continuous "if the surface is smooth" (T001b 00:06:26). Continuity only needs no holes or jumps; a continuous surface can have a sharp edge or corner ($z=|x|+|y|$). Corrected as an exam trap.
5. Class 001: Slide 001·17 says mixed partials "are in general equal". Precisely, Clairaut's theorem needs f_{xy} and f_{yx} continuous near the point; the standard counter-example $\frac{xy(x^2-y^2)}{x^2+y^2}$ has $f_{xy}(0,0)=-1\neq f_{yx}(0,0)=1$ (added for clarity; sympy).
6. Class 001: The lecture asks the class to prove Clairaut's theorem (T001b 00:20:10) and says the proof is short. The proof needs the mean value theorem as well as the limit definition; it is not on the slides and is not reproduced here.
7. Class 001: The lecture's history (T001b 00:10:53) says Leibniz used other letters for partial derivatives in a letter to L'Hôpital. The letters are not on the slide and could not be confirmed from the material; only the slide's facts (Leibniz, Legendre 1786, Jacobi 1841) are kept.
8. Class 001: The lecture's example function (T001a, T001b 00:15:16) adds a term in t that the recognition renders unclearly; W4.2 keeps the clearly spoken terms $x^3\sin y+e^w+\log p$, which show the same point.
9. Class 001: The lecture opens with "Unit Two" (T001a 00:00:01); partial differentiation is Unit 1 of UE26MA141A.
10. Class 001: The deck's motivation slides (001·5–7) carry stock images (a cathode-ray tube, a weather chart, COVID-19 posters); they are described in words, not redrawn.
11. Class 001: Transcript recognition errors were corrected silently: "Clydoids" → Clairaut's, "legendary" → Legendre, "Lobithal" → L'Hôpital, "Ebo" → above, "parable" → parabola, "cathode ratio" → cathode-ray tube, "dou" / "Do" → ∂, "x q" / "XQ" → x^3, "sine 1" → $\sin y$, "topol o topography" → topography, "extending to" → x tending to.
12. Class 002: The deck's title says "Problems on partial derivatives", but the deck has no problem slides, and the transcript defers problems to the next class (T002 00:14:59). Problems were taken from the class-work sheet and ESA papers instead.
13. Class 002: Slide 10 shows an interactive "Second-Order Partial Derivatives" demonstration. The transcript names its source unclearly ("Ultra Am lab"), and the video frame at 00:12:40 does not show a name, so the tool is described without one.
14. Class 002: The link promised "in my next slide" (T002 00:15:16) is not in the deck; slide 11 is the closing slide.
15. Class 002: Slides 4, 5 and 8 use third-party images (a YouTube source is credited on the slide). They were redrawn as Figures G2.1, G3.1, G3.2 and G5.1 rather than copied.
16. Class 002: The UE23MA141A papers are headed "JAN 2023", but the browser print header says Jan 2024. They are cited as Jan 2024.

17. Class 002: Transcript recognition errors were corrected silently: “geometally” → geometrically, “Z1” → z_y, “impartial derivatives” → in partial derivatives, “Claret theorem” → Clairaut's theorem, “jowl” → jolt (another name for jerk).
18. Classes 003–004: Slide 003·6 ($z = x^2y^3$, $x = t^2$, $y = 2t^3$) is wrong: it gives $\partial z/\partial x = 2t^{11}$, $\partial z/\partial y = 9t^{10}$ and $dz/dt = 60t^{12}$. The correct values are $16t^{11}$, $12t^{10}$ and $dz/dt = 104t^{12}$, confirmed by substituting first ($z = 8t^{13}$). The lecture (T003a 00:02:05) used a different example, $x = t$, $y = 2t^2$, giving $64t^7$.
19. Classes 003–004: Slide 003·10 prints $d\vartheta/dt = -25/(300\sqrt{3})$ “$= 0.04831$”, and the lecture repeats 0.04831. The correct value is $-\sqrt{3}/36 \approx -0.0481$ rad/s, which the slide's own last line states.
20. Classes 003–004: Slide 003·8 writes “dy = 0.03”, but the calculation correctly uses $dy = -0.03$ (from -3 to -3.03).
21. Classes 003–004: Slide 003·16 shows $\partial v/\partial y = -3\,\partial v/\partial r + \partial v/\partial s$; the coefficient of $\partial v/\partial s$ should be 3 (since $s = 3y - 4z$). The final result $6v_x + 4v_y + 3v_z = 0$ is right.
22. Classes 003–004: Slide 003·7 calls $x^2/16 - y^2/9 = z$ a paraboloid and the lecture says “like a cup”. With the minus sign it is a hyperbolic paraboloid (a saddle). The numbers are unaffected.
23. Classes 003–004: Slide 004·4 prints $dy/dx = -f_y/f_x$ for a level curve. It should be $-f_x/f_y$, as slide 004·5's branch diagram and slide 004·6 show.
24. Classes 003–004: Slide 004·3's numerator is hard to read in the rendering. The correct dy/dx for $x^3y + 3xy^2 = 3$ is $-3y(x^2 + y)/(x(x^2 + 6y))$.
25. Classes 003–004: In the lecture (T004a 00:18:39) problem 3 of deck 004 is worked at the point (1, 0, 2), giving 0 for both derivatives. That point is not on $z^3 + xy - y^2z = 6$ (it gives 8). The slide's point (0, 1, 2) is on the surface and gives $\partial y/\partial x = 1/4$, $\partial z/\partial x = -1/11$.
26. Classes 003–004: The lecture misspeaks some intermediate values (e.g. “S X is just minus 4”, T003a 00:23:23); the slides' values $r_x = 2$, $s_x = 0$, $t_x = -2$, ... are correct and are used here.
27. Classes 003–004: Transcript recognition errors were corrected silently: “x pi” → xy, “cos by” → cos y, “second square” → $\sec^2$, “phi dy by dx” → find dy/dx, “Gosteep” → cos ϑ, “x incre” → x increases.
28. Class 005: Slide 005·7 (the extension of Euler's theorem) has its equations only as images; the .slides.md text shows just the title. They were read from the rendered slide.
29. Class 005: Slide 005·4 writes $x^3\big(1 + (y/x)^2 + (z/x)^3 + 3\frac{y}{x}\frac{z}{x}\big)$; the second exponent should be 3.
30. Class 005: The lecture (T005 00:07:11) says that a function of degree 0 must be constant. That is false (y/x has degree 0); corrected as an exam trap.
31. Class 005: The lecture (T005 00:11:06) calls homogeneous functions “smooth everywhere”; homogeneity is a scaling property, not smoothness.
32. Class 005: The lecture (T005 00:16:46) swaps the thermodynamics examples: mass is extensive (degree 1); pressure and temperature are intensive (degree 0).
33. Class 005: The lecture's third-order list of partial derivatives (T005 00:12:43) is incomplete; the result $n(n-1)(n-2)u$ is stated with the binomial coefficients added for clarity.
34. Class 005: Biographical remarks about Euler in the lecture are left out (not examinable, and partly inaccurate).

35. Class 005: Class 006 (“Problems on Euler's theorem”, two transcripts) is a separate session and is not included.
36. Class 005: Transcript recognition errors were corrected silently: “oiless theorem” → Euler's theorem, “Euless theorem” → Euler's theorem, “quintiple” → quintic, “y bytes” → y/x.
37. Class 006: Slide 006·3 is titled “Verify Euler's theorem for $\cos^{-1}(\dots)$” but verifies it for $Z = \cos u = \frac{x+y}{\sqrt{x}+\sqrt{y}}$, not for u. For u itself the result is $xu_x + yu_y = -\frac{1}{2}\cot u$ (added for clarity, checked with sympy).
38. Class 006: Slide 006·3's middle factor is printed as $x^2\,\frac{1(1+y/x)}{1+\sqrt{y}/\sqrt{x}}$; the power is $x^{1/2}$, as the next step $\sqrt{x}\,f(y/x)$ shows.
39. Class 006: Slide 006·4 prints the denominator as $\sqrt{x^8+y^8+z}$ in its solution line; it is $\sqrt{x^8+y^8+z^8}$, as in the question.
40. Class 006: Slide 006·5's function is identically 1: $x^4+2x^2y^2+y^4=(x^2+y^2)^2$. The degree-0 answer is right, but the example is trivial; noted as a check.
41. Class 006: Slide 006·7's function simplifies to $\frac{xy}{(x+y)^2}$ (added for clarity), which shows the degree 0 at a glance.
42. Class 006: The lecture reads slide 006·8 as $e^x - e^{-y}$ (T006b 00:04:40); the slide has $e^x - e^y$. Either way the function is not homogeneous.
43. Class 006: The lecture gives the degree on slide 006·16 as “plus 2 ... plus one by two” (T006b 00:12:00); the degree is $\frac{1}{2}$.
44. Class 006: The lecture calls the second-order corollary $n(n-1)u$ an “extension of Euler's theorem” (T006b 00:05:55, 00:07:23). In this pack “corollary” means $n(n-1)u$ and “extension” means the $f(u)$ form $nf(u)/f'(u)$, as on slide 005·7.
45. Class 006: The lecture says the answer to slide 006·9 (i) “becomes minus half ...” (T006b 00:06:59); the slide and the algebra give $-u = -\frac{x-y}{x^2+y^2}$ with no half.
46. Class 006: Slides 006·12–13 ask only for the first-order sum; the second-order result $4u\log u\,(4\log u+3)$ is added for clarity (sympy-checked).
47. Class 006: Transcript part I ends with “Next we will solve some problems approximation” (T006a 00:08:53); that belongs to a later class and is not included.
48. Class 006: The theory of class 005 is in its own pack; the statements used here are repeated without proofs.
49. Class 006: Transcript recognition errors were corrected silently: “Sign in us” / “sine in was” → $\sin^{-1}$, “calc is” → cancelling, “tau” / “tanue” → $\tan u$, “second square” → $\sec^2$, “lambda power x power 3” → $\lambda^6x^3y^3$, “doux” → ∂x, “y'all” → $\sqrt{y}$.
50. Class 007: Slide 007·9's second-order term is garbled in the rendering. It was rebuilt from the transcript (T007 00:18:45) and the Maclaurin form on slide 007·10: $\frac{1}{2!}\{(x-a)^2f_{xx}+2(x-a)(y-b)f_{xy}+(y-b)^2f_{yy}\}$ at (a,b).
51. Class 007: The slides stop at second order. Four of the six ESA questions ask for third-degree terms, so the third-order term $\frac{1}{3!}\{h^3f_{xxx}+3h^2kf_{xxy}+3hk^2f_{xyy}+k^3f_{yyy}\}$ and the operator form are added for clarity.

52. Class 007: Slide 007·3 asks for $\lim_{x\to 0} \frac{\sin x - x}{x^4}$. That limit does not exist (it behaves like $-\frac{1}{6x}$); with x^3 in the denominator the limit is $-\frac{1}{6}$, worked as W1.2.
53. Class 007: Slide 007·3's integrals ($\int e^{x^2} dx$, $\int \cos(x^2) dx$) are partly lost in the rendering; the lecture uses $\int_0^1 \sin(x^2)\, dx$ instead (T007 00:11:51), worked as W1.1.
54. Class 007: Slide 007·7 says "$E = mc^2$ is an approximation for low velocities". More precisely, the Maclaurin expansion of the relativistic energy γmc^2 gives $mc^2 + \frac{1}{2}mv^2$ for $v \ll c$ (added for clarity).
55. Class 007: Slide 007·8's graphs of $\sin x$ and its Maclaurin polynomials have no axis labels; they are redrawn as Figure G1.1.
56. Class 007: Slide 007·11 does not state the expansion point. Its polynomials $1 + x + y$ and $1 + x + xy + \frac{y^2}{2}$ are the Taylor polynomials of $xe^y + 1$ about $(1, 0)$; the lecture's guess that "maybe they have used the Maclaurin series" (T007 00:20:32) is wrong: about the origin they would be $1 + x$ and $1 + x + xy$ (sympy).
57. Class 007: Slide 007·12 is titled "Applications of Euler's Theorem of a multivariate function"; its content is the applications of Taylor's series (a copy error).
58. Class 007: The lecture says $1 + 2 + 2^2 + 2^3 + \ldots$ "gives -1" (T007 00:07:52). The series diverges: $\frac{1}{1-x} = 1 + x + x^2 + \ldots$ holds only for $|x| < 1$. Corrected as an exam trap.
59. Class 007: The lecture's condition "$f(a)$ cannot tend to infinity" (T007 00:10:01) is incomplete: f needs derivatives of every order at the point, and the series must converge to f. The precise condition is added for clarity.
60. Class 007: The lecture's picture of a calculator summing Maclaurin series is a simplification; real calculators use several polynomial and shift-and-add methods. It is kept as the lecture's motivation.
61. Class 007: Class-work problem 26 prints $\tan^{-1}(0.9/1.1) \approx 0.6904$. The second-order Taylor value is 0.6854 (exact 0.6857).
62. Class 007: Class 008 ("Problems on Taylor's and Maclaurin's series") is a separate session and is not included.
63. Class 007: Transcript recognition errors were corrected silently: "McLaren" / "Mechlerin" / "Makelden" / "metogen" → Maclaurin, "McLevins" → Maclaurin, "t-shirts" → Taylor series, "L'opitholes" → L'Hôpital's, "Legrange" → Lagrange, "four-year series" → Fourier series, "sinex" → sin x, "2a into" → 2(x−a)(y−b).
64. Class 008: Slide 008·18 gives $f_{yy}(1, 1) = 3$ for $xy^2 + y\cos(x - y)$; it is 1. Slide 008·17's f_{yy} has the wrong sign on $y\cos(x - y)$: the correct $f_{yy} = 2x + 2\sin(x - y) - y\cos(x - y)$. Slide 008·19's final answer (with $f_{yy} = 1$) is correct.
65. Class 008: Slide 008·12 lists $f_{xy}(0, 0) = 0$ for $e^x \log(1 + y)$; it is 1 ($f_{xy} = \frac{e^x}{1+y}$). Slide 008·13 uses the correct value.
66. Class 008: Slide 008·12 asks for "the first six terms" of $e^x \log(1 + y)$ but slide 008·13 stops after three ($y + xy - \frac{y^2}{2}$). The six terms $y + xy - \frac{y^2}{2} + \frac{x^2 y}{2} - \frac{xy^2}{2} + \frac{y^3}{3}$ are added for clarity (sympy).
67. Class 008: Slide 008·11 computes the third-order derivatives of e^{x-y} but stops the series at second order; the third-degree term $\frac{(x-y)^3}{6}$ is added for clarity.

68. Class 008: Slide 008·9 approximates $\tan^{-1}(0.9 \times (-1.2))$ with first-order terms only: $-\frac{\pi}{4} - 0.05 = -0.8354$. With the second-order terms of slide 008·8 it is -0.8229; the exact value is -0.8238 (sympy). Both are given.

69. Class 008: Slides 008·6, 008·8 and 008·9 are partly garbled in the rendering; they were rebuilt from the transcript (T008 00:10:35, 00:14:35, 00:15:42) and checked by recomputing.

70. Class 008: Slide 008·3 does not state the degree; the lecture says "up to second degree" (T008 00:00:21).

71. Class 008: The lecture twice reads $f_{xx}(1, \frac{\pi}{4})$ and $f_{xy}(1, \frac{\pi}{4})$ as "e divided by two" (T008 00:03:47, 00:04:16); every value on slide 008·3 is $\pm\frac{e}{\sqrt{2}}$.

72. Class 008: The lecture says "$f(0.9, -1.2)$ is approximately equal to $f(1, -1)$" (T008 00:12:10). That only motivates the choice of the expansion point; the approximation is the Taylor polynomial, not $f(1, -1) = -0.785$.

73. Class 008: The lecture works slides 3–9 only (16.2 min). Slides 10–19 (e^{x-y}, $e^x \log(1+y)$, $\sin(xy)$, $xy^2 + y\cos(x-y)$) have no recorded explanation.

74. Class 008: The class 007 theory is in its own pack; the formulas are repeated here without derivation.

75. Class 008: Transcript recognition errors were corrected silently: "McLaren" / "my cleric series" → Maclaurin series, "e bar root" / "e pi root 2" → $e/\sqrt{2}$, "1 comma 5 by 4" → $(1, \frac{\pi}{4})$, "y minus 4" → $y - \frac{\pi}{4}$.

76. Class 009: Slide 009·7 calls "$f_x = 0$, $f_y = 0$ and $rt - s^2 > 0$" the necessary and sufficient condition. Precisely, $f_x = f_y = 0$ is necessary (for a differentiable f) and $rt - s^2 > 0$ is sufficient; neither alone is both. The slide's "$A > 0$" is a garbled "> 0". Corrected for clarity.

77. Class 009: The lecture says that in one variable a point with $\frac{d^2y}{dx^2} = 0$ is "called a saddle point" (T009 00:03:14). It is only an inconclusive case (it may be a maximum, minimum or point of inflection: x^4, $-x^4$, x^3). Corrected as a trap.

78. Class 009: The lecture first says that if $rt - s^2 < 0$ "you cannot conclude anything" (T009 00:08:23) and later that it is a saddle point (T009 00:11:06). The slides (009·7, 009·9) are right: $rt - s^2 < 0 \Rightarrow$ saddle point; only $rt - s^2 = 0$ gives no conclusion.

79. Class 009: The lecture's working rule at T009 00:09:26 says "if this is zero, how do I decide ... maximum or minimum"; it means $rt - s^2 > 0$. Read with the slides, the rule is: > 0 and $r < 0$ maximum, > 0 and $r > 0$ minimum, < 0 saddle, $= 0$ doubtful.

80. Class 009: The lecture says mixed partials are "always equal" (T009 00:07:22); they are equal when they are continuous (Clairaut, class 001).

81. Class 009: The slides give no worked example; the transcript ends with "we'll solve the problem". Problems are in class 010; the short examples W3.1–W3.4 (including the doubtful case $x^4 \pm y^4$) are added for clarity, and W3.5 is CW P31.

82. Class 009: The deck's 3-D pictures are screenshots with no stated functions; Figures G1.1–G1.3 use illustrative functions with the same shapes.

83. Class 009: Transcript recognition errors were corrected silently: “Hesian” → Hessian, “R T minus a s square” → $rt - s^2$, “sin” → sign, “t square y by dx square” → $\frac{d^2y}{dx^2}$, “f x” (for f_{xx}) → f_{xx}, “eight” → (a, b), “sway” → sway.

84. Class 010: Slide 010·4 prints $rt - s^2 = -1600$ at $(0, 0)$ and 356000 at $(\pm\frac{1}{2}, \pm\frac{1}{2})$ for $400xy(1 - x^2 - y^2)$. The correct values are -160000 and 320000 (sympy; also the special case $k = 400$, $a = 1$ of slide 010·10's $-k^2a^4$ and $2k^2a^4$). The conclusions (saddle, maximum 50) are unaffected.

85. Class 010: Slides 010·3–4 and 010·9 miss four stationary points: $(\pm1, 0)$ and $(0, \pm1)$ (generally $(\pm a, 0)$, $(0, \pm a)$), from $y = 0$, $1 - x^2 = 0$ and $x = 0$, $1 - y^2 = 0$. They are saddle points with $T = 0$ (on the circle $z = 0$). The slides also list $(\frac{1}{2}, -\frac{1}{2})$ and $(-\frac{1}{2}, \frac{1}{2})$ but never classify them: they are minima of T (-50, the lowest temperature).

86. Class 010: Slides 010·5–6 and the lecture (T010a 00:10:33) differentiate $A = 2xy + \frac{20}{x} + \frac{10}{y}$ wrongly: $A_y = 2x - \frac{10}{y^2}$, not $2x - \frac{20}{y^2}$, and $A_{yy} = \frac{20}{y^3}$, not $\frac{40}{y^3}$. The printed answer, a cube of side $\sqrt[3]{10}$, is therefore wrong (area 23.21). The correct box has the two edges of the open face $\sqrt[3]{20} \approx 2.714$ m and the third edge half of that, 1.357 m (area $3 \cdot 20^{2/3} \approx 22.10$ m²), which agrees with CW problem 34 ($x = y = (2V)^{1/3}$, $z = x/2$). A cube is the answer for a closed box.

87. Class 010: Slide 010·5 measures the edges in “ft” though the capacity is 10 m³; metres are used here.

88. Class 010: Slide 010·8 writes $rt - s^2 = 36(x - 5)^2 - y^2$; it is $36(x - 5)^2 - 36y^2$ (the values ±36 printed below use the correct form). It also says “the maximum value of f at $(6, 0)$ is 112”; the maximum is at $(4, 0)$ (the lecture repeats the slip, T010b 00:04:42).

89. Class 010: Slide 010·12 gives $r = -2 < 0$ for all four points $(\pm\frac{1}{\sqrt{2}}, 0)$, $(0, \pm\frac{1}{\sqrt{2}})$. At $(\pm\frac{1}{\sqrt{2}}, 0)$, $r = 4$ and $t = -2$; the saddle conclusion ($rt - s^2 = -8 < 0$) is unaffected. The lecture adds “but r is also negative” (T010b 00:09:26); the sign of r plays no part when $rt - s^2 < 0$.

90. Class 010: Slide 010·14 prints $rt - s^2 = \frac{3}{4}$ at $(\frac{\pi}{3}, \frac{\pi}{3})$ for $\sin x + \sin y + \sin(x + y)$; it is $3 - \frac{3}{4} = \frac{9}{4}$ ($s = -\frac{\sqrt{3}}{2}$). The conclusion (maximum $\frac{3\sqrt{3}}{2}$) is unaffected. The slide states no domain; “the only critical point” holds for $0 < x, y < \pi$ (added for clarity).

91. Class 010: Slide 010·9 prints the sphere as $x^2 + y^2 + z^2 = a'$; it is a^2 (slide 010·9's next line and T010b 00:05:40).

92. Class 010: The lecture T010a says “x will be plus or minus 1” (00:02:42) and then corrects itself to $x = \pm y$; and “this should be x equal to 0, not y” (00:03:27) for slide 010·3's f_y line, which indeed should read $x = 0$.

93. Class 010: Class 009 (definitions, necessary conditions, the $rt - s^2$ test) is a separate pack; the test is restated here without proof. Class 011 (Lagrange's multipliers) is also separate; the elimination method of topic 3 is what class 010 uses.

94. Class 010: Transcript recognition errors were corrected silently: “seventy p” → 72, “minus pi square” → $-36y^2$, “115” → 150, “phi” / “5 comma” spoken as five, “plus R minus” → plus or minus, “stattle points” / “settle point” → saddle point(s), “6 comero is 12” → (4, 0) is 112, “RT when C square” → $rt - s^2$, “5 by 3” → $\frac{\pi}{3}$, “P of x y z” → $T(x, y, z)$, “minus x square y square” → $-x^2 - y^2$, “Subtracting 2 from 2” → (2) from (1), “equity” → equation, “dou” → ∂.

95. Class 011: The deck is titled “... for two variables”, but the method, the slides' problems and all four ESA problems use three variables x, y, z and one constraint; the two-variable case ($F_x = F_y = 0$, $\phi = 0$) is the same with z dropped (Jul 2023 Q1.c, W2.4).

96. Class 011: Slide 011·6 prints the third line as $8xzy + 2y^2\lambda = 0$; multiplying $8xy + 2z\lambda = 0$ by z gives $8xyz + 2z^2\lambda = 0$. The conclusion $2x^2\lambda = 2y^2\lambda = 2z^2\lambda$, so $x = y = z$, is right.

97. Class 011: Slide 011·9 says "from (3), we get $z = 0$, $\lambda = 1$"; equation (3), $2z(1 - \lambda) = 0$, gives $z = 0$ or $\lambda = 1$, and $\lambda = -1$ from (1) forces $z = 0$. The lecture likewise says "λ plus or minus 1" (T011a 00:13:10).

98. Class 011: The theory lecture misreads the derivative at T011a 00:07:12 ("yz plus λ into $2z$"); it is $F_x = yz + 2\lambda x$. It also says "x is equal to plus or minus 1" (T011a 00:08:28) for $x = \pm y$, and treats the volume as $8xyz$ and then xyz (both give $x = y = z$).

99. Class 011: Slide 011·4 lists, as a disadvantage, that the method cannot decide the nature of a stationary point. Added for clarity: in examination problems the nature follows from the problem (a largest box, a nearest point), or by comparing the values of f at the stationary points (W2.4 compares 1 and 2).

00. Class 011: CW problem 33's condition is garbled in the text layer; the rendered page reads $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$. Its answer $(a + b + c)^3$ at $(\frac{t}{a}, \frac{t}{b}, \frac{t}{c})$, $t = a + b + c$, is right (sympy, and a numerical check with $a, b, c = 1, 2, 3$).

01. Class 011: No ESA answer key was used; every answer here is recomputed with sympy (May 2024 and Jul 2023 answer PDFs exist in the bucket but were not needed).

02. Class 011: The theory lecture describes "a point in space" as "$x^2 + y^2 + z^2$ ... a sphere can be taken as a point" (T011a 00:09:35). Precisely: the squared distance from the origin is $x^2 + y^2 + z^2$, and minimising it is the same as minimising the distance (added for clarity).

03. Class 011: Transcript recognition errors were corrected silently: "Lagrangius" / "Lagrangian's" / "Slagrange's" / "Lagrangus" / "grandiose" / "grandest" → Lagrange's (auxiliary function), "axillary" → auxiliary, "equity" / "equipment" → equations, "a Q" / "a q" → a cube, "two eight x" → $8xyz$, "Q" → cube, "constant" (for constraint) → constraint, "small problems" → solve problems.

Coverage

Source	Citations	Slides cited (slide × citations)
S001	56	4×1, 5×1, 6×1, 7×1, 8×4, 9×8, 10×5, 11×8, 12×5, 13×3, 14×2, 15×6, 16×3, 17×8
S002	75	3×13, 4×7, 5×5, 6×7, 7×12, 8×12, 9×17, 10×2
S003	81	4×4, 5×10, 6×2, 7×6, 8×7, 9×6, 10×4, 11×3, 12×2, 13×6, 14×3, 15×4, 16×3, 17×3, 18×2, 19×2, 20×1, 21×2, 22×1, 23×1, 24×1, 25×2, 26×3, 27×3
S004	60	3×6, 4×11, 5×7, 6×8, 7×4, 8×3, 9×5, 10×2, 11×1, 12×2, 13×3, 14×2, 15×3, 16×1, 17×2
S005	37	3×6, 4×4, 5×9, 6×10, 7×8
S006	68	3×4, 4×5, 5×7, 6×3, 7×2, 8×5, 9×1, 10×4, 11×5, 12×2, 13×8, 14×2, 15×8, 16×3, 17×9
S007	35	3×2, 4×5, 5×6, 6×1, 7×1, 8×2, 9×6, 10×5, 11×6, 12×1

Source	Citations	Slides cited (slide × citations)
S008	55	3×4, 4×1, 5×3, 6×6, 7×5, 8×1, 9×4, 10×4, 11×4, 12×4, 13×2, 14×4, 15×2, 16×3, 17×3, 18×3, 19×2
S009	50	3×3, 4×4, 5×3, 6×8, 7×12, 8×13, 9×7
S010	52	3×5, 4×3, 5×4, 6×4, 7×6, 8×15, 9×2, 10×2, 11×4, 12×3, 13×2, 14×2
S011	44	3×9, 4×3, 5×4, 6×2, 7×6, 8×3, 9×4, 10×3, 11×2, 12×4, 13×2, 14×2
CW	71	
ESA	90	
Extra	5	
T001a	39	
T001b	44	
T002	69	
T003a	48	
T003b	16	
T004a	35	
T004b	14	
T005	37	
T006a	15	
T006b	33	
T007	49	
T008	36	
T009	37	
T010a	15	
T010b	23	
T011a	27	
T011b	11	
V002	8	

Class 001. Slides 001·1, 001·2 and 001·18 are title, session and closing slides. Slides 001·3–7 are motivation and are summarised in topic 1. The lecture announces the geometric interpretation (T001b 00:20:38); it is class 002.
Class 002. Slides 1, 2 and 11 are the title, session and closing slides and hold no content.

Classes 003–004. Slides 003·1–3, 003·28, 004·1–2 and 004·18 are title, topic-list and closing slides with no content.
Class 005. Slides 005·1, 005·2 and 005·8 are title, session and closing slides. Slide 005·7's equations exist only in the slide image; the text export loses them.
Class 006. Slides 006·1, 006·2 and 006·18 are title, session and closing slides with no content. Every content slide (006·3–17) is worked in full.
Class 007. Slides 007·1, 007·2 and 007·13 are title, session and closing slides with no content.
Class 008. Slides 008·1, 008·2 and 008·20 are title, session and closing slides. The lecture works slides 3–9 only; slides 10–19 have no recorded explanation and are cited from the slides alone.
Class 009. Slides 009·1, 009·2 and 009·10 are title, session and closing slides. The transcript covers all content slides (3–9).
Class 010. Slides 010·1, 010·2 and 010·15 are title, session and closing slides. Lecture T010a (older, 17.3 min) works slides 3–6 (temperature on the unit sphere, open box); lecture T010b (newer, 11.1 min) works slides 7–14. Slide 010·2 says “Session : 9”; the bucket numbers the class 010.
Class 011. Slides 011·1, 011·2 and 011·15 are title, session and closing slides. Lecture T011a (theory, 14.8 min) works slides 3–9; lecture T011b (problems, 10.2 min) works slides 10–14. The deck title says “for two variables”, but every problem uses three variables and one constraint.